

Merged research report for the Surreal project

Omnific Integers and Omnific–Diophantine Geometry

Retractions, rigidity, definability,
and infinite families

Built from seven manuscripts dated September 22 and 23, 2026

Abstract

The omnific integers $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ form a discretely ordered subring of the surreal field whose arithmetic is neither ordinary integer arithmetic nor a routine nonstandard enlargement of it. We organize their theory around the split ring retraction $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ that takes a normal form to its constant coefficient. This gives an exact transfer theorem for polynomial equations with ordinary integer coefficients, a classification of finite quotients, and a complete solution criterion for linear systems.

The Diophantine results distinguish existence from the structure of solutions. At a nonzero constant level, the omnific fiber of a product of affine linear forms is described exactly by a real common kernel; in particular binary forms with at least two distinct projective linear factors, Pell equations and norm forms of finite étale algebras have no new omnific or Gaussian omnific points. Support-preserving Euler derivations prove the same rigidity for every separated equation $ax^m + by^n = c$ with complex $abc \neq 0$ and $m, n \geq 2$, including $y^2 = x^3 + 1$, and for Fermat triples of exponent at least three with all coordinates nonzero that generate the unit ideal. In contrast, every represented nonzero level of a nondegenerate indefinite integral quadratic form in at least three variables carries explicit injective polynomial families through each integer point, and we prove an exhaustive quadratic-level dichotomy that includes degenerate forms. Pell rigidity combined with the three-square or the four-square theorem gives single quartic equations defining the ordinary integers inside $\mathbf{Oz}$. Further results concern non-atomic factorization, full-class common divisors, bounded semialgebraic rigidity, primitive and projective points, a failed existential induction instance, and failures of residue lifting.

The splitting $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ is itself Diophantine: $x^2 = 2y^2$ defines Π with one witness, one order-free system defines $\mathbb{Z}$ in $\mathbf{Oz}$ and $\mathbb{Z}[i]$ in $\mathbf{Oz}[i]$, and $as = \Lambda(t)$, for a classical intersective sextic Λ , is solvable exactly when $\text{ct}(a) \neq 0$. So the graph of ct is Diophantine and homomorphisms preserve constant terms. The multiplier ring of Π recovers $\mathbb{R}$; the pure ring $\mathbf{Oz}$ thus reconstructs the valued field $\mathbf{No}$, and its automorphisms fix $\mathbb{R}$ pointwise, while a phase twist shows that $\mathbf{Oz}[i]$ does not determine the real axis.

Fractions have an exact theory on the points with a unimodular presentation over the real nonnegative-support ring. There the constant coefficients decide whether the scalar multipliers of all omnific presentations form $\mathbf{Oz}$ or Π . So such a point either has unique lowest terms and a principal denominator ideal, or no presentation of it has a gcd. This covers every tuple of real or complex rational functions of one purely infinite parameter. The fibers of the specialization are the orbits of the congruence kernel of $\text{SL}_2(\mathbf{Oz})$, and each of them is dense.

This report merges seven independently written manuscripts; it states which source proved each result, prints shared results once, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 01 (the base text), 02 and 05, pinned to repository commit [2cb9c02](#); 06 and 07, pinned to [71e9606](#); and 08 and 09, pinned to [9a385d3](#). Their titles are in [section 1.2](#). Fiber results of one manuscript of the sibling report, and the direction criterion of another, appear here once. The provenance of every section is recorded in [appendix A](#); non-claims are collected in [appendix B](#). Finite checks accompany the report and check finite identities and examples only.

Contents

1	Scope, results, and conventions	5
1.1	Principal results	5
1.2	Provenance of this report	7
1.3	Class size and foundations	9
1.4	Notation and the exponent convention	10
1.5	Relationship with the repository and literature	11
2	The normal-form rings and the integer part	12
2.1	Normal forms and three related rings	12
2.2	Leading degree, units, and finite elements	13
2.3	Local workspaces	14
2.4	The exact omnific floor	14
3	Divisibility, residues, and quotients	15
3.1	Division by ordinary integers	15
3.2	Completions	16
3.3	Homomorphisms to $\mathbb{Z}$ and to set-sized rings	16
4	Full-class support: common divisors, clearing, factorization	17
4.1	One monomial divides a whole set	17
4.2	Factorization	18
4.3	Clearing arbitrary set-sized families of denominators	19
4.4	Division with remainder is not a terminating Euclidean algorithm	20
4.5	Non-Noetherian and non-normal behavior	20
5	Equational transfer and linear Diophantine equations	21
5.1	Existence is exactly ordinary existence	21
5.2	Why signs and nonvanishing do not transfer	22
5.3	A complete solution criterion for linear systems	22
6	Rigidity of constant products: binary forms, Pell, norms	23
6.1	The constant-product principle	23
6.2	Binary forms, Pell equations, and conics	25
6.3	Number-field and étale norm equations	27
6.4	An algebraic-geometric formulation	29
7	Euler derivations: separated powers and unimodular Fermat	29
7.1	Support-preserving Euler derivations	29
7.2	Separated powers	30
7.3	Unimodular Fermat rigidity	31

8	The quadratic-level dichotomy	33
8.1	Explicit families	36
8.2	What the classification does and does not say	37
9	Bounded geometry, leading forms, and orthogonal matrices	37
9.1	Bounded sets	37
9.2	Orthogonal and symmetric matrices	38
9.3	No real point at infinity	39
10	Defining the ordinary integers	39
10.1	Pell coordinates as finite bounds	39
10.2	The five-auxiliary-variable guard	40
10.3	The four-square definition	41
10.4	Definability consequences	42
10.5	Sums of squares and induction	43
11	Diophantine arithmetic in omnific rings	44
11.1	General nonnegative-support rings	44
11.2	The purely infinite ideal is Diophantine	45
11.3	Pell rigidity and divisible Pell coordinates	46
11.4	An intersective polynomial	47
11.5	One order-free system defining $\mathbb{Z}$ and $\mathbb{Z}[i]$	48
11.6	Single real polynomials and the natural numbers	49
12	The constant term: detector, ideal test and graph	50
12.1	A root detector for augmentations	50
12.2	The largest ideal without a root	52
12.3	The graph of the constant term	52
12.4	Homomorphisms preserve the constant term	53
12.5	Number-field coefficient rings	54
13	Coefficient reconstruction and logical consequences	55
13.1	The multiplier ring of the ideal	55
13.2	Reconstructing the surreal field	57
13.3	Automorphisms of $\mathbf{Oz}$	57
13.4	The Gaussian ring does not determine the real axis	58
13.5	Logical consequences	59
14	Infinite families, primitivity, and projective points	62
14.1	Polynomial curves through integer points	62
14.2	Infinite solutions of homogeneous equations	62

14.3	Unimodular Pythagorean triples	63
14.4	Which real projective directions have primitive representatives?	64
14.5	Finite supports specialize to polynomial arcs	64
15	Fractions, denominator ideals and unimodular presentations	66
15.1	Conventions	66
15.2	Denominator ideals	67
15.3	The multiplier theorem	70
15.4	The unimodular region and its specialization	71
15.5	Rational functions of one parameter	72
15.6	Simultaneous denominators and rational curves	75
15.7	Specialization fibers, congruence orbits and density	76
15.8	The localization at the purely infinite ideal	79
15.9	Polynomial models and scale extensions	80
15.10	Monomial denominators and fixed workspaces	82
15.11	Gaussian omnific fractions	83
15.12	What is and is not classified	84
16	Equations with omnific coefficients	84
16.1	Quadratic equations	84
16.2	Initial forms	85
16.3	Two-term roots and failed lifting	86
17	Fixed Hahn workspaces and a formalization route	87
17.1	Which results are local in the exponent group?	87
17.2	A formalization route	88
18	Questions and synthesis	90
18.1	Questions settled by source 05	90
18.2	Questions answered in the merge of sources 06 and 07	90
18.3	Questions in the merge of sources 08 and 09	91
18.4	Open questions	91
18.5	Synthesis	94
18.6	Relations to neighbouring reports	94
A	Provenance and merge record	96
A.1	Sources	96
A.2	Section map	97
A.3	Where the merge chose	98
A.4	Renamed symbols	101
A.5	Corrections to the sources	103

A.6	Stale repository statements	104
B	Non-claims by source	105
C	Dependencies and verification	113
C.1	Imported inputs	113
C.2	Finite checks	114
D	Repository and literature audit	116
D.1	Repository snapshots	116
D.2	Literature checks	116
D.3	A dated factorization announcement	117
	References	117

1 Scope, results, and conventions

Omnific integers are the surreal numbers with no negative exponents in Conway normal form and with an ordinary integer constant coefficient. The definition is classical, as are their discreteness and integer-part property [1, 2, 3]. This report develops their arithmetic with a particular emphasis on polynomial equations.

Three questions must be kept separate. Does an equation have any solution? Does it have a genuinely infinite solution? Does it have such a solution with a nonvanishing, positivity or primitivity condition? For example, every system of equations over $\mathbb{Z}$ has an omnific solution exactly when it has an ordinary integer solution. Nevertheless, the equation $X^n + Y^n = Z^n$ has positive infinite omnific solutions for every ordinary $n \geq 2$. There is no contradiction: the constant coefficients of these solutions are all zero. On the other hand, every omnific solution of

$$x^2 - 2y^2 = 1, \quad y^2 = x^3 + 17, \quad x^7 - y^5 = 3$$

has ordinary integer coordinates; these conclusions concern the entire solution classes, not just their constant terms.

The central decomposition is

$$\mathbf{Oz} = \mathbb{Z} \oplus \Pi, \quad \text{ct} : \mathbf{Oz} \longrightarrow \mathbb{Z}, \quad \text{ct}(a + b) = \text{ct}(a) + \text{ct}(b), \quad \text{ct}(ab) = \text{ct}(a) \text{ct}(b), \quad (1.1)$$

where Π is the ideal of normal forms supported on strictly positive exponents; coefficients of elements of Π may have either sign. The decomposition is a splitting of additive groups and a ring retraction, not a direct product decomposition of rings. In particular, Π has no identity.

1.1 Principal results

The structural results of Sections 2–4 explain why ordinary congruences see only the constant coefficient, although the fraction field of the full class $\mathbf{Oz}$ is all of $\mathbf{No}$. The main Diophantine statements are the following. Bracketed numbers name the source manuscripts (section 1.2).

- (1) **Equational transfer** [01, 02, 05]. For every finite system over $\mathbb{Z}$, $V(\mathbf{Oz})$ is nonempty exactly when $V(\mathbb{Z})$ is nonempty (theorem 5.1). This does not include inequations or order conditions (proposition 5.3).
- (2) **Constant-product rigidity** [01, 02, 05]. A nonzero constant product of normal forms with no negative exponents has only constant factors. At a nonzero constant level, the omnific points of a product of affine linear forms are exactly the integer points plus purely infinite vectors in the real common kernel of the linear parts (theorem 6.2). Binary forms with two distinct projective factors over $\mathbb{C}$ at a level in $\mathbb{C}^\times$ have only constant points in the complex nonnegative-support ring (theorem 6.7); Pell equations with $D \neq 0$ and number-field norm forms are rigid (corollary 6.9 and theorem 6.14). Source 13 of the sibling report adds the exact fibers over the Gaussian omnific integers $\mathbf{Oz}[i]$ and the converse of the kernel criterion (theorem 6.3 and corollary 6.4); with source 07, norm rigidity holds for finite étale algebras over any base field, in intermediate rings (theorem 6.15).
- (3) **Separated powers and unimodular Fermat** [05]. For complex $a, b, c \neq 0$ and $m, n \geq 2$, every nonnegative-support solution of $ax^m + by^n = c$ is constant (theorem 7.4; source 05 states the real case). For $n \geq 3$, a Fermat triple with all three coordinates nonzero that generates the unit ideal is constant (theorem 7.9); the threshold is sharp.
- (4) **Quadratic-level dichotomy** [01, 02]. For $q \in \mathbb{Z}[X_1, \dots, X_n]$ quadratic homogeneous and $0 \neq c \in \mathbb{Z}$, all solutions are ordinary in the nondegenerate definite case and in

nondegenerate dimension at most two. In every other case each integer point lifts to an injectively parametrized infinite family ([theorem 8.1](#)). For indefinite forms in at least three variables, $(\Pi, +)$ embeds in the kernel of $\mathrm{SO}(q, \mathbf{Oz}) \rightarrow \mathrm{SO}(q, \mathbb{Z})$ ([corollary 8.4](#)).

- (5) **Quartic definitions of $\mathbb{Z}$** [01, 05]. The equation

$$(U^2 - 2V^2 - 1)^2 + \left(U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2 = 0$$

with five existentially quantified auxiliary variables defines exactly $\mathbb{Z}^n$ inside $\mathbf{Oz}^n$ ([theorem 10.3](#)); a variant with six auxiliary variables and four squares defines $\mathbb{Z}$ ([theorem 10.4](#)), and source 07 found the same variant independently. The graph of ct and the ideal Π are first-order definable ([corollary 10.7](#)) and, by item (8), Diophantine; an explicit existential induction instance fails ([theorem 10.10](#)).

- (6) **Primitive and projective phenomena** [01, 02, 05]. There are infinite unimodular Pythagorean triples. A constant real projective direction has a primitive omnific representative precisely when it is rational ([proposition 14.4](#) and [theorem 14.5](#)).
- (7) **Bounded and homogeneous geometry** [01, 02]. Bounded semialgebraic sets have only integer omnific points ([theorem 9.2](#)); a homogeneous system has a nonzero omnific point exactly when it has a nonzero real point, and for integer coefficients this is decidable ([theorem 14.2](#)).
- (8) **Diophantine definitions** [06, 07]. In $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, and in the analogous rings over every ordered exponent group, $\exists y (x^2 = 2y^2)$ defines the purely infinite ideal ([theorem 11.2](#)); one order-free system of three equations with five witnesses defines $\mathbb{Z}$ in $\mathbf{Oz}$ and $\mathbb{Z}[i]$ in $\mathbf{Oz}[i]$, also in intermediate rings ([theorem 11.9](#)); the equation $as = \Lambda(t)$ is solvable exactly when $\mathrm{ct}(a) \neq 0$ ([theorem 12.2](#)), and Π is the largest ideal whose quotient has no root of Λ ([theorem 12.6](#)). The graph of ct is Diophantine ([theorem 12.8](#)), in $\mathbf{Oz}$ by one quartic with six witnesses ([corollary 12.9](#), added by the merge), and every unital homomorphism between such rings preserves ct ([theorem 12.12](#)). The constant definition extends to subrings of number fields ([theorem 12.14](#)).
- (9) **Reconstruction and asymmetry** [06]. The multiplier ring of the ideal recovers the actual coefficient field ([theorem 13.2](#)); from the pure ring $\mathbf{Oz}$ one reconstructs $\mathbf{No}$, its order, finite ring and standard part ([theorem 13.4](#)), and every automorphism of $\mathbf{Oz}$ fixes $\mathbb{R}$ pointwise ([theorem 13.6](#)). A phase twist preserving $\mathbf{Oz}[i]$ shows that neither $\mathbf{No}$ nor conjugation is definable from $\mathbf{Oz}[i]$ ([theorem 13.10](#)).
- (10) **Fractions and denominator ideals** [08, 09]. For a tuple that is unimodular over $\mathcal{B}_{\mathbb{R}}$, the omnific representations are its multiples by $\mathbf{Oz}$ or by Π , according as its constant vector has rational direction or not ([theorem 15.11](#)). So every tuple of rational functions of one parameter $0 \neq t \in \Pi$, and every point of the region $\mathcal{V} \subseteq \mathbb{P}^1(\mathbf{No})$ of points with a $\mathcal{B}_{\mathbb{R}}$ -unimodular presentation, either has unique lowest terms and a principal denominator ideal, or has no presentation with a gcd and a denominator ideal generated by no set ([theorems 15.15](#) and [15.18](#)). The specialization fibers in $\mathcal{V}$ are the orbits of the congruence kernel of $\mathrm{SL}_2(\mathbf{Oz}) \rightarrow \mathrm{SL}_2(\mathbb{Z})$, and all of them are dense ([theorem 15.32](#) and [corollary 15.36](#)). The section adds primitive coordinates on rational curves, a localization with residue field $\mathbb{Q}$, a nonflat integral scale extension with its Tor_1 witness, and the Gaussian form of [theorem 14.5](#) ([corollaries 15.28](#) and [15.55](#), [proposition 15.40](#), and [theorem 15.47](#)).

These statements are proved here, but proving a statement here is not a claim that nobody has previously proved it. The constant-term formalism is classical, and several consequences

are natural applications of it. Each source names its own most distinctive contributions ([appendix B](#)); no exhaustive historical-priority claim is made.

The *universal set-sized quotient theorem*, that every unital ring homomorphism from the full class **Oz** to a set-sized ring is $x \mapsto \text{ct}(x)1$, was proved by source 05 and is printed with its proof in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`. It is quoted here as [cited theorem 3.6](#), together with the elementary special cases proved by source 01.

1.2 Provenance of this report

This report is built from seven manuscripts written independently. Sources 01, 02 and 05, dated September 22, 2026, are items of the batch placed in commit `be06fc8` and pin repository commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33` (107 commits before that placement). Sources 06 and 07, dated September 23, 2026, are items 01 and 07 of the batch placed in commit `cf350b1`; they are numbered 06 and 07 here to follow the earlier sources, as their shipped file prefixes are. Both pin `71e9606`, which precedes the placement `be06fc8`, so neither saw this report or its sibling. Sources 08 and 09, also dated September 23, 2026, are items 01 and 02 of the batch placed in commit `a4dcb91`, numbered by their shipped file prefixes. Both pin `9a385d3`, 35 commits before that placement; they saw this report as assembled from sources 01, 02 and 05, before sources 06 and 07 were integrated. The tag `[13]` refers to manuscript 13 of the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` (item 03 of the batch placed in `cf350b1`, also pinned to `71e9606`); its Gaussian and étale fiber results are printed here once, and it is not counted among this report's sources. The tag `[14]` refers to manuscript 14 of the same report (item 04 of that batch, files prefixed `14-arithmetic-tensors-` there); its criterion for unimodular constant directions over $\mathbb{Z}$ and $\mathbb{Z}[i]$ is printed here once, in [corollary 15.55](#), and it is not counted among this report's sources either.

Source	Manuscript	Principal contribution to this report
01	<i>Omnific Integers and Omnific-Diophantine Geometry</i>	Base text and structure; exhaustive quadratic-level dichotomy; the five-auxiliary three-square guard; rational-direction theorem; no real point at infinity; symmetric-matrix rigidity; elementary uniqueness of ct.
02	<i>Omnific Integers and Diophantine Geometry</i>	Exact decomposable fibers; bounded semialgebraic rigidity; decidable homogeneous existence; quadratic-formula criterion; general pair without a gcd; explicit Lorentz matrix and the unipotent subgroup.
05	<i>Omnific Integers and Diophantine Rigidity</i>	Binary rigidity over $\mathbb{C}$; Euler derivations; separated-power and unimodular Fermat rigidity; the four-square definition of $\mathbb{Z}$; failed existential induction; failed lifting; finite-support specialization; the universal set-sized quotient theorem (printed in the sibling report).
06	<i>Definable Arithmetic and Coefficient Reconstruction in Omnific Integer Rings</i>	Base of sections 11 to 13 ; the one-witness definition of Π ; the Diophantine graph of ct; homomorphisms preserve ct; the multiplier identity and the reconstruction of $\mathbb{R}$, $\mathbf{No}$, order and standard part; automorphisms fix $\mathbb{R}$; the phase twist and the real-Gaussian asymmetry; c.e. sets; the quintic.
07	<i>Defining Arithmetic Inside Omnific Integers</i>	The order-free system in intermediate rings (with 06); the augmentation root detector and the ideal test; support bounds and certificates; number-field coefficient rings; norm rigidity over any base field; recursive saturation; the quartic variant.
08	<i>Fractions of Omnific Integers: Rational Specialization, Lowest Terms, Denominator Ideals, and Scale Extensions</i>	Base of section 15 : the multiplier theorem in any number of coordinates; the classification for rational-function tuples at any $t \in \Pi \setminus \{0\}$; real constants; rational curves; the localization at Π and independent residues; polynomial models, the scale defect, nonflatness and Tor_1 ; monomial denominators; the Gaussian dichotomy.
09	<i>Omnific Fractions: Rationality, Denominator Ideals, and Dense Arithmetic Fibres</i>	The unimodular region $\mathcal{V}$ and its dichotomy; the calculus of denominator ideals and least denominators; fibers as congruence orbits; set-wise focusing, dense fibers and nowhere continuity; the pair version of the multiplier theorem and the one-scale case (special cases of 08).

Where the sources overlap, the statement with the weakest hypotheses is printed and the others are recorded as special cases. Results added by the merge carry the tag [merge] and have complete proofs. [Appendix A](#) records, section by section, which source proved what, where the merge had to choose, and every symbol renamed from a source.

Current review and placement status. Commit `bdd536` assembled sources 01, 02 and 05. The elementary algebra review, its preservation through that assembly, and later proof reviews of the original material in Sections 1–10 and [sections 14](#) and [16](#) are recorded in `RECONCILIATION.md`; assembly is not a review of every later proof, and later proofs and remaining imported foundations still require review. Sources 06 and 07, placed in `cf350b1`, were integrated concurrently with those reviews, as [sections 11](#) to [13](#), together with their material and that of source 13 in [section 6](#) and source 07’s quartic ([remark 10.5](#)). A subsequent pass reviews [sections 11](#) to [13](#), including the distinction between intermediate rings and full positive-support rings, the detector witnesses, coefficient reconstruction and logical consequences. The other newly inserted material remains outside the earlier reviews. Sources 08 and 09, placed in `a4dcb91`, were integrated after those reviews as [section 15](#), together with additions to [section 17](#), one partly answered question and five new questions in [section 18](#); the reviews did not cover this material either. The insertion of [section 15](#) moved the three sections after it by one: the former Sections 15–17 are now [sections 16](#) to [18](#). Nine earlier labels renamed or combined during assembly remain as aliases, so existing references resolve.

1.3 Class size and foundations

The full surreal field and $\mathbf{Oz}$ are proper classes. We work in a customary class theory such as NBG, with the usual background for surreal normal forms; source 02 notes that one may equally read everything relative to universes, with “set-sized” replaced by “small in the specified lower universe”. Each individual surreal has a *set-sized* support. Every polynomial, system, matrix, or tuple in this report has ordinary finite size. Quantifiers over $\mathbf{Oz}$ are class quantifiers expressed through their defining set-theoretic formulas, not an appeal to a set containing all surreals. Sources 06 and 07 state their results first for set-sized exponent groups, where they are ordinary first-order model theory, and read statements about $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ formula by formula ([section 11.1](#)).

Quotients by class ideals are interpreted through their quotient maps and congruence relations. Thus $\mathbf{Oz}/\Pi \cong \mathbb{Z}$ uses the ordinary integer $\text{ct}(x)$ as representative; for $0 \neq n \in \mathbb{Z}$, $\mathbf{Oz}/n\mathbf{Oz}$ uses the ordinary residue of $\text{ct}(x)$ modulo n . Proper-class cosets are not elements of a set. Ideal generation uses finite sums, even for a class of generators. The notation $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ means representation by fractions inside $\mathbf{No}$, not a set of all pairs from $\mathbf{Oz}$.

Choice for sets is used to extend a rational linear functional on a set-sized exponent group ([lemma 7.2](#)). No proof printed in this report uses global choice. Source 05 works in NBG with global choice, which it uses only to pick a cardinal in the proof of the set-sized quotient theorem; the Hartogs ordinal of the target ring serves there without choice, since the pigeonhole step needs only that no injection from that ordinal into the target exists ([remark 3.7](#)).

Most proofs can be carried out in a fixed set-sized Hahn field. We identify the arguments that need the stronger fact that *every set* of surreal exponents has upper bounds and every set of positive surreal exponents has a strictly positive lower bound ([section 4](#)). One must not silently assume these properties for a fixed ordered exponent group ([section 17](#)).

1.4 Notation and the exponent convention

Notation	Meaning in this report
ω^γ	Conway's monomial; $\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$ with surreal addition of exponents.
Π	The ideal of normal forms supported on $\mathbf{No}^{>0}$ (purely infinite elements and 0).
$\mathbf{Oz} = \mathbb{Z} \oplus \Pi$	Omnific integers: nonnegative support, integer constant coefficient.
$\mathcal{B}_{\mathbb{R}} = \mathbb{R} \oplus \Pi$, $\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}}[i]$	Nonnegative-support rings with real or complex coefficients; $\Pi_{\mathbb{C}} = \Pi + i\Pi$.
$\mathcal{B}_{\mathbb{R},H}, \mathcal{B}_{\mathbb{C},H}$	Their subrings with exponents in a set-sized divisible group $H \subseteq \mathbf{No}$ (lemma 2.6).
$\text{ct}(x)$	Coefficient of ω^0 ; not an order standard part.
$\deg_\omega(x), \text{lc}(x)$	Largest exponent in $\text{supp}(x)$ and its coefficient; $\deg_\omega(0) = -\infty$. Not an ordinal and not a birthday.
$\text{Std}(t), \text{Std}_n(\mathbf{x})$	The existential formulas of theorems 10.3 and 10.4 defining $\mathbb{Z}$ and $\mathbb{Z}^n$.
$\partial_\lambda, W_\partial$	Local Euler derivation and two-term Wronskian (section 7).
B, q, G	Symmetric bilinear form, its quadratic form $q(x) = B(x, x)$, and the Lorentz Gram matrix $G = \text{diag}(1, -1, -1)$.
R, S	A generic commutative ring; a set-sized target ring.
$\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$	Gaussian omnific integers (section 6).
$\mathbb{k}, \Gamma, \mathfrak{o}$	A coefficient field of characteristic zero, an ordered abelian exponent group, a unital subring of $\mathbb{k}$ (section 11.1).
$\mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma),$ $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$	Nonnegative-support series, its positive-support ideal, and $\mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma)$; an <i>intermediate ring</i> A satisfies $\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$, $A \cap \mathbb{k} = \mathfrak{o}$.
$\Lambda, \Xi, \Theta, \text{Inf}, \text{CT}$	The intersective polynomial, the order-free definition of $\mathbb{Z}$ and $\mathbb{Z}[i]$, the detector of $\text{ct} \neq 0$, the definition of Π , the graph of ct (sections 11 and 12).
$\text{st}, \mathcal{O}_{\text{fin}}, \mathfrak{m}$	Standard part, finite surreals, infinitesimals (theorem 13.4).
$\mathfrak{D}, \text{sp}, \mathcal{V}, \mathcal{U}, K$	Denominator ideal, specialization, the $\mathcal{B}_{\mathbb{R}}$ - and $\mathbf{Oz}$ -unimodular regions of $\mathbb{P}^1(\mathbf{No})$, the congruence kernel of $\text{SL}_2(\mathbf{Oz})$ (section 15.1).
$\mathbf{Oz}_{\Pi}, \text{res}, \text{Ex}, \mathcal{P}[X]$	Localization at Π , its rational residue, monic monomial exponents, $\mathbb{Z} + X\mathbb{R}[X]$ (section 15).

The calligraphic $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}$ name rings; the italic B is always a bilinear form. The capital Greek letter Π always denotes the ideal; finite products are always written with the operator $\prod$ and its index range. In the material of sources 06 and 07 the coefficient ring is written $\mathfrak{o}$, not D , because D is the Pell parameter; the coefficient field is written $\mathbb{k}$, because k is an ordinary index; and the exponent group is Γ , because G is the Lorentz Gram matrix. The sources' names are listed in [table 1](#).

Sign convention. Exponents are those of Conway normal forms: supports are reverse well ordered, and ω^γ is *infinite* for $\gamma > 0$, even for infinitesimal $\gamma > 0$. Thus

$$\Pi = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\} \quad (\omega\text{-exponents}).$$

In the Hahn convention $t = \omega^{-1}$, which is the convention of the trigonometry report (`docs/surcomplex/trigonometry`, equation `trigonometry:eq:split`), every exponent changes sign: $\omega^\gamma = t^{-\gamma}$, supports become well ordered, the valuation is $v = -\deg_\omega$, and the same class is

$$\Pi = \{x \in \mathbf{No} : \text{supp}_t(x) \subseteq \mathbf{No}^{<0}\} \quad (t\text{-exponents}).$$

So “strictly positive exponents” here and “negative t -exponents” there describe one class, and “no positive exponents” in the t -convention is “no negative exponents” here. The restriction concerns every supported exponent, not only the leading one: $\mathcal{B}_\mathbb{R}$ consists of the forms whose entire t -support is nonpositive. It is not the set of elements with nonpositive valuation; for example $\omega + \omega^{-1}$ has negative valuation but does not lie in $\mathcal{B}_\mathbb{R}$. Nor is $\mathcal{B}_\mathbb{R}$ the valuation ring of finite elements. Using the wrong convention would invalidate the unit computation ([proposition 2.5](#)) and every rigidity proof below.

The repository’s foundations report defines $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ with the same letter Π (`docs/foundations-and-computation/foundations`, label `found:eq:omnific`); the notation here agrees with it. Other repository names for the same class are $\mathbf{No}^\uparrow$ and $\mathbf{No}_{\text{PI}}$ in other reports. The sources used $\mathcal{J}$ (01) and $\mathcal{I}$ (02, 05); the complete per-source rename table is [table 1](#).

1.5 Relationship with the repository and literature

The repository distinguishes normal-form algebra, fixed Hahn workspaces, and the full surreal carrier. At the sources’ pin, the foundations report already defined $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (`found:eq:omnific`), and the trigonometry report already used the omnific integer part, the floor formula and residue arithmetic (`trigonometry:eq:periodclass`); source 01 mentioned only the latter. At the pin there was no report devoted to $\mathbf{Oz}$ and no Lean declaration about it. The collection now contains this report and the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`; there is still no Lean formalization of omnific integers ([section 17](#) and [appendix D](#)). Sources 06 and 07 inspected the repository at the later pin 71e9606, which still preceded both omnific reports; their statements that no explicit Diophantine definition of $\mathbb{Z}$ in $\mathbf{Oz}$ was found are stale, since [theorems 10.3](#) and [10.4](#) are in the collection ([appendix A](#)). Source 06 correctly notes that the surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`) already constructs phase twists that move the real axis while fixing complex coefficients.

The main external sources are Conway and Gonshor for the construction and normal forms [1, 2], L’Innocente–Mantova for modern factorization theory [3], Ehrlich–Kaplan for the initial integer-part context [4], and Jeřábek for the residue theory of $\mathbb{Z}$ -groups [5]. MRDP, Tarski’s quantifier elimination, the three-square and four-square theorems, Shepherdson’s theorem, the Fermat theorem of Wiles and Taylor–Wiles, and Dirichlet’s theorem with quadratic reciprocity are imported with explicit attribution at their points of use. The intersective polynomial of [section 11](#) is classical [22, 23, 24]. Relations to neighbouring reports are listed in [section 18.6](#).

2 The normal-form rings and the integer part

2.1 Normal forms and three related rings

We use the standard normal-form theorem: each surreal has a unique expression

$$x = \sum_{\gamma \in S} r_\gamma \omega^\gamma, \quad r_\gamma \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where $S \subseteq \mathbf{No}$ is a reverse-well-ordered set: every nonempty subset has a greatest member. Conversely such a normal form represents a surreal. Addition is coefficientwise, multiplication is Hahn convolution (the sum of two reverse-well-ordered supports is reverse well ordered, and each exponent of a product receives finitely many contributions), and the sign of a nonzero normal form is the sign of its leading coefficient [1, 2]. The field $\mathbf{No}$ is real closed.

Definition 2.1. Let

$$\begin{aligned} \Pi &= \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}^{>0}\}, \\ \mathcal{B}_{\mathbb{R}} &= \mathbb{R} \oplus \Pi, \\ \mathbf{Oz} &= \mathbb{Z} \oplus \Pi, \\ \mathcal{B}_{\mathbb{C}} &= \mathcal{B}_{\mathbb{R}}[i] \subseteq \mathbf{No}[i]. \end{aligned}$$

The zero series belongs to Π . For an element of any of these rings, $\text{ct}(x)$ denotes its coefficient at exponent zero. For $x = a + ib \in \mathcal{B}_{\mathbb{C}}$, this is $\text{ct}(a) + i \text{ct}(b)$. Put $\Pi_{\mathbb{C}} = \Pi + i\Pi$, so $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$.

Thus $\sqrt{2}\omega$, $\omega/17$, $-\omega^\omega + 3$, $\omega^{1/\omega}$ and $\sum_{k \geq 1} \omega^{1/k}$ are omnific integers. The numbers $\omega + \frac{1}{2}$ and $\omega + \omega^{-1}$ are not. The word “integer” restricts the constant coefficient and excludes infinitesimal tails; it does not require all coefficients to lie in $\mathbb{Z}$. “Purely infinite” is a support condition, not a sign condition: $-\omega \in \Pi$, while $\omega - 3 \notin \Pi$. The normal-form definition of $\mathbf{Oz}$ is equivalent to Conway’s simplicity characterization by the cut $\{x - 1 \mid x + 1\}$ [1, 2]; no proof below uses that characterization.

For every positive surreal exponent γ , even an infinitesimal one, the Conway monomial ω^γ exceeds every ordinary positive real number. This is a normal-form order fact about the omega-map; replacing that map by an unstated analytic expression $\exp(\gamma \log \omega)$ would invalidate the support arguments below.

Proposition 2.2 (Rings and retraction [01, 02, 05]). *$\mathcal{B}_{\mathbb{R}}$, $\mathcal{B}_{\mathbb{C}}$, and $\mathbf{Oz}$ are integral domains, and $\mathcal{B}_{\mathbb{R}}$, $\mathbf{Oz}$ are ordered subrings of $\mathbf{No}$. The class Π is an ideal of $\mathcal{B}_{\mathbb{R}}$ and $\mathbf{Oz}$, and is an $\mathbb{R}$ -vector space; $\Pi_{\mathbb{C}}$ is an ideal of $\mathcal{B}_{\mathbb{C}}$. Constant-coefficient extraction is a surjective ring homomorphism from $\mathcal{B}_{\mathbb{R}}$ to $\mathbb{R}$, from $\mathcal{B}_{\mathbb{C}}$ to $\mathbb{C}$, and from $\mathbf{Oz}$ to $\mathbb{Z}$. In the last case its kernel is Π , the inclusion $\mathbb{Z} \hookrightarrow \mathbf{Oz}$ is a section, and $\mathbf{Oz}/\Pi \cong \mathbb{Z}$.*

Proof. Nonnegative exponents remain nonnegative under addition of exponents. A strictly positive exponent plus a nonnegative exponent is strictly positive. These observations and Hahn convolution prove the ring and ideal claims. In a product of two supported normal forms, exponent zero can arise only as $0 + 0$. Therefore $\text{ct}(xy) = \text{ct}(x)\text{ct}(y)$, and the constant coefficient of a product of elements of $\mathbf{Oz}$ is an integer. Additivity is coefficientwise. Real scalar multiplication does not change a support except when the scalar is zero. Surjectivity and the section are supplied by constant series. All rings embed in a field, so have no zero divisors. $\square$

Remark 2.3 (Two misleading analogies). The map ct is not a ring homomorphism on all of $\mathbf{No}$: $\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$ but $\text{ct}(1) = 1$. The repository records the same warning for

coefficient extraction at zero on **No** (`diff:warn:ct` in the differential-equations report, and `docs/NOTATION.md`); the map `ct` of this report is the restriction of that extraction to **Oz**, where it *is* multiplicative. Also $\mathcal{B}_{\mathbb{R}}$ is not the ring of finite surreals. The latter permits negative exponents but no positive exponents; $\mathcal{B}_{\mathbb{R}}$ has the opposite support restriction. Taking the standard part of a finite element and taking the constant coefficient of an omnific integer are different constructions, although both are multiplicative on their own domains. Finally, `ct` is not order preserving: $\omega - 5 > 0$ but `ct`($\omega - 5$) = -5 .

2.2 Leading degree, units, and finite elements

For nonzero $x \in \mathbf{No}[i]$, combine real and imaginary coefficients and put

$$\deg_{\omega}(x) = \max \text{supp}(x), \quad \text{lc}(x) = [\omega^{\deg_{\omega}(x)}]x, \quad \deg_{\omega}(0) = -\infty.$$

On $\mathcal{B}_{\mathbb{C}} \setminus \{0\}$ the degree lies in $\mathbf{No}^{\geq 0}$.

Lemma 2.4 (Degree [01, 02, 05]). *For nonzero normal forms x, y with real or complex coefficients,*

$$\begin{aligned} \deg_{\omega}(xy) &= \deg_{\omega}(x) + \deg_{\omega}(y), & \text{lc}(xy) &= \text{lc}(x) \text{lc}(y), \\ \deg_{\omega}(x + y) &\leq \max\{\deg_{\omega} x, \deg_{\omega} y\}, \end{aligned}$$

with equality in the last relation when $\deg_{\omega} x \neq \deg_{\omega} y$. An element of $\mathcal{B}_{\mathbb{C}}$ of degree zero is a nonzero complex constant.

Proof. If γ and δ are the respective largest exponents, every contributing exponent of xy is at most $\gamma + \delta$. Equality requires both exponents to be largest: making either smaller makes their sum smaller. Thus the coefficient at $\gamma + \delta$ is the nonzero product of leading coefficients. The statements about sums follow from cancellation at the greatest exponent. Finally, support in $[0, 0]$ contains only zero. $\square$

Proposition 2.5 (Units and finite elements [01, 02, 05]). *The unit groups are*

$$\mathcal{B}_{\mathbb{R}}^{\times} = \mathbb{R}^{\times}, \quad \mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}, \quad \mathbf{Oz}^{\times} = \{1, -1\}.$$

The omnific integers bounded in absolute value by some ordinary real number are exactly the ordinary integers; in particular $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$. Consequently $\mathbf{Oz}$ is discretely ordered, with least positive element 1.

Proof. If $xy = 1$ in $\mathcal{B}_{\mathbb{C}}$, nonnegativity and additivity of degree force both degrees to be zero. This gives the first two unit statements; in **Oz** the constant units are precisely the units of $\mathbb{Z}$.

A nonconstant element of **Oz** has positive leading exponent, so its absolute value exceeds every positive real number by normal-form order. A constant element of **Oz** is an integer. Hence no element lies strictly between 0 and 1, and every positive element is at least 1. $\square$

Notice that $\omega/2$ is an omnific integer even though $1/2$ is not. This is no conflict with discreteness: $\omega/2$ is infinite, and there are still no omnific integers in the interval $(0, 1)$.

The unit computation is stronger than saying that an infinite factor has large real absolute value: complex leading coefficients can cancel in sums, but they cannot cancel the unique leading coefficient in a product. It will do much of the Diophantine work.

2.3 Local workspaces

Lemma 2.6 (Set-sized workspaces [05]). *Any set-sized family of normal forms belongs to a common Hahn field whose exponents lie in a set-sized ordered divisible group $H \subseteq \mathbf{No}$. One can take H to be the rational span of the union of the supports.*

Proof. The union of the supports is a set, and so is its rational span. Normal forms supported in H constitute the real or complex Hahn field with exponents in H . The Hahn field operations preserve this exponent group, including inverses of nonzero series. $\square$

We write $\mathcal{B}_{\mathbb{R},H}$, $\mathcal{B}_{\mathbb{C},H}$ and $\mathbf{Oz}_H$ for the intersections of $\mathcal{B}_{\mathbb{R}}$, $\mathcal{B}_{\mathbb{C}}$ and $\mathbf{Oz}$, respectively, with the workspace; in particular $\mathbf{Oz}_H$ retains the integer constant-coefficient condition. Most proofs below, including all the main equation-rigidity theorems, take place in one such set-sized ring and need no proper-class derivation or basis. By contrast, a positive surreal smaller than every member of a specified set of positive exponents need not belong to a preassigned H ; the results of section 4 genuinely use the full exponent class, and are marked there.

2.4 The exact omnific floor

Every surreal admits a unique decomposition

$$x = x_{>0} + r + \varepsilon, \quad x_{>0} \in \Pi, \quad r \in \mathbb{R}, \quad \text{supp}(\varepsilon) \subseteq \mathbf{No}^{<0}.$$

Here ε is infinitesimal, possibly zero.

Theorem 2.7 (Integer-part formula [01, 02, 05]). *There is exactly one $a \in \mathbf{Oz}$ such that $a \leq x < a + 1$. It is*

$$\lfloor x \rfloor_{\mathbf{Oz}} = \begin{cases} x_{>0} + \lfloor r \rfloor, & r \notin \mathbb{Z} \text{ or } \varepsilon \geq 0, \\ x_{>0} + r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0. \end{cases} \quad (2.2)$$

Proof. For $r \notin \mathbb{Z}$, its distances from the two adjacent integers are strictly positive real numbers, and the infinitesimal ε cannot cross either boundary. If r is an integer and $\varepsilon \geq 0$, the remainder is $\varepsilon \in [0, 1)$. If r is an integer and $\varepsilon < 0$, the remainder after subtracting $x_{>0} + r - 1$ is $1 + \varepsilon \in (0, 1)$. Two possible floors would differ by an omnific integer strictly between -1 and 1 , so would coincide by discreteness. $\square$

For example,

$$\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1, \quad \lfloor \omega + \frac{3}{2} - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega + 1, \quad \lfloor -\omega^{-1} \rfloor_{\mathbf{Oz}} = -1.$$

Merely discarding negative exponents and flooring the constant gives the wrong answer at a negative infinitesimal displacement from an integer.

Provenance. The integer-part assertion is classical. Ehrlich–Kaplan prove the stronger initiality characterization: for an initial subfield K of $\mathbf{No}$, $\mathbf{Oz} \cap K$ is its unique initial integer part [4, Lemma 11.1]. No alternative integer part is introduced. The formula agrees with the trigonometry report’s construction of the integer part, which takes “the appropriate side when f is infinitesimally adjacent to an integer” (text following `trigonometry:eq:split`).

Proposition 2.8 (One-variable rigidity [02, 05]). *If $P \in \mathbb{C}[T]$ is nonzero, every root of P in $\mathcal{B}_{\mathbb{C}}$ belongs to $\mathbb{C}$. In particular, roots in $\mathcal{B}_{\mathbb{R}}$ of a nonzero real polynomial are real, and the roots in $\mathbf{Oz}$ of a nonzero $P \in \mathbb{Z}[T]$ are exactly its ordinary integer roots.*

Proof. If a root x were nonconstant, $\alpha = \deg_\omega(x) > 0$. For nonzero coefficients a_j , the terms $a_j x^j$ have distinct degrees $j\alpha$, so the highest-degree term cannot cancel. Thus $P(x) \neq 0$. A constant root in $\mathbf{Oz}$ is an integer. The real case follows from $\mathcal{B}_\mathbb{R} \cap \mathbb{C} = \mathbb{R}$. $\square$

Source 02 states the integer case and source 05 the real case. The same leading-degree argument proves the complex extension above.

Every infinite omnific integer is therefore transcendental over $\mathbb{R}$. This does not make a tuple of infinite omnific integers algebraically independent: (ω, ω^2) is a counterexample. Infinite coefficients change the problem: $X^2 - \omega^2$ has the infinite roots $\pm\omega$.

3 Divisibility, residues, and quotients

3.1 Division by ordinary integers

Theorem 3.1 (Ordinary moduli [01, 02, 05]). *For every nonzero ordinary integer n and $x \in \mathbf{Oz}$,*

$$n\mathbf{Oz} = \Pi + n\mathbb{Z}, \quad n \mid x \iff n \mid \text{ct}(x), \quad \mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}. \quad (3.1)$$

In particular, for every ordinary prime p ,

$$\bigcap_{n \geq 1} n\mathbf{Oz} = \Pi = \bigcap_{k \geq 1} p^k \mathbf{Oz}. \quad (3.2)$$

If $\text{ct}(x) \neq 0$, the largest ordinary k with $p^k \mid x$ is the p -adic order of $\text{ct}(x)$; if $\text{ct}(x) = 0$, every power of every ordinary prime divides x .

Proof. Necessity in the divisibility criterion follows by applying ct to $x = ny$. Multiplication by a nonzero real scalar preserves Π and is invertible on Π , so $n\Pi = \Pi$. If $x = m + j$ with $m \in \mathbb{Z}$, $j \in \Pi$, then $x/n = m/n + j/n$ lies in $\mathbf{Oz}$ exactly when m/n is an integer. Reduction of ct modulo n is therefore a surjection with kernel $n\mathbf{Oz}$. The only ordinary integer divisible by all positive integers, or by every power of one prime, is zero, which gives (3.2) and the statement on p -adic orders. $\square$

This theorem solves every ordinary congruence. For instance, $\sqrt{2}\omega + 14$ is even and is congruent to 2 modulo 3; ω is divisible by every positive ordinary integer and $\omega + 1$ is odd. The whole infinite part is invisible to these tests.

The statement does *not* extend to arbitrary omnific divisors. For example, $\text{ct}(\omega + 2)$ divides $\text{ct}(2)$ in $\mathbb{Z}$, but $\omega + 2$ does not divide 2 in $\mathbf{Oz}$. More generally:

Proposition 3.2 (Divisors of integers [05]). *If $d \in \mathbf{Oz}$ divides a nonzero ordinary integer N , then d is an ordinary integer divisor of N .*

Proof. Write $de = N$. Neither factor is zero, so $|e| \geq 1$ by discreteness and $|d| \leq |N|$. Hence $d \in \mathbb{Z}$ by proposition 2.5; similarly $e \in \mathbb{Z}$. $\square$

Corollary 3.3 (Ordinary primes and mixed gcds [02, 05]). *Every positive ordinary prime p is a prime element of $\mathbf{Oz}$, and $p\mathbf{Oz}$ is maximal. For $0 \neq n \in \mathbb{Z}$ and $x \in \mathbf{Oz}$,*

$$(n, x) = \gcd(n, \text{ct}(x)) \mathbf{Oz},$$

so an ordinary gcd of n and $\text{ct}(x)$ is a gcd of n and x in $\mathbf{Oz}$. The Chinese remainder theorem for pairwise coprime ordinary moduli holds in $\mathbf{Oz}$ with exactly the ordinary residue rings.

Proof. The quotient by $p\mathbf{Oz}$ is the field $\mathbb{Z}/p\mathbb{Z}$. If $x = m + j$ with $j \in \Pi$, then $x - n(j/n) = m$, so $(n, x) = (n, m)$, which an ordinary Bézout identity identifies. A generator of a two-generated principal ideal is a gcd of its generators in any domain. For coprime moduli, ordinary Bézout coefficients give comaximality, and the quotient descriptions give the Chinese remainder isomorphism. $\square$

Corollary 3.4 (Positive-characteristic quotients [01, 02, 05]). *Every proper ideal $\mathfrak{a} \subset \mathbf{Oz}$ containing a nonzero ordinary integer is $n\mathbf{Oz}$ for a unique ordinary $n \geq 2$; it is prime exactly when n is an ordinary prime. Every unital homomorphism from $\mathbf{Oz}$ to a finite ring kills Π and factors through ct , and every nonzero finite quotient of $\mathbf{Oz}$ is a ring $\mathbb{Z}/n\mathbb{Z}$.*

Proof. If $m \in \mathfrak{a}$ with $0 \neq m \in \mathbb{Z}$, then $\Pi = m\Pi \subseteq \mathfrak{a}$. Thus $\mathfrak{a}$ is the inverse image of an ideal of $\mathbf{Oz}/\Pi = \mathbb{Z}$, namely $n\mathbb{Z}$ with $n > 0$, and properness gives $n \geq 2$. Equation (3.1) identifies its inverse image with $n\mathbf{Oz}$; primality is that of $n\mathbb{Z}$. If a finite target has characteristic m , then each $j \in \Pi$ is $m(j/m)$ and maps to zero. A finite nonzero unital ring has positive characteristic, so the last assertion follows. $\square$

3.2 Completions

All finite-index ideals have just been classified, so the profinite completion, interpreted as the set-sized inverse limit over ordinary $n \geq 1$, is

$$\widehat{\mathbf{Oz}} \cong \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \widehat{\mathbb{Z}}, \quad \varprojlim_k \mathbf{Oz}/p^k\mathbf{Oz} \cong \mathbb{Z}_p. \quad (3.3)$$

The completion maps are the composites of ct with the canonical maps of $\mathbb{Z}$, with kernel Π in both cases by (3.2). Thus these congruence topologies on $\mathbf{Oz}$ are not Hausdorff and $\mathbf{Oz}$ is not residually finite [01, 02, 05]. For example ω , $\sqrt{2}\omega$ and 0 have identical images in every finite quotient. The separated quotient is $\mathbf{Oz}/\Pi = \mathbb{Z}$; completion does not preserve the infinite part.

The first inverse system is indexed by positive integers ordered by divisibility, with reduction maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ for $m \mid n$. The second uses the powers of a fixed ordinary prime p . These are set-sized diagrams of ordinary residue rings, not sets of class ideals or proper-class cosets.

3.3 Homomorphisms to $\mathbb{Z}$ and to set-sized rings

Proposition 3.5 (Uniqueness of the retraction [01]). *The constant-term map is the unique unital ring homomorphism $\mathbf{Oz} \rightarrow \mathbb{Z}$. Every unital endomorphism $\psi : \mathbf{Oz} \rightarrow \mathbf{Oz}$ preserves Π and satisfies $\text{ct} \circ \psi = \text{ct}$.*

Proof. For a homomorphism $h : \mathbf{Oz} \rightarrow \mathbb{Z}$, the integer $h(j)$ is divisible by every positive integer whenever $j \in \Pi$. Hence $h(j) = 0$. Since h fixes ordinary integers, $h = \text{ct}$. An endomorphism preserves divisibility by each ordinary integer, so preserves their common divisibility kernel (3.2). It fixes $\mathbb{Z}$ pointwise, which proves the final identity. $\square$

This does not say that every endomorphism is the identity: the composite of ct with the inclusion $\mathbb{Z} \hookrightarrow \mathbf{Oz}$ is a noninjective unital endomorphism [05].

Provenance. The divisible-kernel and residue-map perspective belongs to classical $\mathbb{Z}$ -group theory; see [5, Sections 2 and 4]. Here it has a concrete ring-theoretic realization because ct is multiplicative. The finite-quotient and completion statements above are proved directly.

Source 05 proved a far-reaching generalization of [proposition 3.5](#) and [corollary 3.4](#). It is proved, in a more general coefficient-ring form and with the sharper theory of set-sized

quotients assembled from several manuscripts, in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`, which also keeps source 05's route; we quote it without proof.

Cited theorem 3.6 (Universal set-sized quotient; proved by source 05, printed in the sibling report). *Let S be a set-sized unital ring, not necessarily commutative, and let $\varphi : \mathbf{Oz} \rightarrow S$ be a unital class ring homomorphism. Then $\varphi(x) = \text{ct}(x) 1_S$ for every $x \in \mathbf{Oz}$. Consequently every set-sized quotient ring of $\mathbf{Oz}$ is isomorphic to $\mathbb{Z}$ (kernel Π) or to $\mathbb{Z}/n\mathbb{Z}$ (kernel $n\mathbf{Oz}$) for an ordinary $n \geq 1$.*

Source 05's proof applies pigeonhole to the monomials $\omega^{\delta/\omega^{\alpha+1}}$ for ordinals α beyond the size of S , and kills a positive monomial ω^δ by the exact Hahn identity $(\omega^a - \omega^b) \sum_{k \geq 0} \omega^{\delta-a-k(a-b)} = \omega^\delta$, whose quotient lies in Π when $0 < b < a \leq \delta/\omega$; it then uses [theorem 4.2](#) below to kill all of Π .

Remark 3.7 (Foundations of the cited theorem). Source 05 chooses an infinite cardinal $\kappa > |S|$, using the global choice of its NBG setting. The Hartogs ordinal of S , the least ordinal admitting no injection into S , can be used instead: the pigeonhole step needs only that the family $(\omega^{\delta/\omega^{\alpha+1}})_\alpha$ indexed by that ordinal does not inject into S . So neither global choice nor choice for sets is needed for [cited theorem 3.6](#).

[Proposition 3.5](#) is the case $S = \mathbb{Z}$ of [cited theorem 3.6](#) together with its consequence for endomorphisms (apply it to $\text{ct} \circ \psi$), and [corollary 3.4](#) is the case of finite S ; the elementary divisibility proofs above are independent second routes. The theorem does not apply to class-sized targets such as $\mathbf{Oz}$ itself, and it is a full-class statement: the subring $\mathbb{Z} + \omega\mathbb{R}[\omega]$ of $\mathbf{Oz}$ admits evaluation at many real values, and such maps do not extend to $\mathbf{Oz}$ [05]. Source 05 also records that, by [cited theorem 3.6](#), the quotient $\mathbf{Oz}/(f)$ by an infinite prime f with $\text{ct}(f) = \pm 1$ (for example the prime of [3, Theorem B] in [section 4.2](#)) is not set-sized; that consequence is likewise printed in the sibling report.

4 Full-class support: common divisors, clearing, factorization

The next results use all surreal exponents, not merely the exponents in one fixed Archimedean or finite-rank group. This is an important source of arithmetic differences between full omnific integers and restricted series rings ([section 17](#)).

4.1 One monomial divides a whole set

Lemma 4.1 (Set-sized exponent bounds [01, 02, 05]). *If $E \subset \mathbf{No}^{>0}$ is a set, there is a surreal $\delta > 0$ such that $\delta < e$ for every $e \in E$. Every set of surreals also has a strict upper bound.*

Proof. A surreal separating the cut $\{0 \mid E\}$ gives the first assertion, and a separator of $\{E \mid \emptyset\}$ the second; their option collections are sets. The assertions also hold for the empty set. These are applications of the set-cut property, not an assertion of an infimum or supremum. No single δ works for the entire proper class $\mathbf{No}^{>0}$. $\square$

Theorem 4.2 (Common monomial divisibility [01, 02, 05]). *For every set $S \subseteq \Pi$ there is $\delta > 0$ such that*

$$\omega^\delta \mid s \quad \text{and} \quad s/\omega^\delta \in \Pi \quad (s \in S).$$

In particular, each nonzero element of Π is a product of two nonzero elements of Π , both nonunits of $\mathbf{Oz}$.

Proof. The union of the supports of the elements of S is a set E of strictly positive surreal exponents. Choose δ as in [lemma 4.1](#). Dividing by ω^δ shifts every exponent e to $e - \delta > 0$, preserving the reverse well-order of each support. The quotient is therefore in Π . For a nonzero s , both factors are nonzero elements of positive degree, and neither is a unit. $\square$

Even the series $\sum_{k \geq 1} \omega^{1/k}$ has such a divisor. The positive exponent δ may be infinitesimal relative to every real $1/k$, for example $\delta = 1/\omega$. This is why the proof cannot be copied into a Hahn ring with exponent group $\mathbb{R}$.

Corollary 4.3 (The purely infinite ideal [01, 02, 05]). *The ideal Π satisfies $\Pi^2 = \Pi$, and each element of Π is a single product of two elements of Π . It is not generated as an ideal by any set of elements; in particular it is not finitely generated. Consequently $\Pi/\Pi^2 = 0$ although $\Pi \neq 0$, $\mathbf{Oz}/\Pi^k \cong \mathbb{Z}$ for every ordinary $k \geq 1$, and the Π -adic completion of $\mathbf{Oz}$ is $\mathbb{Z}$. No nonzero element of Π is a finite product of irreducibles of $\mathbf{Oz}$.*

Proof. The inclusion $\Pi^2 \subseteq \Pi$ follows from the ideal property. For $j \neq 0$, the factorization in [theorem 4.2](#) has both factors in Π , proving the reverse inclusion. Hence $\Pi^k = \Pi$ for all $k \geq 1$ and the inverse system $\mathbf{Oz}/\Pi^k$ is constantly $\mathbb{Z}$.

Suppose a set $S \subseteq \Pi$ generated Π . Choose $\delta > 0$ strictly below every exponent in the supports of its elements. Each finite $\mathbf{Oz}$ -linear combination of elements of S has support strictly above δ : multiplication by an element of $\mathbf{Oz}$ only adds nonnegative exponents. But $\omega^{\delta/2} \in \Pi$ does not have such a support.

Finally, if $j = p_1 \cdots p_k \neq 0$ with every p_i irreducible, then $0 = \text{ct}(j) = \prod_{i=1}^k \text{ct}(p_i)$ would put some p_i in Π , and that factor would be reducible by [theorem 4.2](#). $\square$

The factorization $j = 2(j/2)$ already proves that a nonzero $j \in \Pi$ is reducible, even in a fixed exponent workspace; the stronger content of [theorem 4.2](#) is a common *infinite monomial* divisor for an entire set, with all quotients still in Π . The vanishing of Π/Π^2 does not say that the infinite part is small; it says that this particular quotient cannot detect it. The hypotheses of a finitely generated Nakayama argument are not present [02].

Corollary 4.4 (Nilpotent tests [02]). *Let $\phi : \mathbf{Oz} \rightarrow R$ be a ring homomorphism and $\mathfrak{n} \subseteq R$ a nilpotent ideal with $\phi(\Pi) \subseteq \mathfrak{n}$. Then $\phi(\Pi) = 0$.*

Proof. Choose an ordinary $N \geq 1$ with $\mathfrak{n}^N = 0$. By [corollary 4.3](#), $\Pi^N = \Pi$. The image of every finite sum of N -fold products from Π belongs to $\mathfrak{n}^N = 0$. This proves the claim without assuming that $\phi(\Pi)$ is an ideal of all of R . $\square$

4.2 Factorization

[Corollary 4.3](#) is an explicit failure of atomic factorization, not merely a failure of uniqueness of factorization. It leaves room for nonconstant prime elements having nonzero constant coefficient.

Proposition 4.5 (Irreducibles [01, 02, 05]). *The constant irreducibles of $\mathbf{Oz}$ are exactly the ordinary signed primes. Every nonconstant irreducible has constant coefficient 1 or -1 . The latter condition is not sufficient for irreducibility. In particular, $\mathbf{Oz}$ is neither atomic nor a unique factorization domain.*

Proof. Constant irreducibles are classified by [proposition 3.2](#). If a nonconstant element a has $|\text{ct}(a)| > 1$, choose an ordinary prime p dividing its constant coefficient. Then $a = p(a/p)$

by [theorem 3.1](#) is a factorization into two nonunits: the second factor remains nonconstant. If $\text{ct}(a) = 0$, the factorization $a = 2(a/2)$ has the same property. This leaves only constant coefficients ± 1 . Finally, $(\omega + 1)^2$ has constant coefficient 1 and $\omega^2 - 1 = (\omega - 1)(\omega + 1)$ has constant coefficient -1 ; both are reducible. $\square$

In particular ω has no finite irreducible factorization, not for lack of factors: it is divisible by every ordinary positive integer and by every ω^δ with $0 < \delta < 1$. Genuine infinite primes exist. The series $1 + \sum_{k \geq 1} \omega^{1/k}$ is prime by earlier results reviewed by L’Innocente and Mantova [[3](#), Section 1.3] (irreducibility by Berarducci, primality by Pitteloud, and primality in $\mathbf{Oz}$ by Biljaković–Kochetov–Kuhlmann, as credited there), and $\omega^{\sqrt{2}} + \omega + 1$ is prime by their Theorem B [[3](#), Theorem B], which answers a question of Gonshor. Their general theory uses substantially finer invariants than the degree employed here; no result of this report depends on these primality theorems.

4.3 Clearing arbitrary set-sized families of denominators

Theorem 4.6 (Global monomial clearing [[01](#), [02](#), [05](#)]). *For every set $S \subseteq \mathbf{No}$ there is $h > 0$ such that*

$$\omega^h \in \Pi, \quad \omega^h x \in \Pi \quad (x \in S).$$

Consequently $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, where the equality means that every surreal is a quotient of two omnific integers with nonzero denominator.

Proof. Take the set E obtained by uniting the supports of all elements of S . Choose $h > 0$ strictly greater than every $-e$ for $e \in E$. Multiplication by ω^h shifts every exponent to $h+e > 0$, so all shifted normal forms belong to Π . Applying this to $S = \{x\}$ gives $x = (\omega^h x)/\omega^h$. The reverse inclusion holds because $\mathbf{Oz}$ is a subring of the field $\mathbf{No}$. $\square$

A support unbounded below in a chosen real exponent group is still a set of surreal exponents and can be shifted above zero by a new, larger surreal exponent. The theorem does not assert that the required monomial lies in the original fixed Hahn workspace.

Corollary 4.7 (Projective clearing and common multiples [[01](#), [02](#), [05](#)]). *Every point of $\mathbb{P}^n(\mathbf{No})$ has homogeneous coordinates in Π^{n+1} ; more generally, any set-sized collection of coordinate tuples can be cleared by a single positive monomial, and a common zero of homogeneous polynomials remains a common zero after this rescaling. Every set S of nonzero omnific integers has a nonzero common multiple; in fact some nonzero $M \in \mathbf{Oz}$ lies in $a^k \mathbf{Oz}$ for every $a \in S$ and ordinary $k \geq 1$.*

Proof. Apply [theorem 4.6](#) to the set of all coordinates and multiply all tuples by the resulting nonzero scalar; homogeneity preserves the equations. For the last claim apply [theorem 4.6](#) to the set $\{a^{-k} : a \in S, k \geq 1\}$ and take $M = \omega^h$; then $M/a^k \in \Pi$. $\square$

This is projective *representability*, not primitivity: the coordinates of such a representative always have a common nonunit divisor, and [section 14](#) shows that some real projective directions have no primitive omnific representative at all. The common-multiple statement is an existence theorem, not an algorithm; nor is there one nonzero omnific integer divisible by every member of the proper class of nonzero omnific integers.

4.4 Division with remainder is not a terminating Euclidean algorithm

Proposition 4.8 (Ordered division [01, 02, 05]). *For $a, b \in \mathbf{Oz}$ with $b > 0$, there are unique $q, r \in \mathbf{Oz}$ such that $a = bq + r$ and $0 \leq r < b$, namely $q = \lfloor a/b \rfloor_{\mathbf{Oz}}$ and $r = a - bq$.*

Proof. The floor inequalities of [theorem 2.7](#), multiplied by b , give the remainder bounds. Conversely those bounds force $q \leq a/b < q + 1$, so floor uniqueness determines q and then r . $\square$

Thus the familiar division step exists. Termination of repeated division is the missing ingredient: the positive order on $\mathbf{Oz}$ is not well founded.

Example 4.9 (A Euclidean algorithm that never stops [01, 02, 05]). Let $\rho = \sqrt{2} - 1 \in (0, 1)$ and start with $r_0 = \sqrt{2}\omega$, $r_1 = \omega$. The first quotient is 1, the remainder is $r_2 = \rho\omega$, and every subsequent quotient is 2:

$$r_k = \rho^{k-1}\omega \quad (k \geq 1), \quad r_{k-1} = 2r_k + r_{k+1} \quad (k \geq 2),$$

because $1 - 2\rho = \rho^2$ and $1/\rho = \sqrt{2} + 1 \in (2, 3)$. Every remainder is positive and infinite, so the ordinary finite Euclidean procedure never terminates. (Source 05 starts at $(\omega, \rho\omega)$ and sees the same tail.) A decreasing remainder is not a well-founded measure here.

Theorem 4.10 (Pairs without a gcd [02], generalizing [01]). *Let $0 \neq t \in \Pi$ and let $r \in \mathbb{R} \setminus \mathbb{Q}$. The pair (t, rt) has no greatest common divisor in $\mathbf{Oz}$, in the divisibility sense. In particular, $\mathbf{Oz}$ is neither a GCD domain nor a Bézout domain.*

Proof. Take any common divisor $d \in \mathbf{Oz}$. Since $t \neq 0$, also $d \neq 0$, and there are nonzero $u, v \in \mathbf{Oz}$ with $t = du$ and $rt = dv$. Cancellation in $\mathbf{No}$ gives $v = ru$. Taking constants in $\mathcal{B}_{\mathbb{R}}$ yields $\text{ct}(v) = r \text{ct}(u)$ with both constants ordinary integers, so irrationality of r forces both to be zero: $u, v \in \Pi$.

By [theorem 4.2](#) applied to $\{u, v\}$, choose $\delta > 0$ with $u/\omega^\delta, v/\omega^\delta \in \Pi$. Then $d' = d\omega^\delta$ is another common divisor of t, rt , and $d \mid d'$. But $d' \nmid d$, since the quotient would be $\omega^{-\delta} \notin \mathbf{Oz}$. Thus every common divisor can be strictly enlarged in the divisibility order, and none is a gcd. A Bézout domain would supply a gcd for each pair, namely a generator of the ideal the pair generates. $\square$

Remark 4.11 (A second route for the pair $(\sqrt{2}\omega, \omega)$ [01]). Source 01 proves the instance $t = \omega$, $r = \sqrt{2}$ by degrees alone. Let $d \neq 0$ be a common divisor, $\theta = \deg_\omega(d)$; then $0 \leq \theta \leq 1$. If $\theta = 1$, both quotients are degree-zero omnific integers, hence nonzero ordinary integers, and their ratio would make $\sqrt{2}$ rational. So $\theta < 1$; with $\eta = (\theta + 1)/2$, ω^η is a common divisor (both quotients have exponent $1 - \eta > 0$) that does not divide d , since its degree is larger. This route needs only the exponent $(\theta + 1)/2$, so it works in any workspace with a divisible exponent group, whereas the general [theorem 4.10](#) uses the full-class shift of [theorem 4.2](#).

[Example 4.9](#) shows at once why numerical descent is insufficient and why a division algorithm need not produce a gcd: its input has none.

4.5 Non-Noetherian and non-normal behavior

Proposition 4.12 (Chains and integral closure [01, 02, 05]). *There is a strictly ascending chain of principal ideals*

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \cdots,$$

so $\mathbf{Oz}$ satisfies neither the ascending chain condition on principal ideals nor the Noetherian condition. The full ring $\mathbf{Oz}$ is not integrally closed in its fraction field: both $\sqrt{2}$ and $\sqrt{\omega^2 + 1}$ are integral over $\mathbf{Oz}$, lie in $\text{Frac}(\mathbf{Oz})$, and are not in $\mathbf{Oz}$.

Proof. Each generator is the square of the next, so the inclusions hold. They are strict because the reverse quotients $\omega^{-1/2^{k+1}}$ have negative exponents. The number $\sqrt{2} = (\sqrt{2}\omega)/\omega$ satisfies $T^2 - 2 = 0$ and is not an integer [02, 05]. Since $\mathbf{No}$ is real closed, let $r = \sqrt{\omega^2 + 1} > 0$; it lies in $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and satisfies $T^2 - (\omega^2 + 1) = 0$. If $r \in \mathbf{Oz}$, then $(r - \omega)(r + \omega) = 1$ makes both factors units, hence elements of $\{1, -1\}$ with equal signs; their difference 2ω would be 0, a contradiction [01]. $\square$

In particular the full omnific ring is not a UFD or a PID. Its factorization theory cannot be replaced by an ordinary Euclidean algorithm or by an unstated gcd assumption. The modern theory in [3] is substantially subtler. Conway's refinement question asks whether $ab = cd$ in $\mathbf{Oz}$ always admits $a = ef$, $b = gh$, $c = eg$, $d = fh$ with $e, f, g, h \in \mathbf{Oz}$; this is a refinement property, not atomicity or the existence of gcfs, and none of the preceding arguments refutes it. [Appendix D](#) records why no current open-status claim about it is made.

5 Equational transfer and linear Diophantine equations

5.1 Existence is exactly ordinary existence

For a finite family of polynomials $F_1, \dots, F_r$ over a ring R , write $V_R(F_1, \dots, F_r)$ for its simultaneous zero set in R^n . All exponents of the variables are ordinary nonnegative integers and only finitely many monomials occur; equations with variable exponents, surreal exponentiation, or class-sized sums are not polynomial systems in this sense.

Theorem 5.1 (Retraction theorem for equations [01, 02, 05]). *Let $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$. Coordinatewise constant coefficient gives a retraction, with the inclusion as section,*

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \longrightarrow V_{\mathbb{Z}}(F_1, \dots, F_r).$$

In particular,

$$V_{\mathbf{Oz}}(F_1, \dots, F_r) \neq \emptyset \iff V_{\mathbb{Z}}(F_1, \dots, F_r) \neq \emptyset. \quad (5.1)$$

The same equivalence holds for positive existential formulas in the ring language, evaluated at ordinary integer parameters.

Proof. For every polynomial F over $\mathbb{Z}$ and $\mathbf{x} \in \mathbf{Oz}^n$, $\text{ct}(F(\mathbf{x})) = F(\text{ct}(x_1), \dots, \text{ct}(x_n))$, because ct is a ring homomorphism fixing all coefficients. Thus the constant tuple of an omnific solution is an integer solution. Conversely, an integer solution is already an omnific solution, and applying ct to it does nothing.

A positive existential formula is built from equalities using finite conjunction, finite disjunction, and existential quantification. Ring homomorphisms preserve such formulas, by induction on their formation, with witnesses mapped coordinatewise. Apply this to ct and to the inclusion of $\mathbb{Z}$ in $\mathbf{Oz}$. $\square$

The theorem answers a question about existence, not about cardinalities or complete solution sets. For $\mathbf{a} \in V_{\mathbb{Z}}$, its *constant-coefficient fiber* is

$$V_{\mathbf{a}}(\mathbf{Oz}) = \{\mathbf{a} + \mathbf{u} : \mathbf{u} \in \Pi^n, F_j(\mathbf{a} + \mathbf{u}) = 0 \text{ for all } j\}. \quad (5.2)$$

For $Y = X^2$ the fiber above $(0, 0)$ already contains (t, t^2) for every $t \in \Pi$. These are not infinitesimal deformations: every nonzero coordinate of $\mathbf{u}$ is infinite. We call a system *rigid* when every fiber is its ordinary base point, that is, when $V_{\mathbf{Oz}} = V_{\mathbb{Z}}$ [02].

Corollary 5.2 (Hilbert’s tenth problem [01, 02, 05]). *There is no algorithm deciding, from a finite list of ordinary integer polynomials, whether it has a common zero in $\mathbf{Oz}$. The yes-instances are exactly the ordinary-integer yes-instances of Hilbert’s tenth problem, and form a computably enumerable set that is complete for computably enumerable sets under many-one reductions.*

Proof. Equation (5.1) identifies the two decision problems on exactly the same finite inputs. The undecidability and completeness of the ordinary problem are the Davis–Putnam–Robinson–Matiyasevich (MRDP) theorem and its standard consequences [7, 8, 9, 10]; one passes between the natural-number and integer formulations by the four-square representation of nonnegative integers, and conjunctions are combined as sums of squares. Enumerating ordinary integer tuples semidecides the omnific yes-instances; no enumeration of the proper class $\mathbf{Oz}$ is involved. $\square$

This corollary concerns *ordinary integer coefficients*. A decision problem allowing arbitrary omnific coefficients requires an explicit finite representation scheme and is a different question (section 16).

5.2 Why signs and nonvanishing do not transfer

Consider $(X + 1)^2 - 2(Y + 1)^2 = 0$. It has no solution in $\mathbb{N}^2$, since $X + 1$ and $Y + 1$ would be positive integers with ratio $\sqrt{2}$. However, $X = \sqrt{2}\omega - 1$, $Y = \omega - 1$ is a solution with both coordinates positive in $\mathbf{Oz}$. Its constant tuple is $(-1, -1)$, an ordinary solution of the equation without positivity constraints.

Proposition 5.3 (Retraction closure [01, 02, 05]). *If $D \subseteq \mathbf{Oz}^n$ is defined by a positive existential ring formula with ordinary integer parameters, then $\mathbf{x} \in D$ implies $\text{ct}(\mathbf{x}) \in D$. Consequently none of*

$$\mathbf{Oz} \setminus \{0\}, \quad \mathbf{Oz}^{>0}, \quad \mathbf{Oz}^{\geq 0}$$

is so definable.

Proof. Apply ct to the free variables and to all existential witnesses, as in theorem 5.1. But ω is nonzero and positive while $\text{ct}(\omega) = 0$; and $\omega - 1 > 0$ whereas $\text{ct}(\omega - 1) = -1 < 0$. $\square$

Thus “nonzero” cannot be harmlessly added to an equational transfer statement, and the retraction is not an elementary embedding of ordered rings. Encoding nonvanishing by $XY = 1$ would restrict X to $\{1, -1\}$, not to all nonzero omnific integers. The qualification on coefficients matters as well: a predicate with a fixed infinite parameter need not be preserved by a map that changes that parameter. For coefficients outside $\mathbb{Z}$, specialization gives a necessary residual system whose solvability need not lift (section 16).

5.3 A complete solution criterion for linear systems

Theorem 5.4 (Smith normal form over the omnific integers [01, 02, 05]). *Let $M \in \mathbb{Z}^{m \times n}$ and $\mathbf{b} \in \mathbf{Oz}^m$. Choose unimodular integer matrices U, V with*

$$UMV = \text{diag}(d_1, \dots, d_r, 0), \quad d_i > 0,$$

in rectangular Smith normal form, and put $\mathbf{c} = U\mathbf{b}$. Then $M\mathbf{x} = \mathbf{b}$ is solvable in $\mathbf{Oz}^n$ if and only if

$$d_i \mid \text{ct}(c_i) \quad (1 \leq i \leq r), \quad c_i = 0 \quad (r < i \leq m). \quad (5.3)$$

All solutions are $\mathbf{x} = V\mathbf{y}$, where $y_i = c_i/d_i$ for $1 \leq i \leq r$ and $y_{r+1}, \dots, y_n \in \mathbf{Oz}$ are arbitrary.

Proof. Unimodular integer matrices and their inverses act on omnific tuples. The substitution $\mathbf{x} = V\mathbf{y}$ transforms the system into $\text{diag}(d_1, \dots, d_r, 0)\mathbf{y} = \mathbf{c}$. For each nonzero diagonal entry, division is possible precisely under [theorem 3.1](#). Zero rows require the full equation $c_i = 0$, not merely its constant coefficient. The remaining coordinates are unconstrained. $\square$

Theorem 5.5 (Ordinary right-hand sides [05]). *Let $M \in \mathbb{Z}^{m \times n}$ and $\mathbf{b} \in \mathbb{Z}^m$. All solutions in $\mathbf{Oz}^n$ of $M\mathbf{x} = \mathbf{b}$ are exactly*

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad M\mathbf{a} = \mathbf{b}, \quad \mathbf{u} \in \Pi^n, \quad M\mathbf{u} = 0.$$

If $\mathbf{v}_1, \dots, \mathbf{v}_e \in \mathbb{Q}^n$ is a basis of $\ker(M : \mathbb{Q}^n \rightarrow \mathbb{Q}^m)$, the last condition is equivalent to $\mathbf{u} = \sum_j h_j \mathbf{v}_j$ with uniquely determined $h_j \in \Pi$. In particular, if M has full column rank, all solutions are ordinary.

Proof. Separate each coordinate into its integer constant and its member of Π . The integer and positive-support components of $M\mathbf{x} - \mathbf{b}$ vanish separately. Gaussian elimination over $\mathbb{Q}$ works with entries in the $\mathbb{Q}$ -vector space Π , since nonzero rationals act invertibly on it; applying the same row operations and free-variable parametrization as over $\mathbb{Q}$ gives the basis description and its uniqueness. $\square$

For example, every solution of $2x + 3y = 1$ is $x = -1 + 3t$, $y = 1 - 2t$ with $t \in \mathbf{Oz}$, and every solution of $6x + 10y = 14$ is $x = 4 + 5t$, $y = -1 - 3t$ with $t \in \mathbf{Oz}$. The same formulas as over $\mathbb{Z}$ work, but the parameter class is larger. The equation $2x = \omega + 1$ has no solution, whereas $2x = \omega + 2$ has the unique solution $x = \omega/2 + 1$. A zero row illustrates a different obstruction: $0 \cdot x = \omega$ has no solution, although its constant-term equation is $0 = 0$.

6 Rigidity of constant products: binary forms, Pell, norms

6.1 The constant-product principle

Lemma 6.1 (Constant products [01, 02, 05]). *If $b_1, \dots, b_s \in \mathcal{B}_{\mathbb{C}}$ and $b_1 \cdots b_s = c \in \mathbb{C}^\times$, then every b_i belongs to $\mathbb{C}^\times$.*

Proof. Each b_i has inverse $c^{-1} \prod_{j \neq i} b_j$ in $\mathcal{B}_{\mathbb{C}}$; apply [proposition 2.5](#). Equivalently, their nonnegative degrees have sum zero, so every degree is zero. $\square$

The importance of $\mathcal{B}_{\mathbb{C}}$ is that linear factors may have irrational or complex coefficients. They need not themselves be omnific integers, but they have no negative exponents, so the unit argument applies.

Theorem 6.2 (Exact fibers of decomposable equations [02]; case (b) [01]). *Let*

$$F(\mathbf{X}) = \prod_{j=1}^s L_j(\mathbf{X})^{e_j}, \quad L_j(\mathbf{X}) = \beta_j + \ell_j(\mathbf{X}),$$

where $\beta_j \in \mathbb{C}$, each ℓ_j is a complex linear form in n variables, and each $e_j \geq 1$ is an ordinary integer. A nonzero scalar prefactor may instead be divided into the level. Fix $c \in \mathbb{C}^\times$ and put

$$\mathcal{K} = \{\mathbf{v} \in \mathbb{R}^n : \ell_j(\mathbf{v}) = 0 \text{ for every } j\}.$$

(a) The solutions of $F(\mathbf{x}) = c$ in $\mathbf{Oz}^n$ are exactly

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}^n, \quad F(\mathbf{a}) = c, \quad \mathbf{u} \in \Pi^n, \quad \ell_j(\mathbf{u}) = 0 \text{ for all } j. \quad (6.1)$$

Equivalently, at each exponent the real coefficient vector of $\mathbf{u}$ lies in $\mathcal{K}$; if $\mathbf{v}_1, \dots, \mathbf{v}_e$ is a real basis of $\mathcal{K}$, the fiber above $\mathbf{a}$ is $\mathbf{a} + \{\sum_k t_k \mathbf{v}_k : t_k \in \Pi\}$. If $\mathcal{K} = 0$, all omnific solutions are ordinary.

(b) If the linear parts $\ell_1, \dots, \ell_s$ have column rank n over $\mathbb{C}$, then every solution $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$ of $F(\mathbf{x}) = c$ lies in $\mathbb{C}^n$.

Proof. For $\mathbf{x} \in \mathcal{B}_{\mathbb{C}}^n$, every $L_j(\mathbf{x})$ is in $\mathcal{B}_{\mathbb{C}}$, and if $F(\mathbf{x}) = c \neq 0$, [lemma 6.1](#) makes each $L_j(\mathbf{x})$ constant.

(a) Write $\mathbf{x} = \mathbf{a} + \mathbf{u}$ coordinatewise with $\mathbf{a} \in \mathbb{Z}^n$ and $\mathbf{u} \in \Pi^n$. Then $L_j(\mathbf{x}) = L_j(\mathbf{a}) + \ell_j(\mathbf{u})$, where the first summand is constant and the second lies in $\Pi_{\mathbb{C}}$; uniqueness of normal forms forces $\ell_j(\mathbf{u}) = 0$, and specialization gives $F(\mathbf{a}) = c$. Conversely those conditions give $L_j(\mathbf{a} + \mathbf{u}) = L_j(\mathbf{a})$ and hence $F(\mathbf{a} + \mathbf{u}) = c$. At each exponent of the (set-sized, reverse-well-ordered) union of the coordinate supports, the equations $\ell_j(\mathbf{u}) = 0$ say exactly that the real coefficient vector $\mathbf{u}_{\gamma} \in \mathbb{R}^n$ at that exponent satisfies $\ell_j(\mathbf{u}_{\gamma}) = 0$ for all j , that is, lies in $\mathcal{K}$. Expressing each coefficient vector in the basis $\mathbf{v}_k$ produces t_k whose supports lie in the same union; if $\mathcal{K} = 0$, every coefficient vector vanishes.

(b) Select n of the ℓ_j whose coefficient matrix is invertible over $\mathbb{C}$, subtract the constant terms and solve this ordinary linear system: all coordinates are complex constants. $\square$

The kernel hypothesis in (a) is exactly what matters. A single-factor form can have infinite fibers,

$$(x - y)^2 = 1 \iff x = y \pm 1, \quad y \in \mathbf{Oz},$$

and the two-variable equation $X^d = 1$ forces X to be constant but says nothing about Y . The real kernel in (a) need not equal the complex kernel in (b): $L(X, Y) = X + iY$ has zero kernel on $\mathbb{R}^2$, so its level 1 has the unique omnific point $(1, 0)$. In $\mathcal{B}_{\mathbb{C}}^2$ the same level contains $(1 - it, t)$ for every $t \in \mathcal{B}_{\mathbb{C}}$ (compare [remark 6.5](#)). On the zero level, even independent factors permit infinite points, since the unit argument cannot be applied to a product equal to zero: $(\sqrt{2}\omega)^2 - 2\omega^2 = 0$.

Gaussian variables, the converse, and the kernel criterion. Manuscript 13 of the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` (batch-25 item 03, written [13] here; see [section 1.2](#)) proves the same fiber theorem, with repeated factors in place of the exponents e_j (an equivalent formulation), for the pairs $(\mathbb{Z}, \mathbb{R})$ and $(\mathbb{Z}[i], \mathbb{C})$. Its real case is [theorem 6.2\(a\)](#), printed once above. Its Gaussian case, its converse and its remark on real kernels are printed here, once; source 13's other results are printed in the sibling report. Write

$$\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}} = \{a + ib : a, b \in \mathbf{Oz}\}$$

for the *Gaussian omnific integers*. A finite union of reverse-well-ordered supports is reverse well ordered, so the two descriptions agree, and the constant coefficient of $a + ib$ is $\text{ct}(a) + i \text{ct}(b) \in \mathbb{Z}[i]$. The name refers to this explicitly specified ring, not to another integer part of the surcomplex field [07].

Theorem 6.3 (Exact fibers over the Gaussian omnific integers [13]). *Let $F = \prod_j L_j^{e_j}$, $L_j = \beta_j + \ell_j$ and $c \in \mathbb{C}^{\times}$ be as in [theorem 6.2](#), and put*

$$\mathcal{K}_{\mathbb{C}} = \{\mathbf{v} \in \mathbb{C}^n : \ell_j(\mathbf{v}) = 0 \text{ for every } j\}.$$

The solutions of $F(\mathbf{x}) = c$ in $\mathbf{Oz}[i]^n$ are exactly

$$\mathbf{x} = \mathbf{a} + \mathbf{u}, \quad \mathbf{a} \in \mathbb{Z}[i]^n, \quad F(\mathbf{a}) = c, \quad \mathbf{u} \in \Pi_{\mathbb{C}}^n, \quad \ell_j(\mathbf{u}) = 0 \text{ for all } j. \quad (6.2)$$

Equivalently, at each exponent the complex coefficient vector of $\mathbf{u}$ lies in $\mathcal{K}_{\mathbb{C}}$; if $\mathbf{v}_1, \dots, \mathbf{v}_e$ is a complex basis of $\mathcal{K}_{\mathbb{C}}$, the fiber above $\mathbf{a}$ is $\mathbf{a} + \{\sum_k t_k \mathbf{v}_k : t_k \in \Pi_{\mathbb{C}}\}$. Fibers above distinct $\mathbf{a}$ are disjoint. If $\mathcal{K}_{\mathbb{C}} = 0$, that is, if the linear parts have rank n over $\mathbb{C}$, every solution in $\mathbf{Oz}[i]^n$ lies in $\mathbb{Z}[i]^n$.

Proof. Write $\mathbf{x} = \mathbf{a} + \mathbf{u}$ with $\mathbf{a} \in \mathbb{Z}[i]^n$ and $\mathbf{u} \in \Pi_{\mathbb{C}}^n$. If $F(\mathbf{x}) = c$, [lemma 6.1](#) makes every $L_j(\mathbf{x}) = L_j(\mathbf{a}) + \ell_j(\mathbf{u})$ a complex constant; since $\ell_j(\mathbf{u}) \in \Pi_{\mathbb{C}}$, uniqueness of normal forms gives $\ell_j(\mathbf{u}) = 0$, and applying ct gives $F(\mathbf{a}) = c$. Conversely these conditions give $L_j(\mathbf{x}) = L_j(\mathbf{a})$ for every j , hence $F(\mathbf{x}) = c$. The coefficient of ω^γ in $\ell_j(\mathbf{u})$ is $\ell_j(\mathbf{u}_\gamma)$, where $\mathbf{u}_\gamma \in \mathbb{C}^n$ is the coefficient vector of $\mathbf{u}$ at γ ; so $\ell_j(\mathbf{u}) = 0$ for all j says exactly that every $\mathbf{u}_\gamma$ lies in $\mathcal{K}_{\mathbb{C}}$, and expanding each $\mathbf{u}_\gamma$ in the basis gives series t_k supported in the (set-sized, reverse-well-ordered) union of the coordinate supports. Disjointness holds because the constant coefficient of a normal form is unique. If $\mathcal{K}_{\mathbb{C}} = 0$, then $\mathbf{u} = 0$. $\square$

Corollary 6.4 (The kernel criterion is exact [13]). *Suppose the ordinary fiber $\{\mathbf{a} \in \mathbb{Z}^n : F(\mathbf{a}) = c\}$ is nonempty. Then $F(\mathbf{x}) = c$ has a solution in $\mathbf{Oz}^n$ outside $\mathbb{Z}^n$ if and only if the real kernel $\mathcal{K}$ of [theorem 6.2](#) is nonzero. Likewise, if $F(\mathbf{a}) = c$ for some $\mathbf{a} \in \mathbb{Z}[i]^n$, it has a solution in $\mathbf{Oz}[i]^n$ outside $\mathbb{Z}[i]^n$ if and only if $\mathcal{K}_{\mathbb{C}} \neq 0$.*

Proof. If $\mathbf{v} \neq 0$ lies in the relevant kernel and $\mathbf{a}$ is an ordinary point, then $\mathbf{a} + \mathbf{v}\omega$ is a point of the same fiber with an infinite coordinate. The other implications are [theorem 6.2\(a\)](#) and [theorem 6.3](#). $\square$

So, subject to one ordinary point, the linear-kernel criterion is not merely sufficient: it detects exactly whether a nonzero constant fiber acquires infinite directions [13].

Remark 6.5 (Real and complex kernels [13]). For omnific variables the relevant kernel is $\mathcal{K} = \ker_{\mathbb{C}} \cap \mathbb{R}^n$, which can vanish although the complex rank of the linear parts is smaller than n ; the rank condition of [theorem 6.2\(b\)](#) is sufficient, not necessary, for real variables. Source 13's example is the single form $X + iY$, which is injective on $\mathbb{R}^2$: every omnific solution of $x + iy = c$ is ordinary. For Gaussian variables the criterion is the complex kernel. For instance [merge], $\mathcal{K}_{\mathbb{C}} = \mathbb{C}(1, i)$ for $X + iY$, and $(1 + \omega, i\omega) \in \mathbf{Oz}[i]^2$ solves $X + iY = 1$, since $(1 + \omega) + i \cdot i\omega = 1$.

Example 6.6 (A rank-deficient level [13]). Let $F(X, Y, Z) = (X + Z + \sqrt{2}Y)(X + Z - \sqrt{2}Y) = (X + Z)^2 - 2Y^2$. The common real kernel of the two linear parts is $\mathcal{K} = \mathbb{R}(1, 0, -1)$, so by [theorem 6.2\(a\)](#) the omnific solutions of $F = 1$ are exactly

$$(x, y, z) = (a + u, b, c - u), \quad u \in \Pi, \quad a, b, c \in \mathbb{Z}, \quad (a + c)^2 - 2b^2 = 1.$$

For instance $(3 - \omega, 2, \omega)$ is a nonordinary solution. It is the common kernel of *all* linear parts, not the nonzero level alone, that governs rigidity.

6.2 Binary forms, Pell equations, and conics

Theorem 6.7 (Binary-form rigidity [05], generalizing [01, 02]). *Let $F \in \mathbb{C}[X, Y]$ be homogeneous with at least two distinct projective linear factors over $\mathbb{C}$, and let $c \in \mathbb{C}^\times$. If $x, y \in \mathcal{B}_{\mathbb{C}}$ satisfy $F(x, y) = c$, then $x, y \in \mathbb{C}$. In particular, for $F \in \mathbb{Z}[X, Y]$ and $c \in \mathbb{Z} \setminus \{0\}$,*

$$\{(x, y) \in \mathbf{Oz}^2 : F(x, y) = c\} = \{(x, y) \in \mathbb{Z}^2 : F(x, y) = c\}.$$

Proof. Factor $F = \alpha \prod_{j=1}^d L_j$ over $\mathbb{C}$ with $\alpha \in \mathbb{C}^\times$ and nonzero linear forms repeated according to multiplicity. Two distinct projective factors have linearly independent coefficient vectors, so the linear parts have rank two; apply [theorem 6.2\(b\)](#). Constant omnific coordinates are integers. $\square$

Sources 01 and 02 proved the statement for $F \in \mathbb{Z}[X, Y]$, $c \in \mathbb{Z}$ and $\mathbf{Oz}$ -points; the complex form is 05's. The hypothesis excludes a pure power αL^d , and that exclusion is necessary: fixing $L(x, y)$ to a constant leaves a free infinite direction in the kernel of L . The requirement $c \neq 0$ is also necessary, since zero levels contain lines through the origin. The hypothesis is weaker than squarefreeness in degree at least three: X^2Y qualifies.

Corollary 6.8 (Thue levels and equal powers [01, 02, 05]). *If an integer binary form has at least two distinct projective factors, its nonzero ordinary constant levels have no additional omnific points. This applies to every irreducible binary form over $\mathbb{Q}$ of degree at least two, hence to Thue equations, and to $x^d - y^d = c$ with $d \geq 2$ and $c \in \mathbb{Z} \setminus \{0\}$.*

Proof. In characteristic zero an irreducible binary form of degree at least two has distinct projective roots over $\mathbb{C}$, and $X^d - Y^d$ has distinct linear factors. Apply [theorem 6.7](#). $\square$

The conclusion identifies the omnific solution class with the classical integer solution set; it is not a new proof of ordinary finiteness, gives no bound for the ordinary solutions, and supplies no algorithm for them. No finiteness theorem of Thue or Baker is used.

Corollary 6.9 (Pell rigidity [01, 02, 05]). *For ordinary integers $D \neq 0$ and $c \neq 0$, every solution in $\mathbf{Oz}^2$ of $x^2 - Dy^2 = c$ is an ordinary integer solution. In particular*

$$\{(u, v) \in \mathbf{Oz}^2 : u^2 - 2v^2 = 1\} = \{(u, v) \in \mathbb{Z}^2 : u^2 - 2v^2 = 1\}.$$

Proof. Over $\mathbb{C}$, the factors $X - \sqrt{D}Y$ and $X + \sqrt{D}Y$ are distinct because $D \neq 0$. Apply [theorem 6.7](#). $\square$

Source 02 states this corollary for $D > 0$ only; its proof, like those of 01 and 05, uses only $D \neq 0$, and the conclusion holds whether or not D is a square. Source 13 states the corollary for positive nonsquare D (as $X^2 - dY^2 = m$ with $m \neq 0$); its own fiber theorem, like [theorem 6.2](#), covers every $D \neq 0$, and the statement for $D \neq 0$ stands here. “Rigid” means no new points, not a finite number of old points: the classical Pell equation has ordinary solutions of arbitrarily large finite size, but no single infinite omnific solution, and there is no “Pell solution at index ω ” obtained by substituting an omnific index into a finite recurrence. This is a first warning against treating $\mathbf{Oz}$ as an elementary extension of ordinary integer arithmetic.

Example 6.10 (The full class for $D = 2$, $c = 1$ [05]). Let $X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k$ for ordinary $k \geq 0$. Every omnific solution of $x^2 - 2y^2 = 1$ is $(\pm X_k, \pm Y_k)$ with independent signs. Indeed, [corollary 6.9](#) first makes x, y ordinary. If $y = 0$, then $x = \pm 1$, giving $k = 0$. Otherwise, after changing signs suppose $x, y > 0$; then $y \geq 2$, since $y = 1$ would need $x^2 = 3$. Multiplication by $3 - 2\sqrt{2}$ gives the integer Pell pair $X = 3x - 4y$, $Y = 3y - 2x$ with $X > 0$ (as $x > \sqrt{2}y > 4y/3$) and $0 \leq Y < y$ (as $x^2 = 2y^2 + 1 \leq 9y^2/4$ and $x > y$). Ordinary descent on y reaches $(1, 0)$, and reversing it gives the asserted powers.

In the full surreal field the contrast is visible without solving a recurrence [01; also 07, 13]. For $D > 0$, $c \neq 0$ and $s \in \mathbf{No}^\times$, the pair $x = (s + c/s)/2$, $y = (s - c/s)/(2\sqrt{D})$ satisfies $x^2 - Dy^2 = c$. Taking $s = \omega$ produces negative-exponent terms, so neither coordinate is an omnific integer, and discarding those terms changes the norm from c to zero. A surreal

solution cannot be turned into an omnific solution by unverified truncation. Source 07 makes the same point for any nonconstant monomial of a Hahn field, and source 13 notes that the negative-exponent terms supply exactly the inverses missing from $\mathcal{B}_{\mathbb{R}}$: the rigidity theorems concern the nonnegative-support rings, not non-Archimedean fields in general.

Proposition 6.11 (The zero Pell fiber [01; also 13]). *If D is a positive nonsquare ordinary integer, the solutions of $x^2 - Dy^2 = 0$ in $\mathbf{Oz}^2$ are exactly $(\pm\sqrt{D}t, t)$ with $t \in \Pi$.*

Proof. Factor in $\mathbf{No}$ to obtain $x = \pm\sqrt{D}y$. Write $y = m + j$ with $m \in \mathbb{Z}$ and $j \in \Pi$. The constant coefficient of x is $\pm\sqrt{D}m$, which is an integer only if $m = 0$. Conversely, Π is an $\mathbb{R}$ -vector space, so all displayed pairs belong to $\mathbf{Oz}^2$ and solve the equation. $\square$

Corollary 6.12 (Nonsingular central conics [05]). *Let $F(X, Y) = q(X, Y) + \ell(X, Y) + d \in \mathbb{R}[X, Y]$, where the quadratic form q has rank two, and let $\mathbf{h} \in \mathbb{R}^2$ be the center, so that $F(\mathbf{h} + \mathbf{u}) = q(\mathbf{u}) + F(\mathbf{h})$. If $F(\mathbf{h}) \neq 0$, every solution of $F = 0$ in $\mathcal{B}_{\mathbb{R}}^2$ is constant; for integer coefficients, every omnific solution is ordinary.*

Proof. A rank-two binary quadratic form has two distinct projective factors over $\mathbb{C}$. The translated coordinates $\mathbf{u} = (x, y) - \mathbf{h}$ lie in $\mathcal{B}_{\mathbb{R}}^2$ and satisfy $q(\mathbf{u}) = -F(\mathbf{h}) \neq 0$. Apply [theorem 6.7](#) and translate back. $\square$

This covers ellipses and hyperbolas with a nonzero centered level. Parabolas behave differently: $Y = X^2$ has the infinite family (t, t^2) .

Remark 6.13 (Power differences with a common factor [01]). Source 01 proved that every omnific solution of $x^m - y^n = c \neq 0$ is ordinary when $d = \gcd(m, n) > 1$: the binary form $X^d - Y^d$ in $X = x^{m/d}$, $Y = y^{n/d}$ is rigid by [theorem 6.7](#), and an element of $\mathbf{Oz}$ with an ordinary constant positive power is ordinary (by degree additivity if the power is nonzero, by the domain property otherwise). Source 01 stated explicitly that this argument does not classify coprime cases such as $x^2 - y^3 = 1$. All exponent pairs are covered by [theorem 7.4](#) of source 05.

6.3 Number-field and étale norm equations

Here a norm form is first constructed as a polynomial over $\mathbb{Q}$, then evaluated on omnific coordinates. We do not embed the number field K into $\mathbf{No}$; such an embedding need not exist. Equivalently, form the determinant of the multiplication matrix in a $\mathbb{Q}$ -basis, with indeterminate coordinates. Its complex factorization is used only after this polynomial is formed. The same applies to the algebra E of [theorem 6.15](#), whose norm is factored over an algebraic closure of the coefficient field.

Theorem 6.14 (Norm-form rigidity [01, 02]). *Let $K/\mathbb{Q}$ be a number field of degree d , let $e_1, \dots, e_d$ be a $\mathbb{Q}$ -basis, and let $N(\mathbf{X}) = N_{K/\mathbb{Q}}(\sum_{j=1}^d e_j X_j)$. For every $c \in \mathbb{Q}^\times$, every solution $\mathbf{x} \in \mathbf{Oz}^d$ of $N(\mathbf{x}) = c$ has all coordinates in $\mathbb{Z}$. If the basis consists of algebraic integers (for example a power basis of an algebraic integer θ generating K , or an integral basis), then $N \in \mathbb{Z}[\mathbf{X}]$, and the statement applies to the usual integer norm equations.*

Proof. The d embeddings $\sigma_1, \dots, \sigma_d : K \rightarrow \mathbb{C}$ give

$$N(\mathbf{X}) = \prod_{i=1}^d \left(\sum_{j=1}^d \sigma_i(e_j) X_j \right).$$

The embedding matrix $(\sigma_i(e_j))$ is invertible: for a power basis $1, \theta, \dots, \theta^{d-1}$ of a primitive element of the separable extension $K/\mathbb{Q}$ it is the Vandermonde matrix of the distinct conjugates of θ , and a basis change over $\mathbb{Q}$ preserves invertibility. The forms therefore have zero common kernel and [theorem 6.2](#) applies. If the e_j are algebraic integers, each coefficient of N is fixed by the Galois group of a normal closure, hence rational, and is an integer polynomial in conjugates of the e_j , hence an algebraic integer; so it is an ordinary integer [02]. $\square$

For example, consider the norm form for $K = \mathbb{Q}(\sqrt[3]{2})$:

$$N(X + Y\sqrt[3]{2} + Z\sqrt[3]{4}) = X^3 + 2Y^3 + 4Z^3 - 6XYZ.$$

For every $c \in \mathbb{Z} \setminus \{0\}$, every omnific solution of $X^3 + 2Y^3 + 4Z^3 - 6XYZ = c$ is an ordinary integer triple. This is a three-variable rigidity statement on an unbounded real surface, unlike the indefinite quadratic forms of [section 8](#): dimension alone does not force infinite omnific points. The identity follows from the product over the three conjugates or from the determinant of multiplication by $X + Y\theta + Z\theta^2$; the suites of 01 and 02 check it exactly. Source 13 computes the same determinant (its suite checks it too) and records the Gaussian analogue: every Gaussian omnific solution is a Gaussian integer triple, by [theorem 6.15\(b\)](#) below. No claim is made to enumerate or effectively bound the ordinary solutions of this cubic.

Sources 07 and 13 prove norm rigidity in two further directions: source 07 over an arbitrary base field and in intermediate rings, source 13 for finite étale algebras (finite products of number fields) and with Gaussian omnific variables. The mechanism is that of [theorem 6.14](#); the two generalizations are combined here. The statement uses the general nonnegative-support rings $\mathcal{A}_{\mathbb{k}}(\Gamma)$ and the intermediate rings A of [section 11.1](#): for a unital subring $\mathfrak{o}$ of a field $\mathbb{k}$ of characteristic zero, A is a ring with $\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma)$ and $A \cap \mathbb{k} = \mathfrak{o}$. Examples are $\mathbf{Oz}$ (with $\mathfrak{o} = \mathbb{Z}$, $\mathbb{k} = \mathbb{R}$) and $\mathbf{Oz}[i]$ (with $\mathfrak{o} = \mathbb{Z}[i]$, $\mathbb{k} = \mathbb{C}$).

Theorem 6.15 (Étale norm rigidity in intermediate rings [07, 13] [merge]). *Let $\mathbb{k}$ be a field of characteristic zero, $K \subseteq \mathbb{k}$ a subfield, E a finite étale K -algebra of dimension r , that is, a finite product of finite separable field extensions of K , and $e_1, \dots, e_r$ a K -basis of E . Put $N(\mathbf{X}) = N_{E/K}(\sum_{j=1}^r e_j X_j) \in K[X_1, \dots, X_r]$. Let Γ be an ordered abelian group and A an intermediate ring for $\mathfrak{o} \subseteq \mathbb{k}$. If $\mathbf{x} \in A^r$ and $N(\mathbf{x}) = a \in K^\times$, then every x_j lies in $\mathfrak{o}$. In particular:*

- (a) *for a finite étale $\mathbb{Q}$ -algebra, every solution in $\mathbf{Oz}^r$ of $N(\mathbf{x}) = a \neq 0$ lies in $\mathbb{Z}^r$;*
- (b) *for a finite étale $\mathbb{Q}$ - or $\mathbb{Q}(i)$ -algebra, every solution in $\mathbf{Oz}[i]^r$ of $N(\mathbf{x}) = a \neq 0$ lies in $\mathbb{Z}[i]^r$.*

Proof. Let $\bar{\mathbb{k}}$ be an algebraic closure of $\mathbb{k}$. Since E is étale, there are exactly r homomorphisms of K -algebras $\sigma_1, \dots, \sigma_r : E \rightarrow \bar{\mathbb{k}}$, and together they identify $E \otimes_K \bar{\mathbb{k}}$ with $\bar{\mathbb{k}}^r$. In these coordinates multiplication by $\sum_j e_j X_j$ is diagonal, so taking its determinant gives the polynomial identity

$$N(\mathbf{X}) = \prod_{\ell=1}^r \left(\sum_{j=1}^r \sigma_\ell(e_j) X_j \right) \quad \text{in } \bar{\mathbb{k}}[X_1, \dots, X_r].$$

The matrix $(\sigma_\ell(e_j))$ is invertible: the square of its determinant is the discriminant $\det(\text{Tr}_{E/K}(e_i e_j))$ of the basis, which is nonzero because E is étale. (Source 13 argues with power bases: for a field the matrix is a Vandermonde matrix of distinct conjugates, for a product it is block diagonal, and a change of basis over K preserves invertibility.) Coefficient extension embeds $\mathcal{A}_{\mathbb{k}}(\Gamma)$ in $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$ without changing any support. Each factor $\sum_j \sigma_\ell(e_j) x_j$ lies in $\mathcal{A}_{\bar{\mathbb{k}}}(\Gamma)$, and their product is the nonzero constant a , so every factor is constant by [lemma 11.1](#). Inverting the matrix shows that every x_j is a constant; as x_j has its coefficients in $\mathbb{k}$, it lies in $\mathbb{k}$, hence in $A \cap \mathbb{k} = \mathfrak{o}$. For (a) and (b) note that $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$ and $\mathbf{Oz}[i] \cap \mathbb{C} = \mathbb{Z}[i]$. $\square$

[Theorem 6.14](#) is the case $K = \mathbb{Q}$, E a number field, $A = \mathbf{O}_Z$. Source 07 proved the theorem for a finite separable field extension of an arbitrary subfield $K \subseteq \mathbb{k}$ and intermediate rings; source 13 for finite étale $\mathbb{Q}$ -algebras with $\mathfrak{o} = \mathbb{Z}$ and $\mathfrak{o} = \mathbb{Z}[i]$, and noted that the same proof applies to étale $\mathbb{Q}(i)$ -algebras with Gaussian omnific variables. The combined statement, with its proof above, is the merge's; each ingredient is in one of the two sources. Two cautions from source 07 apply. When some coefficients of N in K do not lie in A , the equation is read in the ambient series ring; to use it literally as a formula with parameters in $\mathfrak{o}$, one assumes that its coefficients and right-hand side lie in $\mathfrak{o}$. The nonzero right-hand side and separability are essential; this is not a theorem that every algebraic variety has only constant points. The norm on the left is evaluation of a finite polynomial, not a newly postulated norm on a class-sized algebra [13]. Pell rigidity in intermediate rings ([corollary 11.5](#)) is the case $E = \mathbb{Q}(\sqrt{2})$.

6.4 An algebraic-geometric formulation

Let V be an affine complex variety whose coordinate ring $\mathbb{C}[V]$ is generated, as a $\mathbb{C}$ -algebra, by units. Every $\mathcal{B}_{\mathbb{C}}$ -point of V is then a constant $\mathbb{C}$ -point [01, 02]. Indeed, a $\mathbb{C}$ -algebra map $\mathbb{C}[V] \rightarrow \mathcal{B}_{\mathbb{C}}$ sends each generating unit to an element of $\mathcal{B}_{\mathbb{C}}^{\times} = \mathbb{C}^{\times}$, so its entire image lies in $\mathbb{C}$. This includes closed subvarieties of an algebraic torus $(\mathbb{C}^{\times})^r$ in their very-affine coordinate presentation. The norm argument is a concrete linear-coordinate instance: a nonzero product relation forces its factors into a constant torus. It is not a claim that every affine variety is constant over $\mathcal{B}_{\mathbb{C}}$; the affine line has the full class $\mathcal{B}_{\mathbb{C}}$ of points.

7 Euler derivations: separated powers and unimodular Fermat

Binary forms can be factored into linear forms, but equations such as $Y^2 = X^3 + 17$ cannot be handled that way. The following local tool, due to source 05, gives short arguments without any finite-support hypothesis.

7.1 Support-preserving Euler derivations

Fix a set-sized ordered divisible group $H \subseteq \mathbf{No}$ and a $\mathbb{Q}$ -linear map $\lambda : H \rightarrow \mathbb{R}$. On the real or complex Hahn field with exponents in H , define

$$\partial_{\lambda} \left(\sum_{h \in H} a_h \omega^h \right) = \sum_{h \in H} \lambda(h) a_h \omega^h. \quad (7.1)$$

This is a local, freely chosen Euler derivation. It is not a canonical surreal derivation and not the Berarducci–Mantova derivation. Under $t = \omega^{-1}$ its rule is $\partial_{\lambda}(t^{\gamma}) = -\lambda(\gamma)t^{\gamma}$; the minus sign comes from reversing exponents. It must not be identified with the differential equations report's $\partial_{\tau} = -\omega\partial$, whose real-exponent rule is $\partial_{\tau}(t^{\gamma}) = \gamma t^{\gamma}$.

Lemma 7.1 (Euler derivations [05]). *The map ∂_{λ} is a derivation, kills every real or complex constant, and never introduces a new supported exponent; so $\deg_{\omega}(\partial_{\lambda}f) \leq \deg_{\omega}(f)$ whenever $\partial_{\lambda}f \neq 0$. It restricts to derivations of $\mathcal{B}_{\mathbb{R},H}$ and $\mathcal{B}_{\mathbb{C},H}$.*

Proof. The new support is a subset of the old one and is thus reverse well ordered. Additivity is immediate, and $\lambda(0) = 0$ kills constants. For a fixed exponent h , the coefficient convolution is finite and $\lambda(a+b) = \lambda(a) + \lambda(b)$, so

$$[\omega^h] \partial_{\lambda}(fg) = \sum_{a+b=h} (\lambda(a) + \lambda(b)) f_a g_b = [\omega^h] ((\partial_{\lambda}f)g + f(\partial_{\lambda}g)).$$

This proves the Leibniz rule coefficientwise; no interchange of conditionally convergent sums is involved. The degree and restriction statements follow from support containment. $\square$

Lemma 7.2 (Detection of nonconstants [05]). *Every nonconstant element of a Hahn field with exponent group H is detected by some ∂_λ . If $f \in \mathcal{B}_{\mathbb{C},H}$ has positive leading exponent α , one may choose $\lambda(\alpha) = 1$; then $\deg_\omega(\partial_\lambda f) = \alpha$.*

Proof. Choose a nonzero supported exponent h . A $\mathbb{Q}$ -linear functional on $\mathbb{Q}h$ taking h to 1 extends to H by extending a rational basis; H is a set, so choice for sets suffices. The derivative has nonzero coefficient at h . Choosing the leading exponent gives the final assertion. $\square$

For $\partial = \partial_\lambda$ define the two-term Wronskian $W_\partial(f, g) = f \partial g - g \partial f$. For $f \neq 0$ it vanishes exactly when $\partial(g/f) = 0$; it need not detect whether g/f is an absolute constant for one fixed ∂ , which is why [lemma 7.2](#) lets the derivation be chosen for the given ratio. For example, on $H = \mathbb{Q} + \mathbb{Q}\sqrt{2}$ the functional $\lambda(a + b\sqrt{2}) = a$ kills the nonconstant $\omega^{\sqrt{2}}$. Detection uses the family of functionals, not a claim that each individual derivation has only scalar constants.

Lemma 7.3 (A strict Wronskian bound [05]). *For nonzero $f, g \in \mathcal{B}_{\mathbb{C},H}$ with $W_\partial(f, g) \neq 0$, $\deg_\omega W_\partial(f, g) \leq \deg_\omega f + \deg_\omega g$. If $\deg_\omega f = \deg_\omega g = \alpha$, the inequality is strict: $\deg_\omega W_\partial(f, g) < 2\alpha$.*

Proof. The weak bound follows from [lemmas 2.4](#) and [7.1](#). If the leading coefficients of f, g are a, b at exponent α , the two possible contributions at 2α are $a\lambda(\alpha)b$ and $b\lambda(\alpha)a$, which cancel; every other contribution has smaller exponent. A nonzero Wronskian has a greatest supported exponent, by reverse well-ordering, so its degree is strictly below 2α ; one does not replace that maximum by an unattained supremum. $\square$

7.2 Separated powers

Theorem 7.4 (Separated-power rigidity [05]). *Let $a, b, c \in \mathbb{C} \setminus \{0\}$ and let $m, n \geq 2$ be ordinary integers. Every solution in $\mathcal{B}_{\mathbb{C}}^2$ of*

$$ax^m + by^n = c \tag{7.2}$$

has $x, y \in \mathbb{C}$. In particular, every solution in $\mathcal{B}_{\mathbb{R}}^2$ is real. Consequently, if a, b, c are nonzero ordinary integers, every solution in $\mathbf{Oz}^2$ is an ordinary integer solution.

Proof. If one coordinate is constant, the other is constant by [proposition 2.8](#). Suppose both are nonconstant, with $\alpha = \deg_\omega x > 0$ and $\beta = \deg_\omega y > 0$. Cancellation of the positive leading degree in (7.2) forces $m\alpha = n\beta$. Take a common set-sized workspace H for x, y ([lemma 2.6](#)) and choose $\partial = \partial_\lambda$ with $\lambda(\alpha) = 1$, so $\partial x \neq 0$. Differentiating gives

$$amx^{m-1}\partial x + bny^{n-1}\partial y = 0, \tag{7.3}$$

and since $x \neq 0$ also $\partial y \neq 0$.

We work in $\mathcal{B}_{\mathbb{C},H}$, in which a, b, c, m, n are units. Modulo the principal ideal (x^{m-1}) , (7.2) says $by^n = c$, so the image of y is a unit; reducing (7.3) then shows that ∂y vanishes in the quotient. Equivalently $\partial y = x^{m-1}q$ with $q \in \mathcal{B}_{\mathbb{C},H}$, and $q \neq 0$ because $\partial y \neq 0$. Nonnegative degree and [lemma 7.1](#) give

$$(m-1)\alpha \leq \deg_\omega(\partial y) \leq \beta = \frac{m}{n}\alpha.$$

Since $\alpha > 0$, this gives $mn - m - n \leq 0$, that is $(m-1)(n-1) \leq 1$, which for $m, n \geq 2$ forces $m = n = 2$. In that case $aX^2 + bY^2$ has two distinct projective linear factors over $\mathbb{C}$ because

$ab \neq 0$, and [theorem 6.7](#) contradicts the assumed nonconstancy. Constant omnific coordinates are integers. $\square$

Source 05 states the real case. The proof above gives its complex extension: it uses nonnegative support degrees and nonzero scalar units, not an order on the coefficients. The real and omnific conclusions follow by intersection with $\mathcal{B}_{\mathbb{R}}$ and $\mathbf{Oz}$.

The quotient-ring step replaces any appeal to unique factorization: if y becomes a unit modulo x^{m-1} , a product of ∂y with a unit can vanish there only if ∂y itself does. The argument is valid in a non-Noetherian ring and for supports of arbitrary set-sized order type.

Remark 7.5 (A direct divisibility identity [merge]). The divisibility $x^{m-1} \mid \partial y$ also follows from an explicit identity. Put $f = ax^m + by^n$. For every derivation ∂ that kills a and b ,

$$ax^{m-1}(nx\partial y - my\partial x) = n f \partial y - y \partial f,$$

as one sees by expanding $\partial f = amx^{m-1}\partial x + bny^{n-1}\partial y$: the right side is $anx^m\partial y + bny^n\partial y - amx^{m-1}y\partial x - bny^n\partial y$. Without that hypothesis, the right side has the additional term $-y((\partial a)x^m + (\partial b)y^n)$. The Euler derivations kill all scalar coefficients, including c . On a solution of (7.2), therefore, $f = c$ and $\partial f = 0$, so $nc\partial y = x^{m-1} \cdot a(nx\partial y - my\partial x)$, and since nc is a unit of $\mathcal{B}_{\mathbb{C},H}$, x^{m-1} divides ∂y . This route avoids the quotient ring. The suite of 05 checks a related identity only for $a = 1$, $b = -1$; the identity above was checked symbolically for general a, b and $2 \leq m, n \leq 7$ when this report was assembled.

Corollary 7.6 (Mordell and Fermat curves [05]). *The following equations have no infinite omnific solutions; their omnific solution classes equal their ordinary integer solution sets:*

$$\begin{aligned} x^m - y^n &= c \quad (m, n \geq 2, c \in \mathbb{Z} \setminus \{0\}), \\ y^2 &= x^3 + k \quad (k \in \mathbb{Z} \setminus \{0\}), \quad x^n + y^n = 1 \quad (n \geq 2). \end{aligned}$$

Proof. Use $(a, b) = (1, -1)$ or $(-1, 1)$ as appropriate in [theorem 7.4](#), and $(a, b, c) = (1, 1, 1)$ for the last equation. $\square$

In particular $Y^2 = X^3 + 1$, which source 01 listed as an open question ([section 18.1](#)), has only its ordinary integer points in $\mathbf{Oz}$. The theorem reduces the omnific problem to ordinary integers; it does not by itself bound or list the integer solutions. A finite computation can verify particular solutions of $y^2 = x^3 + 1$, such as $(-1, 0)$, $(0, \pm 1)$, $(2, \pm 3)$, but a finite search is not a proof that such a list is complete, and the suite of 05 checks only the displayed identities.

The condition $c \neq 0$ cannot be removed: $x = t^n$, $y = t^m$ solves $x^m - y^n = 0$ for every $t \in \mathbf{Oz}$. Nor can an exponent equal one: $y = x^m - c$ is an infinite family. The theorem treats fixed ordinary exponents and says nothing about variable omnific exponents. A general elliptic equation $y^2 = x^3 + ax + b$ with $a \neq 0$ is outside it: the differentiated equation has additional terms, and the divisibility by a high power of x no longer follows. This is a boundary of the proof, not a reason to extend the statement to all elliptic curves.

7.3 Unimodular Fermat rigidity

The homogeneous Fermat equation has abundant infinite solutions ([example 14.3](#)), none of them primitive. An explicit unit-ideal condition excludes nonconstant solutions for every exponent at least three. We use the following terms, which sources 01 and 05 define identically in substance.

Definition 7.7 (Primitive and unimodular [01, 05]). A nonzero tuple $(x_1, \dots, x_r)$ over a commutative ring R is *unimodular* if $\sum_i u_i x_i = 1$ for some $u_i \in R$, that is, $(x_1, \dots, x_r) = R$. Over the domain $\mathbf{Oz}$ it is *primitive* if its entries have no common nonunit divisor.

Unimodularity implies primitivity. The converse is not assumed in the non-PID ring $\mathbf{Oz}$, and no Bézout hypothesis is used; whenever a tuple is claimed unimodular, an explicit identity is given.

Lemma 7.8 (Pairwise comaximality [05]). *Let R be a commutative ring and suppose $x^n + y^n = z^n$ with $n \geq 1$ and $(x, y, z) = R$. Then $(x, y) = (y, z) = (z, x) = R$, and powers of any of these pairs are again comaximal.*

Proof. In $R/(x, y)$ the element z is nilpotent, as $z^n = 0$, and a unit, because a unit-ideal witness reduces to $wz = 1$. The quotient is therefore zero. The other pairs are similar. For ordinary integers $r, s \geq 1$, if $au + bv = 1$, the binomial expansion of $(au + bv)^{r+s-1}$ lies in (u^r, v^s) : each term has either at least r factors u or at least s factors v . $\square$

Theorem 7.9 (Unimodular Fermat rigidity [05]). *Let $n \geq 3$ be an ordinary integer. Suppose $x, y, z \in \mathcal{B}_{\mathbb{C}}$ are nonzero,*

$$x^n + y^n = z^n, \quad (x, y, z) = \mathcal{B}_{\mathbb{C}}.$$

Then $x, y, z \in \mathbb{C}$. In particular, a unimodular Fermat triple in $\mathbf{Oz}$ with all three coordinates nonzero has ordinary integer coordinates.

Proof. First suppose that all ratios among x, y, z are complex constants: $x = \mu z, y = \nu z$ with $\mu, \nu \in \mathbb{C}^\times$. A unit-ideal witness $ux + vy + wz = 1$ gives $z(\mu u + \nu v + w) = 1$, so z is a unit of $\mathcal{B}_{\mathbb{C}}$, and z , hence x, y , are constants by [proposition 2.5](#).

It remains to rule out a nonconstant ratio. Choose a set-sized workspace H containing the supports of x, y, z and of a unit-ideal witness; the ratios $u = x/z$ and $v = y/z$ lie in the Hahn field over H and satisfy $u^n + v^n = 1$. These ratios may have negative exponents; only the original coordinates, their derivatives and the chosen unit-ideal witness are required to lie in $\mathcal{B}_{\mathbb{C}, H}$. In particular that witness establishes unimodularity inside this ring before any comaximality argument is applied. If one of u, v is constant, the other is algebraic over $\mathbb{C}$, hence equal to one of the complex roots of its polynomial, because that polynomial splits over $\mathbb{C}$ and the Hahn field has no zero divisors. So both are nonconstant. By [lemma 7.2](#), choose ∂ with $\partial v \neq 0$; differentiating $u^n + v^n = 1$ gives $\partial u \neq 0$.

All three Wronskians are nonzero. Indeed $W_{\partial}(y, z) = -z^2 \partial v$ and $W_{\partial}(z, x) = z^2 \partial u$. If $W_{\partial}(x, y) = 0$, then $\partial u/u = \partial v/v = h$, say, and differentiating $u^n + v^n = 1$ gives $nh(u^n + v^n) = nh = 0$, so $h = 0$, contrary to $\partial v \neq 0$.

Differentiating the Fermat equation gives $x^{n-1} \partial x + y^{n-1} \partial y = z^{n-1} \partial z$. Multiplying by z and using $z^n = x^n + y^n$, then multiplying by y and using $y^n = z^n - x^n$, gives the two relations

$$x^{n-1} W_{\partial}(z, x) = y^{n-1} W_{\partial}(y, z), \quad -x^{n-1} W_{\partial}(x, y) = z^{n-1} W_{\partial}(y, z).$$

By [lemma 7.8](#), applied in $\mathcal{B}_{\mathbb{C}, H}$, there are r, s with $rx^{n-1} + sy^{n-1} = 1$; multiplying by $W_{\partial}(y, z)$ and using the first relation gives $W_{\partial}(y, z) = x^{n-1}(rW_{\partial}(y, z) + sW_{\partial}(z, x))$. The same argument with the roles of the comaximal pairs exchanged, in both relations, gives

$$x^{n-1} \mid W_{\partial}(y, z), \quad y^{n-1} \mid W_{\partial}(z, x), \quad z^{n-1} \mid W_{\partial}(x, y) \quad (7.4)$$

in $\mathcal{B}_{\mathbb{C}, H}$. Each divisibility has a nonzero quotient, since its Wronskian is nonzero. That quotient has nonnegative degree by the support restriction, which is essential for the next lower bound; divisibility in the ambient field alone would give no such bound.

Let α be the greatest of the three coordinate degrees and β, γ the other two. If $\alpha = 0$, all coordinates are constants. If $\alpha > 0$, the divisibility in (7.4) for the coordinate of degree α and the weak Wronskian bound give

$$(n-1)\alpha \leq \beta + \gamma \leq 2\alpha.$$

For $n > 3$ this is impossible. For $n = 3$ it forces $\beta = \gamma = \alpha$, and the relevant Wronskian then has degree strictly less than 2α by lemma 7.3, contradicting its divisibility by a nonzero element of degree 2α . Hence a nonconstant ratio is impossible. A unit-ideal witness in $\mathbf{Oz}$ is also one in $\mathcal{B}_{\mathbb{C}}$, and constant omnific elements are ordinary integers. $\square$

Corollary 7.10 (An omnific form of Fermat’s Last Theorem [05]). *For every ordinary integer $n \geq 3$ there is no triple of nonzero omnific integers with $x^n + y^n = z^n$ and $ux + vy + wz = 1$ for some $u, v, w \in \mathbf{Oz}$.*

Proof. By theorem 7.9 such a triple is ordinary, and the classical theorem of Wiles and Taylor–Wiles excludes it [15, 16]. Its positive-integer formulation also excludes nonzero signed triples: for even exponents take absolute values, and for odd exponents move negative terms to the other side. $\square$

The rigidity theorem is the argument supplied by source 05; the final nonexistence imports the classical theorem. The unimodular Pythagorean family of proposition 14.4 is infinite, so the threshold $n \geq 3$ is sharp. Zero coordinates are excluded intentionally: $(1, 0, 1)$ is a unimodular constant solution for every n . The theorem uses actual Bézout witnesses and actual quotient rings; it makes no assertion about triples that merely lack a common nonunit divisor (question 18.2).

8 The quadratic-level dichotomy

Binary Pell rigidity might suggest that a nonzero constant quadratic level never contains infinite points. This is false as soon as there is enough indefinite dimension. The following result gives a precise classification.

For a real quadratic form we use the polarization convention

$$B(x, y) = \frac{1}{2}(q(x + y) - q(x) - q(y)), \quad q(x) = B(x, x).$$

Thus a coefficient of $X_i X_j$ for $i \neq j$ is twice the corresponding matrix entry of B . An integral polynomial q can have half-integral bilinear matrix entries. We perform this linear algebra over $\mathbb{R}$, and extend the same finite coordinate expressions to $\mathbf{No}$. This does not assume that 2 is a unit in $\mathbf{Oz}$ or its residue rings. The parameter t in the following families is a free scalar, unrelated to the infinitesimal monomial $t = \omega^{-1}$ used for the change of exponent convention.

Theorem 8.1 (Quadratic-level dichotomy [01]; case (iii) also [02]). *Let $n \geq 1$, let $q \in \mathbb{Z}[X_1, \dots, X_n]$ be a homogeneous quadratic form, and let $c \in \mathbb{Z} \setminus \{0\}$. Write*

$$V_{\mathbf{Oz}} = \{\mathbf{x} \in \mathbf{Oz}^n : q(\mathbf{x}) = c\}, \quad V_{\mathbb{Z}} = \{\mathbf{a} \in \mathbb{Z}^n : q(\mathbf{a}) = c\}.$$

If $V_{\mathbb{Z}}$ is empty, so is $V_{\mathbf{Oz}}$. Otherwise the following alternatives are exhaustive.

- (i) *If q is nondegenerate and is either definite or of dimension at most two, then $V_{\mathbf{Oz}} = V_{\mathbb{Z}}$.*
- (ii) *If q is degenerate, every $\mathbf{a} \in V_{\mathbb{Z}}$ lies on an injective affine-linear family $\mathbf{a} + t\mathbf{v}$, $t \in \Pi$, in $V_{\mathbf{Oz}}$, with every $t \neq 0$ giving an infinite point.*

- (iii) If q is nondegenerate, indefinite, and $n \geq 3$, every $\mathbf{a} \in V_{\mathbb{Z}}$ lies on an injective quadratic polynomial family $\mathbf{a} + t\mathbf{v} + t^2\mathbf{w}$, $t \in \Pi$, in $V_{\mathbf{Oz}}$, with every $t \neq 0$ giving an infinite point. The family is injective on all of $\mathbf{No}$, so every such fiber contains a proper class of points.

In alternatives (ii) and (iii), the constant tuple of every point in the family is $\mathbf{a}$.

We first prove the geometric construction needed in the indefinite case. All vectors and bilinear forms in it are ordinary real objects; only the parameter is later allowed to be surreal.

Lemma 8.2 (A nilpotent quadratic isometry [01, 02]). *Let B be a symmetric bilinear form on $\mathbb{R}^n$, with $q(x) = B(x, x)$. Suppose $e, u \in \mathbb{R}^n$ satisfy*

$$e \neq 0, \quad q(e) = 0, \quad B(e, u) = 0, \quad k = q(u) \neq 0,$$

and put $N(x) = B(x, e)u - B(x, u)e$. Then

$$N^2(x) = -kB(x, e)e, \quad N^3 = 0, \quad B(Nx, y) + B(x, Ny) = 0. \quad (8.1)$$

For a parameter t in any field extension of $\mathbb{R}$, the polynomial map

$$U_t = \text{id} + tN + \frac{t^2}{2}N^2, \quad U_t(x) = x + t(B(x, e)u - B(x, u)e) - \frac{k}{2}t^2B(x, e)e, \quad (8.2)$$

satisfies

$$q(U_t(x)) = q(x), \quad \det U_t = 1, \quad U_s U_t = U_{s+t}, \quad U_t^{-1} = U_{-t}.$$

If $t \in \Pi$, all matrix entries of U_t lie in $\mathbf{Oz}$ and its entrywise constant coefficient is the identity. If $\alpha = B(a, e) \neq 0$, the curve $t \mapsto U_t(a)$ is injective.

Proof. From $B(e, e) = B(e, u) = 0$ one has $N(e) = 0$ and $N(u) = -ke$, which gives the first two identities of (8.1). Directly, $B(Nx, y) = B(x, e)B(u, y) - B(x, u)B(e, y) = -B(x, Ny)$.

For preservation, put $\alpha = B(x, e)$ and $\beta = B(x, u)$. In (8.2) the added vector is $t\alpha u - (t\beta + \frac{k}{2}t^2\alpha)e$. Its scalar product with x , multiplied by two, is $-kt^2\alpha^2$, because the two linear terms cancel. Its own quadratic value is $kt^2\alpha^2$, since e is isotropic and orthogonal to u . Their sum is zero.

Multiplying these polynomials in N and using $N^3 = 0$ gives $U_s U_t = U_{s+t}$ and hence $U_t U_{-t} = \text{id}$ over $\mathbb{R}[t]$. The determinant is therefore a unit of $\mathbb{R}[t]$, a nonzero real constant, and evaluation at $t = 0$ makes it 1. All identities remain valid after substituting a surreal t . Every entry of U_t is an entry of id plus a real combination of t and t^2 ; for $t \in \Pi$ that difference lies in Π .

Finally, for $\alpha = B(a, e) \neq 0$ and $\beta = B(a, u)$,

$$U_t(a) = a + tv + t^2w, \quad v = \alpha u - \beta e, \quad w = -\frac{k\alpha}{2}e.$$

The real vectors u, e are independent (as $q(u) \neq 0 = q(e)$), so v, w are independent: w is a nonzero multiple of e and v has nonzero u -component. Their independence persists over a field extension because some 2×2 real minor is nonzero. If $U_s(a) = U_t(a)$ with $s \neq t$, then $v + (s + t)w = 0$, a contradiction. $\square$

Source 01 wrote this map as a transvection T_t and source 02 as the nilpotent operator U_T with $\text{id} + TN + \frac{T^2}{2}N^2$; they are the same map. No infinite exponential series is used: the expression is a finite polynomial because $N^3 = 0$.

Lemma 8.3 (An isotropic direction visible from a nonzero point [01, 02]). *If q is a nondegenerate indefinite real quadratic form of dimension at least three and $a \in \mathbb{R}^n$ is nonzero, there exist e, u as in lemma 8.2 with $B(a, e) \neq 0$.*

Proof. The real isotropic vectors span $\mathbb{R}^n$. Diagonalize and rescale to an orthogonal real basis with $q(p_i) = 1$ and $q(n_j) = -1$; both types exist by indefiniteness. Since $B(p_i, n_j) = 0$, each $p_i \pm n_j$ is isotropic. Fixing one n_j recovers every p_i as a half-sum, and fixing one p_i recovers every n_j as a half-difference. Since the nonzero functional $B(a, -)$ cannot vanish on a spanning set, some isotropic e has $B(a, e) \neq 0$.

Nondegeneracy gives f_0 with $B(e, f_0) = 1$. Replacing it by $f = f_0 - \frac{1}{2}q(f_0)e$ makes $q(f) = 0$ while retaining $B(e, f) = 1$. The plane spanned by e, f has Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, so it is nondegenerate. Explicitly, $x - B(x, f)e - B(x, e)f$ is orthogonal to both e and f , which gives the direct sum with their orthogonal complement. The complement is nondegenerate: a vector in its radical would be orthogonal to the plane as well, hence to all of $\mathbb{R}^n$, and would vanish. Its dimension is $n - 2 \geq 1$. If q vanished identically there, polarization would make its bilinear form zero, contrary to nondegeneracy in positive dimension. It therefore contains u with $q(u) \neq 0$, and this u is orthogonal to e . $\square$

Source 01 stated this lemma for $q(a) \neq 0$; its proof, like source 02's, uses only $a \neq 0$.

Proof of theorem 8.1. The empty-set assertion is theorem 5.1.

Suppose first that q is definite and $\mathbf{x} \in V_{\mathbf{Oz}}$ has an infinite coordinate. Let $\gamma > 0$ be the largest leading exponent of its nonzero coordinates and r_i the coefficient of ω^γ in x_i (zero when absent). The real vector $\mathbf{r}$ is nonzero, and the coefficient of $\omega^{2\gamma}$ in $q(\mathbf{x})$ is exactly $q(\mathbf{r}) \neq 0$, contradicting $q(\mathbf{x}) = c$. Thus every coordinate is finite and hence an ordinary integer. In dimension two, nondegeneracy means that the binary quadratic has two distinct linear factors over $\mathbb{C}$, so theorem 6.7 applies. A nondegenerate one-dimensional form is definite. This proves (i).

For (ii), choose a nonzero real vector $\mathbf{v} \in \text{rad}(B)$. For every t , $q(\mathbf{a} + t\mathbf{v}) = q(\mathbf{a}) = c$. If $t \in \Pi$, the coordinates of $t\mathbf{v}$ lie in Π , so $\mathbf{a} + t\mathbf{v} \in \mathbf{Oz}^n$ with constant tuple $\mathbf{a}$. The map is injective because $\mathbf{v} \neq 0$, and for $t \neq 0$ some coordinate has positive degree.

For (iii), $\mathbf{a} \neq 0$ since $c \neq 0$. Choose e, u by lemma 8.3 and use lemma 8.2. Each coordinate of $U_t(\mathbf{a})$ is an integer constant plus real multiples of t and t^2 , so it lies in $\mathbf{Oz}$ with constant coefficient a_i for $t \in \Pi$. Injectivity was proved for all surreal t . If $t \neq 0$, its degree is positive, and in a coordinate where $w_i \neq 0$ the quadratic term has degree $2 \deg_\omega(t)$, strictly larger than that of the linear term, so the point is infinite. Since Π contains ω^γ for every surreal $\gamma > 0$, the fiber contains a proper class of points. The same proper-class conclusion holds for the injective affine family in (ii). $\square$

The quadratic family in (iii) also preserves the ideal generated by the coordinates. For $t \in \Pi$, both U_t and U_{-t} have entries in $\mathbf{Oz}$. Writing $\mathbf{x} = U_t \mathbf{a}$ gives $(x_1, \dots, x_n) \subseteq (a_1, \dots, a_n)$, and applying the inverse gives the reverse inclusion. For an ordinary starting tuple with gcd $d > 0$, that ideal is exactly $d\mathbf{Oz}$, by the ordinary integer Bézout identity. In particular, a primitive integer starting point gives an infinite family of unimodular omnific points. This conclusion uses the matrix orbit in (iii), not an arbitrary parametrized family.

8.1 Explicit families

For positive ordinary integers D, D' and $t \in \Pi$, set [01]

$$x = 1 + \frac{D'}{2}t^2, \quad y = \frac{D'}{2\sqrt{D}}t^2, \quad z = t. \quad (8.3)$$

All three coordinates are omnific integers, and $x^2 - Dy^2 - D'z^2 = 1$: the two quartic terms cancel, as do the two quadratic terms. For $t \neq 0$ every coordinate is infinite. The tuple is unimodular, since

$$1 = x - \left(\frac{D'}{2}t\right)z, \quad \frac{D'}{2}t \in \mathbf{Oz}. \quad (8.4)$$

This illustrates a sharp contrast:

$$\begin{aligned} x^2 - Dy^2 &= 1 \text{ has only ordinary points,} \\ x^2 - Dy^2 - D'z^2 &= 1 \text{ has infinite unimodular points.} \end{aligned}$$

There is no appeal to convergence, analytic continuation, or infinite iteration; every expression is a finite polynomial.

For the Lorentz form $q(x, y, z) = x^2 - y^2 - z^2$ with Gram matrix $G = \text{diag}(1, -1, -1)$, take $e = (1, 1, 0)$ and $u = (0, 0, 1)$ [02]. Then $q(u) = -1$, $N(x, y, z) = (z, z, x - y)$, and

$$U_t = \begin{pmatrix} 1 + t^2/2 & -t^2/2 & t \\ t^2/2 & 1 - t^2/2 & t \\ t & -t & 1 \end{pmatrix}, \quad U_t^\top G U_t = G, \quad \det U_t = 1, \quad U_s U_t = U_{s+t}. \quad (8.5)$$

For $t \in \Pi$ this is a one-parameter family of omnific Lorentz matrices with constant coefficient the identity. Acting on $(1, 0, 0)$ gives $(1 + t^2/2, t^2/2, t)$, and replacing t by $2s$ gives the denominator-free identity

$$(1 + 2s^2)^2 - (2s^2)^2 - (2s)^2 = 1,$$

valid for every $s \in \mathbf{Oz}$, with constant fiber $(1, 0, 0)$ whenever $s \in \Pi$. For $s = \omega$ the point is $(1 + 2\omega^2, 2\omega^2, 2\omega)$; the squares contain terms of degrees four, two and zero, and the nonconstant terms cancel exactly. There is no negative-exponent tail. This is concretely how an extra negative square evades the Pell rigidity mechanism.

Corollary 8.4 (An invisible unipotent subgroup [02]). *For every nondegenerate indefinite integral quadratic form q in at least three variables, there is an injective group homomorphism*

$$(\Pi, +) \longrightarrow \ker(\text{SO}(q, \mathbf{Oz}) \xrightarrow{\text{ct}} \text{SO}(q, \mathbb{Z})), \quad t \longmapsto U_t.$$

Proof. Choose a nonzero real test vector a and use lemma 8.3 to construct e, u . Lemma 8.2 gives a homomorphism with entries in $\mathbf{Oz}$, determinant one, preservation of q , and constant coefficient id. If $U_s = U_t$, apply both to a ; injectivity of $t \mapsto U_t(a)$ gives $s = t$. No ordinary point on a prescribed level is needed. $\square$

Here $\text{SO}(q, \mathbf{Oz})$ means determinant-one matrices over $\mathbf{Oz}$ preserving the integral quadratic polynomial q ; entrywise ct lands in $\text{SO}(q, \mathbb{Z})$. This subgroup is invisible to every finite quotient of the coefficient ring (corollary 3.4), yet it is a proper class. It concerns indefinite forms only; for definite forms see corollary 9.4.

8.2 What the classification does and does not say

The classification concerns the presence of genuinely new points on a *represented nonzero level* [01, 02]. It does not decide whether the integer level is represented in the first place. Nor does it classify all infinite points or claim that every omnific point of a higher-dimensional quadric lies on the constructed orbit: a displayed curve is a rigorously constructed subset of the full fiber. Source 02's version covers nondegenerate forms only and excludes degenerate forms and the zero level; source 01's alternative (ii) covers degenerate forms.

For zero levels, homogeneity gives a different mechanism. An indefinite real quadratic form has real isotropic directions, and multiplying any such direction by an element of Π produces omnific points with zero constant tuple. Such points can exist even when there is no nonzero ordinary integer isotropic vector.

9 Bounded geometry, leading forms, and orthogonal matrices

Multiplicative rigidity is not the only mechanism. A different one comes from boundedness and the discreteness of $\mathbf{Oz}$.

9.1 Bounded sets

Proposition 9.1 (Positive bounds [01, 05]). *If $a_1, \dots, a_n$ are positive ordinary reals, $m_1, \dots, m_n \geq 1$ are ordinary integers, $x_1, \dots, x_n \in \mathbf{Oz}$, and*

$$\sum_i a_i x_i^{2m_i} = C \in \mathbb{R},$$

then every x_i is an ordinary integer. In particular, an omnific point on a sphere of ordinary finite radius is an ordinary integer point.

Proof. Each summand is nonnegative and at most C . If $C < 0$ there is no solution; otherwise $|x_i| \leq (C/a_i)^{1/(2m_i)}$, an ordinary nonnegative real bound. [Proposition 2.5](#) then applies. This also handles $C = 0$, when every coordinate is zero. $\square$

Theorem 9.2 (Bounded semialgebraic rigidity [02]). *Let $X \subseteq \mathbb{R}^m$ be a bounded semialgebraic set defined by a finite formula in the language of ordered rings with real parameters, and let $X(\mathbf{No})$ be the class defined by the same formula in $\mathbf{No}$. Then $X(\mathbf{No}) \cap \mathbf{Oz}^m = X \cap \mathbb{Z}^m$.*

Proof. First justify the passage to a set-sized field, including when the defining formula has quantifiers. Any set of surreal parameters is contained in a set-sized real closed subfield: generate a field and take all surreal elements algebraic over it. This is a set because each nonzero polynomial has finitely many roots, ordered by the inherited order. It is real closed because positive square roots and roots of odd-degree polynomials exist in $\mathbf{No}$ and remain algebraic over the generated field.

For any set-sized real closed $K \subseteq \mathbf{No}$, a finite ordered-ring formula with parameters in K has the same truth value in K and in $\mathbf{No}$. Prove this simultaneously for all such K by induction on formulas. Atomic equalities and inequalities agree, and Boolean connectives preserve agreement. For an existential witness in $\mathbf{No}$, enlarge K to a set-sized real closed L containing the witness. The induction hypothesis transfers the subformula satisfied by that witness to L ; Tarski's quantifier elimination makes $K \subseteq L$ elementary, so the existential assertion already holds in K . Conversely, a witness in K works in $\mathbf{No}$ by induction. This uses

model completeness only between set-sized fields [11, 31]; no set of all surreals is made into a model.

Now choose an ordinary real $M > 0$ with $|x_i| \leq M$ for every $\mathbf{x} \in X$. This bound is a first-order implication with real parameters. For a candidate point of $X(\mathbf{No})$, take such a K containing its coordinates and $\mathbb{R}$. Its defining formula holds in K by the preceding argument, and the bound holds there by elementarity of $\mathbb{R} \subseteq K$. Every omnific coordinate is therefore bounded by M , hence an ordinary integer by [proposition 2.5](#). An integer tuple satisfies the defining formula in K exactly when it does in $\mathbb{R}$, again by elementarity. This proves both inclusions, including when X is empty. $\square$

Corollary 9.3 (Definite levels [01, 02]). *For a positive or negative definite real quadratic form q and a real c , every omnific point of $q(\mathbf{x}) = c$ is an ordinary integer tuple.*

Proof. In dimension zero there is only the empty tuple. In positive dimension, every real level set of a definite form is bounded, including an empty level. For a positive definite form, choose $\delta > 0$ with $q(\mathbf{r}) \geq \delta \sum_i r_i^2$ over $\mathbb{R}$, for example the least eigenvalue of its real symmetric matrix. A level $c \geq 0$ then lies in the ball of radius $\sqrt{c/\delta}$, and a negative level is empty. For a negative definite form apply this to $-q$. Now [theorem 9.2](#) applies. Alternatively, the leading-coefficient argument in the proof of [theorem 8.1\(i\)](#) applies verbatim to any real level c [01]. $\square$

For instance, $x^2 + y^2 = 1$ has exactly $(\pm 1, 0), (0, \pm 1)$ as omnific solutions, and so has $x^4 + y^4 = 1$. These concern a fixed finite level; the homogeneous Pythagorean relation lets all coordinates grow together. Positivity does *not* imply being a sum of squares in $\mathbf{Oz}$ ([remark 10.9](#)).

9.2 Orthogonal and symmetric matrices

Corollary 9.4 (Euclidean orthogonal group [01, 02]). *For every ordinary $n \geq 1$, $O_n(\mathbf{Oz}) = O_n(\mathbb{Z})$ is the group of signed permutation matrices, of order $2^n n!$, and $SO_n(\mathbf{Oz})$ consists of the signed permutation matrices of determinant one.*

Proof. For an orthogonal matrix, the sum of the squares in each row is 1, so every entry is ordinary by [proposition 9.1](#). An integer row of squared norm one has exactly one nonzero entry, equal to ± 1 , and orthogonality of different rows forces those entries into different columns. Conversely each signed permutation matrix is orthogonal. $\square$

Requiring *surreal* entries gives the large rotation group of a real closed field, whereas *omnific* entries in standard Euclidean coordinates leave a finite group. This is not a claim that matrix groups over $\mathbf{Oz}$ are finite: $SL_2(\mathbf{Oz})$ contains all $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$, $t \in \mathbf{Oz}$, and indefinite forms have the unipotent subgroup of [corollary 8.4](#). The Euclidean group is rigid because of positivity. More generally, for any definite ordinary real form q , a matrix preserving q has its j th column on the fixed level $q(e_j)$, where e_j is the j th standard basis vector. [Corollary 9.3](#) makes all its omnific entries ordinary integers. Each such real level is bounded, so it contains only finitely many integer columns. Thus the omnific isometry group equals its finite integer isometry group for every definite q , although the signed-permutation description is specific to the standard sum of squares in the chosen coordinates.

Corollary 9.5 (Symmetric matrix rigidity [01]). *Let M be a symmetric matrix over $\mathbf{Oz}$. If $\text{tr}(M^2)$ is an ordinary real constant, every entry of M is an ordinary integer. In particular, this holds if its characteristic polynomial has ordinary integer coefficients.*

Proof. For symmetric M , $\text{tr}(M^2) = \sum_i m_{ii}^2 + 2 \sum_{i < j} m_{ij}^2$, so [proposition 9.1](#) applies. When the size is at least two, $\text{tr}(M^2) = (\text{tr } M)^2 - 2e_2(M)$, where $e_2(M)$, the sum of the principal 2×2 minors, is up to sign a coefficient of the characteristic polynomial, as is $\text{tr } M$; for size one the assertion is immediate, and for size zero there are no entries to check. $\square$

Symmetry is essential. The matrix $\begin{pmatrix} 0 & \omega \\ 0 & 0 \end{pmatrix}$ has square zero and characteristic polynomial T^2 , but has an infinite omnific entry. The trace bound works for symmetric matrices because it is a positive sum of entry squares, not because every matrix with fixed characteristic polynomial has bounded entries.

9.3 No real point at infinity

Theorem 9.6 (Leading-homogeneous obstruction [01]). *Let $F_1, \dots, F_r \in \mathbb{Z}[X_1, \dots, X_n]$ be nonconstant polynomials, and let H_j be the top homogeneous part of F_j . If the H_j have no common nonzero real zero, then every common zero of the F_j in $\mathbf{O}\mathbf{z}^n$ lies in $\mathbb{Z}^n$.*

Proof. Suppose a common zero $\mathbf{x}$ has an infinite coordinate. Let $\gamma > 0$ be the largest leading exponent of its coordinates and $\mathbf{r} \neq 0$ the real vector of coefficients of ω^γ . If $d_j = \deg F_j$, the coefficient at $\omega^{d_j\gamma}$ in $F_j(\mathbf{x})$ is $H_j(\mathbf{r})$: a product attains that exponent only by taking exponent γ in each of d_j factors, and terms of lower ordinary total degree have exponent strictly less than $d_j\gamma$. So every $H_j(\mathbf{r}) = 0$, contrary to the hypothesis. All coordinates are therefore finite and hence ordinary integers. $\square$

The hypothesis is a sufficient absence-of-real-directions-at-infinity condition for the chosen system. Its converse is false: a Pell equation has real directions at infinity when $D > 0$, but its nonzero fibers are rigid by factorization. Initial-form obstructions and constant-product obstructions detect different mechanisms. The chosen generators also matter: the equations $X - Y^2 = 0$ and $X - Y^2 - Y = 0$ have only $(0, 0)$, since subtraction gives $Y = 0$ and then $X = 0$. Their two top forms are both $-Y^2$, which vanish at the nonzero vector $(1, 0)$. Replacing them by the equivalent system $X = Y = 0$ removes this direction. The theorem tests the displayed top forms, not every homogeneous consequence of the defining ideal.

10 Defining the ordinary integers

The retraction `ct` detects ordinary coefficients but is not a symbol of the ring language. Pell rigidity shows that the embedded copy of $\mathbb{Z}$ is nevertheless detected by a single polynomial equation with existential auxiliary variables. Two versions were found independently by sources 01 and 05; source 07 found the second again ([remark 10.5](#)), and [section 11](#) gives an order-free definition that also works in the Gaussian ring.

10.1 Pell coordinates as finite bounds

Define ordinary Pell solutions by

$$X_k + Y_k\sqrt{2} = (3 + 2\sqrt{2})^k, \quad (X_{k+1}, Y_{k+1}) = (3X_k + 4Y_k, 2X_k + 3Y_k), \quad k \geq 0, \quad (10.1)$$

with $(X_0, Y_0) = (1, 0)$. The index k always belongs to the ordinary $\mathbb{N}$; this recurrence does not define witnesses at infinite omnific indices.

Lemma 10.1 (Pell witnesses [01, 05]). *For every ordinary $k \geq 0$, $X_k^2 - 2Y_k^2 = 1$; for $k \geq 1$ both coordinates are positive and $X_k \geq 3^k$. Moreover $X_{2j} \equiv 1$ and $X_{2j+1} \equiv 3 \pmod{8}$, so both parity subsequences are unbounded.*

Proof. Direct expansion gives $(3X + 4Y)^2 - 2(2X + 3Y)^2 = X^2 - 2Y^2$, and ordinary induction proves the identity, positivity and $X_{k+1} \geq 3X_k$. From $X_k = \frac{1}{2}((3 + 2\sqrt{2})^k + (3 - 2\sqrt{2})^k)$ one gets the recurrence $X_{k+2} = 6X_{k+1} - X_k$ with $X_0 = 1$, $X_1 = 3$; modulo 8 it gives the alternating residues $1, 3, 1, 3, \dots$ (as $6 \cdot 3 - 1 \equiv 1$ and $6 \cdot 1 - 3 \equiv 3$). Finally $X_{2j} \geq 9^j$ and $X_{2j+1} \geq 3 \cdot 9^j$, so each residue class used below supplies witnesses above every ordinary integer bound. $\square$

10.2 The five-auxiliary-variable guard

We use the Legendre–Gauss three-square theorem: a nonnegative ordinary integer N is a sum of three integer squares if and only if it is not of the form $4^a(8b + 7)$ with $a, b \geq 0$. It is imported, not reproved; see the exposition [12] and the formal development in Isabelle’s Archive of Formal Proofs [13].

Lemma 10.2 (A three-square gap below a Pell coordinate [01]). *For every ordinary nonnegative integer S , there are ordinary integers u, v, w_1, w_2, w_3 such that*

$$u^2 - 2v^2 = 1, \quad u = S + w_1^2 + w_2^2 + w_3^2.$$

Proof. Choose the residue of u modulo 8 according to the following table; each choice is realized by an unbounded subsequence of the X_k (lemma 10.1).

$S \bmod 8$	0	1	2	3	4	5	6	7
chosen $u \bmod 8$	1	3	3	1	1	3	1	1
$(u - S) \bmod 8$	1	2	1	6	5	6	3	2

Take u in that subsequence so large that $N = u - S > 0$. Its residue is one of $1, 2, 3, 5, 6$, so it is neither divisible by 4 nor congruent to 7 modulo 8, and cannot be $4^a(8b + 7)$. The three-square theorem supplies the w_j , and the corresponding Pell coordinate supplies v . $\square$

For each ordinary $n \geq 1$, define the integer polynomial

$$\Psi_n(\mathbf{X}, U, V, \mathbf{W}) = (U^2 - 2V^2 - 1)^2 + \left(U - \sum_{i=1}^n X_i^2 - W_1^2 - W_2^2 - W_3^2 \right)^2, \quad (10.2)$$

of total degree four in $n + 5$ variables.

Theorem 10.3 (Uniform quartic integrality guard [01]). *For every $\mathbf{x} \in \mathbf{Oz}^n$,*

$$\mathbf{x} \in \mathbb{Z}^n \iff \exists u, v, w_1, w_2, w_3 \in \mathbf{Oz} \quad \Psi_n(\mathbf{x}, u, v, w_1, w_2, w_3) = 0. \quad (10.3)$$

Every coordinate of every zero of Ψ_n in $\mathbf{Oz}^{n+5}$ is an ordinary integer. The number of auxiliary variables is five, independently of n .

Proof. Suppose $\Psi_n = 0$ in $\mathbf{Oz}$. A sum of two squares in an ordered ring vanishes only when both squares vanish, so

$$u^2 - 2v^2 = 1, \quad u = \sum_i x_i^2 + w_1^2 + w_2^2 + w_3^2. \quad (10.4)$$

By Pell rigidity ([corollary 6.9](#)), $u, v \in \mathbb{Z}$. The second equation gives $u \geq 0$ and bounds the square of each x_i and w_j by the ordinary number u . Explicitly, if $z^2 \leq u$, then $|z| < u + 1$; otherwise $z^2 \geq (u + 1)^2 > u$. Thus each coordinate is finite and is an ordinary integer by [proposition 2.5](#). Conversely, for $\mathbf{x} \in \mathbb{Z}^n$ put $S = \sum_i x_i^2$ and choose the witnesses of [lemma 10.2](#). $\square$

For $x = 1$, one witness is $u = 3, v = 2, (w_1, w_2, w_3) = (1, 1, 0)$; for $x = 2, u = 17, v = 12, (w_1, w_2, w_3) = (3, 2, 0)$. The assertions for all integers rely on [lemma 10.2](#), not on examples or a finite computation.

10.3 The four-square definition

Theorem 10.4 (A quartic definition of $\mathbb{Z}$ [05]). *For $t \in \mathbf{Oz}$, the following are equivalent:*

- (i) $t \in \mathbb{Z}$;
- (ii) *there exist $x, y, z_1, z_2, z_3, z_4 \in \mathbf{Oz}$ with*

$$x^2 - 2y^2 = 1, \quad x - t^2 = z_1^2 + z_2^2 + z_3^2 + z_4^2; \quad (10.5)$$

- (iii) *the quartic equation*

$$(x^2 - 2y^2 - 1)^2 + \left(x - t^2 - \sum_{j=1}^4 z_j^2\right)^2 = 0 \quad (10.6)$$

has a solution in its six existential variables over $\mathbf{Oz}$.

Every witness to (10.5) has all six coordinates in $\mathbb{Z}$.

Proof. If (10.5) holds, Pell rigidity makes x, y ordinary, and $0 \leq t^2 \leq x$ bounds t , so $t \in \mathbb{Z}$; each z_j^2 is bounded by the ordinary integer $x - t^2$, so every z_j is ordinary. Conversely, for $t \in \mathbb{Z}$ choose $X_k \geq t^2$ by [lemma 10.1](#); the nonnegative integer $X_k - t^2$ is a sum of four squares by Lagrange’s theorem, which is available in Mathlib as `Nat.sum_four_squares` [14]. In an ordered ring a sum of two squares vanishes exactly when both do, so (10.5) and (10.6) are equivalent. The polynomial has total degree four, integer coefficients and no infinite parameters. $\square$

Write $\text{Std}(t)$ for the existential formula (10.6) and $\text{Std}_n(\mathbf{x})$ for (10.3). Replacing t^2 by $\sum_{i=1}^n t_i^2$ in (10.5) defines $\mathbb{Z}^n$ with six auxiliary variables by the same proof. Here “standard” refers precisely to the embedded $\mathbb{Z}$; it is not an extra predicate of the language.

Remark 10.5 (Source 07’s quartic [07]). Source 07 found, independently, the quartic

$$F_4(x, u, v, \mathbf{a}) = (u^2 - 2v^2 - 1)^2 + \left(u^2 - x^2 - \sum_{j=1}^4 a_j^2\right)^2 \quad (10.7)$$

with the six existential variables $u, v, a_1, \dots, a_4$, and proved that it defines $\mathbb{Z}$ in every ring A with $\mathbb{Z} \subseteq A \subseteq \mathcal{A}_{\mathbb{R}}(\Gamma)$ and $A \cap \mathbb{R} = \mathbb{Z}$ (notation of [section 11.1](#)), every zero in A^7 having all seven coordinates in $\mathbb{Z}$. It differs from (10.6) only in comparing x^2 with the square u^2 of the Pell coordinate instead of t^2 with the coordinate itself, and is printed once, here. The proof is the same; source 07 phrases the bounding step as follows: if a finite sum of squares of elements of $\mathcal{A}_{\mathbb{R}}(\Gamma)$ is a real constant, every summand is constant, because at twice the largest occurring

degree the coefficient is a sum of squares of real leading coefficients, hence positive. Source 07 cites Lagrange’s theorem through [27, Section 6]. Over $\mathbf{Oz}[i]$ the formula accepts every x (take $u = 1$, $v = 0$, $a_1 = ix$, $a_2 = 1$, $a_3 = a_4 = 0$), so it is not asserted there; the order-free definition of $\mathbb{Z}[i]$ in $\mathbf{Oz}[i]$ is [theorem 11.9](#).

Remark 10.6 (Comparison of the two guards). Source 01’s guard is uniform in n with five auxiliary variables and needs the three-square theorem; source 05’s uses one more auxiliary variable and Lagrange’s four-square theorem, and source 01 had noted this simpler variant without writing it out. Neither source claims its count optimal ([question 18.5](#)). For formalization the difference matters: at the Mathlib revision pinned by this repository (81a5d257), `Nat.sum_four_squares` is proved in `Mathlib/NumberTheory/SumFourSquares.lean`, while no three-square theorem is present. Only source 05’s definition therefore rests on a theorem in Mathlib; source 01’s would need the three-square theorem imported or proved first. The four-square theorem is used only on an ordinary nonnegative integer; its analogue for positive omnific integers is false ([remark 10.9](#)). Source 07’s quartic ([remark 10.5](#)) has the same counts as source 05’s, and source 06’s quintic ([remark 11.10](#)), with degree five and seven witnesses, is dominated by both guards.

10.4 Definability consequences

$\text{Std}(\mathbf{Oz}) = \mathbb{Z}$: ordinary integer arithmetic is present not only as the quotient $\mathbf{Oz}/\Pi$ but as an existentially definable subring of $\mathbf{Oz}$. There is no conflict with [proposition 5.3](#), because $\mathbb{Z}$ is stable under ct , whereas the positive cone and the nonzero locus are not.

Corollary 10.7 (Definability of ct and Π [01, 05]). *The graph of ct and the ideal Π are first-order definable in the pure ring language of $\mathbf{Oz}$ without parameters:*

$$\begin{aligned} \text{ct}(a) = m &\iff \text{Std}(m) \wedge \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (a - m = ky)), \\ j \in \Pi &\iff \forall k ((\text{Std}(k) \wedge k \neq 0) \Rightarrow \exists y (j = ky)). \end{aligned}$$

Proof. The right sides say that $m \in \mathbb{Z}$ and that $a - m$ (respectively j) is divisible by every nonzero ordinary integer; by (3.2) this means $a - m \in \Pi$ (respectively $j \in \Pi$), which for $m \in \mathbb{Z}$ is equivalent to $\text{ct}(a) = m$. For an explicit converse in the divisibility test, write $j = p + n$ with $p \in \Pi$ and $n \in \mathbb{Z}$. If $n \neq 0$, take the ordinary positive integer $k = |n| + 1$. The equation $j = ky$ would imply $n = k \text{ct}(y)$, impossible since $0 < |n| < k$. If $n = 0$, division of every coefficient of p by any nonzero ordinary k preserves its positive support and gives the required y . $\square$

All displayed quantifiers range over $\mathbf{Oz}$; Std is an abbreviation for the finite existential formula already given. In particular, the universal divisibility test is one first-order formula, not an infinite conjunction indexed externally by the ordinary integers. The symbol ct occurs only on the left to identify the graph that the ring-language formula defines.

These definitions use a universal quantifier and an implication. Neither source 01 nor source 05 claims an existential (Diophantine) definition of Π or of the graph of ct ([question 18.4](#)). Source 06 gives both: $\Pi = \{x : \exists y x^2 = 2y^2\}$ ([theorem 11.2](#)), and the graph of ct is Diophantine ([theorem 12.8](#)); source 07 adds a Diophantine definition of the complement $\{\text{ct} \neq 0\}$ ([theorem 12.2](#)). The corollary above remains as the first-order route.

Remark 10.8 (An abstract principle [01]). The proof of [theorem 10.3](#) needs less than all surreal numbers. Let R be any discretely ordered ring containing $\mathbb{Z}$ in which every solution of $U^2 - 2V^2 = 1$ is ordinary. Every element of R bounded by an ordinary integer is ordinary, by discreteness and a finite comparison with the integers in the bounding interval; so the same proof shows that Ψ_n defines $\mathbb{Z}^n$ in R^n , and likewise for (10.6). The omnific rings considered here satisfy the Pell hypothesis by [corollary 6.9](#).

10.5 Sums of squares and induction

Remark 10.9 (Squares and Peano arithmetic [01, 02, 05]). The positive omnific integer $\omega - 1$ is not a sum of any finite number of squares in $\mathbf{Oz}$: applying ct to such an identity gives $-1 = \sum_j m_j^2$ with $m_j \in \mathbb{Z}$. So the nonnegative cone of $\mathbf{Oz}$ does not satisfy Peano arithmetic. A direct failed induction instance, independent of formalizing Lagrange’s theorem in arithmetic, is given next. This is compatible with the classical fact that integer parts of real closed fields satisfy open induction [6, 3]; open induction is not full arithmetic.

Theorem 10.10 (A failed existential induction instance [05]). *On the nonnegative part of $\mathbf{Oz}$, consider the formula*

$$\Phi(t) : \quad \exists x, y \geq 0 \ (x^2 = 2y^2 + 1 \wedge x > t \wedge y > 0), \quad (10.8)$$

existential with a quantifier-free ordered-ring matrix. Over $\mathbf{Oz}_{\geq 0}$,

$$\Phi(0), \quad \forall t \geq 0 \ (\Phi(t) \Rightarrow \Phi(t+1)), \quad \neg \Phi(\omega),$$

and $\Phi(t)$ holds exactly for the ordinary nonnegative integers t .

Proof. The pair (3, 2) witnesses $\Phi(0)$. Given witnesses x, y for $\Phi(t)$, set $x' = 3x + 4y$, $y' = 2x + 3y$. The identity of lemma 10.1 gives $(x')^2 = 2(y')^2 + 1$; positivity and discreteness give $x, y \geq 1$, so $x' > x + 1 > t + 1$ and $y' > 0$. This proves the successor step for every omnific t . Pell rigidity makes every witness ordinary, so x cannot exceed an infinite t such as ω . Conversely an ordinary nonnegative t has a larger positive ordinary Pell witness by lemma 10.1. By proposition 2.5 these are the only two kinds of nonnegative omnific integers. $\square$

By Shepherdson’s theorem, an integer part of a real closed ordered field satisfies open induction on its nonnegative cone, that is, induction for quantifier-free ordered-ring formulas with parameters [6, 35]. In a divisible set-sized Hahn workspace H , the ring $\mathbf{Oz}_H$ is an integer part of the real closed field of real Hahn series with exponents in H : the floor formula in theorem 2.7 only truncates support and changes the constant, so stays in this workspace.

Here is the passage to the full $\mathbf{Oz}$ in detail. Fix a quantifier-free formula $\theta(t, \mathbf{a})$ whose base case and successor implication hold in $\mathbf{Oz}_{\geq 0}$, and suppose it fails at some $b \geq 0$. Choose H to contain the supports of the finite parameter tuple $\mathbf{a}$ and of b . Quantifier-free formulas agree between an ordered subring and its ambient ring: polynomial values, equality, order and Boolean operations are preserved. Thus the base case and successor implication hold in $(\mathbf{Oz}_H)_{\geq 0}$, whereas $\theta(b, \mathbf{a})$ still fails there, contradicting Shepherdson’s theorem. This proves each instance of the schema without treating the proper class $\mathbf{Oz}$ as a set-sized model [05]. There is no conflict with theorem 10.10: Φ is existential, not quantifier free. The sentence “every nonnegative t has a larger positive Pell solution” is true over $\mathbb{Z}$ and false over $\mathbf{Oz}$; thus $\mathbf{Oz}$ is not an elementary extension of $\mathbb{Z}$ as an ordered ring.

Remark 10.11 (An induction failure in the pure ring language [merge]). Induction on the nonnegative cone also fails for an existential formula of the pure ring language, without order in its matrix: the existential formula $\text{Std}(t)$ of theorem 10.4 satisfies $\text{Std}(0)$ and $\forall t (\text{Std}(t) \Rightarrow \text{Std}(t+1))$ over $\mathbf{Oz}$ but not $\text{Std}(\omega)$, because it defines $\mathbb{Z}$, which contains 0 and is closed under $t \mapsto t+1$, and $\omega \notin \mathbb{Z}$. The induction variable is still restricted to the nonnegative cone; the displayed successor assertion is stronger, since it holds for every $t \in \mathbf{Oz}$.

One can also keep every existential witness inside that cone. In the language $\{0, 1, +, \cdot\}$ of the nonnegative semiring, use

$$\exists x, y, z_1, z_2, z_3, z_4 \quad x^2 = 2y^2 + 1 \wedge x = t^2 + \sum_{j=1}^4 z_j^2.$$

For each ordinary $t \geq 0$, the Pell sequence and the four-square theorem supply nonnegative ordinary witnesses. Conversely, any witnesses in the nonnegative cone are also witnesses in $\mathbf{Oz}$, so force $t \in \mathbb{Z}$. This order-free existential formula therefore defines precisely $\mathbb{N}$ inside $\mathbf{Oz}_{\geq 0}$, and has the same base case, successor step and failure at ω . No subtraction or order predicate is needed in its matrix.

Separately, over the whole ring, the formula $\text{Std}(t)$ also shows that $\mathbf{Oz}$ is not an elementary extension of $\mathbb{Z}$ in the ring language: the sentence $\forall t \text{ Std}(t)$ holds in $\mathbb{Z}$ and fails in $\mathbf{Oz}$.

11 Diophantine arithmetic in omnific rings

This section and the next two are built from sources 06 and 07, written independently on September 23, 2026 (section 1.2). The two are twins: they share the order-free Gaussian system, the intersective polynomial and the Pell divisibility argument, and each has decisive results the other lacks. Source 06 is the base. It gives the Diophantine definition of Π , the Diophantine graph of ct , the homomorphism theorem and the reconstruction package; source 07 gives the constant-term detector, the ideal test, intermediate rings and number-field coefficients. Shared results are printed once with both sources named. Labels of this material carry the sub-prefix `odg: def: .`

11.1 General nonnegative-support rings

Let $\mathbb{k}$ be a field of characteristic zero and Γ an ordered abelian group. For a set-sized Γ these constructions take place in an ordinary Hahn field. When $\Gamma = \mathbf{No}$, every series still has set-sized support; the same finite algebra is interpreted under the class convention below, rather than treating $\mathbf{No}$ as a set. The notation ω^γ is formal for an abstract group Γ and is a Conway monomial in the actual surreal instances. Consider series $f = \sum_\gamma a_\gamma \omega^\gamma$ with $a_\gamma \in \mathbb{k}$ and reverse-well-ordered support in Γ , added coefficientwise and multiplied by convolution: the Hahn field with the exponent order reversed, written in the ω -convention of section 1.4 [06, 07]. Put

$$\begin{aligned} \mathcal{A}_{\mathbb{k}}(\Gamma) &= \{f : \text{supp}(f) \subseteq \Gamma^{\geq 0}\}, & \Pi_{\mathbb{k}}(\Gamma) &= \{f : \text{supp}(f) \subseteq \Gamma^{>0}\}, \\ \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) &= \mathfrak{o} + \Pi_{\mathbb{k}}(\Gamma) & \text{for a unital subring } \mathfrak{o} \subseteq \mathbb{k}. \end{aligned}$$

As in proposition 2.2, the constant coefficient ct is a ring retraction of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ onto $\mathbb{k}$ with kernel $\Pi_{\mathbb{k}}(\Gamma)$, restricting to a retraction of $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ onto $\mathfrak{o}$. Following source 07, an *intermediate ring* for $\mathfrak{o} \subseteq \mathbb{k}$ is a ring A with

$$\mathfrak{o} \subseteq A \subseteq \mathcal{A}_{\mathbb{k}}(\Gamma), \quad A \cap \mathbb{k} = \mathfrak{o}; \quad (11.1)$$

no inclusion $\text{ct}(A) \subseteq \mathfrak{o}$ is required. This is weaker than being the full ring $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. For example, $A = \mathbb{Z}[\omega + \frac{1}{2}] \subseteq \mathbb{R}[\omega] = \mathcal{A}_{\mathbb{R}}(\mathbb{Z})$ has $A \cap \mathbb{R} = \mathbb{Z}$: a nonconstant polynomial in $\omega + \frac{1}{2}$ still has positive degree. Nevertheless $\text{ct}(\omega + \frac{1}{2}) = \frac{1}{2} \notin \mathbb{Z}$. Thus the constant-term argument for $\mathcal{R}_{\mathfrak{o}}$ must not be silently applied to every intermediate ring. The instances used in this report are

$$\begin{aligned} \mathcal{B}_{\mathbb{R}} &= \mathcal{A}_{\mathbb{R}}(\mathbf{No}), & \mathcal{B}_{\mathbb{C}} &= \mathcal{A}_{\mathbb{C}}(\mathbf{No}), & \Pi &= \Pi_{\mathbb{R}}(\mathbf{No}), & \Pi_{\mathbb{C}} &= \Pi_{\mathbb{C}}(\mathbf{No}), \\ \mathbf{Oz} &= \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \mathbf{No}), & \mathbf{Oz}[i] &= \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \mathbf{No}), \end{aligned}$$

and $\mathbf{Oz}_{\Gamma} = \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$ of section 17. For $\Gamma = \mathbb{Z}$ every reverse-well-ordered subset of $\mathbb{Z}^{\geq 0}$ is finite, so $\mathcal{A}_{\mathbb{k}}(\mathbb{Z}) = \mathbb{k}[\omega]$ and $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \mathbb{Z}) = \mathfrak{o} + \omega \mathbb{k}[\omega]$ [06]. For $\Gamma = \mathbb{R}$ supports such as $\{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ occur, and no proof below uses a least positive exponent. For $\Gamma = \{0\}$ the ideal is zero and $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) = \mathfrak{o}$; the statements in this section include that case.

Lemma 11.1 (Degree, units and algebraic elements [06, 07]). *For nonzero $f, g \in \mathcal{A}_{\mathbb{k}}(\Gamma)$, $\deg_{\omega}(fg) = \deg_{\omega} f + \deg_{\omega} g$, and $\mathcal{A}_{\mathbb{k}}(\Gamma)^{\times} = \mathbb{k}^{\times}$. If a finite product of elements of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ is a nonzero constant, every factor is a nonzero constant. If $Q \in \mathbb{k}[T]$ has degree $n \geq 1$ and $f \in \mathcal{A}_{\mathbb{k}}(\Gamma)$ is nonconstant, then $\deg_{\omega} Q(f) = n \deg_{\omega} f > 0$. Consequently every element of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ algebraic over $\mathbb{k}$ lies in $\mathbb{k}$, and a polynomial over $\mathbb{k}$ with no root in $\mathfrak{o}$ has no root in any intermediate ring for $\mathfrak{o} \subseteq \mathbb{k}$.*

Proof. Write $\alpha = \deg_{\omega} f$ and $\beta = \deg_{\omega} g$. Any exponents contributing to a coefficient of fg satisfy $\gamma \leq \alpha$ and $\delta \leq \beta$. Equality $\gamma + \delta = \alpha + \beta$ forces $\gamma = \alpha$ and $\delta = \beta$, so the coefficient there is the nonzero product of the two leading coefficients. No term of larger degree occurs. If $fg = 1$, the nonnegative degrees α, β have sum zero, hence both vanish. A nonnegative-support form of degree zero is constant; conversely every nonzero constant has a constant inverse. The same degree argument applies to a finite product equal to a nonzero constant.

If f is nonconstant, then $d = \deg_{\omega} f > 0$. For $Q(T) = a_n T^n + \cdots + a_0$ with $a_n \neq 0$, the leading coefficient of $a_n f^n$ is $a_n \text{lc}(f)^n \neq 0$ at exponent nd . Every nonzero term $a_j f^j$ with $j < n$ has degree $jd < nd$, so that coefficient cannot cancel. This proves $\deg_{\omega} Q(f) = nd > 0$ without ordering the coefficient field. An algebraic nonconstant element would make such a nonzero polynomial vanish, a contradiction. A root of a nonconstant polynomial over $\mathbb{k}$ in A must therefore lie in $A \cap \mathbb{k} = \mathfrak{o}$; a nonzero constant polynomial has no root anywhere. $\square$

In particular $\mathbf{Oz}[i]^{\times} = \{\pm 1, \pm i\}$, while the units of $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$ are all of $\mathbb{C}^{\times}$; the larger ring is the one in which the Pell factorization below takes place [06].

Conventions. A subset of A^n is *Diophantine* if it is defined by a formula $\exists \mathbf{y} \bigwedge_{j=1}^m P_j(\mathbf{x}, \mathbf{y}) = 0$ with $P_j \in \mathbb{Z}[\mathbf{X}, \mathbf{Y}]$ [06, 07]. This is the finite-system convention; single-polynomial presentations are stated explicitly, and negative atoms such as $y \neq 0$ are not allowed. The numeral 2 is the ring term $1 + 1$. No formula below names an order, a monomial, $\sqrt{2}$ or a coefficient map. For set-sized Γ the statements are ordinary first-order model theory. For $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ a formula is interpreted by relativizing its quantifiers to the class, formula by formula; no uniform satisfaction predicate for the class structure, no set of all surreals and no class of all class automorphisms is formed, and a statement about arbitrary automorphisms is a statement about each map individually [06, 07].

11.2 The purely infinite ideal is Diophantine

Theorem 11.2 (A quadratic definition of the ideal [06]). *Let $\mathfrak{o} \subseteq \mathbb{k}$ be a subring such that $\sqrt{2} \in \mathbb{k}$ but 2 is not a square in $\text{Frac}(\mathfrak{o})$. For $x \in \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$,*

$$x \in \Pi_{\mathbb{k}}(\Gamma) \iff \exists y \in \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma) \quad x^2 = 2y^2. \quad (11.2)$$

In particular the parameter-free formula $\text{Inf}(x) :\Leftrightarrow \exists y (x^2 = 2y^2)$ defines Π in $\mathbf{Oz}$ and $\Pi_{\mathbb{C}}$ in $\mathbf{Oz}[i]$, and the same formula works for every Γ .

Proof. If $x \in \Pi_{\mathbb{k}}(\Gamma)$, then $y = x/\sqrt{2}$ has the same support, so it lies in $\Pi_{\mathbb{k}}(\Gamma)$ and witnesses the equation. Conversely, apply ct to a solution: $a = \text{ct}(x)$ and $b = \text{ct}(y)$ lie in $\mathfrak{o}$ and $a^2 = 2b^2$. If $b \neq 0$, then $(a/b)^2 = 2$ in $\text{Frac}(\mathfrak{o})$, which is excluded; so $b = 0$ and then $a = 0$, that is, $x \in \Pi_{\mathbb{k}}(\Gamma)$. For $\mathfrak{o} = \mathbb{Z}$ the hypothesis is the irrationality of $\sqrt{2}$; for $\mathfrak{o} = \mathbb{Z}[i]$, a square root of 2 in $\mathbb{Q}(i)$ would lie in $\mathbb{Q}(i) \cap \mathbb{R} = \mathbb{Q}$. $\square$

The full positive-support part is needed for the forward implication: its closure under multiplication by $1/\sqrt{2}$ supplies the witness. The conclusion does not hold for arbitrary

intermediate rings. For example, $\omega \in A \cap \Pi_{\mathbb{R}}(\mathbb{Z})$ for $A = \mathbb{Z}[\omega]$, but no $y \in A$ satisfies $\omega^2 = 2y^2$: degree forces y to be linear and its leading integer coefficient would have square $1/2$. The nonsquare hypothesis also matters: in $\mathcal{R}_{\mathbb{Z}[\sqrt{2}]}(\mathbb{R}, \Gamma)$ the nonzero constant $x = \sqrt{2}$ satisfies the equation with $y = 1$, although it is not in the ideal.

For example $\pi\omega^3 - \sqrt{3}\omega + 7$ does not satisfy Inf , because its constant term is 7, while $\pi\omega^3 - \sqrt{3}\omega$ and $\sum_{n \geq 1} \omega^{1/n}$ do; “purely infinite” includes 0 and negative elements such as $-\omega$ [06]. The formula is existential, has no parameters and one witness, so it answers the first part of [question 18.4](#) ([section 18.2](#)). It is compatible with [proposition 5.3](#): Π is stable under ct , since $\text{ct}(\Pi) = \{0\} \subseteq \Pi$, whereas the nonzero locus is not.

First-order definability of $\mathbb{Z}$ in $\mathbf{Oz}$ is older. In a MathOverflow comment of 6 July 2018 the user *nombre* recognized ordinary integers by the absence of nonzero solutions of $x^2 = 2y^2$ in a bounded interval, and observed the resulting failure of saturation [18]. Source 06 states that the equation of [theorem 11.2](#) belongs to the same obstruction and is not an unrelated discovery; the new point is its positive-existential form.

Corollary 11.3 (Homomorphisms preserve the ideal [06]). *Let $\mathbb{k}, \mathbb{k}'$ be fields of characteristic zero containing $\sqrt{2}$. Every unital ring homomorphism $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}', \Gamma')$ maps $\Pi_{\mathbb{k}}(\Gamma)$ into $\Pi_{\mathbb{k}'}(\Gamma')$, and the same holds between two rings $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}', \Gamma')$. No injectivity or size condition on the target is needed.*

Proof. For x in the source ideal, $x^2 = 2(x/\sqrt{2})^2$ with $x/\sqrt{2}$ in the source ring. A unital homomorphism fixes the integer numeral 2, so its images satisfy the same equation. Apply [theorem 11.2](#) in the target. The homomorphism need not fix the chosen square root, the coefficient field or any monomial: the formula contains none of those parameters. $\square$

Remark 11.4 (Other radicands [06]). If a nonzero $\delta \in \mathfrak{o}$ has a square root in $\mathbb{k}$ but not in $\text{Frac}(\mathfrak{o})$, the same proof shows that $\exists y (x^2 = \delta y^2)$ defines $\Pi_{\mathbb{k}}(\Gamma)$ in $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$; the formula is parameter-free when δ is an integer numeral. The proof uses no ordering of $\mathbb{k}$, no divisibility of Γ and no real closedness.

11.3 Pell rigidity and divisible Pell coordinates

Corollary 11.5 (Pell rigidity in intermediate rings [06, 07]). *Let A be an intermediate ring for $\mathfrak{o} \subseteq \mathbb{k}$. If $u, v \in A$ satisfy $u^2 - 2v^2 = 1$, then $u, v \in \mathfrak{o}$.*

Proof. Extend the coefficients to $\mathbb{k}(\sqrt{2})$; this does not change supports. There $(u + \sqrt{2}v)(u - \sqrt{2}v) = 1$, so both factors are constants by [lemma 11.1](#), and adding and subtracting them shows that u, v are constants. Their coefficients lie in $\mathbb{k}$, so $u, v \in A \cap \mathbb{k} = \mathfrak{o}$. Equivalently, apply [theorem 6.15](#) to $E = \mathbb{Q}(\sqrt{2})$ with basis $1, \sqrt{2}$. $\square$

Source 06 proves this in $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ with $\sqrt{2} \in \mathbb{k}$, source 07 in intermediate rings by coefficient extension; the argument is that of [corollary 6.9](#). It does not estimate u^2 and $2v^2$ separately, so cancellation between their leading terms is harmless, and $\sqrt{2}$ occurs in the proof only, not as a parameter. For $\mathfrak{o} = \mathbb{Z}[i]$ the conclusion is Gaussian constancy, not reality: $(u, v) = (i, i)$ solves $u^2 - 2v^2 = 1$, and the Gaussian construction below uses only Gaussian constancy [06].

Lemma 11.6 (Divisible Pell coordinates [06, 07]). *Let (X_k, Y_k) be the Pell sequence (10.1). For every ordinary $m \geq 1$ there is k with $1 \leq k \leq m^2$ such that $X_k \equiv 1$ and $Y_k \equiv 0 \pmod{m}$; then $Y_k > 0$.*

Proof. Let $M = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$. Its inverse $\begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$ has integer entries, so M permutes $(\mathbb{Z}/m\mathbb{Z})^2$. Among the $m^2 + 1$ points $M^j(1, 0)$ for $0 \leq j \leq m^2$, two coincide, say at $j < \ell$. Applying M^{-j} gives $M^{\ell-j}(1, 0) = (1, 0)$, with $1 \leq k = \ell - j \leq m^2$ [07]. In the ordinary integer recurrence $X_0 = 1, Y_0 = 0$, the first step gives $(3, 2)$ and both coordinates remain positive, so $Y_k > 0$. This also covers $m = 1$, whose residue set has one point. Source 06 argues instead that $3 + 2T$ is a unit of the ring $(\mathbb{Z}/m\mathbb{Z})[T]/(T^2 - 2)$, with inverse $3 - 2T$. Division by the monic polynomial $T^2 - 2$ gives a unique representative $a + bT$, so the ring has m^2 elements; some positive power at most m^2 equals 1. Comparing the two coordinates gives the same congruences. $\square$

No classification of Pell solutions, primitive-divisor theorem or distribution result is used [06]. Pell equations have a long history in Diophantine definability, from Davis's explicit work [20] and Denef's on polynomial rings [21] to Miller and Shlapentokh's on rings of functions [25]. Sources 06 and 07 use only the elementary fact that, after adjoining $\sqrt{2}$, every unit of the nonnegative-support ring is constant; source 07 stresses that no theorem about arbitrary Hahn supports is inferred from theorems about polynomial rings.

Lemma 11.7 (Integers in principal ideals [07]). *Let $\mathfrak{o}$ be a unital subring of a number field. For every nonzero $a \in \mathfrak{o}$, the ideal $a\mathfrak{o}$ contains a nonzero ordinary integer; for a Gaussian integer a one may take $a\bar{a}$.*

Proof. Algebraicity supplies a nonzero integer polynomial vanishing at a after clearing the denominators of a rational relation. Choose one of least degree, $A_n a^n + \cdots + A_1 a + A_0 = 0$. If $A_0 = 0$, cancellation of $a \neq 0$ gives a nonzero relation of smaller degree. Thus $A_0 \neq 0$, and $A_0 = -a(A_1 + A_2 a + \cdots + A_n a^{n-1}) \in a\mathfrak{o}$. The expression uses only integer coefficients and nonnegative powers of a , which belong to the unital subring $\mathfrak{o}$; no algebraic-integrality hypothesis is needed. Changing its sign if necessary gives a positive ordinary integer in the same ideal. $\square$

11.4 An intersective polynomial

A polynomial in $\mathbb{Z}[T]$ is *intersective* if it has a root modulo every positive integer. We use

$$\Lambda(T) = (T^2 - 13)(T^2 - 17)(T^2 - 221) = T^6 - 251T^4 + 6851T^2 - 48841. \quad (11.3)$$

It is a classical example: L   and Spencer list it in [23, Section 2, Example 4] and attribute it to Borevich and Shafarevich [22, p. 3]; Berend and Bilu study the general phenomenon of polynomials with roots modulo every integer [24]. Sources 06 and 07 both reprove the properties they need, because the definitions below use roots modulo *all* integers; the proofs are printed once.

Proposition 11.8 (The polynomial Λ [06, 07]).

- (i) *For every ordinary $N \geq 1$ there is $r \in \mathbb{Z}$ with $\Lambda(r) \equiv 0 \pmod{N}$.*
- (ii) *Λ has no root in $\mathbb{Q}(i)$; hence none in $\mathbb{Z}$ or $\mathbb{Z}[i]$, and by lemma 11.1 none in any intermediate ring for $\mathbb{Z} \subseteq \mathbb{k}$ or $\mathbb{Z}[i] \subseteq \mathbb{k}$.*
- (iii) *For every nonzero $v \in \mathbb{Z}[i]$ there are $s \in \mathbb{Z}[i]$ and $t \in \mathbb{Z}$ with $vs = \Lambda(t)$.*

Proof. (ii) A root in a field satisfies $T^2 = d$ with $d \in \{13, 17, 221\}$. If $(a + bi)^2 = d$ with $a, b \in \mathbb{Q}$, then $2ab = 0$; $a = 0$ gives $d = -b^2 \leq 0$, and $b = 0$ would make d a rational square, which none of the three is.

(i) It suffices to find a root modulo each prime power p^a dividing N and to combine them by the Chinese remainder theorem; the chosen factor of Λ may depend on p . For $N = 1$ every integer works. Let p be odd. If $p \notin \{13, 17\}$, then 13 and 17 are nonzero modulo p ; if neither is a square, their product 221 is, by multiplicativity of the Legendre symbol. For $p = 13$ use $17 \equiv 2^2$, and for $p = 17$ use $13 \equiv 8^2$. In each case some factor $T^2 - d$ has a root $r \not\equiv 0 \pmod{p}$, and it lifts to every p^a : if $r^2 \equiv d \pmod{p^j}$, replacing r by $r + hp^j$ changes $(r^2 - d)/p^j$ by $2rh$ modulo p , and $2r$ is invertible modulo p . For $p = 2$ use 17: the root 1 works modulo 2^a for $a \leq 3$, since $17 \equiv 1 \pmod{8}$; if an odd r satisfies $r^2 \equiv 17 \pmod{2^a}$ with $a \geq 3$, then $(r + 2^{a-1})^2 - r^2 = 2^a r + 2^{2a-2} \equiv 2^a \pmod{2^{a+1}}$, so exactly one of r and $r + 2^{a-1}$ is a root modulo 2^{a+1} . This step constructs a root at each power of 2; it need not preserve the previously chosen residue modulo 2^a , and no compatible sequence of such choices is needed. For each prime-power divisor of N , choose one of the roots just constructed. The Chinese remainder theorem gives an integer congruent to all of them; its Λ -value is divisible by every such prime power, hence by their product N .

(iii) Put $N = v\bar{v} \in \mathbb{Z}_{>0}$, choose t by (i) with $\Lambda(t) = Nq$, and take $s = q\bar{v}$. $\square$

The equation $vs = \Lambda(t)$ is a positive-existential certificate of $v \neq 0$ only when v is already known to be a Gaussian constant. It is not a definition of the nonzero elements: for $v \in \Pi_{\mathbb{C}}$ it has no solution, since applying ct would give a root of Λ in $\mathbb{Z}[i]$ [06].

11.5 One order-free system defining $\mathbb{Z}$ and $\mathbb{Z}[i]$

Let $\mathcal{E}(x; u, v, w, s, t)$ be the system

$$x(u^2 - 2v^2 - 1) = 0, \quad x(v - xw) = 0, \quad x(vs - \Lambda(t)) = 0, \quad (11.4)$$

and put $\Xi(x) :\Leftrightarrow \exists u, v, w, s, t \mathcal{E}(x; u, v, w, s, t)$.

Theorem 11.9 (One system for $\mathbb{Z}$ and $\mathbb{Z}[i]$ [06, 07]). *Let $(\mathfrak{o}, \mathbb{k})$ be $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, let Γ be any ordered abelian group, and let A be an intermediate ring. Then Ξ defines $\mathfrak{o}$ in A . In particular $\Xi(\mathbf{Oz}) = \mathbb{Z}$ and $\Xi(\mathbf{Oz}[i]) = \mathbb{Z}[i]$: the same parameter-free system of three equations, of total degrees 3, 3 and 7, with five witnesses, defines both.*

Proof. Suppose $\mathcal{E}(x; u, v, w, s, t)$ holds with $x \neq 0$. Cancelling x in the domain A gives

$$u^2 - 2v^2 = 1, \quad v = xw, \quad vs = \Lambda(t).$$

By [corollary 11.5](#), $u, v \in \mathfrak{o}$. If $v = 0$, then $\Lambda(t) = 0$ with $t \in A$, which [proposition 11.8\(ii\)](#) excludes. So v is a nonzero constant, and $xw = v$ forces x and w to be constants ([lemma 11.1](#)); hence $x \in A \cap \mathbb{k} = \mathfrak{o}$. If $x = 0$ there is nothing to prove.

Conversely let $x \in \mathfrak{o}$. For $x = 0$ take all witnesses zero. For $x \neq 0$ let $m = |x|$, or $m = x\bar{x}$ in the Gaussian case, so that $m \in x\mathfrak{o}$. By [lemma 11.6](#) choose a Pell pair $(u, v) = (X_k, Y_k)$ with $v > 0$ and $m \mid v$; then $w = v/x \in \mathfrak{o}$. By intersectivity choose $t \in \mathbb{Z}$ with $v \mid \Lambda(t)$ and put $s = \Lambda(t)/v \in \mathbb{Z}$. These witnesses all lie in $\mathfrak{o} \subseteq A$: u, v, s, t are ordinary integers, while w may be a nonreal Gaussian integer. They satisfy the system. $\square$

The proof uses the intermediate-ring condition, not a retraction $A \rightarrow \mathfrak{o}$. It therefore applies, for example, to $\mathbb{Z}[\omega + \frac{1}{2}]$ above even though its constant coefficient need not be an integer. In fact the same proof works with any characteristic-zero coefficient field containing the indicated copy of $\mathbb{Z}$ or $\mathbb{Z}[i]$: coefficient extension proves Pell rigidity, root exclusion uses $A \cap \mathbb{k} = \mathfrak{o}$, and the witnesses are already in $\mathfrak{o}$. The displayed theorem retains the two source pairs; the broader number-field version is stated in [theorem 12.14](#).

Source 07 states the theorem in this form, for both pairs and all intermediate rings. Source 06 proves the identical system (with witnesses h, r for s, t) in $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \Gamma)$ and $\mathbf{Oz}[i]$, noting that the root exclusion takes place after applying ct , so that algebraic closedness of the coefficient field is not needed. The factor x in each equation admits $x = 0$ without a disjunction. For $x = 1 + i$ a witness is $u = 3, v = 2, w = 1 - i, t = 1, s = -21120$, because $\Lambda(1) = -42240$; for $x = \omega + i$ there is none, since a nonzero Gaussian constant is not divisible by $\omega + i$ in $\mathcal{B}_{\mathbb{C}}$ [06, 07]. The formula contains no occurrence of i and no distinguished choice of it.

Why the certificate is needed. The relation $v = xw$ alone would allow $v = 0$ and hence every x . Writing $v \neq 0$ would give an existential formula with an inequation, not a Diophantine one, and replacing it by $vz = 1$ would force v to be a unit, which most Pell coordinates are not. The equation $vs = \Lambda(t)$ excludes $v = 0$ because Λ has no root in A , and accepts every nonzero ordinary v because Λ is interseptive [06, 07]. Over a ring containing i the three equations cannot be combined into one sum of squares ($1^2 + i^2 = 0$), and no single-polynomial Gaussian definition is claimed. In the proof one may take the Pell index at most m^2 and $0 \leq t < v$; these are existence bounds, not claims of optimal or practical complexity [07]. The ring restriction is essential: in a field containing a root r of Λ , $\Xi(x)$ holds for every x , with $u = 1, v = w = s = 0$ and $t = r$; so Ξ defines nothing in $\mathbf{No}$ or $\mathbf{No}[i]$ [07].

11.6 Single real polynomials and the natural numbers

In the real case a conjunction of equations becomes one equation as a sum of squares. The report already prints the quartic guards of sources 01 and 05 (theorems 10.3 and 10.4), and source 07's quartic, which is source 05's up to the substitution described in remark 10.5. Source 06 proposed a quintic as its principal real formula; it is recorded here with the comparison.

Remark 11.10 (Source 06's quintic, and a comparison [06]). Let $\mathbb{k}$ be an ordered field containing $\sqrt{2}$. In $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$, and in particular in $\mathbf{Oz}$,

$$x \in \mathbb{Z} \iff \exists u, v, w, s_1, \dots, s_4 \quad x \left[(u^2 - 2v^2 - 1)^2 + (v - xw)^2 + \left(u - 2 - \sum_{j=1}^4 s_j^2 \right)^2 \right] = 0. \quad (11.5)$$

Indeed, for $x \neq 0$ the three squares vanish; Pell rigidity makes u, v integers, $u = 2 + \sum s_j^2 \geq 2$ excludes $v = 0$, and then $xw = v \neq 0$ forces $x \in \mathbb{Z}$. Conversely, for $0 \neq x \in \mathbb{Z}$ take a Pell pair with $|x| \mid Y_k$ (lemma 11.6), $w = Y_k/x$, and four squares for $X_k - 2 \geq 1$. The polynomial has total degree exactly five (the monomial xu^4) and seven witnesses. Source 06 tabulates complete ordinary witnesses for $x \in \{1, 2, \pm 3, 5, 7\}$, for instance $u = 17, v = 12, w = 4, (s_1, \dots, s_4) = (3, 2, 1, 1)$ for $x = 3$; for $x = \omega$ or $\omega + 1$ there are none in $\mathbf{Oz}$. It is dominated by theorem 10.3 (degree four, five witnesses, uniform in n) and by theorem 10.4 and source 07's (10.7) (degree four, six witnesses); source 06's question on the least degree is thereby partly answered (section 18.2). Its mechanism, divisibility $v = xw$ of a nonzero Pell coordinate, is the one that survives in the Gaussian system theorem 11.9. Source 06 restricts the quintic to the real case, correctly: over $\mathbf{Oz}[i]$ it accepts $x = \omega$ [merge], with $u = 1, v = 0, w = 1, s_1 = i, s_2 = e^{-i\pi/4}\omega^{1/2}, s_3 = s_4 = 0$. Then $u^2 - 2v^2 - 1 = 0, v - xw = -\omega$ and $u - 2 - s_1^2 - s_2^2 = -1 + 1 + i\omega = i\omega$, and $(-\omega)^2 + (i\omega)^2 = 0$. Source 07 records the analogous collapse of its quartic (remark 10.5).

Corollary 11.11 (The natural numbers [06, 07]). *Let Std be any of the Diophantine definitions of $\mathbb{Z}$ in $\mathbf{Oz}$ (theorem 10.4, theorem 11.9, (11.5)). Then $x \in \mathbb{N} \iff \text{Std}(x) \wedge \exists r_1, \dots, r_4 \ x = r_1^2 + r_2^2 + r_3^2 + r_4^2$, where $0 \in \mathbb{N}$.*

Proof. The forward implication is Lagrange's theorem. Conversely $\text{Std}(x)$ makes x an ordinary integer, and a sum of squares in the ordered ring $\mathbf{Oz}$ is nonnegative (alternatively, apply ct). $\square$

Source 07 restricts the r_j by its constant-definition formula as well; in $\mathbf{Oz}$ this is harmless and unnecessary. [Remark 10.11](#) gives a variant in the language $\{0, 1, +, \cdot\}$ of the nonnegative semiring, with every witness in $\mathbf{Oz}_{\geq 0}$, that defines $\mathbb{N}$ inside $\mathbf{Oz}_{\geq 0}$.

12 The constant term: detector, ideal test and graph

12.1 A root detector for augmentations

The following algebraic statement, due to source 07, isolates why a single polynomial can detect a nonzero constant term.

Theorem 12.1 (Augmentation root detector [07]). *Let $\mathcal{H}$ be a commutative unital $\mathbb{k}$ -algebra with a $\mathbb{k}$ -algebra homomorphism $\varepsilon : \mathcal{H} \rightarrow \mathbb{k}$ that is the identity on $\mathbb{k}$. Let $\mathfrak{o} \subseteq \mathbb{k}$ be a unital subring and $A = \mathfrak{o} + \ker \varepsilon$. Let $f \in \mathfrak{o}[T]$ satisfy:*

- (a) *f has no root in $\mathfrak{o}$;*
- (b) *f has a root $r \in \mathbb{k}$;*
- (c) *for every nonzero $c \in \mathfrak{o}$ there is $b \in \mathfrak{o}$ with $f(b) \in c\mathfrak{o}$.*

Then for every $a \in A$,

$$\varepsilon(a) \neq 0 \iff \exists s, t \in A \quad as = f(t). \quad (12.1)$$

No hypothesis that $\mathcal{H}$ is a domain, reduced, complete or Henselian is needed.

Proof. Identify $\mathbb{k}$ with its image in $\mathcal{H}$; the retraction makes this embedding injective. Then $A = \varepsilon^{-1}(\mathfrak{o})$, since every $z \in \mathcal{H}$ decomposes as $\varepsilon(z) + (z - \varepsilon(z))$, with the second summand in the kernel. If $as = f(t)$ and $\varepsilon(a) = 0$, applying ε gives $f(\varepsilon(t)) = 0$ with $\varepsilon(t) \in \mathfrak{o}$, contrary to (a). Conversely let $c = \varepsilon(a) \neq 0$, choose b by (c), and factor $f(T) = (T - r)Q(T)$ in $\mathbb{k}[T]$. Put

$$\alpha = \frac{b - r}{c}, \quad t = r + \alpha a, \quad s = \alpha Q(t). \quad (12.2)$$

Since $t - r = \alpha a$, we get $as = (t - r)Q(t) = f(t)$. Moreover $\varepsilon(t) = r + \alpha c = b \in \mathfrak{o}$ and $c\varepsilon(s) = \varepsilon(as) = f(b) \in c\mathfrak{o}$, so $\varepsilon(s) \in \mathfrak{o}$; thus $s, t \in \mathfrak{o} + \ker \varepsilon = A$. Only the nonzero scalar c is inverted, in the field $\mathbb{k}$. $\square$

The augmentation need not be part of the ring language: the conclusion is a ring-language formula that detects its kernel [07], with the coefficients of f as parameters. In the applications below $f \in \mathbb{Z}[T]$, so these are integer numerals and the formula is parameter-free.

For a concrete example with zero divisors, take $\mathcal{H} = \mathbb{R} \times \mathbb{R}$, embed $\mathbb{R}$ diagonally, and let $\varepsilon(u, v) = u$. Then $A = \mathbb{Z} \times \mathbb{R}$. The zero divisor $a = (1, 0)$ has nonzero augmentation and admits the certificate

$$t = (0, \sqrt{13}), \quad s = (-48841, 0), \quad as = (-48841, 0) = \Lambda(t).$$

Thus invertibility of a is not required; the cancellation in the proof takes place in $\mathbb{k}$, after applying ε .

Theorem 12.2 (Detecting a nonzero constant term [07]). *Let $(\mathfrak{o}, \mathbb{k})$ be $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$ and $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. For every $a \in A$,*

$$\Theta(a) : \Longleftrightarrow \exists s, t \ (as = \Lambda(t)) \quad \Longleftrightarrow \quad \text{ct}(a) \neq 0. \quad (12.3)$$

In particular this holds in $\mathbf{Oz}$ and $\mathbf{Oz}[i]$.

Proof. Apply [theorem 12.1](#) with $\mathcal{H} = \mathcal{A}_{\mathbb{k}}(\Gamma)$, $\varepsilon = \text{ct}$ and $f = \Lambda$. Hypothesis (a) is [proposition 11.8\(ii\)](#); (b) holds with $r = \sqrt{13}$. For (c), given $c \neq 0$ in $\mathfrak{o}$ choose a positive integer $m \in c\mathfrak{o}$ ($|c|$ or $c\bar{c}$) and $b \in \mathbb{Z}$ with $\Lambda(b) \in m\mathbb{Z} \subseteq c\mathfrak{o}$. $\square$

With $r = \sqrt{13}$ the witnesses (12.2) are explicit: $t = r + \alpha a$ and $s = \alpha(2r + \alpha a)(t^2 - 17)(t^2 - 221)$, with $\text{ct}(t) = b$ and $\text{ct}(s) = \Lambda(b)/c$. Neither r nor α need lie in A ; the witnesses do, because their constant coefficients are right [07].

Proposition 12.3 (Support control of the witnesses [07]). *Let $S = \text{supp}(a)$ and $S^* = S \cup \{0\}$. The witnesses above satisfy $\text{supp}(t) \subseteq S^*$ and $\text{supp}(s) \subseteq \bigcup_{j=0}^5 jS^*$, where jS^* is the j -fold sumset and $0S^* = \{0\}$. If a is nonconstant, $\deg_{\omega} t = \deg_{\omega} a$ and $\deg_{\omega} s = 5 \deg_{\omega} a$.*

Proof. The witness t is affine in a and s is a polynomial of degree five in a with constant coefficients, so the support inclusions follow from finite convolution. Finite sumsets of reverse-well-ordered supports are again reverse well ordered, and each fixed exponent has only finitely many contributions in each such convolution. All exponents remain in Γ ; the scalars r, α change coefficients only. Since $b \in \mathfrak{o}$ and $r \notin \mathfrak{o}$, $\alpha \neq 0$; the leading coefficients of the two polynomials in a are α and α^6 , and [lemma 11.1](#) gives the degrees. $\square$

The construction divides no series by a , expands no $(1 + h)^{-1}$ and takes no limit; it applies unchanged to supports accumulating towards zero and to exponent groups of any rank [07].

The predicate Θ detects a nonzero constant term: it rejects the nonzero element ω and accepts the nonunit $1 + \omega$.

Example 12.4 (Certificates [07]). For $a = \omega$ (or $a = \sum_{n \geq 1} \omega^{1/n}$) there is no certificate: a solution of $as = \Lambda(t)$ would give $\Lambda(\text{ct}(t)) = 0$ in $\mathbb{Z}$. For $a = 1 + \omega$ take $c = 1$, $b = 0$, so $\alpha = -\sqrt{13}$ and

$$t = -\sqrt{13}\omega, \quad s = 13(\omega - 1)(13\omega^2 - 17)(13\omega^2 - 221), \quad (1 + \omega)s = \Lambda(t);$$

the constant coefficients are $\text{ct}(t) = 0$ and $\text{ct}(s) = -48841 = \Lambda(0)$, and the irrational coefficient $-\sqrt{13}$ of t is allowed. For $a = 2 + \omega$ one may take $b = 1$, since $\Lambda(1) = -42240$ is even: then $\alpha = (1 - \sqrt{13})/2$, $t = 1 + \alpha\omega$, and $\text{ct}(s) = -21120$.

Remark 12.5 (A root in the coefficient field is needed [07]). Let $A = \mathbb{Z} + \omega\mathbb{Q}[\omega]$, an intermediate ring for $\mathbb{Z} \subseteq \mathbb{Q}$ with $\Gamma = \mathbb{Z}$. The proof of [theorem 11.9](#) still shows that Ξ defines $\mathbb{Z}$ in A (it needs only Pell rigidity and the absence of integer roots of Λ). The detector fails: evaluation at $\omega = -1$ is a surjective homomorphism $A \rightarrow \mathbb{Q}$ (the element $-q\omega$ maps to q) whose kernel is $(1 + \omega)A$, because division by $1 + \omega$ in $\mathbb{Q}[\omega]$ leaves a quotient with the same constant coefficient. So $A/(1 + \omega)A \cong \mathbb{Q}$ has no root of Λ , and $(1 + \omega)s = \Lambda(t)$ has no solution although $\text{ct}(1 + \omega) = 1$. Here $\mathbb{Q}$ contains no root of Λ , so hypothesis (b) of [theorem 12.1](#) fails. The detector is not asserted for arbitrary intermediate rings, which need not contain the polynomial witnesses. For example, take $A = \mathbb{Z}[\omega] \subseteq \mathbb{R}[\omega]$. Here $\text{ct}(A) \subseteq \mathbb{Z}$ and the ambient coefficient field contains $\sqrt{13}$, but a certificate $(1 + \omega)s = \Lambda(t)$ in A would give $\Lambda(t(-1)) = 0$ in $\mathbb{Z}$ upon evaluation at $\omega = -1$. Thus the full pullback condition $A = \mathfrak{o} + \ker \varepsilon$ also matters.

12.2 The largest ideal without a root

Theorem 12.6 (One polynomial tests every ideal [07]). *Let $(\mathfrak{o}, \mathbb{k})$ be $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ and $I = \Pi_{\mathbb{k}}(\Gamma)$. For every ideal J of A ,*

$$(\exists t \in A \quad \Lambda(t) \in J) \iff J \not\subseteq I. \quad (12.4)$$

Thus Λ has no root in A/J exactly when $J \subseteq I$, and I is the unique largest ideal of A whose quotient has no root of Λ .

Proof. If $J \subseteq I$ and $\Lambda(t) \in J$, applying ct gives $\Lambda(\text{ct}(t)) = 0$ in $\mathfrak{o}$, which is impossible. If $J \not\subseteq I$, choose $a \in J$ with $\text{ct}(a) \neq 0$; [theorem 12.2](#) gives $as = \Lambda(t)$, which lies in J . Since $A/I \cong \mathfrak{o}$ has no root, I has the property and contains every ideal that has it. $\square$

For a class ideal of a proper-class ring the displayed equivalence is the primary formulation; no set of quotient classes is formed. The case $J = A$ is allowed: the zero ring has roots of every polynomial [07].

Corollary 12.7 (A universal definition of the ideal [07]). *In the rings of [theorem 12.6](#), $I = \{a : \forall s, t \ (as \neq \Lambda(t))\}$, and $\text{ct}(a) = \text{ct}(b)$ if and only if $\forall s, t \ ((a - b)s \neq \Lambda(t))$.*

Proof. Negate [theorem 12.2](#), and use additivity of ct . $\square$

As a definition of I this universal formula is superseded by the existential one of [theorem 11.2](#); it is kept as the negation of the detector. Together, [theorems 11.2](#) and [12.2](#) make both I and its complement Diophantine. Source 07 also notes the external description $I = \bigcap_{n \geq 1} nA$ for $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$, of which [\(3.2\)](#) is the case of **Oz**. Indeed, writing $x = d + u$ with $d \in \mathfrak{o}$ and $u \in I$, we have $x \in nA$ exactly when $d \in n\mathfrak{o}$: necessity follows by applying ct , and sufficiency uses $u/n \in I$. An ordinary integer divisible by every positive integer is zero; applying this to both coordinates gives the same conclusion in $\mathbb{Z}[i]$. If instead $\mathfrak{o}$ contains $\mathbb{Q}$, then $nA = A$ for every $n \geq 1$, so this intersection does not detect I .

12.3 The graph of the constant term

Theorem 12.8 (The graph of ct is Diophantine [06]). *Let $(\mathfrak{o}, \mathbb{k})$ be $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$ and $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. The formula*

$$\text{CT}(x, n) : \iff \Xi(n) \wedge \exists y \ ((x - n)^2 = 2y^2) \quad (12.5)$$

*defines the graph of $\text{ct} : A \rightarrow \mathfrak{o}$; for each x there is exactly one such n . It is Diophantine, with six witnesses. In **Oz**, Ξ may be replaced by any Diophantine definition of $\mathbb{Z}$.*

Proof. The first clause says $n \in \mathfrak{o}$ ([theorem 11.9](#)); by [theorem 11.2](#) the second says $x - n \in I$. In the decomposition $A = \mathfrak{o} \oplus I$ the two conditions together say exactly $n = \text{ct}(x)$. Existence is explicit: take $n = \text{ct}(x)$, choose the five witnesses for $\Xi(n)$, and put $y = (x - n)/\sqrt{2} \in I$. Uniqueness concerns the output n , not the six existential witnesses: two possible outputs differ by an element of $\mathfrak{o} \cap I = \{0\}$. $\square$

Source 06 proves the theorem with its own predicates, the quintic in the real case and the Gaussian system in the Gaussian case; its single real polynomial $P(n, \dots)^2 + ((x - n)^2 - 2y^2)^2$, with P the quintic of [\(11.5\)](#), has total degree ten and eight witnesses. Using the guard of [theorem 10.3](#) instead gives degree four:

Corollary 12.9 (A quartic for the graph of ct [merge]). *Let $\Psi_1(X, U, V, W_1, W_2, W_3)$ be the guard polynomial (10.2) with $n = 1$. For $x, m \in \mathbf{Oz}$,*

$$m = \text{ct}(x) \iff \exists u, v, w_1, w_2, w_3, y \in \mathbf{Oz} \quad \Psi_1(m, u, v, w_1, w_2, w_3) + ((x - m)^2 - 2y^2)^2 = 0.$$

The polynomial has integer coefficients, total degree four and six witnesses. With the six-witness quartics (10.6) or (10.7) in place of Ψ_1 the same holds with seven witnesses.

Proof. The polynomial is a sum of three squares of integer polynomials, since Ψ_1 is itself a sum of two. In the ordered ring $\mathbf{Oz}$ it vanishes exactly when the three squared polynomials vanish, that is, when $\Psi_1(m, u, v, w_1, w_2, w_3) = 0$ and $(x - m)^2 = 2y^2$. By theorem 10.3, witnesses for the first equation exist exactly when $m \in \mathbb{Z}$, and by theorem 11.2, a witness y for the second exists exactly when $x - m \in \Pi$. Both together say $m = \text{ct}(x)$. Every term has degree at most four, and the coefficient of x^4 is 1, since Ψ_1 does not involve x ; hence the total degree is exactly four. The witnesses are u, v, w_1, w_2, w_3, y . The variant follows in the same way from theorem 10.4 or remark 10.5. $\square$

By remark 10.8 and corollary 11.5 the corollary holds verbatim in $\mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma)$ for every Γ . No single polynomial is claimed in the Gaussian case.

Corollary 12.10 (The canonical splitting is definable [06]). *For the rings of theorem 12.8, the exact sequence $0 \rightarrow I \rightarrow A \xrightarrow{\text{ct}} \mathfrak{o} \rightarrow 0$ has a parameter-free definable retraction and a parameter-free definable section; its kernel and the graph of the retraction are Diophantine.*

Proof. The predicate Ξ defines the embedded copy of $\mathfrak{o}$. On that domain the section is the inclusion, with graph $x = n$. The retraction has graph CT , and its kernel is defined by Inf . These formulas have no parameters. $\square$

The direct-sum decomposition is additive. In the coordinates $x = n + u \leftrightarrow (n, u)$ its multiplication is

$$(n, u)(m, v) = (nm, nv + mu + uv),$$

so the splitting does not assert a direct product of rings. The existence of the sequence is elementary normal-form algebra; the content is the definability of the actual splitting, not merely an interpretation of A/I [06].

Remark 12.11 (Source 07's prenex definitions; a second route [07]). Source 07 defines the graph by $\text{CT}^\forall(a, c) : \Leftrightarrow \Xi(c) \wedge \forall s, t ((a - c)s \neq \Lambda(t))$, with the two prenex forms

$$\exists \mathbf{u} \forall s, t [\mathcal{E}(c; \mathbf{u}) \wedge (a - c)s \neq \Lambda(t)], \quad \forall s, t \exists \mathbf{u} [\mathcal{E}(c; \mathbf{u}) \wedge (a - c)s \neq \Lambda(t)],$$

where $\mathbf{u}$ is a quintuple of witnesses. In the first form the witnesses for $\Xi(c)$ do not occur in the universal inequality; in the second, a single pair, for example $(s, t) = (0, 0)$, already produces witnesses for $\Xi(c)$, and conversely a fixed witness works for all (s, t) . Both are correct by theorem 11.9 and corollary 12.7. Source 07 claimed neither minimality nor the impossibility of an existential definition; source 06's theorem 12.8 supplies one.

12.4 Homomorphisms preserve the constant term

Theorem 12.12 (All homomorphisms respect the constant term [06]). *Let $\mathbb{k}, \mathbb{k}'$ be fields of characteristic zero containing $\sqrt{2}$, and Γ, Γ' ordered abelian groups. Every unital ring homomorphism $f : \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}', \Gamma')$ satisfies $\text{ct}(f(x)) = \text{ct}(x)$. If $i \in \mathbb{k} \cap \mathbb{k}'$ and $f : \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma) \rightarrow \mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}', \Gamma')$ is a unital ring homomorphism, then $\text{ct}(f(x)) = \sigma(\text{ct}(x))$, where σ is the identity or complex conjugation on $\mathbb{Z}[i]$, according as $f(i) = i$ or $f(i) = -i$. No size restriction on the target is imposed.*

Proof. Write $x = n + a$ with $n \in \mathfrak{o}$ and a in the ideal. By [corollary 11.3](#), $f(a)$ lies in the target ideal. A unital homomorphism fixes ordinary integers. In the Gaussian case $f(i)^2 = -1$, and by [lemma 11.1](#) a root of $T^2 + 1$ in the target ring is a constant, hence $\pm i$. Taking constant terms of $f(x) = f(n) + f(a)$ gives the claim. $\square$

Source 06 states the theorem for the rings of its arithmetic sections; the proof uses only that both coefficient fields contain $\sqrt{2}$. It generalizes [proposition 3.5](#) (endomorphisms of $\mathbf{Oz}$) and source 07's statement that ring automorphisms of $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ preserve $\mathfrak{o}$, I and ct up to σ , which source 07 derives from the parameter-free definitions. It does not say that homomorphisms kill the ideal: the identity does not. Preservation of the ideal, annihilation of the ideal and pointwise fixation of the real coefficients are three different assertions [06]. For general homomorphisms the proof uses preservation of equations and their existential witnesses; parameter-free first-order definability alone would only ensure invariance under automorphisms. For set-sized Γ' it is consistent with [cited theorem 3.6](#): every unital homomorphism $\mathbf{Oz} \rightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{R}, \Gamma')$ is then $x \mapsto \text{ct}(x)1$. Source 07's version concerns ring automorphisms of the coefficient-restricted ring; it does not classify field automorphisms of $\mathbf{No}$ or $\mathbf{No}[i]$, nor claim that an arbitrary field automorphism preserves $\mathbf{Oz}$ or $\mathbf{Oz}[i]$.

12.5 Number-field coefficient rings

The polynomial Λ cannot serve every number field, since a field containing $\sqrt{13}$ has a root of it. A tailored polynomial of the same shape exists for each fixed number field [07].

Lemma 12.13 (A tailored intersective polynomial [07]). *For every number field K there are distinct primes p, q such that $\Lambda_K(T) = (T^2 - p)(T^2 - q)(T^2 - pq)$ has no root in K and has a root modulo every positive integer.*

Proof. A number field has only finitely many quadratic subfields: it lies in a finite Galois extension of $\mathbb{Q}$, whose Galois group has finitely many subgroups. By Dirichlet's theorem choose a prime $p \equiv 1 \pmod{8}$ with $\mathbb{Q}(\sqrt{p}) \not\subseteq K$, and then a prime $q \equiv 1 \pmod{8p}$ with $\mathbb{Q}(\sqrt{q}), \mathbb{Q}(\sqrt{pq}) \not\subseteq K$; only finitely many q are excluded, since distinct positive squarefree radicands give distinct quadratic fields. Both arithmetic progressions have residue 1 coprime to their modulus, so Dirichlet supplies infinitely many candidates; also $q \equiv 1 \pmod{p}$ ensures $q \neq p$. Then none of p, q, pq is a square in K . For an odd prime $\ell \notin \{p, q\}$, one of p, q, pq is a nonzero square modulo ℓ , as in [proposition 11.8](#). At $\ell = p$ use $q \equiv 1 \pmod{p}$. At $\ell = q$, quadratic reciprocity and $p \equiv 1 \pmod{4}$ give $(p/q) = (q/p) = (1/p) = 1$. The nonzero roots lift to all odd prime powers, and at 2 the number $p \equiv 1 \pmod{8}$ has a root modulo every 2^a by the lifting step of [proposition 11.8](#). The Chinese remainder theorem finishes the proof. Dirichlet's theorem and reciprocity are classical inputs [26]. $\square$

Theorem 12.14 (Coefficient rings in number fields [07]). *Let $\mathfrak{o}$ be a unital subring of a number field K , let $\mathbb{k} \supseteq K$ be a field of characteristic zero, and let A be an intermediate ring for $\mathfrak{o} \subseteq \mathbb{k}$. Let Ξ_K be Ξ with Λ replaced by Λ_K . Then Ξ_K defines $\mathfrak{o}$ in A . If moreover $A = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ and $\mathbb{k}$ contains a root of Λ_K , then $\text{ct}(a) \neq 0 \Leftrightarrow \exists s, t (as = \Lambda_K(t))$, and the ideal test, the universal definition of the ideal and both prenex definitions of the graph of ct hold with Λ_K .*

Proof. The proof of [theorem 11.9](#) used Pell rigidity ([corollary 11.5](#)), the absence of roots of the polynomial in $\mathfrak{o}$ ([lemmas 11.1](#) and [12.13](#)), a nonzero integer in every $x\mathfrak{o}$ ([lemma 11.7](#)) and roots modulo all integers. More explicitly, for $x \neq 0$ the system forces $u, v \in \mathfrak{o}$ by Pell rigidity and $v \neq 0$ by root exclusion. Then $xw = v$ forces x to have degree zero, hence to lie in $A \cap \mathbb{k} = \mathfrak{o}$. Conversely, if $0 \neq x \in \mathfrak{o}$, choose $m > 0$ in $x\mathfrak{o}$ and ordinary Pell coordinates u, v

with $0 < v \in m\mathbb{Z}$. Put $w = v/x \in \mathfrak{o}$, choose $t \in \mathbb{Z}$ with $\Lambda_K(t) \in v\mathbb{Z}$, and put $s = \Lambda_K(t)/v \in \mathbb{Z}$. These are witnesses already in $\mathfrak{o}$; $x = 0$ is admitted by the factors x in the defining equations. No integrality or finite-generation hypothesis on $\mathfrak{o}$ is used. For the detector, given $c \neq 0$ in $\mathfrak{o}$ choose a positive integer $m \in c\mathfrak{o}$ and $b \in \mathbb{Z}$ with $\Lambda_K(b) \in m\mathbb{Z}$, and apply [theorem 12.1](#) with any root of Λ_K in $\mathbb{k}$. The ideal test and the graph use only the detector and the constant definition. $\square$

The theorem covers subrings of a number field that are not finitely generated. It does not give one polynomial uniform in all number fields, and does not apply as stated to algebraic extensions of $\mathbb{Q}$ of infinite degree: no polynomial of this kind is root-free in a field containing all algebraic numbers [07].

Remark 12.15 (Existential definitions over number-field coefficients [merge]). The formula $\exists y (x^2 = 2y^2)$ of [theorem 11.2](#) fails when $\sqrt{2} \in \text{Frac}(\mathfrak{o})$, for instance for $\mathfrak{o} = \mathbb{Z}[\sqrt{2}]$. In the setting of [theorem 12.14](#) choose a prime δ with $\mathbb{Q}(\sqrt{\delta}) \not\subseteq K$; it exists because K has finitely many quadratic subfields and distinct primes give distinct fields $\mathbb{Q}(\sqrt{\delta})$. Then δ is not a square in $K \supseteq \text{Frac}(\mathfrak{o})$, since $z^2 = \delta$ with $z \in K$ would give $\mathbb{Q}(\sqrt{\delta}) = \mathbb{Q}(z) \subseteq K$. If $\sqrt{\delta} \in \mathbb{k}$, [remark 11.4](#) shows that $\exists y (x^2 = \delta y^2)$ defines $\Pi_{\mathbb{k}}(\Gamma)$ in $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$, and $\Xi_K(n) \wedge \exists y ((x-n)^2 = \delta y^2)$ defines the graph of ct: the first clause says $n \in \mathfrak{o}$, by [theorem 12.14](#) applied to the intermediate ring $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$, and the second says $x - n \in \Pi_{\mathbb{k}}(\Gamma)$. Both definitions are Diophantine; they improve source 07's universal and prenex formulas whenever $\sqrt{\delta} \in \mathbb{k}$, and need no root of Λ_K in $\mathbb{k}$.

In particular, the detector hypothesis in [theorem 12.14](#) already suffices for these existential conclusions. If $\Lambda_K(r) = 0$ in the field $\mathbb{k}$, one of its three factors vanishes, so $r^2 = \delta$ for some $\delta \in \{p, q, pq\}$. Each of these radicands is nonsquare in K , by [lemma 12.13](#). The preceding argument therefore applies with this δ , including the composite choice pq : primality was only a way to find a suitable radicand. Thus the ideal has a one-witness Diophantine definition, its complement has the two-witness detector, and the graph of ct has a six-witness Diophantine definition under exactly the stated detector hypotheses. This is a further consequence of combining sources 06 and 07. The integer δ is chosen once for the fixed coefficient fields; the formulas contain its numeral, not a parameter naming r , and no uniform choice across all number fields is asserted.

13 Coefficient reconstruction and logical consequences

13.1 The multiplier ring of the ideal

Throughout this subsection $\Gamma \neq 0$, $\mathfrak{o} \subseteq \mathbb{k}$ is a unital subring, $\mathcal{R} = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$, $I = \Pi_{\mathbb{k}}(\Gamma)$, $\mathcal{A} = \mathcal{A}_{\mathbb{k}}(\Gamma)$, and $\mathcal{F} = \text{Frac}(\mathcal{R})$, viewed inside the Hahn field of all series with reverse-well-ordered support in Γ [06]. Put $(I : I)_{\mathcal{F}} = \{r \in \mathcal{F} : rI \subseteq I\}$.

Lemma 13.1 ([06]). $\mathcal{A} \subseteq \mathcal{F}$.

Proof. Choose $g \in \Gamma^{>0}$. For $a \in \mathcal{A}$ both ω^g and $a\omega^g$ lie in $I \subseteq \mathcal{R}$, and $\omega^g \neq 0$, so $a = (a\omega^g)/\omega^g \in \mathcal{F}$. This also covers constants $c \in \mathbb{k}$: although c need not belong to $\mathcal{R}$, the fraction $(c\omega^g)/\omega^g$ does. No common positive lower bound on the support of a is needed. $\square$

Theorem 13.2 (The multiplier identity [06]). *Inside $\mathcal{F}$,*

$$(I : I)_{\mathcal{F}} = \mathcal{A}, \quad \mathbb{k} = \{0\} \cup ((I : I)_{\mathcal{F}})^{\times}. \quad (13.1)$$

Proof. A nonnegative and a strictly positive exponent add to a strictly positive exponent, so $\mathcal{A}I \subseteq I$. Conversely let $r \in \mathcal{F} \setminus \mathcal{A}$ and choose a negative exponent h in $\text{supp}(r)$; it need not be the least, and there need not be a least. Then $\omega^{-h} \in I$, and the coefficient of $r\omega^{-h}$ at exponent zero is the nonzero coefficient of ω^h in r , so $r\omega^{-h} \notin I$. Multiplication by a single monomial translates support injectively; no other coefficient contributes at zero and no cancellation can remove it. This witnesses the failure to be a multiplier. The second identity follows from $\mathcal{A}^\times = \mathbb{k}^\times$ (lemma 11.1). $\square$

An integral domain interprets its fraction field by pairs (a, b) with $b \neq 0$, where $(a, b) \sim (c, d)$ means $ad = bc$. Addition and multiplication are represented by $(ad + bc, bd)$ and (ac, bd) , and the original ring by $a \mapsto (a, 1)$. The denominator bd is nonzero because the ring is a domain. For the proper classes “quotient” means this definable presentation, with set-valued representatives supplied by the surreal quotient, and no proper class is used as an element of another [06]. With Inf from theorem 11.2, put

$$\begin{aligned} \text{Mult}(a, b) &: \iff b \neq 0 \wedge \forall x (\text{Inf}(x) \Rightarrow \exists y (\text{Inf}(y) \wedge ax = by)), \\ \text{Coeff}(a, b) &: \iff b \neq 0 \wedge (a = 0 \vee (a \neq 0 \wedge \text{Mult}(a, b) \wedge \text{Mult}(b, a))). \end{aligned}$$

The first says that the fraction a/b multiplies I into I , and is invariant under change of representative. Indeed $ax = by$ is equivalent in $\mathcal{F}$ to $(a/b)x = y$; all x, y in this formula range over the original ring, where Inf defines I . The second formula handles zero separately and otherwise requires both a/b and its inverse b/a to be multipliers, so theorem 13.2 identifies it with $a/b \in \mathbb{k}$. It does not assert that every nonzero multiplier is a coefficient: $\omega^g \in \mathcal{A}$ is not a unit there for $g > 0$.

Corollary 13.3 (Internal reconstruction [06]). *Assume $\Gamma \neq 0$ and the hypotheses of theorem 11.2 (for instance in the real and Gaussian rings). Then $\mathcal{A}$ and the actual embedded coefficient field $\mathbb{k}$ are parameter-free first-order definable in the interpreted fraction field, and the coefficient map $\mathcal{A} \rightarrow \mathbb{k}$ has the definable graph $\{(a, c) : a \in \mathcal{A}, c \in \mathbb{k}, a - c \in I\}$.*

Proof. Use Mult and Coeff in the interpreted fraction field. The last condition means that $a - c$ is the image of an element of the original ring satisfying Inf . Every $a \in \mathcal{A}$ has a unique decomposition $a = c + u$ with $c \in \mathbb{k}$ and $u \in I$, so the displayed relation is exactly the coefficient map. This recovers the embedded field, not just an abstract isomorphic copy. $\square$

The same reconstruction works whenever I has a parameter-free definition in $\mathcal{R}$: the multiplier argument imposes no condition on $\mathfrak{o}$ beyond being a unital subring of $\mathbb{k}$. In particular, for $\Gamma \neq 0$ the integer-radical definitions in remark 12.15 recover the coefficient field and map under the number-field detector hypotheses, even when 2 is a square in $\text{Frac}(\mathfrak{o})$. Substitute that ideal predicate for Inf in the two formulas above; no square root is named in them.

The formulas do not mention monomials; monomials are used only to prove that they define the right sets, and $\sqrt{2}$ is not a hidden parameter. The multiplier formula contains a universal quantifier, so the reconstruction is first order and is not claimed positive existential [06]. Two boundaries are necessary. If $\Gamma = 0$, then $I = 0$ and every fraction multiplies it, so the formula recovers $\text{Frac}(\mathfrak{o})$ rather than $\mathbb{k}$. And $\mathcal{F}$ need not be the whole Hahn field: for $\Gamma = \mathbb{Z}$ it is the rational function field $\mathbb{k}(\omega)$; the multiplier identity holds in $\mathcal{F}$ without any such equality [06]. For example, in $\mathbb{Z} + \omega\mathbb{R}[\omega]$ the ideal $\omega\mathbb{R}[\omega]$ is defined by Inf , the fraction field is $\mathbb{R}(\omega)$, the multiplier ring is $\mathbb{R}[\omega]$ and its units are $\mathbb{R}^\times$; the reconstruction phenomenon is not an artifact of proper-class size. With $\Gamma = \mathbb{R}$ and $f = \sum_{n \geq 1} \omega^{1/n}$, the ideal formula is witnessed by $f/\sqrt{2}$, and the multiplier proof uses one negative exponent, not a least one [06].

13.2 Reconstructing the surreal field

Theorem 4.6 gives $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, with a common monomial denominator for every set of surreals; applied to the set of real and imaginary parts it gives $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$. Source 06 proves the same statement (its common-denominator lemma), printed once as [theorem 4.6](#); it uses bounds for sets of exponents only, and asserts no common denominator for a proper class of inputs.

Theorem 13.4 (Reconstruction of the real structure [06]). *From the pure ring $\mathbf{Oz}$ one can parameter-free interpret the field $\mathbf{No}$ and define in it the canonical subfield $\mathbb{R}$, the order, the ring $\mathcal{O}_{\text{fin}}$ of finite elements, the ideal $\mathfrak{m}$ of infinitesimals and the graph of the standard part $\text{st} : \mathcal{O}_{\text{fin}} \rightarrow \mathbb{R}$.*

Proof. Interpret the fraction field, which is $\mathbf{No}$. By [corollary 13.3](#) the predicate **Coeff** defines its subfield $\mathbb{R}$. Since $\mathbf{No}$ is real closed [1, 2], $x > 0 \Leftrightarrow x \neq 0 \wedge \exists z (z^2 = x)$ defines the order, with $x < y$ obtained by applying it to $y - x$. Here z ranges over the interpreted field, not just $\mathbf{Oz}$. Put $\mathcal{O}_{\text{fin}}(x) :\Leftrightarrow \exists r \in \mathbb{R} (r > 0 \wedge -r \leq x \leq r)$ and $\mathfrak{m}(x) :\Leftrightarrow \forall r \in \mathbb{R} (r > 0 \Rightarrow -r < x < r)$. A nonzero surreal is bounded by a real number exactly when its leading exponent is nonpositive. For a finite surreal, removing the coefficient at zero leaves either zero or a remainder with a largest, strictly negative exponent; that remainder is therefore smaller in absolute value than every positive real. This proves existence of standard part, including zero by taking coefficient zero. It is unique because a nonzero real is not infinitesimal. So $\text{ST}(x, r) :\Leftrightarrow \mathcal{O}_{\text{fin}}(x) \wedge r \in \mathbb{R} \wedge \mathfrak{m}(x - r)$ defines the standard part. $\square$

The two constant-term operations must not be confused ([remark 2.3](#)): **ct** reads the constant term of a number with nonnegative exponents, **st** that of a finite number, with nonpositive exponents. For example $\text{ct}(\omega + 7) = 7$ and $\text{st}(7 + \omega^{-1}) = 7$, but $\omega + 7$ is not finite [06].

Corollary 13.5 (The value group [06]). *The natural ordered value group of $\mathbf{No}$ is interpretable from $\mathbf{Oz}$ as $\mathbf{No}^\times / \mathcal{O}_{\text{fin}}^\times$, ordered by $v(x) \geq v(y) \Leftrightarrow x/y \in \mathcal{O}_{\text{fin}}$.*

Proof. For $x \neq 0$, membership in $\mathcal{O}_{\text{fin}}^\times$ means that both x and x^{-1} are finite. Thus $x \sim y$ iff $x/y \in \mathcal{O}_{\text{fin}}^\times$ is definable, and multiplication descends to the quotient. If the leading exponents of x, y are g, h , then x/y has leading exponent $g - h$. It is finite exactly when $g \leq h$, and a finite unit exactly when $g = h$. Consequently the displayed relation is a well-defined total order on the quotient, compatible with multiplication; in additive valuation notation $v(xy) = v(x) + v(y)$ it corresponds externally to $v(x) = -g$. $\square$

This recovers the value group as an interpreted quotient, not a canonical cross-section of monomials, and does not show that the ω -map, the simplicity order, birthdays or the surreal exponential are definable [06]. The external representative $-g$ also explains the proper-class convention: it is a surreal number representing a valuation class, not a proper class used as an element. Its use to verify the quotient does not put a definable monomial cross-section into the interpreted language. The fraction-field construction can be phrased as a canonical bi-interpretation between the pure ring $\mathbf{Oz}$ and the field $\mathbf{No}$ with a predicate for $\mathbf{Oz}$; the essential content is the reconstruction of the *canonical* coefficient field.

13.3 Automorphisms of $\mathbf{Oz}$

Theorem 13.6 (Automorphisms fix the real coefficients [06]). *Every unital ring automorphism f of $\mathbf{Oz}$ extends uniquely to a field automorphism $\tilde{f}(a/b) = f(a)/f(b)$ of $\mathbf{No}$. It preserves the order, $\mathbb{R}$, $\mathcal{O}_{\text{fin}}$ and $\mathfrak{m}$, fixes every real number, and commutes with **st**.*

Proof. The extension is well defined by cross multiplication and unique because every element of $\mathbf{No}$ is a fraction. It preserves the parameter-free definable sets of [theorem 13.4](#), so it restricts to an order-preserving automorphism of $\mathbb{R}$ fixing $\mathbb{Q}$. If $\tilde{f}(r) \neq r$, a rational strictly between r and $\tilde{f}(r)$ contradicts order preservation. The definable graph of st , with its real coordinate fixed, gives $\text{st} \circ \tilde{f} = \text{st}$ on $\mathcal{O}_{\text{fin}}$. $\square$

The statement concerns automorphisms. A universal multiplier formula need not be preserved by a nonsurjective embedding, so the argument does not apply unchanged to endomorphisms; [theorem 12.12](#) applies to all unital homomorphisms [06].

Example 13.7 (Coefficient rigidity is not total rigidity [06]). The map $S_2(\sum_g a_g \omega^g) = \sum_g a_g \omega^{2g}$ is a field automorphism of $\mathbf{No}$ with inverse $S_{1/2}$, since doubling exponents is an order automorphism of $(\mathbf{No}, +)$; it preserves supports and $\mathbf{Oz}$, fixes real coefficients and sends ω to ω^2 . Nonnegative exponents remain nonnegative, and the coefficient at zero is unchanged, so the inverse also preserves $\mathbf{Oz}$. This is an exponent substitution, not the squaring map $x \mapsto x^2$; for example $S_2(1 + \omega) = 1 + \omega^2$. Thus $\text{Aut}(\mathbf{Oz})$ is not trivial.

The dilations S_a are studied in the repository's surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`), and the single-dilation report (`docs/surcomplex/single-dilation-hahn-support`) defines the coefficient field and the constant coefficient from one dilation of a field; [theorem 13.4](#) starts instead from the pure ring $\mathbf{Oz}$ and its Diophantine ideal. [Theorem 13.6](#) and [example 13.7](#) are also relevant to the questions on automorphisms of $\mathbf{Oz}$ studied in the sibling report `docs/surreal/set-sized-quotients-of-omnific-integers`.

13.4 The Gaussian ring does not determine the real axis

The Gaussian system and the multiplier theorem recover $\mathbf{No}[i] = \text{Frac}(\mathbf{Oz}[i])$ and its actual coefficient field $\mathbb{C}$, but not the real form $\mathbf{No}$ [06].

Proposition 13.8 (All coefficient automorphisms occur [06]). *Every field automorphism σ of $\mathbb{C}$ extends coefficientwise to a field automorphism $\hat{\sigma}(\sum_g c_g \omega^g) = \sum_g \sigma(c_g) \omega^g$ of $\mathbf{No}[i]$ preserving $\mathbf{Oz}[i]$. Every automorphism of $\mathbf{Oz}[i]$ extends to $\mathbf{No}[i]$ and preserves $\mathbb{C}$ setwise, so restriction to $\mathbb{C}$ is surjective and $\sigma \mapsto \hat{\sigma}$ is a composition-preserving section.*

Proof. A coefficient automorphism preserves supports and the finite sums and products defining Hahn convolution; its inverse is $\widehat{\sigma^{-1}}$. The coefficients of each product are finite sums, so no continuity or analytic regularity of σ is needed. It fixes $\mathbb{Z}$ and sends i to $\pm i$, so it preserves $\mathbb{Z}[i]$ and hence $\mathbf{Oz}[i]$. An automorphism of $\mathbf{Oz}[i]$ extends to the fraction field and preserves the first-order coefficient predicate of [corollary 13.3](#). Finally $\widehat{\sigma \circ \rho} = \hat{\sigma} \circ \hat{\rho}$ coefficientwise, and restriction of $\hat{\sigma}$ to $\mathbb{C}$ is σ , giving the stated section. $\square$

For proper-class fields this is a statement about individual maps and their compositions, not a construction of a set of all class automorphisms.

Let $\eta : \mathbf{No} \rightarrow \mathbb{R}$ send a surreal to the coefficient of ω^0 in its normal form. It is additive on $\mathbf{No}$ but not multiplicative ($\eta(\omega) = \eta(\omega^{-1}) = 0$, $\eta(1) = 1$), and is used only as an additive map on the exponent group. Put $\chi(g) = \exp(i\pi\eta(g)/2) \in \mathbb{C}^\times$, the ordinary complex exponential of an ordinary real multiple of i ; then $\chi(g+h) = \chi(g)\chi(h)$, $\chi(0) = 1$ and $\chi(1) = i$.

Lemma 13.9 (A phase twist preserving $\mathbf{Oz}[i]$ [06]). *The map $\tau(\sum_g c_g \omega^g) = \sum_g c_g \chi(g) \omega^g$ is an automorphism of $\mathbf{No}[i]$ that fixes $\mathbb{C}$ pointwise, preserves every support and $\mathbf{Oz}[i]$, and satisfies $\tau(\omega) = i\omega$.*

Proof. All values of χ are nonzero, so supports are unchanged; additivity is clear. If $a + b = g$, the contribution of (a, b) to the coefficient at g is multiplied by $\chi(a)\chi(b) = \chi(g)$, so τ commutes with convolution; χ^{-1} gives the inverse. Constants sit at exponent zero and are fixed. A Gaussian omnific integer keeps its positive support and its Gaussian integer constant term, since $\chi(0) = 1$; the inverse has the same property. These operations are coefficientwise on set-supported normal forms, without an infinite analytic summation. The exponential in χ is only the ordinary complex exponential of a real scalar. $\square$

The character-twist construction and its use to move the real axis while fixing complex coefficients are already in the repository's surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`); source 06 credits it there. The additional fact needed here is that the twist preserves $\mathbf{Oz}[i]$.

Theorem 13.10 (No definable real axis or conjugation [06]). *The subfield $\mathbf{No} \subseteq \mathbf{No}[i]$ is not first-order definable in the structure $(\mathbf{No}[i], \mathbf{Oz}[i])$, even with every complex number named by a constant, and neither is conjugation. The subring $\mathbf{Oz} \subseteq \mathbf{Oz}[i]$ is not definable in the pure ring $\mathbf{Oz}[i]$, even with parameters from $\mathbb{Z}[i]$.*

Proof. The automorphism τ preserves the field, the predicate $\mathbf{Oz}[i]$ and every complex constant, but sends $\omega \in \mathbf{No}$ to $i\omega \notin \mathbf{No}$; definable sets are invariant under automorphisms fixing the parameters. Conjugation does not commute with τ : $\tau(\bar{\omega}) = i\omega$ while $\tau(\omega) = -i\omega$. Finally the restriction of τ to $\mathbf{Oz}[i]$ fixes $\mathbb{Z}[i]$ and moves $\omega \in \mathbf{Oz}$ outside $\mathbf{Oz}$. Naming every element of $\mathbb{C}$ does not affect the first obstruction: τ fixes each of them, and every individual formula uses only finitely many names. For the restriction to $\mathbf{Oz}[i]$, the parameters $\mathbb{Z}[i]$ already lie in that ring; arbitrary complex constants need not. $\square$

There is no conflict with the recovery of $\mathbb{C}$: the twist fixes $\mathbb{C}$ pointwise. The coefficient field and a distinguished real form are different structure. Naming a suitable involution would suffice, since its fixed ring can be taken, but that gives the real structure explicitly instead of reconstructing it [06].

13.5 Logical consequences

The equational transfer of [theorem 5.1](#) is proved again by source 06 (its retraction-transfer lemma), by source 07 (positive-existential conservation) and by source 13 (equation-only specialization and the Hilbert's-tenth-problem transfer), for all rings $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ with parameters in $\mathfrak{o}$ and, in the Gaussian case, with $\mathbb{Z}[i]$ in place of $\mathbb{Z}$. It is printed once, as [theorem 5.1](#) and [corollary 5.2](#); the proof is the same for every $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. Transfer does not make solutions constant: $y = x$ has many nonconstant solutions [07]. Both the parameters and the free tuple must be in the coefficient subring for this two-way transfer. Existential witnesses may range over the entire ring: applying `ct` supplies coefficient witnesses.

Theorem 13.11 (Diophantine subsets of $\mathbb{Z}^n$ [06]). *Let $\mathbb{k}$ be an ordered field containing $\sqrt{2}$. A subset $S \subseteq \mathbb{Z}^n$, regarded as a subset of $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)^n$ (in particular of $\mathbf{Oz}^n$), is Diophantine there with integer coefficients if and only if it is computably enumerable.*

Proof. If S is defined by equations in $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$, its trace on $\mathbb{Z}^n$, which is S , is defined by the same equations over $\mathbb{Z}$, by transfer; enumerating integer witnesses shows that it is computably enumerable. Conversely, by the MRDP theorem [7] (in its integer form, by sign coding and sums of squares) S is defined over $\mathbb{Z}$ by an integer system Φ_S . In $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ impose $\text{Std}(x_1) \wedge \cdots \wedge \text{Std}(x_n) \wedge \Phi_S(\mathbf{x})$, with Std the quintic (11.5); once the free tuple is

standard, transfer makes the remaining condition equivalent to Φ_S over $\mathbb{Z}$. The guards exclude nonstandard free tuples; no guards on the auxiliary witnesses of Φ_S are needed, since ct sends any such witnesses to integer ones. In the real ring the conjunction can be written as one sum of squares. $\square$

Here S is required to consist of standard tuples. No computability classification of arbitrary subsets of the ring is claimed, and arbitrary surreal parameters are not finite inputs to an algorithm [06].

Generality of the classification [merge]. With the finite-system convention for Diophantine definitions, the same equivalence holds in $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ for every characteristic-zero field $\mathbb{k}$, without an ordering or a square root of 2. Apply [theorem 12.14](#) with $\mathfrak{o} = \mathbb{Z}$ and $K = \mathbb{Q}$ to obtain the integer guard $\Xi_{\mathbb{Q}}$; its first assertion needs no root of $\Lambda_{\mathbb{Q}}$ in $\mathbb{k}$. Replace Std by this guard in the proof, and use the same retraction in both directions. All formula coefficients are integers. A single sum-of-squares equation is not claimed over an unordered field. This is a consequence of combining the order-free definition with source 06’s transfer argument.

Corollary 13.12 (Standard arithmetic is definable [06, 07]). *The structures $(\mathbb{Z}, +, \cdot, 0, 1)$ and $(\mathbb{N}, +, \cdot, 0, 1)$ are parameter-free definable in the pure ring $\mathbf{Oz}$, and every arithmetic sentence has an effective translation obtained by relativizing its quantifiers to Std and to [corollary 11.11](#). For a set-sized intermediate ring A for $\mathbb{Z} \subseteq \mathbb{R}$, true arithmetic therefore reduces many-one to $\text{Th}(A)$; so $\text{Th}(A)$ is undecidable and, being complete, is not recursively axiomatizable.*

Proof. The translation replaces $\exists x$ by $\exists x(\text{Std}(x) \wedge \dots)$ and $\forall x$ by $\forall x(\text{Std}(x) \Rightarrow \dots)$; induction on formulas proves it correct. For A use Ξ ([theorem 11.9](#)) in place of Std , and the same four-square definition of $\mathbb{N}$, which is valid in A because sums of squares are nonnegative in the ordered ring $\mathcal{A}_{\mathbb{R}}(\Gamma)$. A recursive axiomatization of a complete theory would decide each sentence by searching for a proof of it or of its negation. $\square$

For $\mathbf{Oz}$ the same holds sentence by sentence under the class convention; no definable global satisfaction predicate for the class ring is asserted. Source 07 credits the non-axiomatizability of omnific arithmetic to Jeřábek’s 2011 MathOverflow answer [19], and source 06 cites the same answer for the logical limitations of $\mathbf{Oz}$; the undecidability of integer-coefficient solvability over $\mathbf{Oz}$ follows from transfer alone ([corollary 5.2](#)). None of these consequences is presented as new, and none is a new undecidability theorem about $\mathbb{Q}$ or about number fields.

Theorem 13.13 (What equations cannot define [06, 07]). *Let $\Gamma \neq 0$, let $\mathfrak{o} \subseteq \mathbb{k}$ be a unital subring and $\mathcal{R} = \mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$. No positive-existential formula with parameters in $\mathfrak{o}$ defines a subset of $\mathcal{R}$ that contains a nonzero element of $\Pi_{\mathbb{k}}(\Gamma)$ and excludes 0. In particular $\mathcal{R} \setminus \{0\}$, $\mathcal{R} \setminus \mathfrak{o}$ and $\Pi_{\mathbb{k}}(\Gamma) \setminus \{0\}$ are not so definable, and in the real case neither is the strictly positive cone.*

Proof. The composite $\mathcal{R} \xrightarrow{\text{ct}} \mathfrak{o} \hookrightarrow \mathcal{R}$ is an endomorphism fixing $\mathfrak{o}$, and positive-existential formulas with parameters fixed by an endomorphism are preserved by it (as in [theorem 5.1](#)). If such a formula held at a nonzero $a \in \Pi_{\mathbb{k}}(\Gamma)$, it would hold at the image $\text{ct}(a) = 0$. Each displayed set contains ω^g for $g \in \Gamma^{>0}$, which is positive in the real case, and excludes 0. $\square$

The parameter restriction is essential to this endomorphism argument: for a fixed nonzero $h \in I$, the equation $x = h$ defines $\{h\}$ with parameter h , which is not fixed by ct . In the real case “strictly positive” means $x > 0$, so zero is excluded. This generalizes [proposition 5.3](#). It is consistent with the Diophantine definitions of Π and of $\{\text{ct} \neq 0\}$: the first contains 0, and

the second contains no element of Π . The detector avoids the obstruction precisely because it defines $\text{ct}(a) \neq 0$, not $a \neq 0$: ω is nonzero but divides no value $\Lambda(t)$ [07]. The set $\mathcal{R} \setminus \{0\}$ has the quantifier-free definition $x \neq 0$; what fails is a positive-existential definition. Source 07 left open whether Π itself has one; [theorem 11.2](#) of source 06 shows that it has.

Proposition 13.14 (A quantifier-free lower bound [06]). *In $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ with $\Gamma \neq 0$, neither $\Pi_{\mathbb{k}}(\Gamma)$ nor $\mathbb{Z}$ is definable by a quantifier-free ring formula in one variable, even with arbitrary parameters; the same holds for the ideal and $\mathbb{Z}[i]$ in $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{k}, \Gamma)$. Hence the complete theory of such a set-sized ring does not eliminate quantifiers in the ring language.*

Proof. In an infinite domain a nonzero one-variable polynomial has finitely many roots. Each atomic equality is therefore true either everywhere (the zero polynomial) or at finitely many points; finite Boolean combinations of finite or cofinite sets remain finite or cofinite. This also covers negated equalities and any finite tuple of ring parameters. Fix $g \in \Gamma^{>0}$. The ideal contains the infinitely many distinct monomials ω^{kg} , $k \geq 1$, and its complement contains the nonzero ordinary integers. The coefficient subring is infinite and its complement contains those same monomials. Neither set is finite or cofinite. If the complete theory eliminated quantifiers, the coefficient-subring definition from [theorem 12.14](#), with $K = \mathbb{Q}$ or $\mathbb{Q}(i)$, would be equivalent to a quantifier-free formula, contradicting what was just proved. $\square$

This is a first lower bound on logical complexity only; it does not show that degree four, or five witnesses, is optimal [06].

Theorem 13.15 (Omitted types [07; also 06]). *Let A be a set-sized intermediate ring for $(\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, or a set-sized ring as in [theorem 12.14](#), and let Ξ_A be the corresponding constant-definition formula. The computable parameter-free type*

$$\{\Xi_A(x)\} \cup \{Q(x) \neq 0 : 0 \neq Q \in \mathbb{Z}[X]\}$$

is finitely satisfiable and omitted; so A is not recursively saturated. For the full rings $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ the same explicit type is finitely satisfiable and omitted.

Proof. Use a fixed canonical syntax for integer polynomials, with signed integer coefficients and no trailing zero coefficient. Testing that a formula is $\Xi_A(x)$ or a displayed inequality for a nonzero polynomial is decidable; this makes the type computable, not just countable. Its only coefficients are integer numerals, including those of the fixed guard Ξ_A . A finite part excludes the roots of finitely many nonzero polynomials. If the sum of their degrees is D , at most D of the ordinary integers $0, 1, \dots, D$ are excluded; at least one satisfies every inequality and Ξ_A . A realization of the whole type would lie in $\mathfrak{o}$, hence be algebraic over $\mathbb{Q}$; clearing denominators in an annihilating rational polynomial gives a nonzero integer polynomial vanishing at it, a contradiction. Recursive saturation would realize this computable finitely satisfiable type over no parameters. $\square$

For $\mathfrak{o} = \mathbb{Z}$ the simpler countable type $\{\text{Std}(x)\} \cup \{x \neq n : n \in \mathbb{Z}\}$ already shows that $\mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ is not countably saturated, and the Gaussian analogue holds with all Gaussian constants [06]. For the full classes this is the class-relative formulation: no set-sized model carrier is assumed. Nonsaturation of $\mathbf{Oz}$ has the 2018 precedent [18]. Even a very rich exponent order does not make these rings recursively saturated; the statement concerns the ring, not its order or a surrounding field [07].

Remark 13.16 (Transfer without elementarity [06; also 13]). For $\Gamma \neq 0$ the inclusion $\mathbb{Z} \hookrightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ is not elementary, although it transfers positive-existential formulas: the sentence $\exists x, y (x^2 = 2y^2 \wedge y \neq 0)$ is false in $\mathbb{Z}$ and in $\mathbb{Z}[i]$ and holds in the ring, with $y = \omega^g$ and

$x = \sqrt{2}\omega^g$ for any $g > 0$ (when $\sqrt{2} \in \mathbb{k}$). This is a second route to the non-elementarity recorded after [theorem 10.10](#) and in [remark 10.11](#). The same non-elementarity conclusion for $\mathbb{Z} \hookrightarrow \mathcal{R}_{\mathbb{Z}}(\mathbb{k}, \Gamma)$ holds even when $\mathbb{k}$ does not contain $\sqrt{2}$: the sentence $\forall x \Xi_{\mathbb{Q}}(x)$, using the guard above, holds in $\mathbb{Z}$ but fails at ω^g in the larger ring. The guard defines $\mathbb{Z}$ in both structures by [theorem 12.14](#); this argument uses no ordering of $\mathbb{k}$. A formula that defines the actual integers in this ring need not select the standard copy in an elementary extension, where it defines an elementary extension of the induced arithmetic; this is compatible with the omitted types above [06].

14 Infinite families, primitivity, and projective points

14.1 Polynomial curves through integer points

Proposition 14.1 (Finite polynomial lifting [01, 02, 05]). *Let $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$, and let $P_1(T), \dots, P_n(T) \in \mathbb{R}[T]$ satisfy*

$$F_j(P_1(T), \dots, P_n(T)) = 0 \quad (1 \leq j \leq r), \quad P_i(0) \in \mathbb{Z} \quad (1 \leq i \leq n).$$

Then $\mathbf{P}(t) \in \mathbf{Oz}^n$ solves the system for every $t \in \Pi$, with constant tuple $\mathbf{P}(0)$. If some P_i is nonconstant, every $t \in \Pi \setminus \{0\}$ produces an infinite point.

Proof. Every nonconstant term of $P_i(t)$ lies in Π , since Π is an ideal and a real vector space, so the constant coefficient is exactly $P_i(0)$. Polynomial identities hold under substitution into any field extension. If P_i has degree $d > 0$ and $\deg_{\omega}(t) = \gamma > 0$, its top term has degree $d\gamma$, strictly larger than every other term's degree. $\square$

No integrality is required of the nonconstant coefficients: for example $(1 + t^2/2, t^2/2, t)$ is omnific for every $t \in \Pi$ although $1/2 \notin \mathbf{Oz}$; it satisfies $X^2 - Y^2 - Z^2 = 1$ by expansion. For $t < 0$, leading degree still proves infinite size, though it need not preserve the eventual signs of the polynomials as $T \rightarrow +\infty$. The finiteness of the polynomials matters, however. Evaluating an infinite real power series at an infinite omnific argument is not licensed by this proposition; the Taylor expansion of $\sqrt{1+T}$ is not an admissible way to assert that $\sqrt{1+\omega}$ lies in $\mathbf{Oz}$. Polynomial maps supply a systematic sufficient condition for nontrivial fibers; they do not exhaust all normal-form constructions, and no such exhaustion is assumed (but see [theorem 14.7](#) for finite supports).

14.2 Infinite solutions of homogeneous equations

Theorem 14.2 (Homogeneous cone criterion [01]; part (b) [02]). *Let $F_1, \dots, F_r \in \mathbb{R}[X_1, \dots, X_n]$ be homogeneous of positive degrees.*

- (a) *Their common zero locus has a nonzero point in $\mathbf{Oz}^n$ if and only if it has a nonzero point in $\mathbb{R}^n$.*
- (b) *If the F_j have ordinary integer coefficients, the equivalence in (a) persists with any fixed combination of coordinate sign and nonvanishing conditions preserved by positive scaling, and these existence problems are decidable.*

Proof. (a) A real nonzero common zero $\mathbf{a}$ gives the omnific common zero $\omega\mathbf{a} \in \Pi^n$ by homogeneity. Conversely, take a nonzero common zero $\mathbf{x} \in \mathbf{Oz}^n$. If every coordinate is constant, it is a real point. Otherwise let $\gamma > 0$ be the maximum of the coordinates' leading

degrees and $r_i = [\omega^\gamma]x_i$; the vector $\mathbf{r}$ is nonzero. For F_j homogeneous of degree d_j , the coefficient at $\omega^{d_j\gamma}$ in $F_j(\mathbf{x})$ is $F_j(\mathbf{r})$, since a product attains that exponent only by taking exponent γ in every factor. So $\mathbf{r}$ is a nonzero real common zero. This route uses no quantifier elimination.

(b) A real solution $\mathbf{r}$ gives $\omega\mathbf{r}$, which has the same signs and zero pattern. Conversely an omnific solution lies, with $\mathbb{R}$, in a set-sized real closed subfield of $\mathbf{No}$ (generate a field and take its real closure inside $\mathbf{No}$). Quantifier elimination makes the inclusion of $\mathbb{R}$ elementary [11], so the existential statement with its finite sign conditions already holds over $\mathbb{R}$; quantifier elimination is an effective decision procedure for this finite formula with integer coefficients. $\square$

The leading-vector route of (a) does not preserve sign or nonvanishing conditions on coordinates of lower degree: for example (ω^2, ω) has both coordinates positive, but its highest-degree coefficient vector is $(1, 0)$. This is why (b) uses Tarski. Any homogeneous solution over $\mathbf{No}$, not only over $\mathbb{R}$, can be multiplied by a common positive monomial into Π^n preserving signs and zero patterns (corollary 4.7) [02, 05]. Neither statement decides affine equations: homogenizing an affine equation and requiring the new coordinate to be nonzero allows division in $\mathbf{No}$, not in $\mathbf{Oz}$. For example $X^2 - 2Z^2 = 0$ has the omnific solution $(X, Z) = (\sqrt{2}\omega, \omega)$ with $Z \neq 0$. Its affine coordinate $X/Z = \sqrt{2}$ is not omnific, and $x^2 = 2$ has no omnific solution by proposition 2.8.

Example 14.3 (Fermat scaling [01, 02, 05]). For every ordinary integer $n \geq 2$ and $0 < t \in \Pi$,

$$x = t, \quad y = t, \quad z = 2^{1/n}t \quad (14.1)$$

is a positive infinite solution of $x^n + y^n = z^n$; for $t = \omega$ this is $\omega^n + \omega^n = (2^{1/n}\omega)^n$. The irrational coefficient is permitted by the definition of $\mathbf{Oz}$. All constant coefficients are zero, consistent with theorem 5.1, whose specialization is $(0, 0, 0)$; it does not refute Fermat's Last Theorem about nonzero ordinary integers. The coordinates are divisible by every ordinary integer and have a common infinite monomial divisor (theorem 4.2), so these solutions are not primitive and must not be cited as resolving a primitive Fermat problem.

Provenance. Source 02 states that the failure of the nonzero analogue of Fermat's Last Theorem for omnific integers "is already known", citing comments by Emil Jeřábek (27 June 2018) and Timothy Chow (16 November 2021) under MathOverflow answer 303688 [17]. This attribution could not be verified when this report was assembled (the site was not reachable), and it is recorded as source 02's, unverified. No source claims novelty for the example.

14.3 Unimodular Pythagorean triples

Proposition 14.4 (Infinite unimodular Pythagorean triples [01, 05]). *For every $t \in \Pi$, put $x = t^2 - 1$, $y = 2t$, $z = t^2 + 1$. Then $x^2 + y^2 = z^2$ and the triple is unimodular. For $t \neq 0$ all three coordinates are infinite, and for $t > 0$ they are positive.*

Proof. The identity is the ordinary polynomial identity $(t^2 - 1)^2 + 4t^2 = (t^2 + 1)^2$. Two unit-ideal witnesses are

$$\frac{t^2}{2}z - \left(\frac{t^2}{2} + 1\right)x = 1 \text{ [01]}, \quad \frac{t}{2}y - x = 1 \text{ [05]}, \quad (14.2)$$

with $t^2/2, t/2 \in \Pi \subseteq \mathbf{Oz}$. Positive leading degree gives the assertion about infinite size. $\square$

The constant tuple is $(-1, 0, 1)$, not the zero tuple; that is exactly where the common-divisor obstruction for Π^3 stops applying. Infinite solutions therefore need not arise by scaling an ordinary tuple, and exponent two differs from exponents at least three ([theorem 7.9](#)). More generally $(u^2 - v^2)^2 + (2uv)^2 = (u^2 + v^2)^2$ for all $u, v \in \mathbf{Oz}$ [02].

14.4 Which real projective directions have primitive representatives?

A point $[r_0 : \cdots : r_n] \in \mathbb{P}^n(\mathbb{R})$ is *rational* if it has homogeneous coordinates in $\mathbb{Q}$, equivalently a nonzero integer coordinate tuple.

Theorem 14.5 (Rationality of primitive constant directions [01]). *For a real projective point $p = [r_0 : \cdots : r_n]$, the following are equivalent:*

- (i) p is rational;
- (ii) p has a unimodular representative in $\mathbf{Oz}^{n+1}$;
- (iii) p has a primitive representative in $\mathbf{Oz}^{n+1}$.

Proof. For (i) $\Rightarrow$ (ii), choose an integer representative with ordinary gcd one; the integer Bézout identity makes it unimodular over $\mathbf{Oz}$. (ii) $\Rightarrow$ (iii) is immediate. For (iii) $\Rightarrow$ (i), write a primitive representative as $z_i = \lambda r_i$ with $\lambda \in \mathbf{No}^\times$ and choose $r_j \neq 0$. Since $z_j \in \mathbf{Oz} \subseteq \mathcal{B}_{\mathbb{R}}$ and division by the nonzero real r_j preserves $\mathcal{B}_{\mathbb{R}}$, $\lambda \in \mathcal{B}_{\mathbb{R}}$. Put $c = \text{ct}(\lambda)$; then $\text{ct}(z_i) = cr_i \in \mathbb{Z}$ for all i . If $c = 0$, every z_i lies in Π , so 2 divides every coordinate, contradicting primitivity. Hence $c \neq 0$ and $(cr_0, \dots, cr_n)$ is a nonzero integer representative of p . $\square$

In fact every primitive representative of a real projective point is an ordinary coprime integer tuple. To see this additional consequence, the theorem first makes the point rational. Fix an ordinary representative $\mathbf{m}$ with gcd one and integers b_i such that $\sum_i b_i m_i = 1$. Every omnific representative has the form $\mathbf{z} = s\mathbf{m}$ for some $s \in \mathbf{No}^\times$, and

$$s = \sum_i b_i z_i \in \mathbf{Oz}, \quad (z_0, \dots, z_n) = s\mathbf{Oz}.$$

Indeed each z_i is a multiple of s , and the displayed Bézout identity puts s in the coordinate ideal. Thus such a representative is primitive if and only if s is a unit, equivalently $s = \pm 1$, and then it is also unimodular. This equivalence is specific to representatives of real directions; no general equivalence of primitivity and unimodularity over $\mathbf{Oz}$ is asserted.

For example, $[1 : \sqrt{2}]$ has omnific representatives such as $(\omega, \sqrt{2}\omega)$ but no primitive, hence no unimodular, representative; source 02 proved the unimodular half of this example directly. The theorem concerns real projective points; it is not a classification of all points of $\mathbb{P}^n(\mathbf{No})$, and neither monomial clearing nor the existence of a homogeneous solution answers the primitive-representative question. [Section 15](#), from sources 08 and 09, extends the classification to all points of $\mathbb{P}^n(\mathbb{R}(t))$ and $\mathbb{P}^n(\mathbb{C}(t))$ for $0 \neq t \in \Pi$, and to the region $\mathcal{V}$ of $\mathbb{P}^1(\mathbf{No})$ ([theorems 15.15, 15.18 and 15.54](#) and [corollary 15.28](#)); the Gaussian form of the theorem is [corollary 15.55](#). A classification of all points of $\mathbb{P}^n(\mathbf{No})$ is still not claimed ([questions 18.12 and 18.13](#)).

14.5 Finite supports specialize to polynomial arcs

The rigidity theorems above do not depend on finite supports. Finite-support expressions nevertheless admit an additional reduction [05].

Lemma 14.6 (Finite ordered specialization [05]). *Let H be a finitely generated subgroup of an ordered abelian group and $E \subseteq H$ finite. There is an additive map $\ell : H \rightarrow \mathbb{Z}$ with $\ell(a) > \ell(b)$ whenever $a, b \in E$ and $a > b$.*

Proof. A finitely generated torsion-free abelian group is isomorphic to $\mathbb{Z}^r$. In such coordinates, form the finite set of vectors $a - b$ for $a, b \in E$ with $a > b$. If this set is empty, $\ell = 0$ works. Otherwise enumerate it as $v_1, \dots, v_s$; each vector represents a positive element in the original group order.

We first show that 0 is not in their ordinary real convex hull. If it were, choose a solution of

$$\sum_j c_j v_j = 0, \quad \sum_j c_j = 1, \quad c_j \geq 0$$

with the smallest possible number of positive coefficients. The augmented columns $(v_j, 1)$ for those coefficients are linearly independent. Otherwise a nonzero dependence d_j has $\sum_j d_j = 0$, hence both positive and negative entries; moving the coefficient vector along this dependence until one positive coefficient reaches zero preserves all constraints and reduces its positive support. Independence gives a square nonsingular minor with integer entries, so the positive c_j are rational by solving that subsystem. Clearing denominators yields a nonempty sum of positive integer multiples of the v_j equal to zero, impossible in an ordered group.

Let C be their convex hull and choose $p \in C$ of least Euclidean norm. Compactness supplies p , and $p \neq 0$. For $v \in C$, minimality along the segment $p + u(v - p)$, $0 \leq u \leq 1$, gives $p \cdot (v - p) \geq 0$. Consequently $p \cdot v_j \geq \|p\|^2 > 0$ for every j . A rational vector sufficiently close to p keeps these finitely many strict inequalities. Multiplying it by a positive common denominator gives an integer vector w , and $\ell(h) = w \cdot h$ is the required additive map. All convexity here is in the ordinary finite-dimensional space $\mathbb{R}^r$, not in a surreal topological space. $\square$

Theorem 14.7 (Finite-support solutions specialize to arcs [05]). *Suppose $\mathbf{x} \in \mathbf{Oz}^n$ has finite normal-form supports and satisfies a finite polynomial system over $\mathbb{R}$. There are real polynomials $f_1(T), \dots, f_n(T)$ satisfying the same polynomial identities with $f_i(0) = \text{ct}(x_i)$, and every nonconstant x_i gives a nonconstant f_i . A prescribed finite list of nonzero real-coefficient polynomial expressions in $\mathbf{x}$ can be kept nonzero, with the same eventual signs as $T \rightarrow +\infty$ as their surreal signs.*

Proof. Let E contain zero, all exponents of the coordinates, and all exponents of the prescribed expressions after expansion; it is finite. Let H be its integer span and choose ℓ by lemma 14.6. The group algebra $\mathbb{R}[H]$ identifies with the finite-support real normal forms on H : distinct monomials are linearly independent, and their multiplication adds exponents. Thus the original equations hold in this algebra. The map $\mathbb{R}[H] \rightarrow \mathbb{R}[T, T^{-1}]$, $\omega^h \mapsto T^{\ell(h)}$, is a ring homomorphism fixing every real coefficient. Positive coordinate exponents map to positive integers, so coordinate images are polynomials with the same constant coefficients; injectivity of ℓ on E prevents cancellation of distinct supported terms, in particular of leading terms. Polynomial identities are preserved, and each prescribed expression keeps its leading coefficient and hence its eventual sign for large positive T . $\square$

Source 05 states the system over $\mathbb{Z}$; fixing real scalars in the same homomorphism proves the extension above. Coefficients of the additional expressions are specified as real for the same reason. The map need not be injective or order preserving on all of H : it preserves the finite list of comparisons used in its construction. For instance, on $H = \mathbb{Z} + \mathbb{Z}\sqrt{2}$, the map $\ell(1) = 1$, $\ell(\sqrt{2}) = 2$ preserves the order of $E = \{0, 1, \sqrt{2}, 1 + \sqrt{2}\}$ but kills the nonzero

group-algebra element $\omega^{\sqrt{2}} - \omega^2$. To preserve that expression as well, its supported exponent 2 must be included in E and a suitable map chosen again. Equations need no such extra condition: every ring homomorphism preserves them, even when it has a kernel.

With [proposition 14.1](#) (substituting $t \in \Pi$ for T), this gives an exact equivalence between the existence of a finite-support nonconstant solution and of a nonconstant real polynomial arc with integer constant coordinates. For the reverse implication choose $t = \omega$: the resulting tuple has finite supports, and each nonconstant polynomial has a nonconstant value. Positive $t \in \Pi \setminus \{0\}$ also preserves the prescribed signs, because its positive leading term makes each polynomial's leading term determine its sign. A single specialization therefore preserves any finite list of polynomial equalities, disequalities and sign conditions. This does not turn infinite-support solutions into polynomial arcs.

Finite-support Bézout certificates can be retained as well. If $x_i, u_i \in \mathbf{Oz}$ all have finite supports and $\sum_i u_i x_i = 1$, apply the theorem to the combined tuple and include this equation. Its images satisfy $\sum_i g_i(T) f_i(T) = 1$, with integer constant coefficients in every f_i and g_i . Evaluating at any $t \in \Pi$ gives an omnific Bézout certificate, so the resulting family is unimodular. This argument requires finite supports for the supplied witnesses u_i , not just for the coordinates x_i ; existence of arbitrary omnific witnesses does not supply that additional hypothesis.

For a finite-support implementation one may store an element as a map from exact exponents to nonzero exact real coefficients; the essential operations are coefficientwise addition, exponent addition in convolution, combination of equal exponents, and deciding integrality of the constant coefficient. Rational exponents with real algebraic coefficients give a practical decidable representation; general symbolic reals or arbitrary surreal exponents need further equality and order oracles. Useful exact certificates are polynomial identities, Bézout witnesses such as [\(14.2\)](#), standard-modulus quotients, and finite normal-form obstructions such as the first negative term in [\(16.2\)](#) [05].

15 Fractions, denominator ideals and unimodular presentations

This section is built from sources 08 and 09 ([section 1.2](#)), with source 08 as the base. [Theorem 4.6](#) makes every surreal a fraction of omnific integers, but it says nothing about lowest terms, and [theorems 4.10](#) and [14.5](#) treat only the constant pairs (t, rt) and the constant real directions. The central theorem below ([theorem 15.11](#)) removes the constancy. For every tuple that is unimodular over $\mathcal{B}_{\mathbb{R}}$, the constant vector alone decides whether the admissible scalar multipliers form $\mathbf{Oz}$ or Π . This classifies every omnific representation of every point of $\mathbb{P}^n(\mathbb{R}(t))$ for $0 \neq t \in \Pi$ ([theorem 15.18](#)). It fills the gap stated after [theorem 14.5](#) and answers [question 18.3](#) in part ([corollary 15.30](#)). Source 09 adds the region $\mathcal{V}$ of points with a $\mathcal{B}_{\mathbb{R}}$ -unimodular presentation. It identifies the specialization fibers with the orbits of the congruence kernel of $\mathrm{SL}_2(\mathbf{Oz}) \rightarrow \mathrm{SL}_2(\mathbb{Z})$ and proves that these fibers are dense. Source 08 adds a localization with residue field $\mathbb{Q}$, polynomial models, the exact denominator defect of a scale extension with its Tor_1 witness, and the Gaussian dichotomy. The Gaussian dichotomy contains the Gaussian unimodular-direction criterion of manuscript 14 of the sibling report ([corollary 15.55](#)). This section is outside the proof reviews recorded in `RECONCILIATION.md`.

15.1 Conventions

Throughout, t denotes a nonzero element of Π , as in [propositions 14.1](#) and [14.4](#). It is not the Hahn variable $t = \omega^{-1}$ of [section 1.4](#). Source 08 writes τ , which here names the phase twist of [lemma 13.9](#); source 09 uses only the monomials $T = \omega^\alpha$ with $\alpha > 0$. Polynomials are written in an indeterminate T , and $p(t)$ is evaluation at t .

Coordinate order. Numerators come first and the denominator last. A tuple $(x_1, \dots, x_n)$ is the point $[x_1 : \dots : x_n : 1]$, and $\mathbb{P}^1(\mathbf{No}) = \mathbf{No} \cup \{\infty\}$ through $[P : Q] \mapsto P/Q$ for $Q \neq 0$ and $[P : 0] \mapsto \infty$. Source 09 uses this order, while source 08 puts the denominator first. The specialized values, such as $f(0)$, are the same in both orders. A point of $\mathbb{P}^1(\mathbf{No})$ other than ∞ is called *affine*. Source 09 calls such points “finite”, a word reserved here for bounded surreals.

Primitivity. “Primitive” and “unimodular” are as in [definition 7.7](#). Over another domain, a tuple is primitive when its only common divisors are units. Source 09’s “divisor-primitive” presentations and “Bézout fractions” are the primitive and the unimodular presentations here. Unimodularity implies primitivity; the converse is not assumed over $\mathbf{Oz}$ and is proved below only for the classified representations.

New symbols. The symbols $\mathfrak{D}$, sp , $\mathcal{V}$, $\mathcal{U}$, $\mathbf{K}$, res , Ex , $\mathcal{P}$, $F_{x,b}$ and $\mathbf{N}_x$ are introduced below. Most of them replace source symbols that collide with symbols of this report ([table 1](#)). In this report Γ is an exponent group, σ an automorphism, $\rho = \sqrt{2} - 1$, D the Pell parameter, G the Gram matrix, U_t the quadratic transvection, A an intermediate ring, $\mathcal{A}_{\mathbf{k}}(\Gamma)$ and $\mathcal{F}$ the rings of [sections 11](#) and [13](#), B a bilinear form, Δ a discriminant and V a zero set. The source symbols are therefore renamed. The calligraphic $\mathcal{V}$ below is a class of projective points, not a zero set.

15.2 Denominator ideals

Definition 15.1 (Denominator ideals [08, 09]). For surreals $x_1, \dots, x_n$ put

$$\mathfrak{D}(x_1, \dots, x_n) = \{b \in \mathbf{Oz} : bx_j \in \mathbf{Oz} \text{ for every } j\}.$$

For a domain R' in a field containing the x_j , $\mathfrak{D}_{R'}$ is defined in the same way with R' in place of $\mathbf{Oz}$. A *representation* of $(x_1, \dots, x_n)$ is a vector $(bx_1, \dots, bx_n, b) \in \mathbf{Oz}^{n+1}$ with $b \neq 0$.

Zero belongs to $\mathfrak{D}(x_1, \dots, x_n)$. The actual common denominators are its nonzero members, and they exist by [theorem 4.6](#). Clearly $\mathfrak{D}(x_1, \dots, x_n) = \bigcap_j \mathfrak{D}(x_j)$. Source 09 writes $D(x)$; the letter D is the Pell parameter here.

Proposition 15.2 (Calculus of denominator ideals [09]). *For every $x \in \mathbf{No}$, $\mathfrak{D}(x)$ is a nonzero ideal of $\mathbf{Oz}$, with $\mathfrak{D}(0) = \mathbf{Oz}$ and $\mathfrak{D}(x) = \mathbf{Oz} \cap x^{-1}\mathbf{Oz}$ for $x \neq 0$. For $z \in \mathbf{Oz}$ and $x \neq 0$,*

$$\mathfrak{D}(x+z) = \mathfrak{D}(x), \quad x\mathfrak{D}(x) = \mathfrak{D}(1/x).$$

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbf{Oz})$ and $cx + d \neq 0$,

$$\mathfrak{D}\left(\frac{ax+b}{cx+d}\right) = (cx+d)\mathfrak{D}(x). \quad (15.1)$$

Proof. Closure under addition and under multiplication by $\mathbf{Oz}$ is direct, and nonzero denominators exist by [theorem 4.6](#). The next identities follow from the definition. For (15.1) put $y = (ax+b)/(cx+d)$. If $h \in \mathfrak{D}(x)$, then $k = (cx+d)h \in \mathbf{Oz}$ and $yk = (ax+b)h \in \mathbf{Oz}$. Conversely let $k \in \mathfrak{D}(y)$ and $\Delta = ad-bc \in \mathbf{Oz}^\times$. Since $a-cy = \Delta/(cx+d)$ and $dy-b = \Delta x/(cx+d)$, the elements $h = \Delta^{-1}(a-cy)k = k/(cx+d)$ and $xh = \Delta^{-1}(dy-b)k$ lie in $\mathbf{Oz}$, so $h \in \mathfrak{D}(x)$ and $k = (cx+d)h$. $\square$

Proposition 15.3 (Least denominators [09]). *A nonzero ideal $\mathfrak{a} \subseteq \mathbf{Oz}$ is principal if and only if its positive members have a least element. Hence $x \in \mathbf{No}$ has a least positive omnific denominator if and only if $\mathfrak{D}(x)$ is principal. If $\mathfrak{D}(x) = b_0\mathbf{Oz}$ with $b_0 > 0$, the representations of x are exactly (a_0h, b_0h) with $a_0 = b_0x$ and $0 \neq h \in \mathbf{Oz}$, and (a_0, b_0) is primitive.*

Proof. For $\mathfrak{a} = b_0 \mathbf{Oz}$ with $b_0 > 0$, the positive members are $b_0 h$ with $h \geq 1$ (proposition 2.5), so b_0 is least. Conversely, if b_0 is the least positive member and $a \in \mathfrak{a}$, ordered division (proposition 4.8) gives $a = qb_0 + r$ with $0 \leq r < b_0$ and $r \in \mathfrak{a}$, so $r = 0$. The description of the representations follows from the definition. If d divides a_0 and b_0 , then $b_0/d \in \mathfrak{D}(x)$, so $b_0/d = b_0 h$ with $h \in \mathbf{Oz}$, and d is a unit. $\square$

Lemma 15.4 (Unimodular fractions [08, 09]). *Let R' be a domain and $(p_1, \dots, p_n, q) \in R'^{n+1}$ unimodular with $q \neq 0$. Then $\mathfrak{D}_{R'}(p_1/q, \dots, p_n/q) = qR'$. Every representation over R' is $h(p_1, \dots, p_n, q)$ for a unique $0 \neq h \in R'$, and h is a gcd of it. For $R' = \mathbf{Oz}$ and $q > 0$, the vector itself is the only primitive representation with positive denominator, and two unimodular representations differ by a sign.*

Proof. Take $u_1 p_1 + \dots + u_n p_n + u_0 q = 1$ in R' . If b is a denominator, then

$$\frac{b}{q} = u_0 b + \sum_j u_j \frac{b p_j}{q} \in R'.$$

Thus $b = qh$, and the numerators are $p_j h$. Conversely every $h \in R'$ gives a denominator. A common divisor of $h(p_1, \dots, p_n, q)$ divides the same linear combination, which is h , and h is a common divisor. For $R' = \mathbf{Oz}$ a representation is primitive exactly when h is a unit, that is, $h = \pm 1$. $\square$

Proposition 15.5 (Ordinary denominators [08, 09]). *Inside $\mathbf{No}$,*

$$\mathbf{Oz}[(\mathbb{Z} \setminus \{0\})^{-1}] = \mathbb{Q} \oplus \Pi, \quad 0 \longrightarrow \Pi \longrightarrow \mathbb{Q} \oplus \Pi \xrightarrow{\text{ct}} \mathbb{Q} \longrightarrow 0$$

with the sequence split exact. The ring $\mathbb{Q} \oplus \Pi$ meets $\mathbb{R}$ in $\mathbb{Q}$, has unit group $\mathbb{Q}^\times$, is not a field, and has $\mathbb{Q}$ as its only subfield. If $x = y + m/n$ with $y \in \Pi$, $n > 0$ and $\gcd(m, n) = 1$, then $\mathfrak{D}(x) = n\mathbf{Oz}$, and $x = (ny + m)/n$ is a unimodular presentation with least positive denominator n : for integers u, v with $um + vn = 1$,

$$u(ny + m) + (v - uy)n = 1, \quad v - uy \in \mathbf{Oz}. \quad (15.2)$$

Proof. If $a = k + y \in \mathbf{Oz}$ with $k \in \mathbb{Z}$, $y \in \Pi$, and $0 \neq n \in \mathbb{Z}$, then $a/n = k/n + y/n \in \mathbb{Q} \oplus \Pi$; conversely $y + m/n = (ny + m)/n$. The real elements and units follow from proposition 2.5 by the same leading-degree argument, and ω lies in the ring while ω^{-1} does not. For $b \in \mathbf{Oz}$ the product bx has no negative exponent, and its constant coefficient is $(m/n) \text{ct}(b)$. So $b \in \mathfrak{D}(x)$ exactly when $n \mid \text{ct}(b)$, which is $b \in n\mathbf{Oz}$ by theorem 3.1. The identity (15.2) is direct, and a positive element of $n\mathbf{Oz}$ is at least n . Finally, a subfield contains $\mathbb{Q}$. A nonreal member of $\mathbb{Q} \oplus \Pi$ is infinite, and its inverse is a nonzero infinitesimal outside the ring. So every member of a subfield is real, hence rational. $\square$

An element of $\mathbf{No}$ algebraic over $\mathbb{Q}$ is a real algebraic number, since $\mathbf{No}$ is an ordered field containing the real closure of $\mathbb{Q}$ in $\mathbb{R}$. Hence the elements of $\mathbb{Q} \oplus \Pi$ algebraic over $\mathbb{Q}$ are exactly the rationals, and those of $\mathbf{Oz}$ are the integers [09]. Fractions that produce $\sqrt{2}$ need infinite omnific denominators. Also $\mathcal{B}_{\mathbb{R}} \cap \mathcal{O}_{\text{fin}} = \mathbb{R}$ and $(\mathbb{Q} \oplus \Pi) \cap \mathcal{O}_{\text{fin}} = \mathbb{Q}$: ct discards the infinite terms and st the infinitesimal ones (remark 2.3) [09]. So there are four rings to keep apart [08]:

$$\mathbb{Q} \subsetneq \mathbb{Q} \oplus \Pi \subsetneq \mathbf{Oz}_{\Pi} \subsetneq \text{Frac}(\mathbf{Oz}) = \mathbf{No}.$$

The second admits only ordinary integer denominators. The third, the localization of $\mathbf{Oz}$ at the prime ideal Π , admits every denominator with a nonzero constant coefficient; it is studied in section 15.8.

Proposition 15.6 (No set of denominators [08]). *No set $S \subseteq \mathbf{Oz} \setminus \{0\}$ supplies a denominator for every surreal: there is $0 < x \in \mathbf{No}$ with $bx \notin \mathbf{Oz}$ for every $b \in S$.*

Proof. By lemma 4.1 choose $\eta > \deg_\omega(b)$ for every $b \in S$, and put $x = \omega^{-\eta}$. Then $bx \neq 0$ has negative leading exponent, so it is a nonzero infinitesimal and does not lie in $\mathbf{Oz}$. $\square$

By contrast, the class $\mathcal{M} = \{\omega^\beta : \beta \geq 0\}$ of monomials does serve: $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$, with the localization read for the class of denominators. This is `osq:prop:fractions(iii)` of the sibling report, which source 09 proves again. A single monomial ω^β fails at $x = \omega^{-\beta-1}$ [09]. Even a set-sized subfield of $\mathbf{No}$ has one common omnific denominator for all its elements (theorem 4.6), but the whole class does not [08].

Theorem 15.7 (Real constants [08, 09]). *For $r \in \mathbb{R}$,*

$$\mathfrak{D}(r) = \begin{cases} q\mathbf{Oz}, & r = p/q \in \mathbb{Q}, \gcd(p, q) = 1, q > 0, \\ \Pi, & r \notin \mathbb{Q}. \end{cases}$$

For irrational r , every representation $r = a/b$ has $a, b \in \Pi$, every nonzero ordinary integer divides both, and the pair has no gcd. For real constants each of the following is equivalent to $r \in \mathbb{Q}$: a primitive representation exists; a unimodular representation exists; $\mathfrak{D}(r)$ is principal; $\mathfrak{D}(r)$ is generated by a set; a least positive omnific denominator exists.

Proof. Write $b = k + y$ with $k \in \mathbb{Z}$ and $y \in \Pi$. Multiplication by a real preserves support, so $rb \in \mathbf{Oz}$ exactly when $rk \in \mathbb{Z}$. For $r = p/q$ in lowest terms this means $q \mid k$, and since $\Pi = q\Pi$ (theorem 3.1) the ideal is $q\mathbf{Oz}$. For irrational r it forces $k = 0$, giving Π .

For irrational r , then $a = rb \in \Pi$ as well. If d is a common divisor of a and b , then $(a/d, b/d)$ is another representation of r , so both quotients again lie in Π and are divisible by 2. Hence $2d$ is a common divisor; it does not divide d , because $1/2 \notin \mathbf{Oz}$. So no common divisor is greatest. Corollary 4.3 excludes a set of generators, and halving shows that there is no least positive denominator. In the rational case the ordinary pair (p, q) is unimodular, and lemma 15.4 gives the remaining assertions. $\square$

Provenance. Source 09 proves $\mathfrak{D}(r) = \Pi$ and the absence of a gcd for irrational r and credits the absence of a gcd to theorem 4.10; source 08 proves the whole theorem and credits the same result. The gcd part is theorem 4.10 for the pair (b, rb) . For irrational r the ideal $\mathfrak{D}(r)$ is the dual $\text{Hom}_{\mathbf{Oz}}(L(M), \mathbf{Oz})$ of the coefficient-lattice module $L(M) = M + \Pi$ with $M = \mathbb{Z} + \mathbb{Z}r$, and the equality $\mathfrak{D}(r) = \Pi$ is the rank-two case of `osq:tn:thm:duals` of the sibling report (manuscript 14 there). Hamkins discusses the failure of lowest terms and the omnific fractions of irrational reals [32], as source 08 records. The theorem does not erase the distinction between $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$, although $\text{Frac}(\mathbf{Oz}) \cap \mathbb{R} = \mathbb{R}$ [08].

Remark 15.8 (The halving route [merge]). Sources 08 and 09 show that no gcd exists by replacing a common divisor d by $2d$. This differs from the shift $d \mapsto d\omega^\delta$ in the proof of theorem 4.10 and from the degree argument of remark 4.11. It needs no new exponent, and so the conclusion of theorem 4.10 holds in every fixed workspace. *Claim:* for every ordered abelian group $\Gamma \neq 0$, every $0 \neq t \in \Pi_\Gamma$ and every $r \in \mathbb{R} \setminus \mathbb{Q}$, the pair (t, rt) has no gcd in the ring $\mathbf{Oz}_\Gamma$ of section 17. Indeed, let d be a common divisor, with $t = du$ and $rt = dv$. Then $v = ru$, and $\text{ct}(v) = r \text{ct}(u)$ with both constants in $\mathbb{Z}$, so $u, v \in \Pi_\Gamma$. Dividing coefficients by 2 preserves supports, so $u/2, v/2 \in \Pi_\Gamma$ and $2d$ is a common divisor. It does not divide d , because $1/2 \notin \mathbf{Oz}_\Gamma$.

15.3 The multiplier theorem

We first isolate the algebraic mechanism. Throughout this subsection, R is a domain containing $\mathbb{R}$ with a ring retraction $c : R \rightarrow \mathbb{R}$ that fixes $\mathbb{R}$. Put $I_R = \ker c$ and $R_{\mathbb{Z}} = c^{-1}(\mathbb{Z}) = \mathbb{Z} \oplus I_R$, and let K be any field containing R , possibly larger than $\text{Frac}(R)$. The main example is $(R, c) = (\mathcal{B}_{\mathbb{R}}, \text{ct})$, with $R_{\mathbb{Z}} = \mathbf{Oz}$ and $I_R = \Pi$. Another is $R = \mathbb{R}[t]$ with evaluation at $T = 0$ (section 15.9).

Definition 15.9 (Rational directions [08]). A nonzero $\mathbf{v} \in \mathbb{R}^{n+1}$ has *rational projective direction* if its line contains a nonzero vector of $\mathbb{Q}^{n+1}$, equivalently of $\mathbb{Z}^{n+1}$. Such a line has a primitive integral generator, unique up to sign.

Lemma 15.10 (The scalar lattice of a real line [08]). For $\mathbf{v} \neq 0$ put $\mathcal{L}_{\mathbf{v}} = \{s \in \mathbb{R} : s\mathbf{v} \in \mathbb{Z}^{n+1}\}$. If the direction of $\mathbf{v}$ is not rational, then $\mathcal{L}_{\mathbf{v}} = \{0\}$. If $\mathbf{v}$ is a primitive integral vector, then $\mathcal{L}_{\mathbf{v}} = \mathbb{Z}$.

Proof. A nonzero s in the first case would make the line rational. In the second case choose integers u_i with $\sum_i u_i v_i = 1$. If every sv_i is an integer, so is $s = \sum_i u_i (sv_i)$. $\square$

Theorem 15.11 (All ambient scalar representations [08]; pairs [09]). Let $\mathbf{P} = (P_0, \dots, P_n) \in R^{n+1}$ be unimodular over R . Its residue vector $\mathbf{v} = (c(P_0), \dots, c(P_n))$ is nonzero, and

$$\{h \in K : hP_i \in R_{\mathbb{Z}} \text{ for all } i\} = \{h \in R : c(h)\mathbf{v} \in \mathbb{Z}^{n+1}\}. \quad (15.3)$$

- (a) If $[\mathbf{v}] \in \mathbb{P}^n(\mathbb{Q})$, multiply $\mathbf{P}$ by a nonzero real scalar so that $\mathbf{v}$ becomes primitive integral. The set (15.3) is then exactly $R_{\mathbb{Z}}$, and the normalized vector is unimodular over $R_{\mathbb{Z}}$.
- (b) If $[\mathbf{v}] \notin \mathbb{P}^n(\mathbb{Q})$, the set (15.3) is exactly I_R , and this does not change under nonzero real rescaling of $\mathbf{P}$.

Proof. Choose $U_i \in R$ with $\sum_i U_i P_i = 1$. Applying c shows $\mathbf{v} \neq 0$. If every hP_i lies in $R_{\mathbb{Z}} \subseteq R$, then

$$h = \sum_i U_i (hP_i) \in R.$$

For $h \in R$, membership of hP_i in $R_{\mathbb{Z}}$ is exactly $c(h)v_i \in \mathbb{Z}$. This proves (15.3), and lemma 15.10 gives the two scalar sets. It remains to show that the normalized vector in (a) is unimodular over $R_{\mathbb{Z}}$, not merely primitive. Choose $u_i \in \mathbb{Z}$ with $\sum_i u_i v_i = 1$, put $S = \sum_i u_i P_i$, so that $c(S) = 1$, and define

$$V_i = u_i + (1 - S)U_i. \quad (15.4)$$

Each V_i has residue $u_i \in \mathbb{Z}$, so $V_i \in R_{\mathbb{Z}}$, and $\sum_i V_i P_i = S + (1 - S) \sum_i U_i P_i = 1$. $\square$

The first identity of the proof forces h back into R even when h is allowed in a much larger field. For $R = \mathcal{B}_{\mathbb{R}}$ this is why numerators and denominators with arbitrary infinite supports, or with scales outside the supports of $\mathbf{P}$, cannot evade the classification; formal reduction of a rational function would not show this [08, 09].

Remark 15.12 (A second route to the lift [09]). Source 09 adjusts constants instead of using (15.4). For a pair (A_1, A_0) normalized to the primitive constant vector (m_1, m_0) with $U_0 A_1 + V_0 A_0 = 1$ over R , the real solutions (y_1, y_0) of $y_1 m_1 + y_0 m_0 = 1$ differ by multiples of $(m_0, -m_1)$. Choose integers u_1, u_0 with $u_1 m_1 + u_0 m_0 = 1$ and the real k with $(u_1, u_0) = (c(U_0), c(V_0)) + k(m_0, -m_1)$. Then $U_0 + kA_0$ and $V_0 - kA_1$ have constants u_1 and u_0 and still form a Bézout row. With $n + 1$ coordinates and constant vector $\mathbf{m}$, let $\mathbf{w} = \mathbf{u} - c(\mathbf{U})$ for an integer row $\mathbf{u}$ with $\mathbf{u} \cdot \mathbf{m} = 1$, and choose a real row $\mathbf{r}$ with $\mathbf{r} \cdot \mathbf{m} = 1$. Add to $\mathbf{U}$ the syzygies $A_i e_j - A_j e_i$ with coefficients $r_i w_j$. Their constant sum is $(\mathbf{r} \cdot \mathbf{m})\mathbf{w} - (\mathbf{w} \cdot \mathbf{m})\mathbf{r} = \mathbf{w}$, since $\mathbf{w} \cdot \mathbf{m} = 0$. Source 08's formula (15.4) is shorter.

Provenance. The ring $R_{\mathbb{Z}}$ is the pullback of c along $\mathbb{Z} \hookrightarrow \mathbb{R}$. Neither source claims the pullback construction or the use of coefficient retractions as new. Source 08 cites the retraction transfer lemma of L’Innocente–Mantova [3, Lemma 9.2.1] as a methodological precedent, and source 09 cites Kława [33] for background on $D + M$ rings. The sibling report credits the $D + M$ context to Dobbs–Papick, and its computation of scalar homomorphisms `osq:tn:thm:hom` (manuscript 14 there) uses the same step: a scalar q with $qm_0 \in B$ satisfies $q = (qm_0)/m_0 \in B$, and then only the constant part matters. Source 09’s pair version (its multiplier theorem, with multipliers in $\Pi \oplus \{c \in \mathbb{R} : cp, cq \in \mathbb{Z}\}$) is the case $n = 1$, $R = \mathcal{B}_{\mathbb{R}}$ of [theorem 15.11](#); its sketch for $n + 1$ coordinates is contained in it.

15.4 The unimodular region and its specialization

Definition 15.13 (The unimodular region [09]). Let $\mathcal{V} \subseteq \mathbb{P}^1(\mathbf{No})$ be the class of points with a presentation $[P : Q]$ such that $P, Q \in \mathcal{B}_{\mathbb{R}}$ and $UP + VQ = 1$ for some $U, V \in \mathcal{B}_{\mathbb{R}}$. For such a presentation put

$$\mathrm{sp}([P : Q]) = [\mathrm{ct}(P) : \mathrm{ct}(Q)] \in \mathbb{P}^1(\mathbb{R}).$$

Let $\mathcal{U} \subseteq \mathcal{V}$ be the class of points with a presentation that is unimodular over $\mathbf{Oz}$.

Lemma 15.14 (Well-defined specialization [09]). *Two $\mathcal{B}_{\mathbb{R}}$ -unimodular presentations of the same point differ by a nonzero real scalar, so $\mathrm{sp} : \mathcal{V} \rightarrow \mathbb{P}^1(\mathbb{R})$ is well defined. Moreover $\mathbb{P}^1(\mathbb{R}) \subseteq \mathcal{V}$ with $\mathrm{sp}(r) = r$, and $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{V}$ with $\mathrm{sp}(x) = \mathrm{ct}(x)$.*

Proof. Applying ct to a Bézout identity shows that the two constants of a unimodular pair are not both zero. If $(P', Q') = h(P, Q)$ with $h \in \mathbf{No}^{\times}$, the Bézout identity of (P, Q) gives $h = UP' + VQ' \in \mathcal{B}_{\mathbb{R}}$, and that of (P', Q') gives $h^{-1} \in \mathcal{B}_{\mathbb{R}}$. So $h \in \mathcal{B}_{\mathbb{R}}^{\times} = \mathbb{R}^{\times}$ ([proposition 2.5](#)), and the constant vectors define the same point. Use the pairs $(r, 1)$, $(1, 0)$ and $(x, 1)$ for the last assertions. $\square$

Theorem 15.15 (Denominator dichotomy on the unimodular region [09]; [08]). *Let $x = P/Q$ be an affine point of $\mathcal{V}$, with (P, Q) unimodular over $\mathcal{B}_{\mathbb{R}}$ and $Q \neq 0$.*

- (i) *If $\mathrm{sp}(x) \in \mathbb{P}^1(\mathbb{Q})$, choose $\lambda \in \mathbb{R}^{\times}$ with $(\lambda \mathrm{ct} P, \lambda \mathrm{ct} Q)$ a primitive integer vector and $\lambda Q > 0$. Then $\mathfrak{D}(x) = \lambda Q \mathbf{Oz}$. The presentation $x = \lambda P / \lambda Q$ is unimodular over $\mathbf{Oz}$ and is the only primitive presentation with positive denominator, and λQ is the least positive element of $\mathfrak{D}(x)$.*
- (ii) *If $\mathrm{sp}(x) \notin \mathbb{P}^1(\mathbb{Q})$, then $\mathfrak{D}(x) = Q\Pi$. This ideal is not generated by any set. Every presentation has both entries in Π , none is primitive, none has a gcd, and every positive denominator can be halved.*

Consequently $\mathcal{U} = \mathrm{sp}^{-1}(\mathbb{P}^1(\mathbb{Q}))$.

Proof. Apply [theorem 15.11](#) with $n = 1$, $(R, c) = (\mathcal{B}_{\mathbb{R}}, \mathrm{ct})$ and $K = \mathbf{No}$. The representations of x are the vectors $h(P, Q)$ with h in the set (15.3), and their denominators are hQ . In case (i), $(\lambda P, \lambda Q)$ is unimodular over $\mathbf{Oz}$, and [lemma 15.4](#) gives the rest. In case (ii) the admissible multipliers form Π , so $\mathfrak{D}(x) = Q\Pi$. Multiplication by Q is an $\mathbf{Oz}$ -linear bijection $\Pi \rightarrow Q\Pi$, so [corollary 4.3](#) excludes a set of generators. Every representation is $h(P, Q)$ with $h \in \Pi$ and $P, Q \in \mathcal{B}_{\mathbb{R}}$, so both entries lie in Π . If d is a common divisor of a representation, the quotient is again a representation, so its entries lie in Π and $2d$ is a common divisor that does not divide d . Halving $h \in \Pi$ keeps it in Π . Finally, the constant vector of an $\mathbf{Oz}$ -unimodular pair is a primitive integer vector (apply ct to its Bézout identity), so $\mathcal{U}$ lies in the rational

fibers. Conversely, (i) gives an $\mathbf{Oz}$ -unimodular presentation of every affine point with rational specialization, and $\infty = [1 : 0]$. $\square$

Remark 15.16 (The exact strength of the dichotomy [09]). On $\mathcal{V}$ the following coincide: rational specialization, unimodularity over $\mathbf{Oz}$, principality of the denominator ideal, a least positive denominator, and a primitive presentation. The equivalence is proved on $\mathcal{V}$ only. It is not claimed for all of $\mathbb{P}^1(\mathbf{No})$, and neither source claims that every surreal lies in $\mathcal{V}$ ([question 18.13](#)). Source 09 notes that L’Innocente–Mantova use “reduced” for a different, support-theoretic notion [3]; the word is not used here for lowest terms. For $n + 1$ coordinates the analogue is [theorem 15.11](#) itself.

15.5 Rational functions of one parameter

Lemma 15.17 (Evaluation at an omnific parameter [08]; monomials [09]). *Evaluation at t embeds $\mathbb{R}[T]$ in $\mathcal{B}_{\mathbb{R}}$ and extends to an embedding $\mathbb{R}(T) \hookrightarrow \mathbf{No}$. For every polynomial p , $\text{ct}(p(t)) = p(0)$, and for $p \neq 0$, $\deg_{\omega}(p(t)) = (\deg p) \deg_{\omega}(t)$.*

Proof. $\text{ct}(t) = 0$ gives the constant coefficient. The nonzero terms $a_j t^j$ have distinct leading exponents $j \deg_{\omega}(t)$, so the term of largest degree cannot cancel; this is the argument of [proposition 2.8](#). So no nonzero polynomial vanishes at t . $\square$

Theorem 15.18 (Representations of rational-function tuples [08]). *Let $p_1, \dots, p_n, q \in \mathbb{R}[T]$ with $q \neq 0$ and $\gcd(p_1, \dots, p_n, q) = 1$, put $x_j = p_j(t)/q(t)$ and*

$$\text{sp}(\mathbf{x}) = [p_1(0) : \dots : p_n(0) : q(0)] \in \mathbb{P}^n(\mathbb{R}).$$

This point is defined. Every representation of $\mathbf{x} = (x_1, \dots, x_n)$ is uniquely of the form

$$h(p_1(t), \dots, p_n(t), q(t)), \quad h \neq 0,$$

and:

- (a) *If $\text{sp}(\mathbf{x}) \in \mathbb{P}^n(\mathbb{Q})$, rescale the polynomials by a common nonzero real scalar so that their constant vector is primitive integral. Then h ranges over $\mathbf{Oz} \setminus \{0\}$, $\mathfrak{D}(\mathbf{x}) = q(t) \mathbf{Oz}$, and the polynomial vector at t is unimodular over $\mathbf{Oz}$.*
- (b) *If $\text{sp}(\mathbf{x}) \notin \mathbb{P}^n(\mathbb{Q})$, no normalization is needed. Then h ranges over $\Pi \setminus \{0\}$, and $\mathfrak{D}(\mathbf{x}) = q(t) \Pi$, an ideal with no set of generators.*

These statements cover representations with arbitrary surreal supports.

Proof. The polynomial gcd gives a Bézout identity $\sum_j U_j(T)p_j(T) + U_0(T)q(T) = 1$ in $\mathbb{R}[T]$. By [lemma 15.17](#) its evaluation shows that the vector at t is unimodular over $\mathcal{B}_{\mathbb{R}}$, with residue vector the constant vector of the polynomials. Apply [theorem 15.11](#) with $(R, c) = (\mathcal{B}_{\mathbb{R}}, \text{ct})$ and $K = \mathbf{No}$. Every representation has $h = b/q(t)$, and every admissible h gives a representation. The ideals are the sets of last coordinates, and multiplication by $q(t)$ is an $\mathbf{Oz}$ -linear bijection $\Pi \rightarrow q(t)\Pi$, so [corollary 4.3](#) excludes a set of generators. $\square$

Provenance. Source 09 proves the case $n = 1$, $t = \omega^{\alpha}$ (its one-scale classification), including the formula $\text{sp}(P(\omega^{\alpha})/Q(\omega^{\alpha})) = [P(0) : Q(0)]$ and the membership of these points in $\mathcal{V}$. It is the special case of [theorem 15.18](#) and [theorem 15.15](#) and is not printed separately. Source 08’s theorem has any number of coordinates and any $t \in \Pi \setminus \{0\}$; a nonzero t that is merely infinite, with a nonintegral constant, is not allowed [08]. The coprimality over $\mathbb{R}[T]$ is necessary: an uncanceled pair such as $(T(T + r), T(T + 1))$ has constant vector zero [09].

Corollary 15.19 (Lowest terms are all or nothing [08]; [09]). *In [theorem 15.18\(a\)](#) every representation has a gcd, namely its multiplier h up to sign. The only primitive representations are the two signs of the normalized polynomial vector, and both are unimodular. The least positive common denominator is $|q(t)|$. In case (b) no representation is primitive or has a gcd, every nonzero ordinary integer divides every coordinate of every representation, every common divisor can be replaced by a strictly larger one in the divisibility order, and there is no least positive common denominator.*

Proof. Case (a) is [lemma 15.4](#) with $\mathbf{Oz}^\times = \{\pm 1\}$, and $|h| \geq 1$ for $0 \neq h \in \mathbf{Oz}$. In case (b) every coordinate is h times an element of $\mathcal{B}_\mathbb{R}$ with $h \in \Pi$, hence lies in Π . If d is a common divisor, dividing by d gives another representation, whose multiplier h/d lies in Π by [theorem 15.18](#); so $2d$ is a common divisor not dividing d . Every positive denominator can be halved. $\square$

This strengthens [theorem 4.10](#): there the pair is the constant point (t, rt) , while here the absence of a gcd holds for every representation of every point with irrational specialization, for example $h(t + r, t + 1)$.

Corollary 15.20 (Leading exponents of denominators [08]). *The leading exponents of the nonzero common denominators of $\mathbf{x}$ in [theorem 15.18](#) are exactly the surreals $\deg_\omega q(t) + \eta$ with $\eta \geq 0$ in case (a), and with $\eta > 0$ in case (b).*

Proof. Use [lemma 2.4](#) and the two ideal formulas. The nonzero elements of $\mathbf{Oz}$ have every nonnegative leading exponent, and those of Π every positive one; ordinary integers and monomials ω^η realize them. $\square$

This measures the leading scale, not the support. In case (b) the endpoint is missing although denominators exist at every larger leading exponent [08].

Corollary 15.21 (One rational function [08, 09]). *Let $f = p/q \in \mathbb{R}(T)$ be reduced. Then $f(t) \in \mathcal{V}$ with $\text{sp}(f(t)) = f(0)$, read in $\mathbb{R} \cup \{\infty\}$. It has a primitive omnific representation if and only if $f(0) \in \mathbb{Q} \cup \{\infty\}$; lowest terms are then unique once the denominator is positive, and otherwise no representation has a gcd. The criterion does not depend on t .*

Proof. For a reduced pair $p(0)$ and $q(0)$ are not both zero, and $[p(0) : q(0)]$ is rational exactly when $f(0) \in \mathbb{Q} \cup \{\infty\}$. Apply [theorem 15.18](#) and [corollary 15.19](#). $\square$

Remark 15.22 (A formal specialization [08, 09]). The value $f(0)$ is a substitution in an ordinary rational function before evaluation at t . It is not a surreal value of the infinite number t , nor the valuation residue at infinity. There is no operation of “setting $\omega = 0$ ” on arbitrary normal forms, and no homomorphism $\mathbf{No} \rightarrow \mathbb{R}$ extending the substitution: a unital field homomorphism extending $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ would send the invertible element ω to 0. The criterion belongs to the rational-function subfield and the retraction on $\mathcal{B}_\mathbb{R}$. The map sp is not a field homomorphism either: its domain is only $\mathcal{V}$, its values are projective, and field operations can make both constants of a presentation vanish [09].

Example 15.23 (Denominator ideals [08, 09]). Let $t > 0$ in Π and let r be an irrational real. Generators are given up to sign.

x	$\text{sp}(x)$	$\mathfrak{D}(x)$	Lowest terms	Source
$m/n, \gcd(m, n) = 1, n > 0$	m/n	$n\mathbf{Oz}$	yes	08, 09
r	r	Π	no	08, 09
$t + r$	r	Π	no	08
$rt + 3/5$	$3/5$	$5\mathbf{Oz}$	yes	09
$1/t$	∞	$t\mathbf{Oz}$	yes	08
r/t	∞	$(t/r)\mathbf{Oz}$	yes	08
$(rt + 1)/(t + 1)$	1	$(t + 1)\mathbf{Oz}$	yes	08
$(2t + 3)/(5t + 7)$	$3/7$	$(5t + 7)\mathbf{Oz}$	yes	09
$(rt + 2)/(t + 3)$	$2/3$	$(t + 3)\mathbf{Oz}$	yes	09
$(t + r)/(t + 1)$	r	$(t + 1)\Pi$	no	08, 09
$(t^2 + 1)/(t^2 + r)$	$1/r$	$(t^2 + r)\Pi$	no	09

All displayed polynomial pairs are coprime because r is irrational. The function $(rt + 1)/(t + 1)$ has irrational standard part r but unique lowest terms, while $(t + r)/(t + 1)$ has standard part 1 and none. So the standard part does not decide cancellation [08].

The class of surreals with a primitive omnific representation is not a subring [08]. Both t and $\sqrt{2}/t$ have one, but their product $\sqrt{2}$ does not. Both $1/t$ and $\sqrt{2} - 1/t = (\sqrt{2}t - 1)/t$ specialize to ∞ , but their sum $\sqrt{2}$ does not. These are statements about primitive representations, not about the existence of fractions.

Example 15.24 (Nearly equal values, opposite denominators [09]). Let r be irrational and $t > 0$ in Π , and put

$$x_+ = r + \frac{1}{t} = \frac{rt + 1}{t}, \quad x_- = r + \frac{1}{t + 1} = \frac{rt + r + 1}{t + 1}.$$

Then $\text{sp}(x_+) = \infty$, and t is the least positive denominator of x_+ , with the certificate

$$(1 - rt)(1 + rt) + (r^2t)t = 1 \tag{15.5}$$

over $\mathbf{Oz}$. But $\text{sp}(x_-) = r + 1 \notin \mathbb{Q}$, so $\mathfrak{D}(x_-) = (t + 1)\Pi$ and no presentation of x_- has a gcd. Yet $x_+ - x_- = 1/(t(t + 1))$ and $\text{st}(x_+) = \text{st}(x_-) = r$. The standard part does not detect the obstruction, even at the next infinitesimal scale.

Remark 15.25 (A certificate procedure [08, 09]). For $f_1, \dots, f_n \in \mathbb{R}(T)$ the proofs give a finite procedure. (1) Choose a common denominator, divide out the polynomial gcd, and compute a polynomial Bézout identity. (2) Evaluate at 0 and decide whether the direction is rational. (3) If it is not, return $\mathfrak{D} = q(t)\Pi$. (4) If it is, normalize to a primitive integer vector, compute an integer Bézout identity, and apply (15.4) (or remark 15.12) to get an omnific Bézout certificate. (5) Choose the sign so that $q(t) > 0$, from the leading coefficient of q and the sign of t . The certificate is valid for every nonzero purely infinite parameter and, by the ambient-scalar identity, governs representations with arbitrary supports. The procedure is effective only over coefficient fields in which equality, polynomial gcds and rationality are decidable, for example real algebraic number fields; it is not a uniform algorithm for reals given by approximation oracles, and no factorization in $\mathbf{Oz}$ is used [08, 09].

Proposition 15.26 (Prescribed specialization near a given function [09]). *Give $\mathbb{R}(T)$ its topology at infinity: g is N -close to f if $\deg(g - f) < -N$, where $\deg$ is the degree of a rational function ($\deg 0 = -\infty$). For every $f \in \mathbb{R}(T)$, every ordinary $N > 0$ and every $r \in \mathbb{R}$ there is $g \in \mathbb{R}(T)$, regular at 0, with $g(0) = r$ and $\deg(g - f) < -N$.*

Proof. Choose an ordinary $L > 0$ larger than the pole order of f at 0 and put

$$H = (1 + T^L)^{-M}, \quad g = (1 - H)f + rH.$$

Since $1 - H$ vanishes to order L at 0, g is regular at 0 with $g(0) = r$. Moreover $g - f = H(r - f)$. If $f = r$ this is 0; otherwise its degree is $\deg(r - f) - LM$, which is less than $-N$ for large M . $\square$

Source 09 states the conclusion “unless $g = f$ exactly”; the clause is not needed, since $g = f$ gives degree $-\infty$. This topology is not the surreal order topology: no ordinary power t^{-N} lies below every positive infinitesimal. [Theorem 15.35](#) removes this restriction by choosing a new monomial after the target interval [09].

15.6 Simultaneous denominators and rational curves

Example 15.27 (No least common denominator [08]). Let $t > 0$ in Π . Each of $x_1 = 1/t$ and $x_2 = \sqrt{2}/t$ has unique lowest terms, with $\mathfrak{D}(x_1) = t \mathbf{Oz}$ and $\mathfrak{D}(x_2) = (t/\sqrt{2}) \mathbf{Oz}$. But

$$\mathfrak{D}(x_1, x_2) = t \mathbf{Oz} \cap \frac{t}{\sqrt{2}} \mathbf{Oz} = t \Pi. \quad (15.6)$$

This ideal has no set of generators and no least positive member, so the two canonical denominators have no least common multiple in the divisibility order.

Proof. The common polynomial vector is $(1, \sqrt{2}, T)$, with specialization $[1 : \sqrt{2} : 0]$, which is not rational. [Theorem 15.18](#) gives (15.6). An lcm would generate the intersection of the two principal ideals. The individual assertions are [corollary 15.21](#). $\square$

The two fractions are already reduced; the obstruction is the irrational relation between their residues. Every finite tuple in $\mathbb{R}(t)$ has either a principal common denominator ideal or a scalar copy of Π , with no intermediate case [08]. For monomial $t = \omega^g$, the identity (15.6) is the intersection formula $X^g A \cap c X^g A = X^g \Pi$ of `osq:tn:thm:syzygy(iii)` in the sibling report, with $c = 1/\sqrt{2}$.

Corollary 15.28 (Primitive coordinates on rational curves [08]). *Let $p_0, \dots, p_n \in \mathbb{R}[T]$ have gcd one, and let $\varphi = [p_0 : \dots : p_n]$. For every $0 \neq t \in \Pi$, the point $\varphi(t)$ has primitive omnific homogeneous coordinates if and only if $\varphi(0) \in \mathbb{P}^n(\mathbb{Q})$. Such coordinates are then unimodular and, after the common real normalization of the p_i , equal to $\pm(p_0(t), \dots, p_n(t))$. Otherwise no omnific coordinate vector of $\varphi(t)$ has a gcd.*

Proof. The polynomial gcd gives a Bézout identity, so the vector at t is $\mathcal{B}_{\mathbb{R}}$ -unimodular with residue vector $(p_0(0), \dots, p_n(0))$. Apply [theorem 15.11](#) and the argument of [corollary 15.19](#). No affine chart is needed. $\square$

For constant p_i this is [theorem 14.5](#) of source 01. The corollary replaces a constant vector by any reduced polynomial parametrization; the criterion concerns the specialization at $T = 0$, not the direction at infinity [08]. Source 08’s illustration, the conic $(1 - t^2, 2t, 1 + t^2)$ with certificate $(1 + \frac{t^2}{2})(1 - t^2) + \frac{t^2}{2}(1 + t^2) = 1$ and specialization $[1 : 0 : 1]$, is [proposition 14.4](#) with source 01’s witness (14.2), up to the sign of the first coordinate; source 08 presents it as an illustration, not as a new construction.

Example 15.29 (A curve without primitive coordinates [08]). The vector $(1, \sqrt{2} + t, t^2)$ has specialization $[1 : \sqrt{2} : 0]$, which is not rational. Its entries are not all omnific, but $h(1, \sqrt{2} + t, t^2)$ is omnific for every $0 \neq h \in \Pi$, and by [theorem 15.11](#) these are all the omnific coordinate vectors of the point. None is primitive, although the first unscaled coordinate is 1: the unscaled vector does not lie in $\mathbf{Oz}^3$.

Corollary 15.30 (Solutions on rational curves [08] [merge]). *Let $F_1, \dots, F_r \in \mathbb{R}[X_0, \dots, X_n]$ be homogeneous, and let $p_0, \dots, p_n \in \mathbb{R}[T]$ have gcd one, satisfy $F_j(p_0(T), \dots, p_n(T)) = 0$ for all j , and have $[p_0(0) : \dots : p_n(0)] \in \mathbb{P}^n(\mathbb{Q})$. After the normalization of [corollary 15.28](#), for every $0 \neq t \in \Pi$ the vector $(p_0(t), \dots, p_n(t))$ is a unimodular omnific solution of the system, and it has an infinite coordinate when some p_i is nonconstant. If instead $[p_0(0) : \dots : p_n(0)] \notin \mathbb{P}^n(\mathbb{Q})$, no point $\varphi(t)$ of the curve with $0 \neq t \in \Pi$ has a primitive, in particular a unimodular, omnific coordinate vector.*

Proof. The identities $F_j(p_0, \dots, p_n) = 0$ persist under evaluation. Apply [corollary 15.28](#) and [lemma 15.17](#). $\square$

This answers [question 18.3](#) in part: a homogeneous system whose projective zero locus contains such a rational curve has unimodular omnific solutions, and along the curve the points with primitive coordinates are exactly those of rational specialization. It says nothing about points off rational curves, nor about the dependence on an integral model asked there ([section 18.3](#)). The corollary is the merge's formulation of source 08's [corollary 15.28](#). [Proposition 14.1](#) asks for integer constants of the p_i and gives omnific solutions, not unimodularity; here the constant vector need only be rational, and the output is unimodular.

15.7 Specialization fibers, congruence orbits and density

Apply ct entrywise to matrices. It gives a split exact sequence

$$1 \longrightarrow \mathbf{K} \longrightarrow \text{SL}_2(\mathbf{Oz}) \xrightarrow{\text{ct}} \text{SL}_2(\mathbb{Z}) \longrightarrow 1, \quad \mathbf{K} = \ker(\text{ct} : \text{SL}_2(\mathbf{Oz}) \rightarrow \text{SL}_2(\mathbb{Z})),$$

split by the integer matrices. The kernel of $\text{ct} : \text{SL}_2(\mathcal{B}_{\mathbb{R}}) \rightarrow \text{SL}_2(\mathbb{R})$ is the same group $\mathbf{K}$: a matrix over $\mathcal{B}_{\mathbb{R}}$ with constant matrix I has integer constants, hence entries in $\mathbf{Oz}$ [09]. Source 09 writes Γ for $\mathbf{K}$.

Lemma 15.31 (Equivariance [09]). *The class $\mathcal{V}$ is stable under $\text{SL}_2(\mathcal{B}_{\mathbb{R}})$, and $\text{sp}(gx) = \text{ct}(g) \text{sp}(x)$. In particular $\mathbf{K}$ preserves sp .*

Proof. An invertible matrix carries a unimodular column to a unimodular column: apply its inverse to transport a Bézout row. Taking constants gives the formula. $\square$

Theorem 15.32 (Specialization fibers are congruence orbits [09]). *For every $r \in \mathbb{P}^1(\mathbb{R})$, $\text{sp}^{-1}(r) = \mathbf{K} \cdot r$. Hence*

$$\mathcal{V} = \coprod_{r \in \mathbb{P}^1(\mathbb{R})} \mathbf{K} \cdot r, \quad \mathcal{U} = \coprod_{r \in \mathbb{P}^1(\mathbb{Q})} \mathbf{K} \cdot r,$$

disjoint unions of classes, and every $\mathbf{K}$ -orbit in $\mathcal{V}$ contains exactly one real point.

Proof. One inclusion is [lemma 15.31](#). For the other let (P, Q) be a $\mathcal{B}_{\mathbb{R}}$ -unimodular column representing x , with $UP + VQ = 1$, and put

$$M = \begin{pmatrix} P & -V \\ Q & U \end{pmatrix} \in \text{SL}_2(\mathcal{B}_{\mathbb{R}}), \quad M_0 = \text{ct}(M) \in \text{SL}_2(\mathbb{R}), \quad g = MM_0^{-1}.$$

Then $M\infty = x$, $\text{ct}(g) = I$, so $g \in \mathbf{K}$, and $g(M_0\infty) = x$ with $M_0\infty = \text{sp}(x)$. Distinct real points have distinct specializations, and the description of $\mathcal{U}$ follows from [theorem 15.15](#). $\square$

Corollary 15.33 (Modular equivalence [09]). *For $r, s \in \mathbb{P}^1(\mathbb{R})$, $s \in \text{SL}_2(\mathbf{Oz}) \cdot r$ if and only if $s \in \text{SL}_2(\mathbb{Z}) \cdot r$. Moreover $\mathcal{U} = \text{SL}_2(\mathbf{Oz}) \cdot \infty$.*

Proof. If $s = gr$, then $s = \text{sp}(s) = \text{ct}(g)r$ by [lemma 15.31](#); the converse uses integer matrices. An $\mathbf{Oz}$ -unimodular column completes to a determinant-one matrix over $\mathbf{Oz}$ as in the proof of [theorem 15.32](#). $\square$

The orbits carry a canonical real label, and the rational labels select exactly the fibers whose fractions have lowest terms [09].

Lemma 15.34 (Focusing matrices [09]). *For $x \in \mathbf{No}$ and $b \in \Pi$ with $bx, bx^2 \in \Pi$ put*

$$\mathbf{F}_{x,b} = \begin{pmatrix} 1+bx & -bx^2 \\ b & 1-bx \end{pmatrix} = I + b\mathbf{N}_x, \quad \mathbf{N}_x = \begin{pmatrix} x & -x^2 \\ 1 & -x \end{pmatrix}.$$

Then $\mathbf{N}_x^2 = 0$, $\mathbf{F}_{x,b} \in \mathbf{K}$, $\mathbf{F}_{x,b}$ fixes x , $\mathbf{F}_{x,b}\mathbf{F}_{x,c} = \mathbf{F}_{x,b+c}$ when both are defined, $\mathbf{F}_{x,b}^{-1} = \mathbf{F}_{x,-b}$, and

$$\mathbf{F}_{x,b}(z) = x + \frac{z-x}{1+b(z-x)}, \quad \mathbf{F}_{x,b}(\infty) = x + \frac{1}{b}, \quad (15.7)$$

the first formula where its denominator is nonzero.

Proof. The entries other than the constant diagonal lie in Π , and $\det \mathbf{F}_{x,b} = (1+bx)(1-bx) + b^2x^2 = 1$. The identities follow from $\mathbf{N}_x^2 = 0$, and (15.7) from subtracting x from the fractional-linear action. $\square$

Source 09 calls $\mathbf{F}_{x,b}$ an omnific transvection. It is unrelated to the quadratic isometry of [lemma 8.2](#).

Theorem 15.35 (Set-wise focusing [09]). *Let $S \subseteq \mathbb{P}^1(\mathbf{No})$ be a set, $x \in \mathbf{No}$ and $\varepsilon > 0$. There is a monomial $b = \omega^\beta$ with $\beta > 0$ such that $\mathbf{F}_{x,b} \in \mathbf{K}$ and*

$$\mathbf{F}_{x,b}(S) \subseteq (x - \varepsilon, x + \varepsilon).$$

In particular no point of S is sent to ∞ .

Proof. The set $\{1, x, x^2\}$ can be cleared by one monomial ([theorem 4.6](#)), and any larger exponent also clears it. The positive surreals $2/\varepsilon$ and $2/|z-x|$, for affine $z \in S \setminus \{x\}$, form a set, so by [lemma 4.1](#) we may also take β larger than all their leading exponents. Then $b, bx, bx^2 \in \Pi$, $b > 2/\varepsilon$, and $b|z-x| > 2$ for those z . For such z put $w = z-x$; then $|1+bw| \geq |bw| - 1 > \frac{1}{2}|bw| > 0$ and

$$|\mathbf{F}_{x,b}(z) - x| = \frac{|w|}{|1+bw|} < \frac{2}{b} < \varepsilon.$$

The point x is fixed, and ∞ goes to $x + 1/b$, in the same interval. $\square$

The conclusion holds for every set-sized family, not only for finite ones. It fails for the proper class $\mathbb{P}^1(\mathbf{No})$, since an invertible matrix acts bijectively; the restriction to sets is essential [09].

Corollary 15.36 (Dense fibers [09]). *Every $\mathbf{K}$ -orbit in $\mathbb{P}^1(\mathbf{No})$ is dense in the order topology. In particular every fiber $\mathrm{sp}^{-1}(r)$, $r \in \mathbb{P}^1(\mathbb{R})$, is dense, and $\mathcal{V}$ contains continuum many pairwise disjoint dense fibers.*

Proof. Apply [theorem 15.35](#) to a singleton and a nonempty interval; every nonempty open set of $\mathbb{P}^1(\mathbf{No})$ contains an interval. Then use [theorem 15.32](#). $\square$

Corollary 15.37 (Both denominator types in every interval [09]). *Every nonempty open interval of $\mathbf{No}$ contains a fraction with a least positive omnific denominator and a unimodular presentation, and a fraction whose denominator ideal is not generated by any set and none of whose presentations has a gcd. The first can be taken in any prescribed rational fiber, the second in any prescribed irrational fiber.*

Proof. Combine [corollary 15.36](#) and [theorem 15.15](#). $\square$

Explicitly [09], for $x \in \mathbf{No}$ and $\varepsilon > 0$ choose b as in the proof of [theorem 15.35](#) with $b > 1/\varepsilon$. Then

$$y = x + \frac{1}{b} = \frac{1 + bx}{b}, \quad (1 - bx)(1 + bx) + (bx^2)b = 1,$$

so b is the least positive denominator of y , and $|y - x| < \varepsilon$. For an irrational real r , the point $y = F_{x,b}(r) = x + (r - x)/(1 + b(r - x))$ has $\mathfrak{D}(y) = (1 + b(r - x))\Pi$, since the transformed $\mathcal{B}_{\mathbb{R}}$ -unimodular column has constant vector $(r, 1)$. For $r = x$ this is $y = x$.

Corollary 15.38 (Nowhere continuity [09]). *The map $\mathrm{sp} : \mathcal{V} \rightarrow \mathbb{P}^1(\mathbb{R})$ is continuous at no point, for the relative order topology on $\mathcal{V}$ and the usual topology on $\mathbb{P}^1(\mathbb{R})$.*

Proof. At $y \in \mathcal{V}$ choose $r \neq \mathrm{sp}(y)$ and a neighbourhood of $\mathrm{sp}(y)$ omitting r . Every neighbourhood of y meets the dense fiber $\mathrm{sp}^{-1}(r)$. $\square$

Denominator arithmetic is thus invisible to order approximation of any accuracy. This does not conflict with the continuity of the field operations. The dense fibers are invariant arithmetic classes; they are not open and not separated components [09].

Remark 15.39 (Elementary matrices do not suffice [merge]). The matrices $\mathbf{N}_x$ are the transposes of the matrices Q_α of the non-elementary unipotents in `docs/surreal/omnific-groups-and-lattices` (`ogl:el:thm:none1`), at $\alpha = x$. That report proves that $h \cdot \infty$ has rational standard part whenever $h \in E_2(\mathbf{Oz})$ and $h \cdot \infty$ is finite (`ogl:el:prop:cuspresidue`), where E_2 is the group generated by elementary matrices. Two consequences follow.

- (i) If x is finite with $\mathrm{st}(x) \notin \mathbb{Q}$, for example an irrational real, then $F_{x,b} \notin E_2(\mathbf{Oz})$ for every b as in [lemma 15.34](#) with $b \neq 0$. Indeed $F_{x,b}(\infty) = x + 1/b$ is finite with standard part $\mathrm{st}(x)$.
- (ii) The orbit $E_2(\mathbf{Oz}) \cdot \infty$ is not dense. It misses the interval $(\sqrt{2} - \omega^{-1}, \sqrt{2} + \omega^{-1})$, whose points are finite with standard part $\sqrt{2}$. But $\mathbf{K} \cdot \infty = \mathrm{sp}^{-1}(\infty)$ is dense ([corollary 15.36](#)), and $\mathcal{U} = \mathrm{SL}_2(\mathbf{Oz}) \cdot \infty$ strictly contains $E_2(\mathbf{Oz}) \cdot \infty$. For example $\sqrt{2} + \omega^{-2}$ lies in that interval and in $\mathcal{U}$, with the unimodular pair $(1 + \sqrt{2}\omega^2, \omega^2)$ and the certificate $(1 - \sqrt{2}\omega^2)(1 + \sqrt{2}\omega^2) + 2\omega^2 \cdot \omega^2 = 1$.

So the density of the orbit of ∞ needs matrices outside $E_2(\mathbf{Oz})$, and source 09 rightly works with $\mathbf{K}$ rather than with the elementary kernel of that report. Source 09 never claims $\mathbf{K} \subseteq E_2(\mathbf{Oz})$; the remark adds the comparison.

15.8 The localization at the purely infinite ideal

The ideal Π is prime, since $\mathbf{Oz}/\Pi \cong \mathbb{Z}$ is a domain. Let

$$\mathbf{Oz}_\Pi = (\mathbf{Oz} \setminus \Pi)^{-1} \mathbf{Oz} = \left\{ \frac{a}{b} : a, b \in \mathbf{Oz}, \text{ct}(b) \neq 0 \right\} \subseteq \mathbf{No},$$

the localization at Π , not the quotient $\mathbf{Oz}/\Pi$. Source 08 writes $\mathcal{A}$ and ρ for $\mathbf{Oz}_\Pi$ and res .

Proposition 15.40 (A local ring with residue field $\mathbb{Q}$ [08]). *The map $\text{res} : \mathbf{Oz}_\Pi \rightarrow \mathbb{Q}$, $\text{res}(a/b) = \text{ct}(a)/\text{ct}(b)$, is a well-defined surjective ring homomorphism. The ring $\mathbf{Oz}_\Pi$ is local with maximal ideal $\ker \text{res} = (\mathbf{Oz} \setminus \Pi)^{-1} \Pi$, its real elements are exactly $\mathbb{Q}$, and it is neither a field nor a valuation ring of $\mathbf{No}$.*

Proof. If $a/b = a'/b'$ with $\text{ct}(b), \text{ct}(b') \neq 0$, then $ab' = a'b$; taking constant coefficients proves well-definedness, and the fraction rules give a surjective homomorphism. If $\text{res}(a/b) \neq 0$, then $\text{ct}(a) \neq 0$ and $b/a \in \mathbf{Oz}_\Pi$, so a/b is a unit; an element of $\ker \text{res}$ is not a unit. For $r \in \mathbb{R}$, a representation with denominator outside Π exists exactly when $\mathfrak{D}(r)$ has an element with nonzero constant, which by [theorem 15.7](#) happens exactly for $r \in \mathbb{Q}$. So neither $\sqrt{2}$ nor $1/\sqrt{2}$ lies in $\mathbf{Oz}_\Pi$, although $\text{Frac}(\mathbf{Oz}_\Pi) = \mathbf{No}$; a valuation ring contains every nonzero element or its inverse. Finally $t \in \ker \text{res}$ is a nonzero nonunit. $\square$

This is consistent with the sibling report's list of the prime ideals with set-sized quotient: Π is prime with $\mathbf{Oz}/\Pi = \mathbb{Z}$ and fraction field $\mathbb{Q}$ (`osq:cor:smallprimes`), and res restricts to ct on $\mathbf{Oz}$.

Proposition 15.41 (Rational functions in the localization [08]). *Under $\mathbb{R}(T) \cong \mathbb{R}(t)$,*

$$\mathbf{Oz}_\Pi \cap \mathbb{R}(t) = \{f(t) : f \in \mathbb{R}(T) \text{ regular at } 0, f(0) \in \mathbb{Q}\},$$

and $\text{res}(f(t)) = f(0)$ on this ring.

Proof. Write $f = p/q$ in reduced form. If $f(0)$ is irrational, every denominator lies in $q(t)\Pi \subseteq \Pi$ ([theorem 15.18](#)). If f has a pole at 0, then $\mathfrak{D}(f(t)) = \lambda q(t)\mathbf{Oz}$ with $q(0) = 0$, and again all denominators lie in Π . In both cases $f(t) \notin \mathbf{Oz}_\Pi$. If f is regular at 0 with $f(0) \in \mathbb{Q}$, normalize $(p(0), q(0))$ to a primitive integer pair; its second entry is nonzero, so the normalized denominator lies outside Π , and $\text{res}(f(t)) = f(0)$. $\square$

Theorem 15.42 (Independent real and rational residues [08]). *Let $\mathbf{Oz}_\Pi^{\text{fin}} = \mathbf{Oz}_\Pi \cap \mathcal{O}_{\text{fin}}$. The map $(\text{st}, \text{res}) : \mathbf{Oz}_\Pi^{\text{fin}} \rightarrow \mathbb{R} \times \mathbb{Q}$ is surjective, its two kernels are comaximal, and $\mathbf{Oz}_\Pi^{\text{fin}}/(\ker \text{st} \cap \ker \text{res}) \cong \mathbb{R} \times \mathbb{Q}$. For $r \in \mathbb{R}$, $q \in \mathbb{Q}$ and $t > 0$ in Π , the element*

$$x_{r,q} = \frac{rt + q}{t + 1} = r + \frac{q - r}{t + 1}$$

lies in $\mathbf{Oz}_\Pi^{\text{fin}}$ with $\text{st}(x_{r,q}) = r$ and $\text{res}(x_{r,q}) = q$.

Proof. With $q = m/n$, $x_{r,q} = (nrt + m)/(nt + n)$ has numerator and denominator in $\mathbf{Oz}$ and denominator constant $n \neq 0$, so $\text{res}(x_{r,q}) = m/n$; the second expression differs from r by an infinitesimal. The elements $u = t/(t + 1)$ and $v = 1/(t + 1)$ lie in $\mathbf{Oz}_\Pi^{\text{fin}}$, satisfy $u + v = 1$, and have $(\text{st}, \text{res})(u) = (1, 0)$ and $(\text{st}, \text{res})(v) = (0, 1)$. So the kernels are comaximal, and the quotient is the Chinese remainder isomorphism. $\square$

The standard part sees the behaviour at an infinite parameter and res the constant-coefficient specialization; on finite elements they agree only on $\mathbb{Q}$ [08]. Neither map is a retraction of $\mathbf{No}$, and their domains differ. The map res is not order preserving: $u > 0$ but $\text{res}(u) = 0$, and $r > 0 > q$ gives a positive element with negative rational residue. For reduced $f = p/q$, $f(t)$ is finite exactly when $\deg p \leq \deg q$, with standard part 0 or the ratio of leading coefficients. So $\mathbf{Oz}_{\Pi}^{\text{fin}} \cap \mathbb{R}(t)$ consists of the rational functions regular at 0 and at ∞ with a rational value at 0 and any real value at ∞ . The surjection onto $\mathbb{R} \times \mathbb{Q}$ is a set-sized quotient of the class ring $\mathbf{Oz}_{\Pi}^{\text{fin}}$. That ring meets $\mathbf{Oz}$ in $\mathbb{Z}$ and contains no nonzero element of Π , so the sibling report's `osq:prop:fractions(iv)`, about localizations in which an element of Π becomes a unit, does not apply. There is no contradiction.

15.9 Polynomial models and scale extensions

For an indeterminate X write $\mathcal{P}[X] = \mathbb{Z} + X\mathbb{R}[X]$; source 08 writes A_{τ} , A_0 , A_m and source 09 writes $\mathcal{O}_{\mathbb{Z}}$. These are set-sized rings.

Proposition 15.43 (The omnific part of a one-parameter field [08]). $\mathcal{B}_{\mathbb{R}} \cap \mathbb{R}(t) = \mathbb{R}[t]$ and $\mathbf{Oz} \cap \mathbb{R}(t) = \mathbb{Z} + t\mathbb{R}[t]$, the image $\mathcal{P}_t$ of $\mathcal{P}[T]$ under $T \mapsto t$. Moreover $\text{Frac}(\mathcal{P}_t) = \mathbb{R}(t)$.

Proof. Write a rational function as a polynomial plus a strictly proper part. At t a nonzero strictly proper part is infinitesimal, and $\mathcal{B}_{\mathbb{R}}$ has no nonzero infinitesimal, so an element of $\mathcal{B}_{\mathbb{R}} \cap \mathbb{R}(t)$ is a polynomial in t ; it lies in $\mathbf{Oz}$ exactly when its constant coefficient is an integer. The fraction field contains t and every real $r = (rt)/t$. $\square$

Real constants are absent from $\mathcal{P}_t$ except for $\mathbb{Z}$ but reappear in its fraction field. With $(R, c) = (\mathbb{R}[t], \text{evaluation at } T = 0)$, [theorem 15.11](#) gives, in the notation of [theorem 15.18](#),

$$\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) = q(t) \mathcal{P}_t \quad \text{or} \quad q(t) \cdot t\mathbb{R}[t], \quad (15.8)$$

according as $\text{sp}(\mathbf{x})$ is rational or not, with the same normalization. The primitive or nonprimitive status is the same in the model and in $\mathbf{Oz}$, although the ideal of all denominators becomes much larger.

Proposition 15.44 (Generator counts [08]). *The ideal $t\mathbb{R}[t]$ of $\mathcal{P}_t$ needs exactly $\mathfrak{c} = |\mathbb{R}|$ generators, and so does every $q(t)t\mathbb{R}[t]$ with $q \neq 0$. The corresponding ideal $q(t)\Pi$ of $\mathbf{Oz}$ has no set of generators.*

Proof. With $I = t\mathbb{R}[t]$, $I^2 = t^2\mathbb{R}[t]$ and $I/I^2 \cong (\mathbb{R}, +)$ through the coefficient of t . The action of $\mathcal{P}_t$ on I/I^2 factors through $\mathcal{P}_t/I = \mathbb{Z}$, and an abelian group generated by κ elements has cardinality at most $\max(\kappa, \aleph_0)$; so $\mathfrak{c}$ generators are needed. The set $\{rt : r \in \mathbb{R}\}$ generates I . Multiplication by $q(t)$ is an isomorphism onto the scaled ideal. The last assertion is [theorem 15.18](#). $\square$

For an ideal J of $\mathcal{P}_t$ let $J\mathbf{Oz}$ be its extension to $\mathbf{Oz}$.

Theorem 15.45 (Enlargement defect [08]). *In the notation of [theorem 15.18](#), $\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) \mathbf{Oz} = q(t) \mathbf{Oz} = \mathfrak{D}(\mathbf{x})$ in the rational case, and $\mathfrak{D}_{\mathcal{P}_t}(\mathbf{x}) \mathbf{Oz} = q(t)t\mathcal{B}_{\mathbb{R}} \subsetneq q(t)\Pi = \mathfrak{D}(\mathbf{x})$ otherwise.*

Proof. Only the second case needs an argument. The extension $(t\mathbb{R}[t])\mathbf{Oz}$ contains rt for real r and tu for $u \in \Pi$, hence $t\mathcal{B}_{\mathbb{R}}$; conversely a product of an element of $t\mathbb{R}[t]$ and an omnific integer lies in $t\mathcal{B}_{\mathbb{R}}$. For strictness choose $0 < \varepsilon < \deg_{\omega} t$: then $\omega^{\varepsilon} \in \Pi$, but ω^{ε}/t has negative leading exponent, so it is a nonzero infinitesimal outside $\mathcal{B}_{\mathbb{R}}$. Multiplication by $q(t)$ preserves the strict inclusion. $\square$

Enlarging the available scales and recomputing a denominator ideal can thus reveal denominators not generated by those of the smaller model. The primitive or nonprimitive answer is stable, but the module of all denominators is not obtained by extending scalars [08].

In the rest of this subsection $m \geq 2$, the rings are $\mathcal{P}[T] \subseteq \mathcal{P}[U]$ with $T = U^m$, embedded in $\mathbf{Oz}$ by $T = \omega$ and $U = \omega^{1/m}$, and $I_T = T\mathbb{R}[T]$, $I_U = U\mathbb{R}[U]$.

Theorem 15.46 (Scale-extension defect [08]). *Let $p_1, \dots, p_n, q \in \mathbb{R}[T]$ have gcd one, $q \neq 0$, $\mathbf{f} = (p_j/q)_j$ and $\mathbf{s} = [p_1(0) : \dots : p_n(0) : q(0)]$. If $\mathbf{s} \in \mathbb{P}^n(\mathbb{Q})$, then $\mathfrak{D}_{\mathcal{P}[U]}(\mathbf{f}(U^m)) = \mathfrak{D}_{\mathcal{P}[T]}(\mathbf{f}(T))\mathcal{P}[U]$. If $\mathbf{s} \notin \mathbb{P}^n(\mathbb{Q})$, then*

$$\mathfrak{D}_{\mathcal{P}[U]}(\mathbf{f}(U^m)) = q(U^m)U\mathbb{R}[U], \quad \mathfrak{D}_{\mathcal{P}[T]}(\mathbf{f}(T))\mathcal{P}[U] = q(U^m)U^m\mathbb{R}[U],$$

and division by $q(U^m)$ induces an isomorphism of the quotient of the first by the second with $U\mathbb{R}[U]/U^m\mathbb{R}[U]$, which has the real basis $U, U^2, \dots, U^{m-1}$ and real dimension $m - 1$.

Proof. A polynomial Bézout identity persists under $T = U^m$, and the constant vector is unchanged; apply [theorem 15.11](#) in both polynomial rings. In the rational case both ideals are the same principal ideal. Otherwise they are $q(T)I_T$ and $q(U^m)I_U$, and $I_T\mathcal{P}[U] = U^m\mathbb{R}[U]$: the inclusion $\subseteq$ is by degrees, and every rU^m comes from rT , while multiplication by U^k gives the higher monomials. $\square$

The dimension is that of a real coefficient space. It is not a length of a $\mathcal{P}[U]$ -module, since $\mathcal{P}[U]$ does not contain the real constants [08].

Theorem 15.47 (An integral extension that is neither finite nor flat [08]). *For every $m \geq 2$ the inclusion $\mathcal{P}[T] \hookrightarrow \mathcal{P}[U]$ is integral, but $\mathcal{P}[U]$ is neither finitely generated nor flat as a $\mathcal{P}[T]$ -module.*

Proof. Each rU^k with $k \geq 1$ is a root of $X^m - r^mT^k$ over $\mathcal{P}[T]$, and sums of integral elements are integral. If $\mathcal{P}[U]$ were a finite $\mathcal{P}[T]$ -module, then $\mathcal{P}[U]/I_T\mathcal{P}[U] \cong \mathbb{Z} \oplus \bigoplus_{j=1}^{m-1} \mathbb{R}U^j$ would be finitely generated over $\mathcal{P}[T]/I_T = \mathbb{Z}$, which it is not. For flatness put $r = \sqrt{2}$ and consider the row $(rT, T) : \mathcal{P}[T]^2 \rightarrow \mathcal{P}[T]$. Its kernel is $\{(h, -rh) : h \in I_T\}$, because $rh \in \mathcal{P}[T]$ forces $\text{ct}(h) = 0$. Over $\mathcal{P}[U]$ the kernel of the same row is $\{(h, -rh) : h \in I_U\}$, while the image of the old relations is only $\{(h, -rh) : h \in I_T\mathcal{P}[U] = U^m\mathbb{R}[U]\}$. The relation $(U, -rU)$ is not in this image. A flat module preserves the exactness of such sequences [34, Tag 00HK], so $\mathcal{P}[U]$ is not flat. $\square$

Proposition 15.48 (The defect as a Tor group [08]). *Let $J = (\sqrt{2}T, T) \subseteq \mathcal{P}[T]$ and $C = \mathcal{P}[T]/J$. Then $\text{Tor}_1^{\mathcal{P}[T]}(C, \mathcal{P}[U]) \cong U\mathbb{R}[U]/U^m\mathbb{R}[U]$ as $\mathcal{P}[U]$ -modules.*

Proof. From $0 \rightarrow J \rightarrow \mathcal{P}[T] \rightarrow C \rightarrow 0$ and the freeness of $\mathcal{P}[T]$, $\text{Tor}_1(C, \mathcal{P}[U])$ is the kernel of $J \otimes \mathcal{P}[U] \rightarrow \mathcal{P}[U]$. Presenting J by the row $(\sqrt{2}T, T)$ with its relation module, right exactness identifies this kernel with the kernel of the evaluated row modulo the extended old relations, both computed in the proof of [theorem 15.47](#); the first coordinate gives the isomorphism. $\square$

Remark 15.49 (Scope of the scale computations [08]; [merge]). Integrality, nonflatness and non-finiteness of pullback extensions are familiar; the point is the exact denominator-sensitive defect realized by a finer surreal monomial, and the same $m - 1$ layers witness the failure of flatness. All tensor products and Tor groups here are over set-sized rings [08]. The tower $\mathcal{P}[T] \subseteq \mathcal{P}[T^{1/2}] \subseteq \mathcal{P}[T^{1/6}] \subseteq \dots$ (exponents $1/m!$) has union the set-sized core $A_1 \cong T_{\mathbb{Q}}$ of the sibling report (`osq:hd:lem:cores`), with $(D, K) = (\mathbb{Z}, \mathbb{R})$. The steps have exponent group $\mathbb{Z}$ and are not among the cores T_{Γ} there, which need a divisible Γ ; their ideals I_T are not idempotent ($I_T/I_T^2 \cong \mathbb{R}$), whereas $\Pi_{\Gamma}^{\text{fin}}$ is. No statement of that report is contradicted.

15.10 Monomial denominators and fixed workspaces

For $x \in \mathbf{No}$ let $\text{Ex}(x) = \{\beta \in \mathbf{No} : \beta \geq 0, \omega^\beta x \in \mathbf{Oz}\}$, the class of exponents of monic monomial denominators. Its elements have coefficient 1; ordinary integers such as 2 are not of this form. Source 08 writes $M(x)$ and source 09 $E(x)$.

Proposition 15.50 (Monomial denominators [08, 09]). *Let $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma \neq 0$ and $r_\eta = 0$ for $\eta \notin S$. Then $\beta \in \text{Ex}(x)$ if and only if $\beta \geq 0$, $\gamma + \beta \geq 0$ for all $\gamma \in S$, and $r_{-\beta} \in \mathbb{Z}$. The class $\text{Ex}(x)$ is nonempty and upward closed, and $\omega^{\beta'} x \in \Pi$ whenever $\beta' > \beta \in \text{Ex}(x)$. It has a least member exactly when $x \in \mathbf{Oz}$ (least member 0) or S has a negative least element a with $r_a \in \mathbb{Z}$ (least member $-a$). Also $\text{Ex}(0) = \mathbf{No}^{\geq 0}$.*

Proof. Multiplication by ω^β shifts supports by β and keeps coefficients, which gives the criterion; nonemptiness is [theorem 4.6](#), and a strictly larger exponent makes every shifted exponent positive. Let $\beta > 0$ be least. If the shifted support were entirely positive, a positive ε below β and below every shifted exponent, which exists by [lemma 4.1](#), would make $\beta - \varepsilon$ admissible. So 0 is in the shifted support and is its minimum; hence $-\beta$ is a negative least element of S with integral coefficient. Conversely such an element makes $-a$ admissible and forbids smaller exponents. The case $\beta = 0$ is $x \in \mathbf{Oz}$. $\square$

Example 15.51 (Monomial and algebraic minimization differ [08, 09]). (i) The rational $1/2$ has least omnific denominator 2, but $\text{Ex}(1/2) = \{\beta : \beta > 0\}$ has no least member [08]. (ii) The number $x = \frac{1}{2}\omega^{-1} = 1/(2\omega)$ has least positive denominator 2ω , by the unimodular pair $(1, 2\omega)$, but $\text{Ex}(x) = \{\beta : \beta > 1\}$, since at $\beta = 1$ the numerator is $1/2$ [09]. (iii) For irrational r , $\text{Ex}(r) = \{\beta : \beta > 0\}$, and here even unrestricted denominators have no least member ([theorem 15.7](#)) [09]. (iv) The geometric series $x = 1/(\omega - 1) = \sum_{n \geq 1} \omega^{-n}$ has least positive denominator $\omega - 1$, but $\text{Ex}(x)$ is the class of β exceeding every ordinary integer, with no least member [08]. Straightforward monomial clearing therefore needs an exponent above every integer, where a two-term denominator of leading exponent 1 suffices; support bounds cannot replace denominator-ideal arithmetic, and a preferred monomial denominator need not be the arithmetically preferred one.

Example 15.52 (One fixed workspace [08, 09]). Let $K = \mathbb{R}((\omega^{-1}))$, the Laurent series in ω^{-1} . A reverse-well-ordered set of nonnegative integers is finite, so $K \cap \mathbf{Oz} = \mathcal{P}_\omega = \mathbb{Z} + \omega\mathbb{R}[\omega]$, and

$$\{\omega^n : n \in \mathbb{N}\}^{-1} \mathcal{P}_\omega = \mathbb{R}[\omega, \omega^{-1}] \subsetneq \text{Frac}(\mathcal{P}_\omega) = \mathbb{R}(\omega) \subsetneq K.$$

The first inclusion is strict because $1/(\omega - 1) = \sum_{n \geq 1} \omega^{-n}$ is not a Laurent polynomial [09]. The second is strict by either of two elements.

- (i) $E = \sum_{n \geq 0} \omega^{-n}/n!$ [08]. If $\sum_n z^n/n!$ were rational, a nonzero $\sum_{j=0}^d q_j z^j$ would give $\sum_j q_j/(n-j)! = 0$ for all large n ; multiplying by $n!$ gives $\sum_j q_j n(n-1) \cdots (n-j+1) = 0$ for infinitely many n , and falling factorials of distinct degrees are linearly independent.
- (ii) $F = \sum_{n \geq 1} \omega^{-n!}$ [09]. If $F = A(u)/B(u)$ in $u = \omega^{-1}$ with $B(0) \neq 0$, comparing coefficients in $BF = A$ gives a linear recurrence of fixed length d for all large indices, and a block of d zero coefficients then forces all later ones to vanish; the factorial gaps give such blocks followed by a coefficient 1. A constant B would make F a polynomial.

In $\mathbf{No}$ the monomial ω^ω clears F , with exponents $\omega - n! > 0$, but $\omega^\omega \notin K$ [09].

The example is consistent with source 06's non-claim that the fraction field of a fixed Hahn ring need not be the whole Hahn field ($\Gamma = \mathbb{Z}$ gives $\mathbb{k}(\omega)$), and with the cofinality criterion

of L’Innocente–Mantova [3, Proposition 2.4.5]. The small common divisor of [theorem 4.2](#) may likewise need an exponent outside a workspace. Even where the needed shifts exist, the dichotomy in a workspace requires a unimodular pair over its nonnegative-support ring; the polynomial Bézout certificate of [theorem 15.18](#) supplies it, and a formal cancellation in an ambient field does not [09].

15.11 Gaussian omnific fractions

Here $\mathcal{B}_{\mathbb{C}} = \mathbb{C} \oplus \Pi_{\mathbb{C}}$ and $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$ ([definition 2.1](#)), with $\text{ct} : \mathcal{B}_{\mathbb{C}} \rightarrow \mathbb{C}$ a retraction and $\mathbf{Oz}[i] = \text{ct}^{-1}(\mathbb{Z}[i])$.

Proposition 15.53 (Gaussian fractions [08]; [09]). $\mathbf{Oz}[i][(\mathbb{Z} \setminus \{0\})^{-1}] = \mathbb{Q}(i) \oplus \Pi_{\mathbb{C}}$. The units of $\mathbf{Oz}[i]$ are $\pm 1, \pm i$, and $\Pi_{\mathbb{C}}$ is divisible by every nonzero ordinary integer and not generated by any set as an ideal of $\mathbf{Oz}[i]$.

Proof. Clearing the two rational constants gives the first identity. By [lemma 2.4](#) a unit of $\mathcal{B}_{\mathbb{C}}$ is a complex constant, and the units of $\mathbb{Z}[i]$ are the four displayed. Division by a nonzero integer preserves positive support, and the support-margin argument of [corollary 4.3](#) applies to the union of the real and imaginary supports. $\square$

The identity $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$, proved by sources 08 and 09 through clearing the real and imaginary parts with one monomial, is already stated with source 06 in [section 13](#), and the units of $\mathbf{Oz}[i]$ are recorded in [section 11.1](#). Source 09 gives no complex lowest-term classification; source 08 does.

Theorem 15.54 (Gaussian representation dichotomy [08]). Let $t \in \Pi$ be nonzero and let $p_1, \dots, p_n, q \in \mathbb{C}[T]$, $q \neq 0$, have gcd one. [Theorem 15.18](#) holds with $(\mathbb{R}, \mathbb{Z}, \mathbb{Q}, \mathcal{B}_{\mathbb{R}}, \mathbf{Oz}, \Pi)$ replaced by $(\mathbb{C}, \mathbb{Z}[i], \mathbb{Q}(i), \mathcal{B}_{\mathbb{C}}, \mathbf{Oz}[i], \Pi_{\mathbb{C}})$. In particular primitive homogeneous coordinates exist exactly when $[p_1(0) : \dots : p_n(0) : q(0)] \in \mathbb{P}^n(\mathbb{Q}(i))$; they are then unimodular and unique up to a Gaussian unit. Otherwise no representation has a gcd, and the denominator ideal is $q(t)\Pi_{\mathbb{C}}$, with no set of generators.

Proof. The evaluated vector is unimodular over $\mathcal{B}_{\mathbb{C}}$ by a complex polynomial Bézout identity, and the identity forces every ambient scalar giving Gaussian omnific coordinates into $\mathcal{B}_{\mathbb{C}}$. For a nonzero complex residue vector $\mathbf{v}$, the complex s with $s\mathbf{v} \in \mathbb{Z}[i]^{n+1}$ are 0 unless the line is defined over $\mathbb{Q}(i)$. In that case choose a primitive Gaussian integral generator; since $\mathbb{Z}[i]$ is Euclidean it has a Gaussian Bézout identity, and as in [lemma 15.10](#) the admissible constant scalars form $\mathbb{Z}[i]$. Formula (15.4) gives a Gaussian omnific Bézout certificate. The scalar sets are therefore $\mathbf{Oz}[i]$ or $\Pi_{\mathbb{C}}$, the replacement of d by $2d$ excludes a gcd in the second case, and the generator assertion is [proposition 15.53](#). $\square$

Corollary 15.55 (Constant directions over $\mathbb{Z}$ and $\mathbb{Z}[i]$ [08, 14]). Let (D, k, A_D) be $(\mathbb{Z}, \mathbb{R}, \mathbf{Oz})$ or $(\mathbb{Z}[i], \mathbb{C}, \mathbf{Oz}[i])$. For a point $[r_0 : \dots : r_n] \in \mathbb{P}^n(k)$ the following are equivalent:

- (i) it lies in $\mathbb{P}^n(\text{Frac } D)$;
- (ii) it has a unimodular representative in A_D^{n+1} ;
- (iii) it has a primitive representative in A_D^{n+1} .

The primitive representatives are then the unit multiples of one primitive tuple in D^{n+1} .

Proof. Apply [theorem 15.54](#) (or [theorem 15.18](#) for $D = \mathbb{Z}$) to the constant polynomials $r_0, \dots, r_n$, which have gcd one; after normalization the vector is a primitive tuple in D^{n+1} . $\square$

Provenance. The real case is [theorem 14.5](#) of source 01. Manuscript 14 of the sibling report proves the criterion (i) $\Leftrightarrow$ (ii) for both pairs (D, k) , using only that D is a principal ideal domain: taking constants in a Bézout identity makes the constant multiplier nonzero, and a primitive tuple over D has a Bézout identity over D . That report records that the Gaussian and principal-ideal form is printed here with [theorem 14.5](#); this corollary is that statement. Manuscript 14 notes that the criterion concerns coordinate representatives, not all points of a projective scheme over a ring, and it credits only the example $[1 : \sqrt{2}]$ of the present report, not [theorem 14.5](#).

For constants $z \in \mathbb{C}$, ordinary Gaussian rationality $z \in \mathbb{Q}(i)$ is thus exactly the existence of a primitive Gaussian omnific fraction, although Gaussian omnific fractions produce every surcomplex number [08]. The localization of $\mathbf{Oz}[i]$ at the elements with nonzero constant has residue field $\mathbb{Q}(i)$, and on its finite part the real and rational residues are prescribed independently in $\mathbb{C} \times \mathbb{Q}(i)$ by $(zt + w)/(t + 1)$ with $z \in \mathbb{C}$ and $w \in \mathbb{Q}(i)$, using a common ordinary denominator for the two parts of w ; the proof is that of [theorem 15.42](#) coordinatewise [08].

15.12 What is and is not classified

The slogan that all surreals are rational over the omnific integers is correct only for membership in the fraction field. It does not give ordinary rationality, lowest terms, or a rational specialization map [08]. The results classify every tuple in $\mathbb{R}(t)$ and $\mathbb{C}(t)$ for $0 \neq t \in \Pi$, and every point of $\mathcal{V}$. They do not classify $\mathfrak{D}(x)$ for arbitrary $x \in \mathbf{No}$, and they do not show that $\mathcal{V} = \mathbb{P}^1(\mathbf{No})$; density is weaker than equality, as the disjoint dense fibers show [09]. A multivariate polynomial gcd of one does not give a Bézout identity (T_1 and T_2 have gcd one in $\mathbb{R}[T_1, T_2]$ but generate a proper ideal), so the proofs do not extend directly to several parameters [08]. The scale computation is one explicit test case, not a general flatness or normalization theorem for Hahn integer parts [08]. The questions left by the two sources are [questions 18.12](#) to [18.16](#).

16 Equations with omnific coefficients

For $P \in \mathbf{Oz}[X_1, \dots, X_n]$ let $\bar{P} \in \mathbb{Z}[X_1, \dots, X_n]$ apply ct to every coefficient. Then [02, 05]

$$\text{ct}(P(\mathbf{x})) = \bar{P}(\text{ct } x_1, \dots, \text{ct } x_n), \quad P(\mathbf{x}) = 0 \implies \bar{P}(\text{ct } \mathbf{x}) = 0.$$

The reverse implication need not hold, because the coefficients are no longer fixed by ct.

16.1 Quadratic equations

Theorem 16.1 (Square-discriminant criterion [02]). *For $b, c \in \mathbf{Oz}$, the equation $x^2 + bx + c = 0$ has a root in $\mathbf{Oz}$ if and only if $\Delta = b^2 - 4c$ is a square in $\mathbf{Oz}$. If $y^2 = \Delta$ with $y \in \mathbf{Oz}$, its roots are $(-b \pm y)/2$, and both are omnific.*

Proof. A root gives $(2x + b)^2 = \Delta$. Conversely, suppose $y^2 = \Delta$. Specializing and reducing modulo 4 gives $\text{ct}(y)^2 \equiv \text{ct}(b)^2 \pmod{4}$. Integer squares are 0 or 1 modulo 4, so $\text{ct}(y)$ and $\text{ct}(b)$ have the same parity, and [theorem 3.1](#) shows that $-b \pm y$ are divisible by two in $\mathbf{Oz}$. The

usual identity verifies the roots. Conversely every root determines one of these two square roots by $y = 2x + b$, so the list is exhaustive. When $\Delta = 0$ the two expressions coincide; otherwise they are distinct. $\square$

Monicity is essential to the criterion as stated. For a general quadratic $ax^2 + bx + c$ with $a \neq 0$ in $\mathbf{Oz}$, a root is equivalent to an element $y \in \mathbf{Oz}$ satisfying both $y^2 = b^2 - 4ac$ and $y - b \in 2a\mathbf{Oz}$. Indeed a root gives $y = 2ax + b$; conversely write $y - b = 2ax$ and use $y^2 - (b^2 - 4ac) = 4a(ax^2 + bx + c)$, cancelling $4a \neq 0$. A square discriminant alone does not ensure this divisibility: the polynomial $6x^2 + x - 1 = (3x - 1)(2x + 1)$ has discriminant 25, but its two field roots $1/3$ and $-1/2$ are not omnific. For a monic quadratic, the parity argument above supplies exactly the missing divisibility by two.

Testing whether a nonnegative surreal has an omnific square root is not the same as testing whether it has a surreal square root: real closedness gives the latter, while nonnegative support and an integral constant must still be checked for the former.

Example 16.2 (Residual and leading tests that do not lift [01, 02, 05]). The equation $x^2 = \omega^2 + 1$ has no solution in $\mathbf{Oz}$ (proposition 4.12). Nevertheless its residual equation $\text{ct}(x)^2 = 1$ is solvable, and the leading terms allow degree 1 and leading coefficient ± 1 . The positive root in $\mathbf{No}$ has the formal binomial expansion

$$\sqrt{\omega^2 + 1} = \omega + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-3} + \frac{1}{16}\omega^{-5} - \dots,$$

whose negative exponents exclude it from $\mathbf{Oz}$. This is a support-controlled Hahn expansion, not a claim of convergence in the surreal order topology. In contrast, $\omega^2 + 2\omega + 1 = (\omega + 1)^2$ is an omnific square.

The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` prints a later manuscript's image-gap theorem at a fresh scale (its files carry the prefix `16-fresh-scale-`): if f has degree at least two and coefficients in the Hahn field K_G with exponents in a set-sized group G , and $t = \omega^b$ with $b > G$, then $f(\mathcal{B}_{\mathbb{R}}) \cap (f(t) + K_G) = \{f(t)\}$, and likewise over $\mathcal{B}_{\mathbb{C}}$; for ordinary f one may take $G = \{0\}$ and any $b > 0$. The unsolvability of $x^2 = \omega^2 + 1$ in example 16.2 is the case $f = X^2$, $t = \omega$, and the two-term equations of proposition 16.4 are the cases $f = aY^n$, $t = \omega^{\gamma/n}$, after the substitution $X = a^{1/n}Y$ in $\mathcal{B}_{\mathbb{R}}$. The short proofs here are kept; the general statement is printed there.

16.2 Initial forms

Proposition 16.3 (Initial forms of an omnific solution [01, 02]). *Let $P = \sum_{\alpha} a_{\alpha} \mathbf{X}^{\alpha} \in \mathbf{Oz}[X_1, \dots, X_n]$ be nonzero, let $\mathbf{x} \in \mathbf{Oz}^n$ have all coordinates nonzero, and write $\gamma_i = \deg_{\omega}(x_i)$, $r_i = \text{lc}(x_i)$. Discarding zero coefficients, put*

$$W = \max_{\alpha} \left(\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i \right).$$

If $P(\mathbf{x}) = 0$, then

$$\sum_{\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i = W} \text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i} = 0, \quad (16.1)$$

and in particular the maximum is attained by at least two monomials.

Proof. By lemma 2.4, each evaluated monomial has degree $\deg_{\omega}(a_{\alpha}) + \sum_i \alpha_i \gamma_i$ and leading coefficient $\text{lc}(a_{\alpha}) \prod_i r_i^{\alpha_i}$. The coefficient of $P(\mathbf{x})$ at ω^W is exactly the sum in (16.1), and it must vanish; a single maximizer would contribute a nonzero product. $\square$

The maximum is taken over the finite nonempty polynomial support, so it exists even when the surreal weights have arbitrary rank. The exponent W is a candidate degree before cancellation, not an assertion that $\deg_\omega(P(\mathbf{x})) = W$. Several maximal terms can cancel; their leading coefficients, including their signs, must actually sum to zero. Merely having two terms of maximal weight is insufficient: for $P(X) = X^2 + \omega^2$ and a candidate with $\deg_\omega(x) = 1$, the initial equation is $r^2 + 1 = 0$, which has no real solution.

Variables assigned zero can first be substituted out. If this makes the polynomial identically zero, it imposes no initial-form condition; otherwise apply the proposition to the remaining nonzero coordinates. For ordinary coefficients this is a weighted initial polynomial over $\mathbb{R}$ with surreal weights. The constant-coefficient equation and the initial equation constrain opposite ends of a normal form; they are complementary necessary tests, not a lifting criterion, as [example 16.2](#) shows: the missing information lies in the full tail of a potential root.

16.3 Two-term roots and failed lifting

Proposition 16.4 (A two-term root obstruction [05]). *Let $a \in \mathbb{R}_{>0}$, $\gamma > 0$, $b \in \mathbb{Z}$, and let $n \geq 2$ be ordinary. The equation $X^n = a\omega^\gamma + b$ has a solution in $\mathbf{Oz}$ if and only if $b = 0$. When $b = 0$ its positive solution is $a^{1/n}\omega^{\gamma/n}$, together with its negative when n is even.*

Proof. When $b = 0$, put $u = a^{1/n}\omega^{\gamma/n} \in \Pi$. Then $u^n = a\omega^\gamma$, and strict monotonicity of positive integer powers on the positive cone gives the stated list of roots.

For $b \neq 0$, the right side is still positive and infinite. Suppose an omnific root exists, replacing it by its absolute value if n is even; the resulting root x is positive. With the same positive u , factor

$$b = x^n - u^n = (x - u)S, \quad S = \sum_{j=0}^{n-1} x^{n-1-j} u^j.$$

Both factors belong to $\mathcal{B}_\mathbb{R}$, so [lemma 6.1](#) would make S a nonzero real constant. But every summand is positive and $S \geq u^{n-1}$ is infinite, a contradiction. This finite-algebra proof uses only the previously established unit computation. $\square$

Source 05 instead proves the obstruction by displaying the positive field root. Its formal binomial expansion is

$$a^{1/n}\omega^{\gamma/n} \sum_{k=0}^{\infty} \binom{1/n}{k} \left(\frac{b}{a}\right)^k \omega^{-k\gamma}. \quad (16.2)$$

For every positive surreal γ , the exponents $\gamma/n - k\gamma$ form a reverse-well-ordered set: the largest exponent of any nonempty subset corresponds to its least natural index. Each coefficient in an n -fold product involves finitely many index tuples with a fixed sum. Hence the formal power-series binomial identity evaluates coefficientwise to the n th-root identity in a set-sized Hahn workspace containing γ/n . The leading coefficient is positive, so this is the positive field root; the only other possible root is its negative for even n .

If $b \neq 0$, the $k = 1$ term has nonzero coefficient $(b/n)a^{1/n-1}$ at exponent $\gamma/n - \gamma < 0$, which no other term has. Thus the expansion identifies the forbidden term for both possible signs. No convergence of partial sums in the full surreal order topology is used, and no assumption that the multiples of γ are cofinal among positive surreals is needed.

Example 16.5 (A simple residue root need not lift [05]). Let $f(T) = T^2 - T - \omega$. Modulo Π it becomes $T(T - 1)$, and the residue roots 0 and 1 are simple, with derivatives -1 and 1 ,

respectively, both units in $\mathbb{Z}$. Nevertheless f has no root in $\mathbf{Oz}$: its discriminant $1 + 4\omega$ is not an omnific square by [proposition 16.4](#), so [theorem 16.1](#) applies. Its roots in $\mathbf{No}$ are

$$\frac{1}{2} \pm (\omega^{1/2} + \frac{1}{8}\omega^{-1/2} - \frac{1}{128}\omega^{-3/2} + \dots),$$

with constant coefficient $\frac{1}{2}$ and negative exponents. Hensel-style intuition fails for $\mathbf{Oz} \rightarrow \mathbb{Z}$: the kernel Π is not contained in the Jacobson radical, since $-\omega \in \Pi$ but $1 + \omega$ is not a unit, and $\Pi^2 = \Pi$, so there is no decreasing nilpotent-error filtration. This is a failure of the simple-root lifting property itself.

This example concerns the constant-term map on the omnific ring. The standard-part map on the valuation ring of finite surreals has a different domain and an infinitesimal kernel. In particular ω is not finite, so the displayed polynomial does not even have all its coefficients in that valuation ring. Simple-residue-root lifting there, as used in the trigonometry report (`trigonometry:lem:hensel`), makes no lifting claim for $\mathbf{Oz} \rightarrow \mathbb{Z}$. The two uses of residue must be kept separate.

For general coefficients, a viable approach must inspect actual normal-form supports. Matching constant terms is necessary, and occasionally useful, but is not a sufficient solvability criterion ([question 18.6](#)).

17 Fixed Hahn workspaces and a formalization route

17.1 Which results are local in the exponent group?

For a set-sized ordered abelian group Γ , work in a Hahn field whose monomials are written ω^γ with reverse-well-ordered supports, and define [01, 02]

$$\mathbf{Oz}_\Gamma = \mathbb{Z} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}, \quad \mathcal{B}_\Gamma = \mathbb{R} + \left\{ \sum_{\gamma > 0} r_\gamma \omega^\gamma \right\}.$$

The displayed sums are elements of the ambient Hahn field with the specified support restrictions; this is a convention, not an identification of every abstract group element with a surreal exponent. When an ordered embedding of Γ into $\mathbf{No}$ is fixed, it gives the corresponding surreal workspace.

Local results. The ring decompositions, the unit lemma, the retraction, finite quotients, equational transfer, the Smith criterion, constant-product and decomposable rigidity, binary, Pell and norm rigidity, the quadratic constructions, the quartic guards, and all explicit polynomial identities apply to $\mathbf{Oz}_\Gamma$ [01, 02]. Their proofs use finite algebra, leading exponents and real coefficients, not the availability of a new exponent outside Γ . The derivation arguments of [section 7](#) take place in the workspace of the rational span of the relevant supports, so they apply to $\mathbf{Oz}_\Gamma$ after passing to the divisible hull of Γ [05]. If $\Gamma \neq 0$, its positive cone provides nonconstant parameters for the infinite-family statements; the case $\Gamma = 0$ is $\mathbf{Oz}_\Gamma = \mathbb{Z}$. The Lorentz family uses only t , t^2 and real scalars, so one positive monomial already produces examples. If the coefficient field is restricted below $\mathbb{R}$, the general indefinite lifting additionally needs the chosen real vectors e, u to be available, or an explicit scalar extension [02]. The definability results of sources 06 and 07 ([sections 11](#) and [12](#)) and the multiplier identity are proved for an arbitrary ordered exponent group, so they apply to $\mathbf{Oz}_\Gamma = \mathcal{R}_\mathbb{Z}(\mathbb{R}, \Gamma)$ as stated; the multiplier identity needs $\Gamma \neq 0$. Only $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and the reconstruction of $\mathbf{No}$ ([theorem 13.4](#)) use the full class, and the fraction field of $\mathbf{Oz}_\Gamma$ need not be the whole Hahn field [06].

Full-class results. The common-divisor theorem for an arbitrary set of purely infinite elements, the monomial clearing theorem, $\Pi^2 = \Pi$, the non-generation of Π by a set, $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, the general pair without a gcd ([theorem 4.10](#)) and the set-sized quotient theorem ([cited theorem 3.6](#)) require bounds that may not exist in Γ . For instance, in $\Gamma = \mathbb{R}$ the set $\{1/k : k \geq 1\}$ has no strictly positive real lower bound, whereas in $\mathbf{No}$ it has one. A support in Γ may also be unbounded below in Γ . These conclusions are not asserted for $\mathbf{Oz}_\Gamma$; in particular one must not infer $\Pi_\Gamma^2 = \Pi_\Gamma$ by copying the proof for $\mathbf{No}$ [01, 02, 05]. Proofs that stay inside a workspace must be distinguished from proofs that enlarge it. The conclusion of [theorem 4.10](#), though not the proof printed with it, does hold in every $\mathbf{Oz}_\Gamma$, by the halving route of sources 08 and 09 ([remark 15.8](#)).

Fractions. In [section 15](#) the multiplier theorem is ring-theoretic and holds for any retraction ([theorem 15.11](#)); the polynomial models, the scale defect and the Tor_1 computation concern set-sized rings; and the classification of representations, primitivity and least denominators uses only Bézout identities and constant coefficients [08]. The non-generation of the denominator ideals $q(t)\Pi$, the absence of a set of denominators for all surreals, the orbit density and the focusing theorem use the full class [08, 09]. Sources 08 and 09 show in one workspace that the fraction field of its omnific ring is smaller than the Hahn field ([example 15.52](#)).

17.2 A formalization route

No part of this report has been formalized. At the placement commit the repository contains no Lean declaration about omnific integers. The sources proposed overlapping routes; the following merges them. All module names are suggestions, not claims that such files or declarations exist.

Proposed layer	Main obligations
<code>Omnific.Basic</code> [01, 02, 05]	In a real Hahn field over a set-sized ordered divisible group: the nonnegative-support subring, the integer-constant subring, the retraction ct from the zero-exponent convolution formula, and the finite-element criterion.
<code>Omnific.Residue</code> [01, 02]	$n\mathbf{Oz} = \Pi + n\mathbb{Z}$, finite quotients, preservation of Π ; source 02 suggests proving these once for any ring retraction onto $\mathbb{Z}$ with an integer-divisible kernel.
<code>Omnific.Rigidity</code> [01, 02, 05]	Additive leading degree, units of the real and complex support rings, the exact decomposable fibers (finite linear algebra plus coefficient extensionality), binary and norm rigidity.
<code>Omnific.Derivation</code> [05]	Euler derivations from a $\mathbb{Q}$ -linear functional (Leibniz from finite convolution fibers), the quotient step for separated powers, comaximality of powers, and Wronskian degree bounds.
<code>Omnific.Quadratic</code> [01, 02]	The finite bilinear identities for U_t over any field of characteristic zero, the isotropic-vector choices, and the level classification.
<code>Omnific.IntegralityGuard</code> [01, 05]	Pell rigidity with <code>Nat.sum_four_squares</code> (source 05's six-auxiliary form); source 01's five-auxiliary form needs a three-square theorem, which Mathlib lacks.
<code>Omnific.Definability</code> [06, 07]	Units of $\mathcal{A}_{\mathbb{k}}(\Gamma)$ on the large-monomial side (a small-monomial Hahn implementation must reverse the exponent sign), Pell rigidity by coefficient extension, the finite Pell-divisibility argument, intersectivity of Λ with every prime power including 2, 13 and 17 (a finite modular computation is no substitute), the order-free system, the augmentation root detector (a statement about augmented commutative algebras, independent of surreals), the multiplier identity, and then the transport to $\mathbf{Oz}$ and $\mathbf{Oz}[i]$.
<code>Omnific.Global</code> [01, 02, 05]	The small-cut theorem for support bounds, common monomial divisibility and global clearing, kept separate and with explicit smallness (universe) hypotheses; the upper bound for clearing and the positive lower bound for idempotence are distinct lemmas.
<code>Omnific.Fractions</code> [09]	Pullback algebra for a domain with a split map to a field (the multiplier theorem and the Bézout lift); ordered denominator ideals in a discretely ordered domain with integer-part division (principal ideals and least members, covariance); then the normal-form instantiation; last, the matrices $F_{x,b}$ and focusing, with an explicit smallness hypothesis on the input family.

Source 05 points to the normal-form bridge (`docs/NORMAL_FORM_BRIDGE.md`) and the modules `SmallNormalFormFieldEquiv.lean`, `SmallNormalFormHahnEmbedding.lean`,

`SmallNormalFormWorkspace.lean` and `SignSequenceWorkspace.lean` under `Surreal1/Foundations/` as the transfer route to the actual surreal carrier; all four exist at the placement commit (with `SignSequenceWorkspaceExponents.lean`), and their signatures must be checked before use. The imported logical theorems (Tarski’s quantifier elimination, MRDP) remain separate dependencies; formalizing the retraction does not formalize them. The transvection identity is finite bilinear algebra with no dependence on birthday recursion, and the only hard number-theoretic input to the guards is the three- or four-square theorem. The order-free system needs only elementary local arithmetic of Λ and the finite Pell argument; the number-field extension also needs Dirichlet’s theorem and quadratic reciprocity, and the reconstruction of **No** the real closedness of **No**. Sources 06 and 07 propose the order followed in the row `Omnific.Definability`, keeping the class-sized instantiation separate from the set-sized Hahn results; this is a proposal, not a report of completed formalization.

18 Questions and synthesis

The sources list continuation problems that their proofs do not decide. None of them claims that these problems are open in the literature, or that the list is exhaustive. Two were settled inside the batch of manuscripts from which sources 01, 02 and 05 come. In the merge of sources 06 and 07 one more was answered and two were answered in part. Source 08 answers [question 18.3](#) in part, and sources 08 and 09 add five questions ([section 18.3](#)).

18.1 Questions settled by source 05

Coprime exponents and $Y^2 = X^3 + 1$ (source 01). Source 01 asked which curves such as $Y^2 = X^3 + 1$ have genuinely infinite omnific points, and proved rigidity of $x^m - y^n = c$ only for $\gcd(m, n) > 1$ ([remark 6.13](#)). [Theorem 7.4](#) of source 05 answers this for every separated equation $ax^m + by^n = c$ with $abc \neq 0$ and $m, n \geq 2$; in particular $Y^2 = X^3 + 1$ and $x^2 - y^3 = 1$ have only ordinary omnific points ([corollary 7.6](#)). The general part of source 01’s question, a criterion for arbitrary affine curves, remains open as [question 18.1](#).

Unimodular Fermat (sources 01 and 02). Source 01 asked for the classification of primitive or unimodular omnific solutions of $X^n + Y^n = Z^n$ with nonzero coordinates, $n \geq 3$; source 02 asked which homogeneous systems admit nonzero omnific solutions generating **Oz**. [Theorem 7.9](#) and [corollary 7.10](#) of source 05 settle the unimodular Fermat case: there are none. The primitive case (no common nonunit divisor) remains open as [question 18.2](#), and source 02’s question for other homogeneous systems remains open as [question 18.3](#).

18.2 Questions answered in the merge of sources 06 and 07

An existential definition of Π (source 01; source 07). Source 01 asked for an existential definition of Π over **Oz** ([question 18.4](#)), and source 07 closed by asking whether the purely infinite ideal, or the graph of ct , has a positive-existential definition. Source 06 answers both: $\exists y (x^2 = 2y^2)$ defines Π , with one witness and no parameters, in **Oz**, in **Oz**[i] and in every ring $\mathcal{R}_{\mathfrak{o}}(\mathbb{k}, \Gamma)$ with $\sqrt{2} \in \mathbb{k}$ and $\sqrt{2} \notin \text{Frac}(\mathfrak{o})$ ([theorem 11.2](#)), and the graph of ct is Diophantine ([theorem 12.8](#)). Source 07’s detector makes the complement $\{\text{ct} \neq 0\}$ Diophantine too ([theorem 12.2](#)). The second part of [question 18.4](#), which fibers or subideals of Π are Diophantine, remains open.

The least degree of a definition of $\mathbb{Z}$ (source 06). Source 06 asked for the least total degree, and the best trade-off between degree and witnesses, of a single integer polynomial defining $\mathbb{Z}$ in $\mathbf{Oz}$, offering a quintic with seven witnesses. The guards of sources 01 and 05 (theorems 10.3 and 10.4), which source 06 did not see, reach degree four with five and six witnesses. This answers the question in part; the least degree and the trade-off remain open and are merged into question 18.5, together with source 06’s question on a single Gaussian polynomial and source 07’s on better quantifier and degree bounds.

18.3 Questions in the merge of sources 08 and 09

Primitive homogeneous solutions (source 02). Source 02 asked which homogeneous systems admit a nonzero omnific solution whose coordinates generate $\mathbf{Oz}$ (question 18.3). Source 08’s rational-curve criterion answers it in part (corollaries 15.28 and 15.30). If the projective zero locus contains a curve $[p_0 : \cdots : p_n]$ given by real polynomials with gcd one and rational specialization at $T = 0$, then after a real normalization every $0 \neq t \in \Pi$ gives a unimodular omnific solution $(p_0(t), \dots, p_n(t))$. Along such a curve, the points with primitive coordinates are exactly those of rational specialization. Points off rational curves and the dependence on the integral model remain open, so the question stays open, re-scoped. Neither source addresses question 18.2: for $n \geq 3$ the Fermat curve has positive genus and contains no nonconstant rational curve, so the criterion does not apply to it.

New questions (sources 08 and 09). Sources 08 and 09 end with questions about denominator ideals beyond one parameter, the extent of the unimodular region, orbits outside it, several parameters and other scale extensions. They are questions 18.12 to 18.16 below; source 08’s first question and source 09’s second are merged into question 18.12. The sources present them as questions of their articles, not as certified open problems.

18.4 Open questions

Question 18.1 (Rigid affine varieties and curves [01, 02, 05]). For which affine varieties X over $\mathbb{Z}$ does $X(\mathbf{Oz}) = X(\mathbb{Z})$ hold? For which affine curves does every omnific point have ordinary coordinates? A natural next test is $Y^2 = X^3 + aX + b$ with $a \neq 0$, outside theorem 7.4.

Theorems 6.2, 6.7 and 7.4 answer this for nonzero decomposable levels with trivial common kernel and for separated equations, theorems 9.2 and 9.6 for bounded loci and systems with no real point at infinity, and theorem 8.1 and polynomial arcs give large opposing classes. A satisfactory criterion should separate polynomial parametrization, torus rigidity and genuinely algebraic branches [01]; it needs either a new degree inequality or a structural analysis of possible supports [05]. No claim is made that all other fibers are rigid, or that all nontrivial fibers come from real polynomial curves [02]. The report docs/surreal/omnific-groups-and-lattices answers the question for closed subgroup schemes of GL_N over $\mathbb{Z}$ (and over $\mathbb{Z}[i]$ for $\mathbf{Oz}[i]$); general affine varieties and curves remain open.

Question 18.2 (Primitive Fermat triples [01, 05]). For $n \geq 3$, can the unit-ideal hypothesis of theorem 7.9 be weakened to “no common nonunit divisor”? That is, are there primitive, non-unimodular omnific solutions of $X^n + Y^n = Z^n$ with nonzero coordinates?

The answer cannot be read off from ordinary integer intuition; it needs an explicit relation between common divisors, ideal generation and the applicable factorization theory. Theorem 7.9 deliberately avoids any such relation, and the binary nonzero-fiber theorem does not directly answer a three-variable homogeneous question [01].

Question 18.3 (Primitive homogeneous solutions [02]). Which homogeneous systems admit a nonzero omnific solution whose coordinates generate $\mathbf{Oz}$, and how does this depend on the chosen integral model of the projective variety?

The real-solvability criterion [theorem 14.2](#) omits unimodularity, and [theorem 14.5](#) exhibits the gap. A gcd normalization cannot be assumed, by [theorem 4.10](#). Fermat equations of exponent at least three have none with all coordinates nonzero ([corollary 7.10](#)), although $(1, 0, 1)$ is a unimodular solution with a zero coordinate; the exponent-two Pythagorean and indefinite quadratic cases have many ([proposition 14.4](#), (8.4)). Source 08 answers the question in part ([section 18.3](#)): a rational curve of real polynomials with gcd one in the zero locus gives unimodular omnific solutions exactly when its specialization at $T = 0$ is rational ([corollary 15.30](#)). Points off rational curves, and the dependence on the integral model, remain open.

Question 18.4 (An existential definition of Π [01]; first part answered by [06]). Is there an existential definition of Π over $\mathbf{Oz}$ with ordinary integer parameters? More generally, which fibers or subideals of Π are Diophantine? Retraction stability is necessary but far from sufficient.

The first question is answered: $\exists y (x^2 = 2y^2)$ defines Π ([theorem 11.2](#), [section 18.2](#)). The second remains open. Retraction stability is still necessary ([theorem 13.13](#)): a Diophantine set containing a nonzero element of Π contains 0.

Question 18.5 (Complexity of the integrality guard [01, 06, 07]). Can the number of auxiliary variables be reduced below five, with a single quartic equation, or can a meaningful lower bound be proved? What is the least total degree of a single integer polynomial defining $\mathbb{Z}$ in $\mathbf{Oz}$, and what is the best trade-off between degree and witnesses [06]? Can the Gaussian three-equation system of [theorem 11.9](#) be replaced by a single polynomial of controlled degree [06], and can its quantifier and degree bounds be improved without an order [07]? The present constructions give explicit upper bounds only (five in [theorem 10.3](#), six in [theorem 10.4](#), seven in the quintic (11.5)); other rigid arithmetic families might serve as finite bounds.

Any Gaussian sum-of-squares replacement must address isotropy ($1^2 + i^2 = 0$) rather than assume it away [06]. The quantifier-free obstruction [proposition 13.14](#) is only a first lower bound on logical complexity; it does not make any of the present counts optimal.

Question 18.6 (Roots with omnific coefficients [01, 02, 05]). For polynomials with finitely represented omnific coefficients, which root questions in a chosen support class are decidable? Can one give tractable necessary and sufficient support criteria for a root of a polynomial in $\mathbf{Oz}[T]$ to be omnific?

[Theorem 16.1](#) reduces the monic quadratic case to square membership, [proposition 16.4](#) settles two-term binomials, and [example 16.5](#) rules out a naive lifting theorem; multiterm cancellation needs separate analysis. Any effective claim must specify the coefficients, the exponent representation, and whether workspace enlargement is permitted. An unrestricted algorithm for ordinary-coefficient affine existence would contradict [corollary 5.2](#).

Question 18.7 (The exact size boundary [05]). Which set-sized exponent groups or universe-relative constructions admit analogues of the common-monomial and set-sized quotient theorems?

The sibling report `docs/surreal/set-sized-quotients-of-omnific-integers` contains cardinal-threshold results on this question for specific set-sized integer parts.

Question 18.8 (Effective finite-support discovery [05]). How far can a search for polynomial arcs and exact Bézout certificates, suggested by [theorem 14.7](#), go before transfinite support constructions are needed? It can find witnesses and expose obstructions, but it cannot become a complete decision procedure for integer-coefficient systems, by [corollary 5.2](#).

Question 18.9 (Coefficient recovery of lower complexity [06]). Can the canonical $\mathbb{R}$ or $\mathbb{C}$ in the interpreted fraction field be recovered with weaker quantifier complexity than the multiplier formula of [corollary 13.3](#), keeping the integer-part ring as the only named structure?

The pure field and the field with its integer-part predicate must be distinguished in formulating this. The present formulas are first order in the interpreted ring structure; they give no Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ and no solution of Hilbert's tenth problem over $\mathbb{Q}$ [06].

Question 18.10 (Which real structure must be added? [06]). What additional natural structure on $\mathbf{Oz}[i]$ rules out the coefficient-fixing twists of [lemma 13.9](#) and determines the real omnific subring, without simply naming conjugation or $\mathbf{Oz}$?

Naming all complex constants does not suffice ([theorem 13.10](#)); naming a suitable involution does, but gives the real structure explicitly rather than reconstructing it [06].

Question 18.11 (Infinite algebraic extensions [07]). Which coefficient subrings of algebraic extensions of $\mathbb{Q}$ of infinite degree admit a root-free intersective polynomial, or another positive-existential certificate of a nonzero constant, so that [theorem 12.14](#) extends to them?

No polynomial of the shape used in [lemma 12.13](#) is root-free in a field containing all algebraic numbers, so a different certificate would be needed there [07].

Question 18.12 (Denominator ideals beyond one parameter [08, 09]). Which surreal tuples have denominator ideals generated by a set? Must such an ideal be principal, or can a finitely or set-generated nonprincipal denominator ideal occur [08]? For arbitrary $x \in \mathbf{No}$, does a principal $\mathfrak{D}(x)$ force a unimodular presentation; which nonprincipal denominator ideals occur outside $\mathcal{V}$; and can a primitive presentation exist there without a Bézout identity [09]?

[Theorems 15.15](#) and [15.18](#) answer this for tuples in $\mathbb{R}(t)$ and on $\mathcal{V}$ only; they do not classify $\mathfrak{D}(x)$ for all $x \in \mathbf{No}$.

Question 18.13 (The unimodular region [09]). Which points of $\mathbb{P}^1(\mathbf{No})$ lie in $\mathcal{V}$? Give a support-theoretic criterion for a fraction to admit a unimodular presentation over $\mathcal{B}_{\mathbb{R}}$, or an explicit obstruction.

Every rational function of one parameter lies in $\mathcal{V}$, even when its normal form has infinitely many terms. The dense fibers do not imply $\mathcal{V} = \mathbb{P}^1(\mathbf{No})$ [09].

Question 18.14 (Orbits outside the region [09]). By [corollary 15.36](#), every K-orbit is dense, including any orbit outside $\mathcal{V}$. What complete invariants distinguish such orbits, if they exist?

Within $\mathcal{V}$ the invariant is exactly one real projective point ([theorem 15.32](#)).

Question 18.15 (Several parameters [08]). Is there a similarly explicit classification for fractions in $\mathbb{R}(t_1, \dots, t_k)$ with suitably independent purely infinite parameters? What local or projective data replace the single specialization at zero?

The proof does not repeat directly: a multivariate gcd of one does not give a Bézout identity, as T_1, T_2 in $\mathbb{R}[T_1, T_2]$ show, and [theorem 15.11](#) needs unimodularity [08].

Question 18.16 (Other scale extensions [08]). For inclusions of more general set-sized omnific Hahn models, which extensions commute with denominator ideals, and can their relation defects be expressed in terms of the newly admitted support scales?

[Theorem 15.46](#) and [proposition 15.48](#) settle one explicit case; they give no general flatness or normalization theorem for Hahn integer parts [08].

18.5 Synthesis

The arithmetic of $\mathbf{Oz}$ is governed by three simultaneous facts:

$$\mathbf{Oz}/\Pi = \mathbb{Z}, \quad \mathbf{Oz}^\times = \{\pm 1\}, \quad \text{Frac}(\mathbf{Oz}) = \mathbf{No}.$$

The first makes ordinary congruences and equation-existence questions look classical. The second imposes powerful rigidity on nonzero product fibers. The third makes projective representation flexible while saying little about primitive representatives. Growth degree, support and chosen Euler derivations then govern whether individual solutions escape the ordinary integers [05].

For Diophantine equations the key divide is not only between finite and infinite numbers. It is between *equation existence*, which the retraction controls; *multiplicative, differential or bounded rigidity*, which can force every coordinate to be ordinary; and *polynomial flexibility*, which creates infinite points without any infinitesimal tail [02]. The difference between a Pell conic and a three-variable indefinite quadric is structural: binary factorization forces constants, while an additional indefinite dimension supports finite polynomial isometries. The guards use the rigid side of this dichotomy to recover the ordinary integers inside $\mathbf{Oz}$. Omnific integers preserve much more of ordinary equational existence than their size suggests, and much less of ordinary quantified arithmetic than their discreteness suggests [05]. None of this requires unique factorization, a Euclidean termination principle, or Conway’s refinement property.

Sources 06 and 07 show that the decomposition itself is visible from inside the ring. The ideal Π is the solution set of $x^2 = 2y^2$, its complement is the set of divisors of values of Λ , and $\mathbb{Z}$ is cut out by a Pell equation, a divisibility and an intersective certificate; so the retraction ct , although not a symbol of the language, has a Diophantine graph. The multiplier ring of Π then recovers the real coefficients, and with them the valued field $\mathbf{No}$, from the pure ring $\mathbf{Oz}$, while the Gaussian ring $\mathbf{Oz}[i]$ recovers $\mathbb{C}$ but not its real form. Arithmetic recovery and recovery of a real form are different tasks [06].

Sources 08 and 09 say how much the third fact leaves of ordinary fractions. Wherever a point has a unimodular presentation over $\mathcal{B}_{\mathbb{R}}$, and in particular for all rational functions of one purely infinite parameter, the retraction decides lowest terms: a rational constant vector gives a principal denominator ideal and a unique primitive presentation, and an irrational one gives a scalar copy of Π and no gcd. This arithmetic is invisible to the order: both kinds of point occur in every interval, and the specialization is continuous nowhere.

18.6 Relations to neighbouring reports

- `docs/foundations-and-computation/foundations`: defines $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (`found:eq:omnific`, subsection `found:sub:omnific`) with the same letter Π ; this report develops the arithmetic of that ring.
- `docs/surcomplex/trigonometry`: writes the purely infinite part with t -exponents of the opposite sign (`trigonometry:eq:split`; translation in [section 1.4](#)); its integer-part

construction agrees with [theorem 2.7](#), and its identity $2\pi\mathbf{Oz} = \Pi + 2\pi\mathbb{Z}$ ([trigonometry: eq:periodclass](#)) rests on the same fact $r\Pi = \Pi$ for real $r \neq 0$ as [theorem 3.1](#).

- [docs/surcomplex/gamma-and-zeta-functions](#): the parity statements of [gz:cor: canonical-zeros](#) ($-\omega/2 \in \mathbf{Oz}$, $-\omega - 1 \notin 2\mathbf{Oz}$) are instances of [theorem 3.1](#); its statement that the internal integers of an ultrapower are not $\mathbf{Oz}$ is consistent with [theorem 10.10](#) and [remark 10.11](#).
- [docs/surreal/transcendence-over-bounded-support](#): its hypothesis [bst:hyp:gcd](#) that the bounded-support ring is a GCD domain concerns a different ring, whose units are the nonzero monomials; [theorem 4.10](#) concerns $\mathbf{Oz}$, whose units are ± 1 . Neither statement bears on the other.
- [docs/surreal/set-sized-quotients-of-omnific-integers](#): prints the universal set-sized quotient theorem ([cited theorem 3.6](#)) with source 05's proof and further threshold results; its preliminaries overlap [sections 2 to 4](#). Its manuscript 13 proves the Gaussian fiber theorem, the kernel converse and étale norm rigidity printed here once in [section 6](#); its manuscript 16 proves the fresh-scale image-gap theorem of which the equations of [example 16.2](#) and [proposition 16.4](#) are special cases. Its manuscript 14 proves the constant-direction criterion over $\mathbb{Z}$ and $\mathbb{Z}[i]$ printed here as [corollary 15.55](#); its coefficient-lattice results contain the rank-two case of [theorem 15.7](#) ([osq:tn:thm:duals](#)), the intersection formula of [example 15.27](#) for monomial t ([osq:tn:thm:syzygy](#)), and the scalar step of [theorem 15.11](#) ([osq:tn:thm:hom](#)). Localization at the positive monomials gives $\mathbf{No}$ ([osq:prop:fractions](#)), and the residue field $\mathbb{Q}$ of $\mathbf{Oz}_\Pi$ agrees with its list of small primes ([osq:cor:smallprimes](#)).
- [docs/surreal/omnific-groups-and-lattices](#): its non-elementary unipotents and cusp-residue obstruction ([ogl:el:thm:none1](#), [ogl:el:prop:cuspresidue](#)) show that the focusing matrices of [lemma 15.34](#) are not products of elementary matrices at irrational centres, and that the density of [corollary 15.36](#) needs such matrices ([remark 15.39](#)). That report also cites [theorems 4.10](#) and [14.5](#), and it answers [question 18.1](#) for closed subgroup schemes of GL_N over $\mathbb{Z}$ (and over $\mathbb{Z}[i]$ for $\mathbf{Oz}[i]$), not for general affine varieties or curves. [Theorem 12.12](#) concerns homomorphisms into class-sized Hahn rings as well and is consistent with the quotient theorem there.
- [docs/surcomplex/surcomplex-field-automorphisms](#): constructs the coefficient-fixing phase twists and the dilations S_a ; source 06 credits it and adds that the twist of [lemma 13.9](#) preserves $\mathbf{Oz}[i]$, whence [theorem 13.10](#).
- [docs/surcomplex/single-dilation-hahn-support](#): shows that the single automorphism S_2 of a Hahn field defines its coefficient field, constant coefficient and valuation ring in the language of fields with one automorphism; [theorems 13.2](#) and [13.4](#) recover the coefficient field, order and valuation ring from the pure ring $\mathbf{Oz}$ and its Diophantine ideal instead.
- [docs/surreal/euclidean-three-space](#): its theorem that class homomorphisms from $\mathrm{SO}(3, \mathbf{No})$ to set-sized groups factor through the standard part ([e3:cut:thm:smallquotient](#)) is an analogous size phenomenon for groups, not the same theorem; by [corollary 9.4](#), $\mathrm{SO}(3, \mathbf{Oz})$ is finite.

A Provenance and merge record

A.1 Sources

The report is built from seven manuscripts. Sources 01, 02 and 05 are three of a batch of eleven placed in commit `be06fc8`, and their numbers are the batch item numbers; they are dated September 22, 2026, and the repository commit they pin is `2cb9c0200af749fbbd27796c97d017e7f16fbf33`. Sources 06 and 07 are items 01 and 07 of a batch of eight placed in commit `cf350b1`, numbered here after the earlier sources, as their shipped file prefixes are; they are dated September 23, 2026, and the commit they pin is `71e9606d297c9b98c732070dc462682cc6948981` (“Prove Prony’s main theorem, holonomic ODE rigidity and eight more results”), which precedes `be06fc8`. None of the five was aware of the others. Item 03 of the second batch is manuscript 13 of the sibling report; its Gaussian and étale fiber results (its Sections 8 and 9) are printed here, tagged [13], and its remaining results there. The manuscripts themselves are not shipped. Their code and data are shipped under the prefixes `01-diophantine-geometry-`, `02-diophantine-`, `05-diophantine-rigidity-`, `06-definability-reconstruction-` and `07-defining-arithmetic-`; source 02’s provenance note, source 05’s build note and source 06’s source audit are shipped as `02-diophantine-PROVENANCE.md`, `05-diophantine-rigidity-BUILD.md` and `06-definability-reconstruction-SOURCE_AUDIT.md`. Source 01 delivered a checksum manifest, which is not shipped because it describes the delivered file names; the delivered descriptions of sources 06 and 07 are not shipped either. Source 13’s files are shipped with the sibling report.

Sources 08 and 09 are items 01 and 02 of a batch of nine placed in commit `a4dcb91`, numbered here by their shipped file prefixes. Both are dated September 23, 2026, and both pin `9a385d3957bdfe3d9ea79f9a524751c90bd2c894` (“Merge the rewritten Diophantine report and refresh its pending source index”), 35 commits before that placement. Neither was aware of the other. Their code and data are shipped under the prefixes `08-fractions-` and `09-fraction-fibres-`, and source 08’s source audit as `08-fractions-SOURCE_AUDIT.md`. Their texts, PDFs and delivered READMEs are not shipped, nor is source 09’s checksum manifest, which lists the delivered file names. Manuscript 14 of the sibling report is item 04 of the batch placed in `cf350b1`; its files are shipped there with the prefix `14-arithmetic-tensors-`.

A.2 Section map

Section	Sources
1 Scope and conventions	01 base; 02 and 05 results; new notation and provenance text
2 Rings and integer part	01 base; 05 degree of sums, workspaces, univariate roots; 02 examples
3 Residues and quotients	01 base; 02 mixed gcds and completions; 05 p -adic orders, divisors of integers, CRT; 05's set-sized quotient theorem cited
4 Full-class support	01 base; 02 nilpotent tests, common multiples, general no-gcd pair; 05 witness $\sqrt{2}$ and prime examples
5 Transfer and linear equations	01 base; 02 fibers and examples; 05 rational-basis form
6 Constant products	02 exact fibers; 05 binary form over $\mathbb{C}$, $D = 2$ classification, conics; 01 zero Pell fiber, $\gcd > 1$ powers, norm forms with any $\mathbb{Q}$ -basis; 13 Gaussian fibers, converse, kernel remark, rank-deficient example, étale norms; 07 norms over any base field; merged étale theorem
7 Derivations	05, extended to $\mathbb{C}$ by the review; merge remark 7.5
8 Quadratic dichotomy	01 base; 02 skewness, determinant, Lorentz matrix, unipotent subgroup
9 Bounded geometry	02 semialgebraic theorem; 01 positive bounds, symmetric matrices, no real point at infinity; 05 even powers
10 Defining $\mathbb{Z}$	01 three-square guard; 05 four-square definition and induction; merge remark 10.11 ; 07 quartic variant

Section	Sources
11 Diophantine arithmetic	06 base: setting, ideal, Pell, Λ , quintic remark; 07 intermediate rings, integer multiples, the system in its general form (with 06)
12 Constant term	07 detector, support bounds, ideal test, universal forms, number fields; 06 graph of ct, splitting, homomorphisms; merge quartic graph and number-field remark
13 Reconstruction and logic	06 multiplier, reconstruction, automorphisms, twist, c.e. sets, QF bound; 07 recursive saturation and arithmetic; 06 and 07 collapse obstruction
14 Families and primitivity	01 base; 02 sign conditions and decidability; 05 witness and finite-support specialization
15 Fractions	08 base: multiplier theorem, rational-function classification, real constants, curves, localization, models, scale defect, Tor_1 , monomials, Gaussian dichotomy; 09 region $\mathcal{V}$, denominator calculus, least denominators, orbits, focusing, density, nowhere continuity, examples; 14's Gaussian direction criterion credited; merge remarks 15.8 , 15.39 and corollary 15.30
16 Omnific coefficients	02 quadratic criterion and initial forms (also 01); 05 binomials and failed lifting
16 (continued)	pointer to the image-gap theorem of the sibling report
17 , 18	All seven; settled questions recorded in sections 18.1 to 18.3

A.3 Where the merge chose

1. *Base text.* Source 01, which has the weakest hypotheses on most shared statements (Pell with $D \neq 0$, norm forms for any $\mathbb{Q}$ -basis and $c \in \mathbb{Q}^\times$, the only exhaustive quadratic dichotomy, the guard uniform in n , the QE-free homogeneous criterion, the localization discussion).
2. *Decomposable fibers.* Source 02's exact-fiber theorem is printed, with source 01's rank- n theorem as case (b) of [theorem 6.2](#).
3. *Binary forms.* Source 05's complex form is printed; sources 01 and 02 proved the integer case.
4. *Power equations.* Source 05's separated-power theorem replaces source 01's $\text{gcd} > 1$ corollary, which is kept as [remark 6.13](#).
5. *Isotropic vectors.* Source 02's hypothesis $a \neq 0$ replaces source 01's $q(a) \neq 0$ in [lemma 8.3](#).
6. *Pairs without gcd.* Source 02's general theorem is printed; source 01's degree proof is kept as a second route ([remark 4.11](#)) because it localizes to divisible workspaces.
7. *Homogeneous systems.* Source 01's elementary proof and source 02's Tarski proof are both kept, since only the latter handles sign conditions and decidability.

8. *Uniqueness of ct*. Source 01's divisibility proof is printed; source 05 derives it from the set-sized quotient theorem, which is quoted and whose proof belongs to the sibling report.
9. *Guards*. Both are printed: source 01's (five auxiliaries, three squares) and source 05's (six auxiliaries, four squares).
10. *Additions*. Two remarks carry the tag [merge]: [remark 7.5](#) (a direct identity giving the divisibility in the separated-power proof) and [remark 10.11](#) (existential induction on the nonnegative cone fails in the pure ring language). Both have complete proofs.
11. *Definability base*. Source 06 is the base of [sections 11](#) to [13](#): its existential definitions of Π and of the graph of ct supersede source 07's universal and prenex ones, which are kept as [corollary 12.7](#) and [remark 12.11](#), and its homomorphism theorem covers all unital homomorphisms, where source 07's covers automorphisms.
12. *The order-free system*. The two sources' systems are identical; source 07's statement (both pairs, intermediate rings) is printed, with source 06 credited ([theorem 11.9](#)).
13. *Real single polynomials*. The quartics of sources 01 and 05 stay the printed versions. Source 07's quartic is source 05's up to the substitution of [remark 10.5](#) and is printed there once; source 06's quintic, dominated by both, is a remark with the comparison ([remark 11.10](#)).
14. *Pell divisibility*. Source 07's permutation argument is printed with source 06's finite-ring argument as a second route ([lemma 11.6](#)).
15. *Norms*. Source 07's theorem (any base field, intermediate rings) and source 13's (étale algebras, Gaussian variables) are combined in [theorem 6.15](#); [theorem 6.14](#) of sources 01 and 02 is its special case.
16. *Fibers*. Source 13's real fiber theorem is [theorem 6.2\(a\)](#), printed once with source 02 as base; its Gaussian case, converse, kernel remark and rank-deficient example are added.
17. *Transfer and obstructions*. Sources 06, 07 and 13 re-prove the transfer theorem, printed once ([theorem 5.1](#)); the collapse obstructions of sources 06 and 07 generalize [proposition 5.3](#) and are printed as [theorem 13.13](#). Source 07's recursive-saturation statement is printed, with source 06's countable type as the simpler instance.
18. *Fraction fields*. Source 06's common-denominator lemma is [theorem 4.6](#), printed once; the Gaussian consequence is stated in [section 13](#). Source 06's remark distinguishing ct from st joins [remark 2.3](#).
19. *Four squares*. Source 06's appendix proof of Lagrange's theorem is not reprinted; the theorem is cited ([14] for Mathlib, source 07's [27]).
20. *Further additions*. [Corollary 12.9](#) (a quartic with six witnesses for the graph of ct , improving source 06's degree-ten polynomial), [remark 12.15](#) (existential definitions over number-field coefficients), the Gaussian example in [remark 6.5](#), the collapse of the quintic over $\mathbf{Oz}[i]$ in [remark 11.10](#), and the combined [theorem 6.15](#) carry the tag [merge] and have complete proofs.
21. *Fractions base*. Source 08 is the base of [section 15](#). Its multiplier theorem has the weaker hypotheses: any domain with a retraction onto $\mathbb{R}$, any number of coordinates, and any $t \in \Pi \setminus \{0\}$. Source 09 proves the pair version over $\mathcal{B}_{\mathbb{R}}$ and the rational functions of one monomial $t = \omega^\alpha$. So source 09's multiplier theorem and one-scale classification are special cases of [theorems 15.11](#) and [15.18](#) and are not printed separately (credited in [sections 15.3](#)

- and 15.5). Source 09's proofs use only $\mathcal{B}_{\mathbb{R}}$ -unimodularity and full-class clearing, so they carry over unchanged to source 08's setting.
22. *The region $\mathcal{V}$.* Source 09's formulation on the region $\mathcal{V}$, with $\mathcal{U} = \text{sp}^{-1}(\mathbb{P}^1(\mathbb{Q}))$, is kept as [theorem 15.15](#), proved from [theorem 15.11](#).
 23. *Real constants and ordinary denominators.* Source 08's five characterizations of rationality and source 09's example are printed once ([theorem 15.7](#)). Source 08's localization identity and source 09's statements on units, subfields and Bézout pairs are combined in [proposition 15.5](#).
 24. *Second routes.* The two Bézout lifts are kept: source 08's ([15.4](#)) and source 09's adjustment of constants ([remark 15.12](#)). The halving argument of both sources for the absence of a gcd is kept next to the proofs of [theorem 4.10](#) and [remark 4.11](#) ([remark 15.8](#)). The two non-rational elements of one workspace (source 08's exponential series and source 09's factorial-gap series) and the monomial examples of both sources are all printed ([examples 15.51](#) and [15.52](#)).
 25. *Tables.* The two sources' example tables are merged into one ([example 15.23](#)); the rows m/n , r and $(t+r)/(t+1)$ occur in both.
 26. *Questions.* Source 08's question on set-generated denominator ideals and source 09's on principal ideals outside $\mathcal{V}$ are merged into [question 18.12](#); the other three are printed as stated.
 27. *Credits added.* Source 08's conic example is [proposition 14.4](#) with source 01's witness and is not reprinted. Its example without an lcm is, for monomial t , the sibling report's `osq:tn:thm:syzygy(iii)`, and its real-constant ideal is, for irrational r , `osq:tn:thm:duals`. Source 09's localization $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$ is `osq:prop:fractions(iii)`. Source 08's Gaussian dichotomy contains manuscript 14's direction criterion, which is printed once, crediting both ([corollary 15.55](#)).
 28. *Further additions.* [Remark 15.8](#) (the conclusion of [theorem 4.10](#) in every $\mathbf{Oz}_{\Gamma}$), [corollary 15.30](#) (the formulation of source 08's curve criterion for homogeneous systems) and [remark 15.39](#) (comparison with the elementary group of the omnific groups report) carry the tag `[merge]` and have complete proofs.

Printed once. The floor formula, retraction, congruences, finite quotients and completions, transfer, Hilbert's tenth problem, the Smith criterion, units, the Euclid example, non-atomicity, the ACC chain, $\Pi^2 = \Pi$ and non-generation, $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, the Pythagorean family, the Fermat scaling, $x^2 = \omega^2 + 1$, the orthogonal group, the nilpotent isometry, the torus remark, and the failure of four squares for $\omega - 1$ appear in two or three of sources 01, 02 and 05 and are printed once, with all sources named. From sources 06, 07 and 13: the units lemma, Pell rigidity (with the Gaussian solution (i, i)), the Pell divisibility lemma, the intersective polynomial and its proof, the order-free system and its Gaussian witness for $1 + i$, the transfer theorem and Hilbert's tenth problem, the collapse obstruction, $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$, the zero Pell fiber, the cubic norm, the negative-exponent Pell solutions and the omitted types are each printed once, with every proving source named. Sources 08 and 09 both reprove the rings and their units, the common monomial divisor, the non-generation of Π and $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ (credited by both to Conway and to L'Innocente–Mantova [3, Proposition 2.4.5]); source 09 also reproves the floor, ordered division and $\Pi^2 = \Pi$. Both recover the pair (t, rt) without a gcd and the rational-direction theorem as special cases and credit them. These are printed once, in [sections 2, 4](#) and [14](#). Within [section 15](#), the unimodular-fraction lemma, the real constants, ordinary denominators, the one-function criterion, the monomial criterion and the Gaussian fraction field are printed once with both sources named.

A.4 Renamed symbols

Table 1: Symbols renamed from each source. No normalization changed except where stated.

Source	Source symbol	This report
01	A (omnific integers)	$\mathbf{Oz}$
01	$\mathcal{J}$ (purely infinite ideal)	Π
01	$\text{Int}, \text{Int}(x)$	Std_n, Std
01	G_n (guard polynomial)	Ψ_n (G is the Lorentz Gram matrix)
01	A_Γ, B_Γ	$\mathbf{Oz}_\Gamma, \mathcal{B}_\Gamma$
01	P (purely infinite part in the floor)	$x_{>0}$
01	ϵ (common-divisor exponent)	δ
01	$\alpha = \sqrt{2} - 1$	ρ
01	$\delta = \deg_\omega d$ (gcd proof)	θ
01	E (hyperboloid coefficient)	D'
01	T_t (transvection)	U_t
01	$\lambda_\alpha, c_\alpha, M$ (initial forms)	$\deg_\omega a_\alpha, \text{lc } a_\alpha, W$
01	λ (scalar in decomposable forms)	divided into the nonzero level
01	f (endomorphism)	ψ
02	$\mathcal{I}, \mathcal{I}_\mathbb{C}$	$\Pi, \Pi_\mathbb{C}$
02	$\mathcal{A}_\mathbb{R}, \mathcal{A}_\mathbb{C}$	$\mathcal{B}_\mathbb{R}, \mathcal{B}_\mathbb{C}$
02	$d(f)$	$\deg_\omega f$
02	Q, B (quadratic and bilinear form)	q, B
02	$a, q = Q(a)$ (lemma vectors)	$u, k = q(u)$
02	u (integer point)	$\mathbf{a}$
02	U_T, T, S	U_t, t, s
02	$J = \text{diag}(1, -1, -1)$	G
02	J (index set)	sets S of elements
02	J (ideal)	$\mathbf{a}$ or $\mathbf{n}$
02	A (integer matrix)	M
02	K (common real kernel)	$\mathcal{K}$
02	η (infinitesimal part), P (purely infinite part)	$\varepsilon, x_{>0}$
02	$s = \sqrt{2} - 1$	ρ
02	T (in the no-gcd theorem)	t
05	$\mathcal{I}$	Π
05	$\mathcal{B}, \mathcal{B}_H, \mathcal{B}_{\mathbb{C},H}$	$\mathcal{B}_\mathbb{R}, \mathcal{B}_{\mathbb{R},H}, \mathcal{B}_{\mathbb{C},H}$
05	A (generic ring; set-sized target)	$R; S$
05	D_λ, W_D	$\partial_\lambda, W_\partial$
05	D, N (Pell)	D, c
05	p (Fermat exponent)	n (p is reserved for primes)
05	a, b (ratio constants in the Fermat proof)	μ, ν
05	$\deg_\omega, \lfloor \cdot \rfloor_{\mathbf{Oz}}, X_k, Y_k, \rho, \text{Std}, \Phi, x_{>0}$	unchanged
06	G, k, D (exponent group, coefficient field and ring)	$\Gamma, \mathbb{k}, \mathbf{o}$ (G is the Gram matrix, k an index, D the Pell parameter)

Source	Source symbol	This report
06	$X^g; d(f); \mathbf{K} = \mathbf{No}(i)$	$\omega^g; \deg_\omega f; \mathbf{No}[i]$
06	$\mathcal{A}_k(G), \mathcal{I}_k(G), B_D(k, G); \mathcal{I}_{\mathbb{R}}, \mathcal{I}_{\mathbb{C}}$	$\mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{R}_{\mathbf{o}}(\mathbb{k}, \Gamma); \Pi, \Pi_{\mathbb{C}}$
06	H (intersective polynomial); witnesses h, r	Λ (H is a workspace group); s, t
06	P and its parts A, B, C (quintic); Std_D	the display (11.5); Ξ or any definition of $\mathbb{Z}$
06	$(u_m, v_m); F = \text{Frac } B; \mathcal{O}$	$(X_k, Y_k); \mathcal{F}; \mathcal{O}_{\text{fin}}$
06	T, T_2 (twist, dilation); ℓ $(\omega^0\text{-coefficient on } \mathbf{No})$	$\tau, S_2; \eta$ (ℓ_j are linear forms)
06	$\text{Inf}, \text{CT}, \text{Mult}, \text{Coeff}, \text{ST}, \chi, \text{st}, \mathbf{m}$	unchanged
07	Γ, k, D, Ω (large monomial), c_0	$\Gamma, \mathbb{k}, \mathbf{o}, \omega, \text{ct}$
07	$H_k^+(\Gamma), I_k^+(\Gamma), R_D(\Gamma)$	$\mathcal{A}_{\mathbb{k}}(\Gamma), \Pi_{\mathbb{k}}(\Gamma), \mathcal{R}_{\mathbf{o}}(\mathbb{k}, \Gamma)$
07	P, P_K (intersective polynomials)	Λ, Λ_K (P names polynomials in general)
07	Φ, Φ_K (constant definitions); $\mathcal{I}, \mathcal{C}$	Ξ, Ξ_K (Φ is the induction formula of theorem 10.10); the formula of corollary 12.7, $\text{CT}^{\vee}$
07	B, M, F (augmented algebra, kernel, polynomial)	$\mathcal{H}, \ker \varepsilon, f$
07	$(u_n, v_n); L/K$ (norm)	$(X_k, Y_k); E/K$
07	$\mathcal{E}, \Theta, F_4, \varepsilon$	unchanged
13	$J_k, J_{\mathbb{R}}, J_{\mathbb{C}}; P_k; A_D$	$\Pi_{\mathbb{k}}(\mathbf{No}), \Pi, \Pi_{\mathbb{C}}; \mathcal{A}_{\mathbb{k}}(\mathbf{No}); \mathcal{R}_{\mathbf{o}}(\mathbb{k}, \mathbf{No})$ (its P_k is a ring)
13	ℓ_j (affine forms), A (matrix of linear parts), $W, J(W)$	L_j , the linear parts $\ell_j, \mathcal{K}$ or $\mathcal{K}_{\mathbb{C}}$, the coefficient-vector description
13	d, m (Pell); n, r (norm)	$D, c; r, a$
08	$\mathcal{B}, \text{Deg}$	$\mathcal{B}_{\mathbb{R}}, \deg_\omega$
08	τ (purely infinite parameter)	t (τ is the phase twist)
08	$B, c, I, A = c^{-1}(\mathbb{Z}), F$ (abstract pullback)	$R, c, I_R, R_{\mathbb{Z}}, K$ (B is a bilinear form, A an intermediate ring)
08	$L_{\mathbf{v}}; P$ (projective specialization)	$\mathcal{L}_{\mathbf{v}}; \text{sp}(\mathbf{x})$
08	$[q : p_1 : \cdots : p_n]$ (denominator first)	$[p_1 : \cdots : p_n : q]$ (denominator last)
08	$\Delta = \text{Deg } q(\tau)$	$\deg_\omega q(t)$ (Δ is a discriminant)
08	$\mathcal{A} = \mathbf{Oz}_{\Pi}, S = \mathbf{Oz} \setminus \Pi, \rho,$ $\mathcal{F} = \mathcal{A} \cap \mathcal{O}, \mathcal{O}$	$\mathbf{Oz}_{\Pi}, \mathbf{Oz} \setminus \Pi, \text{res}, \mathbf{Oz}_{\Pi}^{\text{fin}}, \mathcal{O}_{\text{fin}}$
08	$A_{\tau}, I_{\tau}; A_0, A_m, I_0, I_m$	$\mathcal{P}_t, t\mathbb{R}[t]; \mathcal{P}[T], \mathcal{P}[U], I_T, I_U$
08	$C_0, M(x), \mathbf{c}$	$C, \text{Ex}(x), \mathbf{c}$
08	$\mathbf{Oz}_{\mathbb{C}}, \mathcal{B}_{\mathbb{C}}$	$\mathbf{Oz}[i], \mathcal{B}_{\mathbb{C}}$
08	$\Pi, \mathfrak{D}, \text{ct}, \text{st}$	unchanged
09	$\mathcal{B}$	$\mathcal{B}_{\mathbb{R}}$
09	$T = \omega^\alpha$	t (any $t \in \Pi \setminus \{0\}$ in the general statements)
09	$D(x)$	$\mathfrak{D}(x)$ (D is the Pell parameter)
09	$\mathcal{A}_{\mathbb{Q}} = S^{-1}\mathbf{Oz}$	$\mathbb{Q} \oplus \Pi$
09	σ	sp (σ is an automorphism)
09	$\Pi, \mathcal{V}, \mathcal{U}, \deg_\omega, \text{ct}, \text{st}, \mathcal{O}_{\text{fin}}$	unchanged
09	Γ (congruence kernel)	$\mathbf{K}$ (Γ is an exponent group)
09	$G_{x,b}, N_x$	$F_{x,b}, \mathbf{N}_x$ (G is the Gram matrix, U_t the quadratic transvection)

Source	Source symbol	This report
09	$\Lambda(p, q)$	the set (15.3) (Λ is the intersective polynomial)
09	$E(x), O_{\mathbb{Z}}, M$ (monomials)	$\text{Ex}(x), \mathcal{P}_{\omega}, \mathcal{M}$
09	“finite” point; “divisor-primitive”; “Bézout fraction”	affine point; primitive; unimodular presentation
09	fibre	fiber

A.5 Corrections to the sources

1. Source 02 states Pell rigidity for $D > 0$ (in its abstract, its introduction, its catalogue table and its Pell corollary); its proof uses only $D \neq 0$, as in sources 01 and 05. [Corollary 6.9](#) states $D \neq 0$.
2. Source 01’s abstract says that nonzero fibers of “nondegenerate binary forms” are rigid. Its theorem requires, and proves, the weaker hypothesis of at least two distinct projective linear factors, which in degree at least three is weaker than squarefreeness (X^2Y qualifies). The abstract of this report states the theorem’s hypothesis.
3. Source 02 cites Matiyasevich’s translation as *Soviet Math. Dokl.* 11 (1970), 354–358, and source 05 as 354–357. The range 354–357 is the one given, with MathSciNet number MR 258744, in the bibliography of Bhatt and Poonen’s notes *Diophantine sets*, and in the title of the review of the paper in the *Journal of Symbolic Logic*; it is used here.
4. Source 05’s proof of the set-sized quotient theorem invokes global choice to pick a cardinal; the Hartogs ordinal suffices ([remark 3.7](#)).
5. Source 01’s isotropic-vector lemma assumes $q(a) \neq 0$ where $a \neq 0$ suffices ([lemma 8.3](#)).
6. Source 05’s verification script checks the separated-power identity only for $a = 1, b = -1$; the general identity of [remark 7.5](#) was checked separately (symbolically, for $2 \leq m, n \leq 7$) when this report was assembled. The shipped script is unchanged.
7. Source 01’s questions on $Y^2 = X^3 + 1$ and on unimodular Fermat triples, and the Fermat case of source 02’s question on unimodular homogeneous solutions, are recorded as settled by source 05 ([section 18.1](#)); the remaining parts are re-scoped as open questions.
8. Source 02’s attribution of the nonzero omnific Fermat failure to MathOverflow comments could not be verified; it is printed as source 02’s attribution, marked unverified ([example 14.3](#)).
9. Source 06 presents its quintic (degree five, seven witnesses) as its principal real Diophantine formula. The guards of sources 01 and 05, which it did not see, have degree four with five and six witnesses; the quintic is printed as a remark with that comparison ([remark 11.10](#)), and source 06’s question on the least degree is recorded as answered in part ([section 18.2](#)).
10. Source 06’s single real polynomial for the graph of ct has degree ten and eight witnesses; [corollary 12.9](#) gives degree four and six witnesses.
11. Source 07’s universal definition of Π and its $\exists^5 \forall^2$ and $\forall^2 \exists^5$ definitions of the graph of ct are superseded as definitions by source 06’s existential ones; they are kept as the negation of the detector and as a second route ([corollary 12.7](#) and [remark 12.11](#)), and source 07’s closing question on positive-existential definitions is recorded as answered ([section 18.2](#)).

12. Source 13 states Pell rigidity for positive nonsquare d ; its own fiber theorem covers every $D \neq 0$, which [corollary 6.9](#) states.
13. Sources 06 and 07 credit the interseptive polynomial to Lê–Spencer (and 07 also to Berend–Bilu). Lê and Spencer attribute it to Borevich and Shafarevich [22, p. 3], which is added as the classical source.
14. Source 06 cites Matiyasevich’s translation with pages 354–358, as source 02 does; the range 354–357 is used here, as explained above.
15. The formula $\exists y (x^2 = 2y^2)$ fails over number-field coefficient rings containing $\sqrt{2}$ in their fraction field; source 06’s remark on other radicands supplies the replacement used in [remark 12.15](#).
16. Source 09 says “for finite x ” for points $x \neq \infty$ of $\mathbb{P}^1(\mathbf{No})$; “finite” means bounded here, so such points are called affine ([section 15.1](#)).
17. Source 09’s local density proposition concludes “unless $g = f$ exactly”; the clause is unnecessary, since $g = f$ gives degree $-\infty$ ([proposition 15.26](#)).
18. Source 09’s introduction speaks of “a unique positive least denominator”; a least positive denominator is unique by definition, and the statement is [theorem 15.15\(i\)](#).
19. Source 08 writes $A = c^{-1}(\mathbb{Z})$ for its abstract pullback and $\mathcal{A}$ for the localization at Π , and orders projective coordinates with the denominator first, source 09 with the numerator first. One order is fixed here and both symbols are renamed ([table 1](#)).
20. Source 09’s lemma on the matrices $F_{x,b}$ is called “lem:transvection” in its source, the name of a different lemma here ([lemma 8.2](#)); it is [lemma 15.34](#).
21. Source 08 does not name this report’s Pythagorean witness ([14.2](#)), which was in the report at its pin, although it disclaims novelty for the conic. The sibling report’s coefficient-lattice results and its localization $\mathcal{M}^{-1}\mathbf{Oz} = \mathbf{No}$ were written after the pin of sources 08 and 09. These credits are added (merge choices above).

No mathematical error was found in any main proof of the seven sources, or in the printed results of sources 13 and 14. The later proof review of [sections 5 to 7](#) (recorded in `RECONCILIATION.md`) corrects the coefficient-derivation hypothesis in the merge’s own [remark 7.5](#): the derivation must kill a and b , and its use at level c needs $\partial c = 0$; the Euler derivations used in the proof satisfy both. The complex extensions of one-variable and separated-power rigidity ([proposition 2.8](#) and [theorem 7.4](#)) are explained after their proofs. A further review of [sections 8 to 10](#) expands the polarization argument and the transfer in [theorem 9.2](#), which now covers quantified ordered-ring formulas explicitly; it adds that the quadratic orbits preserve the coordinate ideal, and it makes the nonnegative induction domain and the order-free semiring formula of [remark 10.11](#) explicit. It did not cover [remark 10.5](#).

A.6 Stale repository statements

Sources 01, 02 and 05 inspected the repository through the GitHub connector, without a checkout or build, at the commit `2cb9c0200af749fbbd27796c97d017e7f16fbf33` named above; sources 06 and 07 inspected it at `71e9606` in the same way, and source 06 also read the opening of the surcomplex automorphism report. Sources 08 and 09 inspected it at `9a385d3`, also through the connector.

1. Source 01 says that the repository’s trigonometry material already uses the omnific integer part. At the pin this was true but incomplete: the foundations report already defined $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ (`found:eq:omnific`). The collection now also contains this report and the sibling report on set-sized quotients.
2. All three sources correctly found no report devoted to $\mathbf{Oz}$ and no Lean declaration about it at the pin. The first half is no longer true; the second still is.
3. Sources 02 and 05 describe `docs/NORMAL_FORM_BRIDGE.md` and the workspace modules; these descriptions are still accurate at the placement commit.
4. The status of Conway’s refinement conjecture is recorded as of September 22, 2026 ([appendix D](#)); the transcendence-over-bounded-support report takes the same stance.
5. Sources 06 and 07 say that the explicit Diophantine definitions of $\mathbb{Z}$ in $\mathbf{Oz}$ they give were not found in the repository. At their pin 71e9606 this was true; since the placement be06fc8 the collection contains the quartic guards of this report ([theorems 10.3](#) and [10.4](#)). Source 13’s statement that the exact fiber formulation was not found is stale in the same way ([theorem 6.2](#)). None of the three saw this report or its sibling.
6. Source 06’s descriptions of the surcomplex automorphism report (phase twists that move the real axis while fixing $\mathbb{C}$) and of the catalogue entry on coefficient recovery by a monomial dilation, and source 07’s description of the formalization ledger as separating source claims from Lean declarations, are accurate at the placement commit cf350b1. There is still no Lean declaration about omnific integers.
7. Sources 08 and 09 inspected the repository through the GitHub connector at 9a385d3, without a checkout or build. Both cite the rational-direction criterion as “Theorem 11.5” of this report; it was 11.5 at their pin and is now [theorem 14.5](#), since sources 06 and 07 were inserted as [sections 11](#) to [13](#). Likewise source 09’s “§§4,11” for the absence of lowest terms are now [sections 4](#) and [14](#). Their other citations ([theorems 4.2](#), [4.6](#) and [4.10](#) and [corollary 4.3](#), and the floor of [section 2](#)) are accurate.
8. Source 09 describes this report as merged from manuscripts dated September 22, 2026. That was true at its pin, when the report had three sources; it now has seven, dated September 22 and 23. Its statement that the omnific-integer results are pending in Lean is still accurate.
9. Source 08’s bibliography gives the sibling report the title *The Universal Set-Sized Quotient of the Omnific Integers*. That was its title at the pin; it is now *Set-Sized Quotients of the Omnific Integers*. Source 08 uses that report only for comparison, not as a proof dependency.
10. The sibling report states that manuscript 14’s Gaussian and principal-ideal form of the direction criterion is printed in this report with [theorem 14.5](#). Before this merge it was not; it is now [corollary 15.55](#).

B Non-claims by source

Every limitation, disclaimer and priority caveat stated by a source is kept. Most also appear at their point of use in the text; this appendix is the per-source record.

Common to sources 01, 02 and 05. No independent peer review; no Lean or other proof-assistant verification; the finite scripts check finite identities and examples, not the theorems; the repository was not checked out or built; the Conway refinement announcement of September 18, 2026 was not audited, is not used, and is neither assumed nor claimed settled; no result assumes refinement, GCD or UFD properties; no priority claim is made.

Source 01 (24 items).

1. Proving a statement is not a claim that nobody proved it before; no exhaustive historical-priority claim; the quadratic classification and the guard are named as its most distinctive constructions, found in the inspected sources in no such form.
2. Familiar mechanisms (coefficient retractions, unit arguments, polynomial parametrizations, nilpotent quadratic isometries) are not claimed as new; a focused search does not establish priority.
3. The script is a supplement, not a proof; it does not implement the surreal numbers.
4. Transfer does not include inequations or order conditions.
5. Transfer concerns existence, not cardinalities or complete solution sets.
6. The Hilbert's-tenth-problem statement concerns ordinary integer coefficients; omnific coefficients are a different question.
7. Thue levels: absence of additional omnific solutions only; no new proof of finiteness and no algorithm for ordinary solutions.
8. The $\gcd > 1$ power argument does not classify coprime cases such as $x^2 - y^3 = 1$ (now covered by source 05).
9. The torus remark does not say that every affine variety is constant.
10. The quadratic classification concerns represented nonzero levels; it does not decide representability or classify all infinite points.
11. The guard's five auxiliary variables are not claimed optimal.
12. The first-order definition of Π is not an existential definition.
13. Polynomial lifting does not license evaluating infinite power series.
14. Fermat scaling solutions are not primitive and resolve no primitive Fermat problem.
15. The rational-direction theorem concerns real projective points only.
16. Initial equations are necessary, not sufficient.
17. The converse of the no-real-point-at-infinity theorem is false.
18. Common divisors, clearing, $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and non-generation of Π are full-class and not asserted for fixed workspaces; clearing is representability, not primitivity.
19. The three-square theorem, MRDP and the normal-form foundations are imported, not reproved.
20. L'Innocente–Mantova's Theorem B is not used in any proof.
21. None of its arguments refutes the refinement property.
22. Proposed module names are proposals; no Lean files; not checked against a built repository.
23. A missing repository search match is not proof of absence.
24. Its questions are not all claimed open in the literature.

Source 02 (22 items).

1. The definition, integer-part property and pathological factorization belong to established theory; the retraction and many consequences should not be advertised as new.
2. The literature search does not establish priority for its principal formulations (exact decomposable fibers, uniform quadratic lifting); independent rediscovery is likely; absence of an identical formulation is no evidence of priority.
3. AI-assisted; no peer review; no complete Lean verification; no Lean build.
4. The checks do not prove transfinite-support results, the Diophantine claims, or the imported quantifier-elimination and undecidability theorems.
5. Transfer says nothing about how many points lie above an integer point.
6. The common-multiple theorem is an existence theorem, not an algorithm; no nonzero element is divisible by every nonzero omnific integer.
7. $\Pi/\Pi^2 = 0$ does not mean the infinite part is small.
8. The L’Innocente–Mantova prime is not reproved.
9. No effective enumeration of surreals is assumed; arbitrary class coefficients have no implicit finite encoding.
10. Nonvanishing and order do not transfer; the retraction is not an elementary embedding; residual solvability need not lift for non-integer coefficients.
11. The failure of four squares does not contradict open induction.
12. Residual and initial tests are not a complete lifting criterion.
13. The homogeneous theorem does not decide affine omnific equations.
14. The Fermat example is not a refutation of Fermat’s Last Theorem and not a novelty claim.
15. Clearing and homogeneous solvability do not answer the primitive-representative question.
16. Polynomial maps are not assumed to exhaust all constructions.
17. Binary and norm rigidity give no effective bounds for ordinary points.
18. Not every omnific linear group is finite.
19. The quadratic classification concerns the appearance of infinite points, conditional on ordinary solvability; not every point lies on the orbit; degenerate forms and the zero level are excluded from its statement.
20. Do not infer $\Pi_G^2 = \Pi_G$ for fixed workspaces; restricted coefficient fields need the vectors e, u or a scalar extension.
21. Its questions: no claim that other fibers are rigid or come from curves; no general root-membership algorithm (one would contradict H10).
22. Formalization stages are proposals; formalizing the retraction does not formalize Tarski or MRDP; the repository review was targeted and the normal form theorems are used as mathematical inputs, not as an audit of the implementation.

Source 05 (26 items).

1. No general solution of omnific Diophantine equations and no effective presentation of every surreal.

2. No new proof of the theorems of Lagrange, Matiyasevich, Shepherdson or Wiles.
3. No historical-priority claim for the derived results.
4. Not peer reviewed; not formally verified; finite checks are not substitutes for proofs and do not check arbitrary supports or the class-theoretic pigeonhole argument.
5. The normal-form bridge implementation is not independently certified; the Lean implementation was not built or audited.
6. The substantial factorization theory of L’Innocente–Mantova is a separate achievement, not replaced by elementary arguments.
7. Transfer does not make fibers singletons.
8. The computational statement does not cover omnific coefficients.
9. Projective representability does not give unimodular representatives.
10. The Euler derivations are local and not canonical, and are not the Berarducci–Mantova derivation.
11. Full-class results are not asserted for a preassigned workspace.
12. The set-sized quotient theorem does not apply to class targets such as $\mathbf{Oz}$, and it does not say that endomorphisms are the identity.
13. The size conclusion for $\mathbf{Oz}/(f)$ concerns the class ring, not a set-sized quotient.
14. There is no Pell solution at index ω .
15. No Thue finiteness proof is contained in the reduction.
16. The conic statement covers central conics, not every quadratic equation.
17. Separated-power rigidity gives no bound or complete list of integer solutions; the script certifies no completeness.
18. $y^2 = x^3 + ax + b$ with $a \neq 0$ is outside the method, a genuine boundary.
19. Only fixed ordinary exponents are treated, not variable omnific exponents.
20. The Fermat theorem says nothing about triples that merely lack a common nonunit divisor, and assumes no gcd equivalence.
21. The first-order definition of ct is not existential.
22. The four-square theorem is used only on ordinary nonnegative integers.
23. Transcendence of infinite omnific integers does not imply algebraic independence of tuples.
24. Finite-support specialization does not turn infinite-support solutions into arcs; finite-support search is not a complete decision procedure; matching constant terms is not a sufficient solvability criterion.
25. Its questions are not an exhaustive list of established open problems.
26. Its formalization route is a proposal; declarations were not invented; the size-sensitive layer must not be generalized to fixed Hahn groups.

Common to sources 06 and 07. Unrefereed research drafts with written proofs; no Lean files and no Lean-verified result; the repository was not built; the finite checks are exact but do not certify quantified statements over Hahn supports or the class interpretation; the literature and repository comparisons were focused, not exhaustive, and a missing search hit is not evidence of absence; no named classical factorization conjecture, and no named longstanding

conjecture, is claimed to be resolved; historical priority is not asserted. Their pin 71e9606 predates both omnific reports.

Source 06 (25 items).

1. First-order definability of $\mathbb{Z}$ in $\mathbf{Oz}$ and the nonsaturation of $\mathbf{Oz}$ are not new (*nombre*, 2018); the ideal equation belongs to the same obstruction and is not presented as an unrelated discovery.
2. Undecidability of integer-coefficient solvability follows from the retraction and is not a new consequence; the logical limitations of $\mathbf{Oz}$ were discussed by Jeřábek in 2011.
3. The intersective polynomial and its local-root property are classical; Pell methods in definability have a long history [20, 21]; phase twists are an existing repository construction, reproved with attribution.
4. No optimality assertion for degree five, seven witnesses or the coefficient sizes of the quintic, nor for the Gaussian system.
5. The Gaussian predicate is a system; no single-polynomial Gaussian claim is made, since a sum of squares does not preserve zero sets over a ring containing i .
6. The certificate $vs = \Lambda(t)$ is not a global definition of nonzeroness.
7. The multiplier and coefficient-field formulas are first order, not positive existential; the finite valuation ring is defined in the interpreted fraction field, not as a subset of $\mathbf{Oz}$.
8. The value group is recovered as an interpreted quotient, not a monomial cross-section; no definability of the ω -map, the simplicity order, birthdays or the exponential is claimed.
9. Pointwise fixation of $\mathbb{R}$ is proved for automorphisms only, not for nonsurjective embeddings.
10. The homomorphism theorem does not say that homomorphisms kill the ideal; preservation, annihilation and pointwise fixation are different assertions.
11. $\Gamma \neq 0$ is necessary for coefficient reconstruction, and the fraction field of a fixed Hahn ring need not be the whole Hahn field ($\Gamma = \mathbb{Z}$ gives $\mathbb{k}(\omega)$).
12. Full-field reconstruction uses surreal bounds for sets of exponents and the real closedness of $\mathbf{No}$; no common denominator for a proper class is asserted; pointwise fixation uses the density of $\mathbb{Q}$ in $\mathbb{R}$, and a general ordered coefficient field can have nontrivial automorphisms.
13. The c.e. classification concerns standard-supported sets and integer coefficients; arbitrary surreal parameters are not finite inputs to an algorithm.
14. The quantifier-free obstruction does not prove a lower bound on degree.
15. A formula defining the standard integers in this ring need not select a standard copy in an elementary extension.
16. No convergence, analytic continuation, canonical complex logarithm or infinite-height trigonometry is used; the only exponential is the ordinary complex exponential of a real multiple of i .
17. Class statements are formula by formula: no uniform truth predicate, no set of all surreals, no class of all class automorphisms.
18. Recovering $\mathbb{C}$ in the Gaussian case does not recover the real form or conjugation; naming an involution gives the real structure explicitly rather than reconstructing it.
19. The coefficient-recovery formulas give no Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ and no solution of Hilbert's tenth problem over $\mathbb{Q}$.

20. No named classical factorization conjecture is addressed.
21. No Lean files, no kernel-checked theorem, no repository rebuild; the checks guard against mistakes and do not certify quantified statements.
22. Qualified novelty only; the comparison was focused, and a missing search hit was not used as evidence of mathematical nonexistence.
23. Its formalization order is a proposal, not a report of completed work.
24. Its questions are not claimed to be recognized open problems.
25. Lagrange's four-square theorem is classical; its appendix proof is not a new theorem (and is not reprinted here).

Source 07 (19 items).

1. The normal-form machinery, the Pell method, the intersective polynomial, the four-square theorem and the undecidability background are established and credited; the literature check is targeted, not exhaustive.
2. No prime-element or factorization theorem is rediscovered, and the historical conjecture list is not treated as a list of current open problems; the ideal characterization proves no factorization, unique factorization or normality property.
3. No single-equation claim in the Gaussian case; the real quartic is not asserted over $\mathbf{Oz}[i]$, where it accepts every x .
4. Two universal variables, five existential witnesses and degree four are upper bounds only.
5. The $\exists^5\forall^2$ and $\forall^2\exists^5$ bounds are not claimed minimal, and no claim is made that the graph of ct has no existential definition.
6. The detector needs a root of Λ in the coefficient field; it fails for $\mathbb{Z} + \omega\mathbb{Q}[\omega]$ and is not asserted for all intermediate rings.
7. The number-field theorem is not uniform in K and does not apply as stated to infinite algebraic extensions.
8. Its interpretation of $\mathbb{Z}[i]$ is not identified with a Diophantine definition of $\mathbb{Z}$ inside $\mathbb{Z}[i]$; no such result is needed.
9. Undecidability and non-axiomatizability are consequences of established results (Jeřábek 2011, DPRM), not new undecidability theorems about $\mathbb{Q}$ or number fields.
10. It does not show that Π lacks a positive-existential definition (and source 06 shows that it has one).
11. Its automorphism corollary concerns ring automorphisms of the coefficient-restricted ring, not field automorphisms of $\mathbf{No}$ or $\mathbf{No}[i]$, and does not say that an arbitrary field automorphism preserves $\mathbf{Oz}$ or $\mathbf{Oz}[i]$.
12. Norm rigidity needs a nonzero right side and separability; it does not say that every variety has only constant points; coefficients outside A are read in the ambient ring.
13. $I = \bigcap nA$ must not be transferred to coefficient rings containing $\mathbb{Q}$.
14. The results are not claimed for the full surreal or surcomplex fields without the omnific restriction.
15. No efficient algorithm; witness existence, decision complexity and computability of representations are different questions, and the witness recipe assumes access to coefficient data.

16. Saturation and $\text{Th}(A)$ are asserted for set-sized rings; class statements are formula by formula, without a global satisfaction predicate.
17. The checks do not verify arbitrary Hahn supports, the all-moduli proof, prime existence for number fields, the model-theoretic claims or the class interpretation; the script must not be run with `python -O`.
18. Its formalization route is a proposal; no Lean module is supplied.
19. Its questions are not claimed to be previously catalogued open problems.

Source 13, for the results printed here (7 items). Its other non-claims are recorded in the sibling report.

1. The fiber theorem classifies the infinite points above ordinary points only; it gives no Thue or Baker finiteness and no enumeration or bound of the ordinary solutions, including those of the cubic norm equation.
2. The zero level is excluded, and the exclusion is sharp.
3. The fiber is a disjoint union of constant-coefficient fibers, with no assertion that the total solution class is a set.
4. The norm is evaluation of a finite polynomial, not a new norm on a class; nonordinary surreal solutions with negative exponents exist, so the theorem is about the nonnegative-support ring.
5. Equational transfer and the Hilbert’s-tenth-problem statement transfer the classical theorem and do not reprove it; inputs are finite integer polynomials, no encoding of surreal parameters is postulated, and no complexity claim is made for Gaussian or number-field problems.
6. The exact-fiber formulation is a proposed contribution, not a certified priority claim; the search was bounded.
7. Unrefereed, not checked in Lean; the finite script does not verify arbitrary Hahn sums; no claim is made about the status of Conway’s refinement conjecture.

Common to sources 08 and 09. AI-assisted research drafts with written proofs; not refereed; no Lean files and no Lean-verified result; the repository was inspected through the GitHub connector at 9a385d3 and not checked out or built; the SymPy checks are exact but finite and verify identities and examples, not the theorems; historical priority is not asserted, and each names its candidate contributions without certifying them.

Source 08 (25 items).

1. $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is classical (Conway; L’Innocente–Mantova, Proposition 2.4.5) and not claimed as new.
2. Hamkins’s observations on the failure of lowest terms and on irrational reals as omnific fractions are not new.
3. The repository’s constant-term arithmetic, set-uniform clearing and common divisors, the pair (t, rt) without a gcd and the constant-direction criterion are credited; the needed arguments are reproved, not imported.
4. The pullback construction and the use of coefficient retractions are classical; L’Innocente–Mantova’s Lemma 9.2.1 is a methodological precedent.

5. Integrality, nonflatness and non-finiteness of such extensions are classical phenomena; the equational flatness criterion (Stacks 00HK) is standard.
6. The novelty assessment is qualified: the search was targeted, not exhaustive; no priority is claimed; a result may be a specialization of known algebra.
7. No named longstanding conjecture is claimed to be solved.
8. The parameter t must be nonzero and purely infinite; merely infinite elements with a nonintegral constant are excluded.
9. The polynomial vector needs gcd one in one variable; a multivariate gcd of one does not give a Bézout identity, and the several-parameter case is open.
10. Rational residues require a common real normalization to a primitive integer vector before the principal generator is stated.
11. Primitivity and unimodularity are shown to coincide only for the classified representations, not over all of $\mathbf{Oz}$.
12. Denominator ideals contain 0; representations need a nonzero multiplier.
13. The scale-defect dimension is a real dimension, not a module length.
14. Tensor products and Tor are taken only over set-sized rings.
15. res and st have different domains, and neither is a retraction of $\mathbf{No}$ onto a smaller field.
16. There is no homomorphism $\mathbf{No} \rightarrow \mathbb{R}$ that sets $\omega = 0$; $f(0)$ is a substitution in a rational function.
17. Monic monomial denominators are more restrictive than arbitrary ones ($1/(\omega - 1)$).
18. The class of surreals with primitive fractions is not a subring; this concerns primitive representations, not the existence of fractions.
19. The classification covers tuples in $\mathbb{R}(t)$ and $\mathbb{C}(t)$ only; it does not classify all $\mathfrak{D}(x)$.
20. There is no general flatness or normalization theorem for Hahn integer parts.
21. The certificate procedure is effective only over exact coefficient fields; arbitrary reals are not assumed computable.
22. The conic example illustrates the classification; it is not a new construction of Pythagorean triples.
23. The checks do not verify the normal-form theorem, arbitrary supports, class-size claims or universal quantifiers, and they are not a proof assistant.
24. The repository audit read catalogues and guides through a connector whose responses were sometimes truncated; theorem numbers are cited from the catalogue, and an empty code search was not taken as proof of absence.
25. Its questions are not assertions that the problems are unstudied.

Source 09 (20 items).

1. $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is Conway's theorem and not new.
2. The repository's constant-term arithmetic and floor, small common divisors, $\Pi^2 = \Pi$ and non-generation, the fraction field, the pair without a gcd and the rational-direction theorem are not relabelled as new.

3. It does not claim that every surreal lies in $\mathcal{V}$, that divisor-coprime pairs are unimodular, or that $\mathbf{Oz}$ is a GCD or Bézout domain.
4. The five-fold equivalence is proved on $\mathcal{V}$ only.
5. It does not claim that a fixed Hahn field has the full-class clearing properties; in a workspace a formal cancellation does not give a unimodular pair.
6. No theorem needs Conway’s refinement conjecture or any announced proof of it.
7. sp is not a field homomorphism, does not extend ct , and is defined only on $\mathcal{V}$.
8. The dense fibers are not open and not separated; focusing fails for proper classes.
9. The one-scale test at $T = 0$ is a formal specialization, not a surreal value and not a valuation residue.
10. The topology at infinity on $\mathbb{R}(T)$ is not the surreal order topology.
11. The certificate procedure needs exact coefficients and is not uniform for reals given by oracles; the checks are not a surreal computer-algebra system.
12. It gives no complex lowest-term classification (source 08 does).
13. “Reduced” in L’Innocente–Mantova is a support notion and is not used for lowest terms.
14. The pullback and matrix-completion methods are ordinary algebra; Klawia is cited for background only; no invention of pullbacks, projective coordinates or unipotent matrices is claimed.
15. The candidate contributions are its multiplier and dichotomy theorems, the one-scale classification, and the orbit and focusing theorems; no priority is established.
16. The literature search was keyword-based; a failed match is not evidence of absence.
17. The checks do not prove well-ordering of supports, proper-class separation, non-generation of ideals, the theorems or novelty.
18. Its formalization guide is a proposed order, not code; writing **Set** in a statement does not discharge a universe condition.
19. The repository was not cloned or built, and only selected sources were reviewed, not every manuscript in its history.
20. Its questions are the article’s own, not certified open problems.

Source 14, for the result printed here (2 items). Its other non-claims are recorded in the sibling report.

1. The unimodular-direction criterion concerns coordinate representatives, not all points of a projective scheme over a ring.
2. The example $[1 : \sqrt{2}]$ is not new; it predates manuscript 14.

C Dependencies and verification

C.1 Imported inputs

The foundation imported from surreal theory is the existence, uniqueness, order and multiplication of set-supported Conway normal forms, the set-cut property, and the real closedness

of **No** [1, 2]. The other imported theorems are: MRDP (corollary 5.2); Tarski’s quantifier elimination (theorem 9.2, theorem 14.2(b)); the three-square theorem (lemma 10.2); Lagrange’s four-square theorem (theorem 10.4, corollary 5.2); Shepherdson’s theorem (open induction); the Fermat theorem of Wiles and Taylor–Wiles (corollary 7.10 only, after the rigidity theorem has reduced to ordinary integers); the algebraic closedness of $\mathbb{C}$ (to factor binary forms; the algebraic closedness of a class-sized surcomplex field is not used); and, as context only, the primes of [3] and [4, Lemma 11.1]. The material of sources 06 and 07 imports in addition MRDP (theorem 13.11), Lagrange’s theorem (corollary 11.11 and remarks 10.5 and 11.10), Dirichlet’s theorem on primes in progressions and quadratic reciprocity (lemma 12.13 only), the existence of algebraic closures and the discriminant criterion for étale algebras (theorem 6.15), and the real closedness of **No** (theorem 13.4); the local arithmetic of Λ , Hensel lifting and the Chinese remainder theorem are proved in the text. The material of sources 08 and 09 imports only the normal-form foundations, the set-cut property, ordinary polynomial and integer Bézout identities, the Euclidean property of $\mathbb{Z}[i]$, the equational criterion of flatness [34, Tag 00HK] (theorem 15.47) and standard tensor exact sequences (proposition 15.48), and it uses remark 15.39’s citation of the omnific groups report. All other deductions are proved here from these and finite algebra.

The principal dependency chains are:

normal forms $\implies$ ct is a retraction $\implies$ transfer and finite quotients;
normal forms $\implies$ constant units in $\mathcal{B}_{\mathbb{C}}$ $\implies$ decomposable, binary and norm rigidity;
local Euler derivations + binary rigidity $\implies$ separated powers and unimodular Fermat;
Pell rigidity + three- or four-square theorem $\implies$ quartic definitions of $\mathbb{Z}$;
real quadratic linear algebra + support-ring algebra $\implies$ quadratic-level dichotomy;
normal forms + set-cut bounds $\implies$ common divisors and clearing;
Pell rigidity and divisibility + intersective Λ $\implies$ order-free definition of $\mathbb{Z}$, $\mathbb{Z}[i]$;
 $\text{ct} + \sqrt{2} \notin \text{Frac}(\mathfrak{o}) \implies$ the Diophantine ideal $\implies$ graph of ct and homomorphisms;
augmentation + root of Λ in $\mathbb{k} \implies$ detector and ideal test;
Diophantine ideal + one negative exponent $\implies$ multiplier identity;
multiplier identity + real closedness of **No** $\implies$ reconstruction of $\mathbb{R}$ and **No**;
retraction + Bézout identities $\implies$ multiplier theorem $\implies$ denominator dichotomy;
matrix completion + set-cut bounds $\implies$ congruence orbits, focusing, dense fibers.

The source-05 proofs that most deserve independent review are, by its own account, the separated-power theorem (divisibility in a quotient ring without unique factorization), the unimodular Fermat theorem (strict leading-term cancellation in the equal-degree Wronskian), and the set-sized quotient theorem (positivity of every exponent in the geometric quotient; printed in the sibling report).

C.2 Finite checks

Seven scripts are shipped in `code/`, each with its recorded output in `data/`. The first three need Python 3.9 or later and SymPy (tested with SymPy 1.14.0); source 06’s needs Python 3.10 or later and SymPy, and source 07’s Python 3.10 or later and the standard library only; source 08’s needs Python 3.10 or later and source 09’s Python 3.9 or later, both with SymPy. They check finite identities and examples only.

- `01-diophantine-geometry-verification.py`: scalar identities (Pythagorean and hyperboloid identities and their Bézout certificates, the cubic norm by resultant, the Euclid

ratio), 20 quadratic-isometry cases in dimensions 3 to 7, Pell indices 0 to 30, and 256 guard witnesses. It writes a JSON report, by default `verification_report.json` in the working directory, so it should be run with `-output` pointing elsewhere or on a copy; the shipped record is `01-diophantine-geometry-verification_report.json`.

- `02-diophantine-verify_examples.py`: 13 exact identity checks (Pythagorean, cubic norm two ways, $N^3 = 0$, skewness, Lorentz preservation, determinant, group law and inverse, the hyperboloid family, binomial coefficients, the Euclid recurrence) and 49 finite-support products; output only to the terminal, recorded in `02-diophantine-verification.txt`.
- `05-diophantine-rigidity-verify_examples.py`: nine groups (Pell recurrence, Pythagorean witness, the ρ identity, 64 separated-power identities, Fermat Wronskian identities for $2 \leq p \leq 10$, 16 finite telescoping identities, binomial root coefficients, quartic-definition certificates for $|t| \leq 40$ by default, and five points on $y^2 = x^3 + 1$); output to the terminal, recorded in `05-diophantine-rigidity-verification.txt`.
- `06-definability-reconstruction-verify.py`: the degrees of the quintic (five, seven witnesses) and of source 06's degree-ten polynomial for the graph of `ct`, the displayed real witnesses, six Gaussian examples of the three-equation system, the Pell unit order for moduli 1 to 500, roots of Λ modulo 1 to 2000, the four-square product identity, and finite-support ideal and twist identities. It writes a JSON report only to the path given with `-output` and otherwise prints it; the shipped record is `06-definability-reconstruction-verification.json`.
- `07-defining-arithmetic-verify.py`: 13 groups (the expansion of Λ , the universal witness identity of [theorem 12.1](#) symbolically, the witness degrees, the certificate for $1 + \omega$, roots of Λ modulo 1 to 4096, the 2-adic lifting step through 2^{33} , Pell identities to index 300, the modular return for moduli 1 to 500, witnesses of the system for $-6 \leq x \leq 6$ and six Gaussian samples, quartic witnesses for $-100 \leq x \leq 100$, the degree of F_4 and its Gaussian collapse); output to the terminal only, recorded in `07-defining-arithmetic-verification.txt`. It uses assertions and must not be run with `python -O`.
- `08-fractions-verify.py`: 833 exact assertions in 13 categories, among them 40 random polynomial vectors (seed 23092026) with polynomial and integer Bézout identities, the lifted certificates ([15.4](#)), the rational or irrational classification over $\mathbb{Q}(\sqrt{2})$, sample scalar memberships, the examples, the substitution $T = U^m$ and the defect layers, the nonflat relation, integrality equations, the residue identities and falling-factorial independence. It writes a JSON report, by default `verification.json` next to the script, so it should be run with `-output` pointing elsewhere or on a copy; the shipped record is `08-fractions-verification.json`.
- `09-fraction-fibres-verify_examples.py`: 92 exact assertions (the matrices $F_{x,b}$: nilpotence, determinant, composition, inverse, action, fixed centre and image of ∞ ; the covariance ([15.1](#)); Bézout certificates such as ([15.5](#)); nine one-scale examples with coprimality, specialization, direction type and lifted certificates; the local density construction; five finite focusing samples over $\mathbb{Q}$ and the image of ∞). It always writes `verification_report.json` next to the script, so it should be run on a copy; the shipped record is `09-fraction-fibres-verification_report.json`.

When this report was assembled the first three passed with Python 3.14.4 and SymPy 1.14.0, and source 05's script also passed with `-standard-radius 200`. The outputs of sources 01 and 02 differ from the shipped records only in the recorded Python version (3.13.5 there); the output of source 05 is identical to its record. The scripts of sources 06 and 07 also passed

with Python 3.14.4 (and SymPy 1.14.0 for source 06), during placement on copies and again when this report was assembled, with source 06’s report written outside this directory; that report differs from its shipped record only in the recorded Python version, and source 07’s output is identical to its record. The degrees in [corollary 12.9](#) and the collapse of the quintic over $\mathbf{Oz}[i]$ in [remark 11.10](#) were checked separately when this report was assembled; the shipped scripts are unchanged. The scripts of sources 08 and 09 passed with Python 3.14.4 and SymPy 1.14.0, on copies, during placement and again when [section 15](#) was written. Source 08’s report (833 assertions) has the same content as its shipped record, which carries no timestamp; source 09’s (92 assertions) differs from its record only in the generation time and the recorded Python version (3.13.5 there). The certificates of [remarks 15.12](#) and [15.39](#) and the table of [example 15.23](#) were checked separately with SymPy when the section was written; the shipped scripts are unchanged.

D Repository and literature audit

D.1 Repository snapshots

Sources 01, 02 and 05 pin `2cb9c0200af749fbbd27796c97d017e7f16fbf33` (“Merge origin/main after the inverse-sine series formalization”). Source 01 read the tree, the root README, the documentation index, the trigonometry listing and its period-arithmetic source audit; source 02 read the README, the documentation index and the opening of the normal-form bridge note; source 05 read the documentation index and the bridge note. Each source says that this was a focused inspection through the GitHub connector, not a checkout, build, or review of every manuscript and declaration. Sources 06 and 07 pin `71e9606`. Source 06 read the root README, the documentation index, the opening of `docs/FORMALIZATION.md` and the first 235 source lines of the surcomplex automorphism report; source 07 read the root README, the documentation index and the opening of `docs/FORMALIZATION.md`. Both call this a focused comparison, not an exhaustive audit, and neither modified or built the repository. Sources 08 and 09 pin `9a385d3`. Source 08 read the root README, the tree, the documentation catalogue, the guides of this report and of the sibling report, and source 06’s source audit; source 09 read this report’s article and README, the root README and the collection guide. Both call this a selective inspection through the connector, without a checkout or build. The checks of all these repository statements against the placement commits are in [appendix A](#).

D.2 Literature checks

- L’Innocente–Mantova, arXiv:1710.07304v5: Theorem B states that $\omega^{\sqrt{2}} + \omega + 1$ is prime in $\mathbf{Oz}$, answering a question of Gonshor; their Section 1.3 reviews the earlier results on $1 + \sum_k \omega^{1/k}$; their introduction records, citing Shepherdson, that the nonnegative part of $\mathbf{Oz}$ is a model of open induction. The citations of all three sources agree with this.
- Shiu (*Math. Gazette* 104 (2020)) and the Archive of Formal Proofs entry of Danilkin and Chevalier (2023) for the three-square theorem, and Ehrlich–Kaplan Lemma 11.1 (as also cited in the trigonometry report), were confirmed during placement.
- Jeřábek’s MLQ article on rigid models of Presburger arithmetic (source 01) is cited for background; it was not re-read for this report.
- Source 02’s MathOverflow attribution could not be checked ([example 14.3](#)).

- Matiyasevich’s page range is corrected as described in [appendix A](#).
- Source 06’s attribution to *nombre* was confirmed during placement through the Stack Exchange interface: a comment of 6 July 2018 on MathOverflow question 304290 defines the ordinary integers in **Oz** by excluding nonzero solutions of $x^2 = 2y^2$ between $-|n|$ and $|n|$, and observes that **Oz** is not saturated [18].
- Jeřábek’s answer of 15 August 2011 to MathOverflow question 72691, cited by sources 06 and 07, was confirmed: it notes that the theory of **Oz** is a complete extension of Robinson’s arithmetic and hence not recursively axiomatizable [19].
- Lê–Spencer, arXiv:1309.2259v1, Section 2, Example 4 lists Λ and attributes it to Borevich–Shafarevich, page 3; Berend–Bilu is *Proc. Amer. Math. Soc.* 124 (1996), 1663–1671. Miller–Shlapentokh, arXiv:1706.03302, cited by source 07 as related Pell-method precedent for function rings, has the stated title [25]. These were confirmed during placement; Borevich–Shafarevich itself was not re-read.
- The following were confirmed during placement for sources 08 and 09. In the arXiv version 1710.07304v5 of L’Innocente–Mantova, Proposition 2.4.5 states a cofinality criterion for the fraction field of $\mathbb{Z} + K((G^{<0}))$ to be $K((G))$, and records $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ as Conway’s Theorem 32; their Lemma 9.2.1 is a primality transfer along an algebra retraction, and “reduced” is a support notion in their paper. Hamkins’s post of 4 November 2025 [32] says in its public part that lowest terms fail and that $\sqrt{2}$ and π are omnific fractions; it does not discuss gcds there. Klawns’s arXiv:2211.13412v2 [33] exists with the stated title and is cited by source 09 for background only. Stacks Tag 00HK is the equational criterion of flatness [34].

D.3 A dated factorization announcement

On September 18, 2026 Dan Abramov announced a Lean proof of Conway’s refinement conjecture [29, 30], noting explicitly the absence of independent mathematical verification. None of the sources audited that code or used the announced result. Accordingly this report does not repeat older descriptions of the refinement conjecture as a verified *currently open* problem, does not claim to settle it, and none of its proofs depends on its status.

References

- [1] John H. Conway. *On Numbers and Games*. Academic Press, 1976; second edition, A K Peters, 2001.
- [2] Harry Gonshor. *An Introduction to the Theory of Surreal Numbers*. London Mathematical Society Lecture Note Series 110. Cambridge University Press, 1986. doi:10.1017/CBO9780511629143.
- [3] Sonia L’Innocente and Vincenzo Mantova. A factorisation theory for generalised power series and omnific integers. *Advances in Mathematics* **442** (2024), 109513. doi:10.1016/j.aim.2024.109513. arXiv:1710.07304v5, January 22, 2024.
- [4] Philip Ehrlich and Elliot Kaplan. Surreal ordered exponential fields. *Journal of Symbolic Logic* **86**(3) (2021), 1066–1115. doi:10.1017/jsl.2021.59.
- [5] Emil Jeřábek. Rigid models of Presburger arithmetic. *Mathematical Logic Quarterly* **65**(1) (2019), 108–115. doi:10.1002/malq.201800019.

- [6] John C. Shepherdson. A non-standard model for a free variable fragment of number theory. *Bulletin de l'Académie Polonaise des Sciences, Série des Sciences Mathématiques, Astronomiques et Physiques* **12** (1964), 79–86.
- [7] Yuri V. Matiyasevich. Enumerable sets are Diophantine. *Soviet Mathematics Doklady* **11** (1970), 354–357 (MR 258744); translation of *Doklady Akademii Nauk SSSR* **191** (1970), 279–282. Reprinted in *Mathematical Logic in the 20th Century*, doi:10.1142/9789812564894_0013.
- [8] Yuri V. Matiyasevich. *Hilbert's Tenth Problem*. MIT Press, 1993. Author-hosted supplementary text, Section 5.7: https://logic.pdmi.ras.ru/~yumat/H10Pbook/par_5_7.htm.
- [9] Bjørn Poonen. Undecidability in number theory. *Notices of the American Mathematical Society* **55**(3) (2008), 344–350.
- [10] Yuri V. Matiyasevich. Matiyasevich theorem. *Scholarpedia* **3**(7) (2008), 7095. doi:10.4249/scholarpedia.7095.
- [11] Alfred Tarski. *A Decision Method for Elementary Algebra and Geometry*, second edition, University of California Press, 1951. Reprinted in *Quantifier Elimination and Cylindrical Algebraic Decomposition*, Springer, 1998, pp. 24–84.
- [12] Peter Shiu. The three-square theorem of Gauss and Legendre. *The Mathematical Gazette* **104**(560) (2020), 209–214. doi:10.1017/mag.2020.42.
- [13] Anna Danilkin and Loïc Chevalier. Three Squares Theorem. *Archive of Formal Proofs*, May 3, 2023. https://isa-afp.org/entries/Three_Squares.html.
- [14] The Mathlib community. `Mathlib/NumberTheory/SumFourSquares.lean`, theorem `Nat.sum_four_squares`.
- [15] Andrew Wiles. Modular elliptic curves and Fermat's Last Theorem. *Annals of Mathematics* **141** (1995), 443–551.
- [16] Richard Taylor and Andrew Wiles. Ring-theoretic properties of certain Hecke algebras. *Annals of Mathematics* **141** (1995), 553–572.
- [17] Emil Jeřábek, comment of 27 June 2018, and Timothy Chow, comment of 16 November 2021, on *Does Fermat's last theorem hold in the Grothendieck ring of the ordinals?*, MathOverflow, <https://mathoverflow.net/a/303688>. Cited by source 02; not verified for this report.
- [18] *nombre* (MathOverflow user), comment of 6 July 2018 on *For which theories does ZFC without global choice prove the existence of a proper class monster model?*, MathOverflow question 304290, <https://mathoverflow.net/questions/304290>. Cited by source 06; confirmed for this report.
- [19] Emil Jeřábek. Answer of 15 August 2011 to *Can we axiomatize Omnific Integers without the Surreal Number system?*, MathOverflow question 72691, <https://mathoverflow.net/questions/72691>. Cited by sources 06 and 07.
- [20] Martin Davis. An explicit Diophantine definition of the exponential function. *Communications on Pure and Applied Mathematics* **24** (1971), 137–145. doi:10.1002/cpa.3160240205.
- [21] Jan Denef. The Diophantine problem for polynomial rings and fields of rational functions. *Transactions of the American Mathematical Society* **242** (1978), 391–399.
- [22] Zenon I. Borevich and Igor R. Shafarevich. *Number Theory*. Academic Press, 1966, p. 3. Cited as the source of the intersective polynomial by Lê–Spencer [23].
- [23] Thai Hoang Lê and Craig V. Spencer. Intersective polynomials and Diophantine approximation, II. [arXiv:1309.2259v1](https://arxiv.org/abs/1309.2259v1), 2013, Section 2, Example 4.
- [24] Daniel Berend and Yuri Bilu. Polynomials with roots modulo every integer. *Proceedings of the American Mathematical Society* **124**(6) (1996), 1663–1671. doi:10.1090/S0002-9939-96-03210-8.

- [25] Russell Miller and Alexandra Shlapentokh. On existential definitions of c.e. subsets of rings of functions of characteristic 0. [arXiv:1706.03302](https://arxiv.org/abs/1706.03302). Cited by source 07 as precedent for Pell methods in polynomial and function rings.
- [26] Tom M. Apostol. *Introduction to Analytic Number Theory*. Springer, 1976, Chapters 7 and 9. [doi:10.1007/978-1-4757-5579-4](https://doi.org/10.1007/978-1-4757-5579-4).
- [27] Keith Conrad. *Proofs by descent*, expository notes, Section 6. <https://kconrad.math.uconn.edu/blurbs/ugradnumthy/descent.pdf>. Cited by source 07 for Lagrange’s theorem.
- [28] Vladimir Reshetnikov and contributors. *Surreal*: repository and research documentation. <https://github.com/VladimirReshetnikov/Surreal>. Snapshot 2cb9c0200af749fbbd27796c97d017e7f16fbf33.
- [29] Dan Abramov. *How I Vined a Proof of Conway’s Conjecture*. September 18, 2026. <https://overreacted.io/how-i-vined-a-proof-of-conways-conjecture/>. Cited only for the existence and qualified status of the announcement.
- [30] Dan Abramov. *A Proof of Conway’s Refinement Conjecture*, <https://github.com/gaearon/conway-refinement>. README inspected by source 02 on September 22, 2026; not rebuilt or audited.
- [31] David Marker. Introduction to model theory. In *Model Theory, Algebra, and Geometry*, MSRI Publications **39** (2000), 15–35. Theorem 3.5 and Proposition 4.1. <https://library.slmath.org/books/Book39/files/marker.pdf>.
- [32] Joel David Hamkins. *The omnific integers are strange*. Infinitely More, November 4, 2025. <https://www.infinitelymore.xyz/p/omnific-integers-not-like-integers-after-all>. Cited by source 08 for its publicly accessible part.
- [33] Hannah Klawns. *Global perinormality in a generalized $D + M$ construction*. [arXiv:2211.13412v2](https://arxiv.org/abs/2211.13412v2), January 18, 2023. Cited by source 09 for background on pullback constructions.
- [34] The Stacks Project Authors. *Flat modules and flat ring maps*, Section 10.39, Lemma 10.39.11 (equational criterion of flatness), Tag 00HK. <https://stacks.math.columbia.edu/tag/00HK>.
- [35] Jana Glivická and Petr Glivický. *Shepherdson’s theorems for fragments of open induction*. [arXiv:1701.02001](https://arxiv.org/abs/1701.02001) (2017), Sections 2.1, 2.3 and Theorem 1. <https://arxiv.org/abs/1701.02001>.